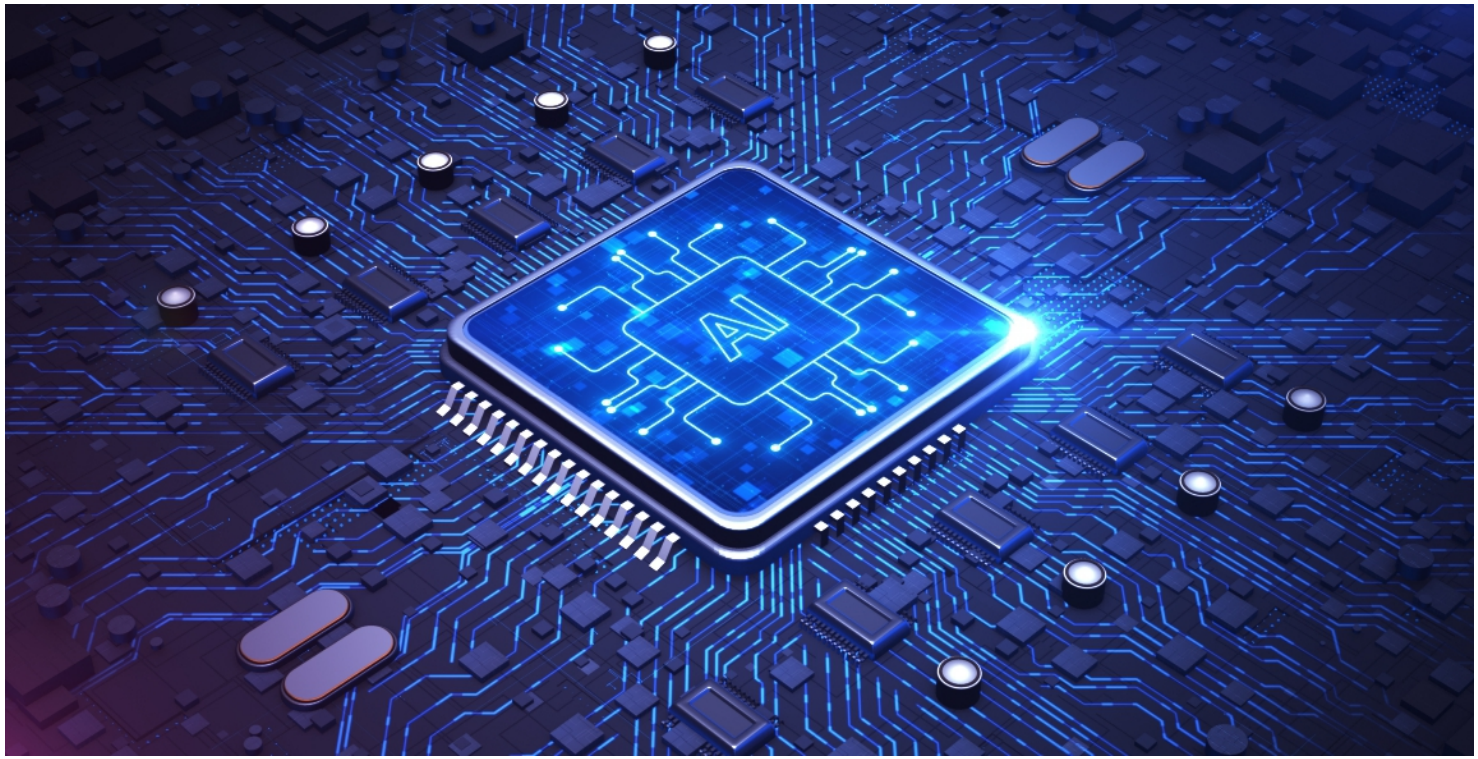
APAC Tech Strategy

Sector Keys

October v3: More Mixed Tech Sentiment, Amidst AI Strength



Equities

Global

Technology

Nicolas Gaudois
Analyst
nicolas.gaudois@ubs.com
+65-6495 5148

Jimmy Yoon
Analyst
jimmy.yoon@ubs.com
+65-6495 4617

Luke Yoo
Analyst
luke.yoo@ubs.com
+82-2-3702 8132

This report has been prepared by UBS Securities Pte. Ltd. (Reg. No. 198500648C) **ANALYST CERTIFICATION AND REQUIRED DISCLOSURES, INCLUDING INFORMATION ON THE QUANTITATIVE RESEARCH REVIEW PUBLISHED BY UBS, BEGIN ON PAGE 216.**

APAC Tech Strategy: Key Debates

Q #1: Is the AI cycle sustainable?

UBS View: Yes. Visibility into 2028E at least

- Key upstream enablers – TSMC, Samsung, SK Hynix – continue to see medium term demand well supported
- Agentic AI is taking off and increasingly viewed as the major inflexion point in AI adoption
- As a result, 2027E demand upside continues for server CPUs (recently: AMD Venice and Nvidia Vera) as well as AI accelerators (notably ASICs)
- Overall monetization improving, increased OCF forecasts for hyperscalers/neoclouds by 30% in '27E
- Funding gap between OCF and capex remains the No. 1 risk factor; AI credit spreads in our view key to watch

Q #2: Will the memory upcycle continue for longer?

UBS View: Yes. We now see the DRAM upcycle extending towards 2Q28, NAND 4Q27, with possible upside

- Server DDR and SSD demand keeps strengthening; near term pricing trending better than expected
- DDR demand upside has led to further constraints on HBM supply, and hence, de-specing
- The main risk to us is overall affordability as memory industry revenues exceed US$1.6tn in '27E, >50% of this to be consumed by hyperscalers/neoclouds
- Investors' key focus will increasingly move to longer term FCFs and shareholders returns

Q #3: Beyond Memory, what do we like?

UBS View: Our sector pecking order is Memory > Semiscap > MLCC > Select Taiwan Semis

- **Semiscap:** Noticeable uptick in Advanced logic WFE demand; Increased capex forecasts for Korea; Testers supported by CPU/AI ASIC demand & testing cycles. HBM hybrid bonding adoption delayed
- **Taiwan Semis:** TSMC has AI/HPC visibility into '29/'30; ASE LEAP well on track; MTK TPU v9/v10 promising
- **MLCCs:** AI server demand continues to strengthen and takes out capacity elsewhere
- **IC substrates:** Continuing upside in pricing, ABF demand well supported by server CPUs/AI accelerators
- **Top picks regionally:** Key Call Buy Delta, MediaTek, Naura, Samsung, Tokyo Electron

Primary Stock Coverage

	Rating	Price	Price Target	Upside/Downside
Nicolas Gaudois				
Samsung Electronics	Buy	Won 232,000	Won 535,000	130.6%
SK Hynix	Buy	Won 1,419,000	Won 3,000,000	111.4%
Eugene Technology	Buy	Won 149,100	Won 182,000	22.1%
Nanya Tech	Buy	NT$ 481	NT$ 640	33.2%
SemiFive	Buy	Won 18,610	Won 38,000	104.2%
Wonik IPS	Buy	Won 121,500	Won 155,000	27.6%
*ASML	Buy	€ 1,602	€ 2,400	49.9%
*CXMT	Buy	Rmb 46.	Rmb 80.00	73.9%
*TSMC	Buy	NT$ 2,490	NT$ 3,650	46.6%
*Xiaomi	Neutral	HK$ 26.38	HK$ 30.4	15.2%
Jimmy Yoon				
D.I Corp	Buy	Won 41,300	Won 50,000	21.1%
Hanmi Semiconductor	Sell	Won 217,500	Won 170,000	-21.8%
Hansol Chemical	Buy	Won 248,500	Won 350,000	40.8%
ISU Petasys	Buy	Won 111,200	Won 190,000	70.9%
LG Display	Neutral	Won 7,610	Won 11,600	52.4%
LG Electronics	Neutral	Won 177,400	Won 168,000	-5.3%
LG Innotek	Buy	Won 441,500	Won 940,000	112.9%
SEMCO	Buy	Won 1,374,000	Won 2,700,000	96.5%
Simmtech	Buy	Won 141,000	Won 195,000	38.3%
Soulbrain	Buy	Won 378,000	Won 540,000	42.9%
Wonik Materials	Buy	Won 38,850	Won 59,000	51.9%
YC Corp	Buy	Won 16,100	Won 20,500	27.3%
Luke Yoo				
Partron	Neutral	Won 4,925	Won 5,700	15.7%
SFA	Buy	Won 30,350	Won 39,000	28.5%

Source: UBS estimates (priced as of October 13th, 2026)
*ASML, CXMT, TSMC, Xiaomi – Co-covered stocks

APAC Tech Market, Sector and Stock Preferences

UBS APAC Tech Most/Least Preferred List

Company	Ticker	Analyst	Rating	Price	PT	'27E P/E	'28E P/E	'27E P/BV	'28E P/BV
Most preferred									
Accton	2345.TW	Jerry Su	Buy	NT$2040	NT$3688	17.0	12.1	8.95	6.40
Advantest	JP6857	Atsuhiro Kinoshita	Buy	¥40400	¥43600	42.2	34.2	20.43	14.28
AEM Holdings	AEM.SI	Joseph Koh	Buy	S$10.02	S$16	22.0	16.7	4.45	3.65
All Ring	6187.TWO	Sunny Lin	Buy	NT$1600	NT$1700	33.5	20.4	13.08	9.56
ASE	3711.TW	Sunny Lin	Buy	NT$740	NT$835	24.1	19.2	6.31	5.45
Delta Electronics	2308.TW	Randy Abrams	Buy	NT$1775	NT$2750	26.4	19.8	9.26	7.00
D.I Corp	003160.KS	Jimmy Yoon	Buy	Won42650	Won50000	14.4	10.9	4.33	3.58
Hon Hai Precision	2317.TW	Randy Abrams	Buy	NT$240	NT$340	10.4	8.8	1.61	1.46
ISU Petasys	007660.KS	Jimmy Yoon	Buy	Won117200	Won190000	18.4	14.6	5.68	4.21
JCET Group	600584.SS	Jimmy Yu	Buy	Rmb60.16	Rmb135	25.6	18.6	3.02	2.65
King Yuan	2449.TW	Sunny Lin	Buy	NT$305	NT$325	20.1	15.7	5.01	4.32
MediaTek	2454.TW	Sunny Lin	Buy	NT$4525	NT$7300	25.2	12.3	12.39	8.57
Murata Mfg.	JP6981	Shingo Hirata	Buy	¥7582	¥13200	23.1	17.1	4.36	3.79
NAURA Technology	002371.SZ	Jimmy Yu	Buy	Rmb591	Rmb905	36.7	24.6	7.76	6.02
Samsung Electronics	005930.KS	Nicolas Gaudois	Buy	Won232000	Won535000	3.8	3.4	1.58	1.20
SEMCO	009150.KS	Jimmy Yoon	Buy	Won1363000	Won2700000	24.5	15.4	6.77	4.80
Simmtech	222800.KQ	Jimmy Yoon	Buy	Won150500	Won195000	14.5	11.3	4.59	3.27
SK Hynix	000660.KS	Nicolas Gaudois	Buy	Won1443000	Won3000000	2.8	2.3	2.06	1.46
Tokyo Electron	JP8035	Atsuhiro Kinoshita	Buy	¥11555	¥18720	23.8	19.7	9.76	8.57
TSMC	2330.TW	Sunny Lin	Buy	NT$2490	NT$3650	16.7	12.3	6.15	4.46
Least preferred									
Asustek	2357.TW	Randy Abrams	Neutral	NT$966	NT$910	14.1	13.8	2.40	2.28
Compal	2324.TW	Randy Abrams	Neutral	NT$35.8	NT$40	13.1	10.5	1.16	1.10
Desay Battery	000049.SZ	Edward Liu	Sell	Rmb21.54	Rmb21	15.4	11.8	1.09	1.02
Hanmi Semiconductor	042700.KS	Jimmy Yoon	Sell	Won229000	Won170000	41.0	34.3	17.36	13.64
INNOLUX Corporation	3481.TW	Jerry Su	Sell	NT$50	NT$26	84.3	72.5	1.76	1.78
Largan	3008.TW	Randy Abrams	Sell	NT$7275	NT$5000	38.2	31.3	4.26	3.94
LG Display	034220.KS	Jimmy Yoon	Neutral	Won7560	Won11600	6.5	4.7	0.51	0.47
LG Electronics	066570.KS	Jimmy Yoon	Neutral	Won178000	Won168000	10.1	9.2	0.92	0.83
Maxscend	300782.SZ	Jimmy Yu	Neutral	Rmb60.11	Rmb80.2	-131.7	115.8	2.68	2.62
Pegatron	4938.TW	Randy Abrams	Neutral	NT$102.5	NT$95	14.0	11.7	1.20	1.12
Realtek	2379.TW	Sunny Lin	Neutral	NT$758	NT$690	19.3	18.0	7.59	6.90
Xiaomi	1810.HK	Jimmy Yu	Neutral	HK$25.04	HK$30.4	19.5	13.4	1.97	1.73

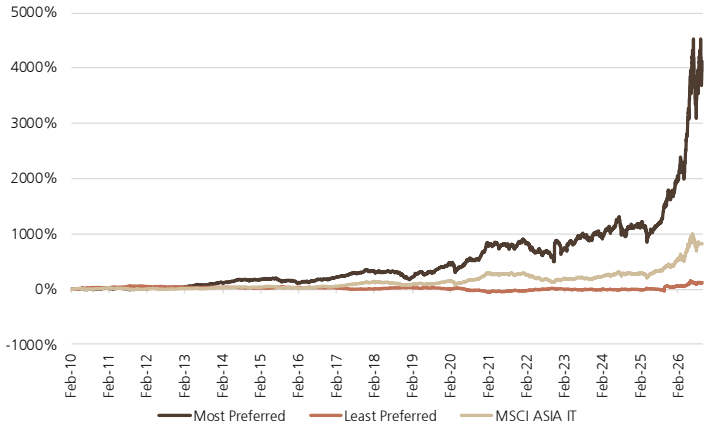
Market/Subsector Preferences

	Overweight	Neutral weight	Underweight
Markets	Korea, Taiwan	Japan	China
Sub-sectors	IC Substrates	Fabless ICs	Displays
	Leading Edge Foundry	Silicon wafers	PCs
	Memory Semis	Trailing edge foundries	Smartphones
	MLCCs		
	OSATs		
	Server ODMs		
	Wafer Fab Equipment		

Model Portfolio Performance

Absolute Returns	YTD	2025	Since inception
Most Preferred	144.8%	51.9%	4513.2%
Least Preferred	55.7%	-31.3%	113.5%
Overall Portfolio	**23.5%**	**10.3%**	**2313.4%**

Model Portfolio relative performance vs sector



Source: UBS estimates (priced as of October 13th, 2026), Bloomberg.
Model Portfolio Date of Inception: February 1st, 2010. Past performance is not indicative of future results.

APAC Tech Valuation – 1

	Rtg	Mcap (USD m)	Price	Target	EV/Sales			EV/EBITDA			P/E			P/BV		
Analog / Diversified					**26E**	**27E**	**28E**	**26E**	**27E**	**28E**	**26E**	**27E**	**28E**	**26E**	**27E**	**28E**
China Resources Microelectr	B	9,885	Rmb 49.82	Rmb 81.3	4.5	3.7	3.2	22.3	18.2	15.2	55.8	33.4	25.0	2.75	2.55	2.34
Maxscend Microelectronics	N	5,163	Rmb 60.11	Rmb 80.2	9.0	7.5	5.9	185.9	52.3	24.4	NA	NA	119.8	2.72	2.77	2.71
Omnivision	B	14,314	Rmb 75.96	Rmb 116.5	3.0	2.5	2.1	18.1	13.5	10.7	28.5	20.8	16.2	2.70	2.42	2.14
Wuxi NCE Power	B	2,903	Rmb 46.78	Rmb 82.2	7.2	5.4	4.2	28.4	19.6	15.0	38.0	26.0	20.2	4.10	3.64	3.19
Renesas Electronics	B	38,673	¥ 3349	¥ 5800	3.4	2.9	2.5	8.4	7.1	6.2	10.9	9.7	9.1	1.78	1.60	1.44
Rohm	B	11,174	¥ 4581	¥ 8300	3.3	3.1	2.8	21.9	14.9	13.0	49.5	28.7	24.9	2.28	2.17	2.04
Silergy Corporation	B	5,684	NT$ 471	NT$ 700	7.7	5.9	4.8	42.6	20.3	13.3	40.5	24.6	15.7	4.47	3.93	3.30
StarPower Semiconductor	B	2,656	Rmb 74.23	Rmb 142.8	3.9	3.0	2.5	26.1	17.9	14.3	86.2	45.5	33.1	2.51	2.42	2.30
Sector Average					**5.0**	**4.1**	**3.3**	**23.2**	**15.6**	**13.6**	**36.0**	**26.2**	**20.1**	**2.89**	**2.66**	**2.40**
Logic																
ASMedia Technology Inc	B	3,587	NT$ 1545	NT$ 1700	9.6	8.3	6.7	32.1	24.6	18.4	14.9	13.5	11.2	2.67	2.43	2.17
ASPEED Technology Inc	B	20,390	NT$ 17350	NT$ 22000	38.7	22.2	15.4	67.8	37.9	26.2	91.9	52.9	37.0	58.88	35.80	24.52
MediaTek Inc.	B	222,855	NT$ 4475	NT$ 7300	10.4	5.6	3.3	52.4	20.4	10.1	66.3	25.3	12.3	16.03	12.39	8.57
Montage Technology - H	B	37,657	HK$ 241.8	HK$ 370	30.8	20.1	13.2	77.7	43.4	26.4	71.2	44.4	28.9	11.58	9.96	8.08
Novatek Microelectronics Cc	N	10,580	NT$ 560	NT$ 540	2.6	2.4	2.3	11.9	11.6	11.9	16.0	16.3	16.9	5.14	4.93	4.77
Parade Technologies	N	1,343	NT$ 547	NT$ 565	1.9	1.7	1.4	10.1	8.2	5.6	17.9	13.3	9.1	1.87	1.73	1.56
Realtek Semiconductor Corp	N	12,389	NT$ 778	NT$ 690	2.3	1.9	1.7	16.1	13.7	12.8	23.1	19.3	18.0	8.67	7.59	6.90
Sector Average					**4.1**	**4.0**	**3.1**	**17.6**	**22.8**	**15.9**	**18.0**	**26.4**	**19.1**	**4.59**	**5.33**	**5.34**
Memory																
CXMT Corporation	B	476,542	Rmb 46.99	Rmb 80	8.1	4.4	3.1	9.0	4.7	3.3	15.0	8.7	6.7	9.84	5.05	3.20
Kioxia Holdings Corp	B	152,603	¥ 14690	¥ 33000	2.1	1.2	1.1	2.4	1.4	1.4	3.5	2.2	2.7	4.01	1.85	1.25
Nanya Technology	B	51,474	NT$ 480.5	NT$ 640	3.3	1.8	1.2	4.1	2.1	1.5	6.3	3.7	3.6	3.26	1.98	1.39
Samsung Electronics	B	1,025,926	Won 232000	Won 535000	1.7	0.8	0.5	2.8	1.2	0.7	5.5	3.1	2.7	2.21	1.37	1.02
SK Hynix	B	771,699	Won 1419000	Won 3000000	2.7	1.4	1.0	3.2	1.6	1.2	3.7	2.4	2.0	2.92	1.78	1.26
Sector Average					**3.6**	**1.9**	**1.4**	**4.3**	**2.2**	**1.6**	**6.8**	**4.0**	**3.5**	**4.45**	**2.40**	**1.62**
Foundries																
Hua Hong Grace Semicondu	N	18,843	HK$ 85.1	HK$ 148	6.2	5.2	4.8	20.0	15.3	12.6	108.1	75.0	53.9	2.42	2.35	2.26
Powerchip Semiconductor M	B	9,312	NT$ 71	NT$ 100	4.1	2.4	2.1	7.5	5.0	4.1	9.8	7.3	6.3	2.82	2.20	1.78
SMIC	B	56,723	HK$ 55.55	HK$ 102.4	5.1	4.0	3.4	9.1	7.0	5.5	45.5	30.4	22.6	2.12	1.98	1.82
Taiwan Semiconductor Mani	B	2,004,905	NT$ 2490	NT$ 3650	11.4	8.0	5.8	15.4	10.9	7.9	22.8	16.7	12.3	8.49	6.15	4.46
UMC	B	55,501	NT$ 142	NT$ 230	6.0	4.5	3.4	12.8	9.0	6.5	18.4	17.9	11.9	3.68	3.45	2.99
Vanguard	N	10,158	NT$ 175	NT$ 168	5.6	4.1	3.4	15.3	10.3	8.3	28.6	21.8	16.0	3.53	3.27	2.90
WIN Semiconductors	B	7,328	NT$ 556	NT$ 725	11.2	8.4	7.1	38.3	26.1	21.0	69.1	40.2	30.7	5.25	4.84	4.44
Sector Average					**5.4**	**5.2**	**4.3**	**16.9**	**11.9**	**9.4**	**25.0**	**22.4**	**22.0**	**4.04**	**3.46**	**2.95**

Source: UBS estimates (priced as of October 13th, 2026)

APAC Tech Valuation – 2

	Rtg	Mcap (USD m)	Price	Target	EV/Sales			EV/EBITDA			P/E			P/BV		
					26E	27E	28E	26E	27E	28E	26E	27E	28E	26E	27E	28E
OSAT / Testing																
ASE Industrial	B	103,165	NT$ 747	NT$ 835	4.3	3.4	3.0	18.5	11.1	8.7	40.5	24.7	19.7	7.50	6.31	5.45
Chipbond Technology	B	5,116	NT$ 221	NT$ 290	6.1	5.2	4.1	21.4	17.0	13.2	39.2	26.9	20.0	3.36	3.16	2.94
ChipMOS Technologies	B	2,559	NT$ 117	NT$ 166	2.8	2.5	2.2	10.1	8.4	7.4	26.0	18.0	14.7	3.10	2.79	2.51
DI Corporation	B	803	Won 41300	Won 50000	1.8	1.4	1.0	9.3	7.4	5.3	18.0	14.4	10.9	5.22	4.33	3.58
Inari Amertron Berhad	N	2,516	RM 2.71	RM 2.3	4.6	6.0	5.1	20.1	22.9	18.8	38.8	40.0	32.3	2.69	3.68	3.60
King Yuan Electronics	B	11,257	NT$ 296.5	NT$ 325	8.2	6.0	5.2	15.5	10.8	9.0	32.5	20.0	16.2	5.86	4.98	4.35
JCET Group	B	16,082	Rmb 60.16	Rmb 135	2.5	2.2	1.9	16.8	13.3	10.8	40.0	26.2	19.1	3.45	3.10	2.72
Powertech Technology	N	7,036	NT$ 298.5	NT$ 260	2.9	2.8	2.6	9.6	8.5	7.4	23.7	19.6	16.8	3.53	3.19	2.88
YC Corporation	B	986	Won 16100	Won 20500	3.6	2.6	2.1	20.8	14.2	10.5	33.9	25.4	18.0	3.20	2.86	2.50
Sector Average					**4.0**	**3.6**	**3.1**	**15.3**	**12.6**	**10.3**	**34.6**	**25.7**	**20.4**	**4.16**	**3.79**	**3.38**
IC Design House / IP																
Alchip Technologies	B	9,950	NT$ 3940	NT$ 6600	4.2	2.3	1.7	18.8	12.1	9.7	27.8	17.5	13.5	4.89	4.10	3.46
Faraday Technology	N	1,707	NT$ 211	NT$ 190	4.0	2.6	1.9	24.4	12.1	8.3	61.1	24.3	15.0	3.94	3.50	3.05
Global Unichip	B	34,291	NT$ 8245	NT$ 8660	16.5	6.7	5.0	114.2	48.9	34.7	141.6	59.3	42.4	61.16	33.12	19.95
SemiFive	B	471	Won 18610	Won 38000	3.1	1.6	1.0	NA	30.0	9.0	NA	107.3	17.1	2.88	2.80	2.42
Sector Average					**3.8**	**3.3**	**2.4**	**21.6**	**25.8**	**15.4**	**27.8**	**33.7**	**22.0**	**3.90**	**3.47**	**2.98**
Semiscap / Display equipment																
ACM Research (Shanghai)	B	19,320	Rmb 267.98	Rmb 376	14.5	10.0	7.4	70.0	41.0	27.4	74.0	43.6	29.9	8.67	7.41	6.15
Advantest	B	184,720	¥ 40380	¥ 43600	16.9	13.3	10.6	36.8	30.6	23.8	51.2	42.2	34.2	30.64	20.43	14.28
Advanced Micro-Fabrication	B	43,551	Rmb 307.69	Rmb 446	16.5	10.9	7.7	88.1	45.3	29.3	57.8	45.1	31.0	10.56	8.76	6.99
All Ring Tech	B	4,851	NT$ 1600	NT$ 1700	16.6	10.4	6.6	45.8	26.8	16.0	54.1	34.4	21.0	16.71	13.08	9.56
ASMPT Limited	B	8,366	HK$ 158.4	HK$ 200	3.1	2.6	2.2	16.4	12.2	9.7	31.5	21.4	17.2	3.58	3.28	2.72
Chroma ATE	B	29,722	NT$ 2250	NT$ 2960	17.6	13.4	11.3	42.5	29.3	23.8	49.6	36.1	29.6	27.38	19.95	15.49
Disco	N	39,820	¥ 58100	¥ 71200	10.6	8.6	7.5	22.8	18.3	15.7	34.6	28.0	24.2	9.04	7.57	7.95
Eugene Technology	B	2,552	Won 149100	Won 182000	6.9	4.8	3.8	32.8	17.1	12.5	25.9	22.9	17.3	5.79	4.69	3.73
Frontken Corporation	B	2,032	RM 5.09	RM 5.7	10.7	8.2	6.3	23.8	18.1	12.9	44.6	35.4	26.0	7.28	5.54	4.56
Grand Process Technology	B	2,144	NT$ 2360	NT$ 4300	8.4	5.4	3.5	36.2	18.8	11.6	41.2	24.7	14.9	24.88	15.54	9.61
Greatech Technology	B	1,581	RM 2.58	RM 3.3	6.1	4.1	3.7	26.5	16.4	14.7	40.9	23.6	22.1	5.58	4.51	3.75
Gudeng Precision Industrial	B	1,651	NT$ 553	NT$ 710	6.1	4.6	3.8	24.3	16.2	13.0	37.3	27.7	20.5	4.05	3.81	3.57
Hanmi Semiconductor	S	15,486	Won 217500	Won 170000	26.7	16.5	13.4	54.2	32.1	25.9	64.5	41.0	34.3	23.97	17.36	13.64
Kelington Group Berhad	B	1,780	RM 8.5	RM 10.8	4.4	2.5	2.2	28.0	14.7	13.6	41.3	21.5	20.2	8.81	7.21	6.33
Kokusai Electric	B	13,161	¥ 8918	¥ 11500	6.1	4.8	4.1	21.5	16.2	13.9	36.3	26.2	22.2	7.94	6.46	5.30
Lasertec Corporation	N	25,695	¥ 45370	¥ 47200	13.2	9.7	8.1	27.8	20.2	16.8	37.4	27.6	23.6	11.41	8.06	5.99
SCREEN Holdings Co., Ltd.	N	18,129	¥ 15170	¥ 15500	3.6	3.0	2.6	18.4	14.4	12.0	29.7	23.2	19.6	5.68	4.93	4.25
Tokyo Electron	B	161,452	¥ 11240	¥ 18720	7.4	5.7	4.9	22.5	17.1	14.5	32.7	23.8	19.8	11.25	9.76	8.57
Tokyo Seimitsu	N	4,997	¥ 19485	¥ 19500	4.0	3.5	3.3	14.3	11.7	10.8	23.0	18.6	17.4	3.72	3.31	2.96
Ulvac Inc.	N	2,651	¥ 8531	¥ 10100	1.4	1.2	1.1	9.4	7.7	7.2	19.4	17.0	16.1	1.64	1.55	1.46
Wonik IPS	B	4,416	Won 121500	Won 155000	4.8	3.5	2.8	29.7	18.0	12.8	27.6	26.9	19.6	5.02	4.30	3.57
Sector Average					**5.5**	**5.1**	**4.7**	**27.5**	**21.3**	**16.2**	**37.6**	**29.4**	**23.0**	**6.37**	**5.92**	**5.55**

Source: UBS estimates (priced as of October 13th, 2026)

APAC Tech Valuation – 3

	Rtg	Mcap (USD m)	Price	Target	EV/Sales			EV/EBITDA			P/E			P/BV		
					26E	27E	28E	26E	27E	28E	26E	27E	28E	26E	27E	28E
Silicon Wafers																
GlobalWafers	B	15,087	NT$ 1015	NT$ 1750	8.8	6.0	3.8	30.2	13.6	7.3	56.1	26.3	13.5	4.98	4.33	3.55
Shin-Etsu Chemical	B	73,105	¥ 6175	¥ 8160	3.6	3.4	3.3	10.3	9.3	8.5	20.5	17.4	15.2	2.49	2.46	2.33
SUMCO Corporation	B	6,804	¥ 3079	¥ 4000	2.8	2.3	1.9	9.2	6.5	4.5	56.4	15.3	7.3	1.88	1.68	1.47
Sector Average					**5.1**	**3.9**	**3.0**	**16.6**	**9.8**	**6.8**	**44.3**	**19.6**	**12.0**	**3.12**	**2.83**	**2.45**
Semis Materials																
Hansol Chemical	B	2,017	Won 248500	Won 350000	3.0	2.5	2.1	12.6	9.5	7.8	16.2	12.9	10.4	2.29	2.01	1.76
Soulbrain	B	2,197	Won 378000	Won 540000	2.2	1.8	1.4	10.2	7.6	6.1	15.0	13.3	10.7	2.36	2.05	1.75
Tekscend Photomask Corp	B	2,361	¥ 3760	¥ 5700	2.3	1.8	1.5	5.1	4.4	3.8	15.5	13.2	11.4	1.92	1.72	1.53
Wonik Materials	B	366	Won 38850	Won 59000	1.3	1.1	0.9	4.8	3.9	3.5	8.3	6.6	5.7	0.85	0.76	0.68
Sector Average					**2.0**	**1.6**	**1.4**	**7.8**	**6.4**	**5.6**	**13.4**	**11.7**	**9.9**	**1.66**	**1.48**	**1.30**
Japan Semis					**3.4**	**4.1**	**3.6**	**16.2**	**12.8**	**10.9**	**29.2**	**20.9**	**17.7**	**4.02**	**4.10**	**3.69**
Korea Semis					**3.1**	**2.1**	**1.7**	**14.0**	**13.0**	**8.7**	**17.1**	**16.9**	**13.5**	**3.27**	**2.70**	**2.23**
Taiwan Semis					**5.2**	**4.5**	**3.6**	**21.9**	**16.7**	**12.2**	**29.8**	**24.3**	**17.6**	**4.55**	**3.93**	**4.18**
China Semis					**5.1**	**4.3**	**3.7**	**20.2**	**19.8**	**14.9**	**38.1**	**30.8**	**24.5**	**4.18**	**4.14**	**3.46**
OEMs																
Asustek Computer Inc.	N	22,833	NT$ 990	NT$ 910	0.7	0.6	0.5	10.9	11.9	11.5	13.1	14.1	13.8	2.50	2.40	2.28
Lenovo Group	B	47,466	HK$ 30.4	HK$ 42	0.6	0.4	0.4	10.9	7.5	5.1	28.0	19.6	11.3	6.25	4.71	3.27
LG Electronics	N	23,863	Won 177400	Won 168000	0.4	0.3	0.3	4.8	4.2	3.9	13.7	10.1	9.2	1.01	0.92	0.83
TCL Tech	B	13,862	Rmb 4.46	Rmb 5.6	1.4	1.3	1.2	7.5	6.5	6.0	11.6	9.0	8.6	1.46	1.31	1.19
Shenzhen Transsion Holdings	N	8,306	Rmb 48.29	Rmb 60	0.7	0.6	0.5	12.4	12.3	8.6	17.9	14.6	11.0	2.57	2.33	2.06
Xiaomi	N	86,466	HK$ 26.38	HK$ 30.4	1.3	1.1	1.0	18.0	13.1	9.2	25.5	20.1	13.8	2.15	1.97	1.73
Sector Average					**0.9**	**0.7**	**0.6**	**10.7**	**9.3**	**7.4**	**18.3**	**14.6**	**11.3**	**2.66**	**2.27**	**1.89**
ODMs & Motherboard																
Compal Electronics	N	5,085	NT$ 37	NT$ 40	0.2	0.2	0.2	9.5	8.5	8.0	17.6	13.2	10.5	1.23	1.16	1.10
Giga-Byte Technology	N	7,405	NT$ 356	NT$ 390	0.4	0.4	0.4	7.6	7.8	7.4	10.7	10.8	9.6	3.25	2.90	2.55
Hon Hai Precision	B	105,008	NT$ 243	NT$ 340	0.3	0.2	0.2	7.0	5.5	4.4	13.1	10.4	8.8	1.80	1.61	1.46
Inventec	N	6,494	NT$ 58.3	NT$ 65	0.2	0.1	0.1	9.2	5.4	4.0	16.8	13.8	10.8	2.73	2.47	2.11
Micro-Star International	N	4,171	NT$ 159	NT$ 155	0.5	0.4	0.4	7.6	8.0	7.4	12.9	14.7	13.2	2.17	2.05	1.89
Pegatron	N	8,298	NT$ 100	NT$ 95	0.2	0.1	0.1	6.5	5.1	4.3	17.2	14.1	11.8	1.29	1.20	1.12
Quanta	B	38,656	NT$ 323	NT$ 450	0.3	0.3	0.2	9.4	7.2	6.6	12.0	9.4	7.6	5.24	4.03	3.20
Wistron	B	17,605	NT$ 187	NT$ 230	0.3	0.2	0.1	6.1	5.0	4.1	10.8	7.2	5.3	2.62	1.77	1.23
Wiwynn	B	36,041	NT$ 2115	NT$ 2470	0.9	0.5	0.4	12.4	8.6	7.3	17.4	12.0	10.3	9.85	6.67	5.08
Sector Average					**0.4**	**0.3**	**0.2**	**8.4**	**6.8**	**5.9**	**14.3**	**11.7**	**9.8**	**3.35**	**2.65**	**2.19**

Source: UBS estimates (priced as of October 13th, 2026)

APAC Tech Valuation – 4

	Rtg	Mcap (USD m)	Price	Target	EV/Sales			EV/EBITDA			P/E			P/BV		
					26E	27E	28E	26E	27E	28E	26E	27E	28E	26E	27E	28E
MLCC																
Murata Manufacturing	B	86,276	¥ 7501	¥ 13200	6.4	5.4	4.6	20.5	13.8	10.5	37.6	23.1	17.1	4.83	4.36	3.79
Samsung Electro-Mechanics	B	76,666	Won 1374000	Won 2700000	7.3	5.2	3.9	33.2	15.5	10.3	64.9	24.9	15.7	9.11	6.77	4.80
Taiyo Yuden	N	9,029	¥ 10970	¥ 17700	3.7	3.3	3.0	17.1	11.8	9.7	46.1	24.5	18.6	3.78	3.39	2.96
TDK	B	37,805	¥ 3152	¥ 4950	2.2	2.0	1.8	10.1	8.2	7.1	22.2	16.2	13.9	2.51	2.29	2.04
Sector Average					**4.9**	**4.0**	**3.3**	**20.2**	**12.3**	**9.4**	**35.3**	**22.2**	**16.3**	**5.06**	**4.20**	**3.40**
Diversified / Components																
Accton Technology Corporat	B	35,716	NT$ 2050	NT$ 3688	2.6	1.9	1.4	17.2	12.1	8.5	23.5	17.0	12.1	12.74	8.95	6.40
Alps Alpine	N	2,657	¥ 2155.5	¥ 2350	0.3	0.4	0.4	4.5	4.5	5.0	11.6	12.8	13.6	0.78	0.90	0.87
Delta Electronics	B	143,963	NT$ 1785	NT$ 2750	5.8	4.1	3.1	26.2	16.9	12.7	43.2	26.4	19.8	12.74	9.26	7.00
Delta Electronics (Thailand)	N	97,295	Bt 262	Bt 320	11.2	8.3	6.1	69.7	52.5	38.6	105.3	76.9	55.1	26.68	21.06	16.21
Shenzhen Desay Battery Tec	S	1,238	Rmb 21.54	Rmb 21	0.4	0.4	0.4	7.8	7.2	5.8	15.6	15.7	12.0	1.17	1.11	1.04
Genius Electronic Optical	B	3,728	NT$ 1065	NT$ 1250	4.1	3.6	3.2	11.3	10.1	8.7	27.8	24.2	21.6	4.18	3.80	3.46
Hirose Electric	N	5,201	¥ 25155	¥ 31100	2.8	2.6	2.5	10.2	9.4	9.0	21.9	19.4	18.1	2.09	1.99	1.89
Kyocera	B	31,256	¥ 3810	¥ 4620	2.2	2.1	1.9	15.8	13.3	12.0	25.9	22.8	20.6	1.48	1.46	1.40
Largan Precision	S	28,052	NT$ 6880	NT$ 5000	12.8	11.7	9.8	25.9	22.8	19.0	43.0	38.2	31.3	4.53	4.26	3.94
LG Innotek	B	7,806	Won 441500	Won 940000	0.5	0.5	0.4	5.2	4.7	4.3	13.7	9.9	8.1	1.55	1.36	1.18
Lite-On Technology	B	20,137	NT$ 283	NT$ 400	2.8	2.1	1.7	16.7	11.8	9.5	26.1	17.7	14.1	6.55	5.50	4.72
Niterra	N	4,784	¥ 3837	¥ 7400	1.0	1.0	0.8	3.9	3.5	3.1	6.6	6.3	5.9	0.93	0.85	0.78
Nidec	N	14,052	¥ 1940	¥ 2400	1.1	0.9	0.8	NA	8.9	8.0	NA	16.3	12.9	3.56	2.39	2.07
Partron Co Ltd	N	182	Won 4925	Won 5700	0.2	0.1	0.1	3.3	3.4	2.8	11.7	15.5	10.1	0.46	0.45	0.44
Sunny Optical Technology	N	8,156	HK$ 58.85	HK$ 63	1.1	0.9	0.8	8.3	7.2	5.3	15.4	13.6	10.9	1.75	1.57	1.41
Universal Scientific Industrial	B	8,030	Rmb 22.5	Rmb 36.8	0.7	0.6	0.5	13.5	10.3	7.8	25.4	18.3	13.5	2.45	2.22	1.97
Sector Average					**2.0**	**2.0**	**2.1**	**13.0**	**9.9**	**9.9**	**22.4**	**18.5**	**17.3**	**2.48**	**3.00**	**2.51**
PCB / Substrates																
Avary Holding Shenzhen	B	27,887	Rmb 80.53	Rmb 130	3.9	3.1	2.3	19.6	13.6	10.0	37.2	24.7	17.2	4.28	3.76	3.20
Ibiden	B	37,271	¥ 10495	¥ 10250	11.1	8.7	7.1	30.0	20.7	14.7	74.7	50.2	35.7	9.47	8.08	6.67
ISU Petasys	B	5,254	Won 111200	Won 190000	4.2	3.0	2.3	18.0	11.1	8.6	27.0	18.4	14.6	7.92	5.68	4.21
Kinsus Interconnect Technolc	B	15,258	NT$ 1050	NT$ 800	9.3	7.1	4.9	29.7	19.3	16.4	89.0	47.9	28.2	12.77	10.69	8.44
Nan Ya Printed Circuit Board	N	28,690	NT$ 1430	NT$ 1150	15.6	10.0	7.9	51.9	27.4	21.8	89.0	39.5	30.9	16.74	12.59	10.08
Shengyi Technology	B	44,469	Rmb 122.54	Rmb 209	6.7	4.1	3.1	28.3	17.7	13.4	38.3	23.4	17.8	13.92	10.55	8.41
Simmtech	B	3,933	Won 141000	Won 195000	2.8	2.1	1.7	16.7	10.0	7.7	26.1	14.5	11.3	6.68	4.59	3.27
Unimicron Technology	B	56,551	NT$ 1185	NT$ 1300	9.7	6.6	4.9	33.8	17.3	13.1	49.7	30.3	19.8	13.41	10.07	7.33
Sector Average					**6.6**	**5.8**	**4.3**	**25.7**	**17.4**	**13.6**	**35.7**	**33.0**	**22.6**	**7.09**	**5.53**	**6.25**

Source: UBS estimates (priced as of October 13th, 2026)

APAC Tech Valuation – 5

	Rtg	Mcap (USD m)	Price	Target	EV/Sales			EV/EBITDA			P/E			P/BV		
					26E	27E	28E	26E	27E	28E	26E	27E	28E	26E	27E	28E
Panel makers																
AUO Corp	N	8,522	NT$ 36.25	NT$ 21.9	1.1	1.0	0.9	11.8	10.0	8.6	20.9	39.5	46.0	1.55	1.53	1.52
INNOLUX Corporation	S	12,620	NT$ 50.9	NT$ 26	1.5	1.4	1.3	11.2	11.4	11.2	17.1	84.4	72.5	1.73	1.76	1.78
LG Display	N	2,842	Won 7610	Won 11600	0.7	0.6	0.5	3.5	3.1	2.7	NA	6.5	4.7	0.56	0.51	0.47
Sharp	S	2,378	¥ 579.6	¥ 470	0.4	0.4	0.4	9.4	7.8	6.9	12.4	8.9	8.9	1.21	1.07	0.95
Sector Average					**0.9**	**0.9**	**0.8**	**9.0**	**8.1**	**7.3**	**16.8**	**18.3**	**19.9**	**1.26**	**1.22**	**1.18**
Japan Hardware					**2.2**	**2.7**	**2.3**	**13.5**	**10.2**	**8.6**	**23.1**	**20.0**	**16.5**	**3.06**	**2.68**	**2.34**
Asia ex-Japan Hardware					**2.4**	**2.2**	**1.9**	**14.4**	**10.8**	**9.5**	**21.9**	**19.7**	**16.1**	**3.33**	**3.10**	**3.12**
Korea					**2.5**	**1.8**	**1.5**	**13.1**	**8.0**	**6.3**	**18.4**	**15.9**	**11.9**	**3.67**	**2.78**	**2.13**
Taiwan					**2.6**	**2.4**	**2.2**	**15.0**	**12.0**	**10.0**	**21.8**	**22.4**	**17.7**	**3.42**	**3.61**	**3.75**
China					**1.9**	**1.4**	**1.1**	**14.3**	**10.5**	**7.8**	**23.9**	**17.2**	**12.5**	**2.87**	**2.44**	**2.66**
APAC Tech					**3.5**	**3.2**	**2.8**	**17.2**	**13.9**	**11.1**	**26.9**	**22.4**	**17.9**	**3.86**	**3.57**	**3.37**
Japan Tech					**2.9**	**3.5**	**3.1**	**15.3**	**11.8**	**10.0**	**27.2**	**20.5**	**17.3**	**3.61**	**3.53**	**3.15**
Asia ex-Japan Tech					**3.7**	**3.2**	**2.7**	**17.7**	**14.5**	**11.3**	**26.8**	**23.0**	**18.1**	**3.93**	**3.58**	**3.43**
Korea					2.8	2.0	1.6	13.6	10.9	7.7	17.6	16.4	12.8	3.45	2.73	2.18
Taiwan					3.9	3.5	3.0	18.7	14.6	11.3	26.1	23.5	17.7	4.07	3.78	3.98
China					3.8	2.9	2.7	17.8	16.5	12.5	31.6	25.7	20.2	3.73	3.62	3.22

Source: UBS estimates (priced as of October 13th, 2026)

Tech Crowding Scores

Korea	
Samsung	13.1
SK Hynix	15.8
SEMCO	4.2
LG Electronics	1.0
Hanmi	-4.4
LG Innotek	-2.0
Wonik IPS	-2.6
ISU Petasys	0.3
Simmtech	-3.6
Eugene	-0.2

Taiwan	
TSMC	19.3
MediaTek	5.2
Delta	5.5
Hon Hai	2.2
ASE	5.5
UMC	3.8
Nanya Tech	1.4
Accton	3.0
Quanta	-2.1
Wiwynn	-8.8
Chroma	4.0
Nan Ya PCB	0.8
GlobalWafers	-10.4
Aspeed	4.2
Global Unichip	-0.3
Asustek	-1.1
Largan	-4.9
Wistron	-1.1
Innolux	-8.5
King Yuan	-6.5
Kinsus	-0.7
Alchip	-0.3

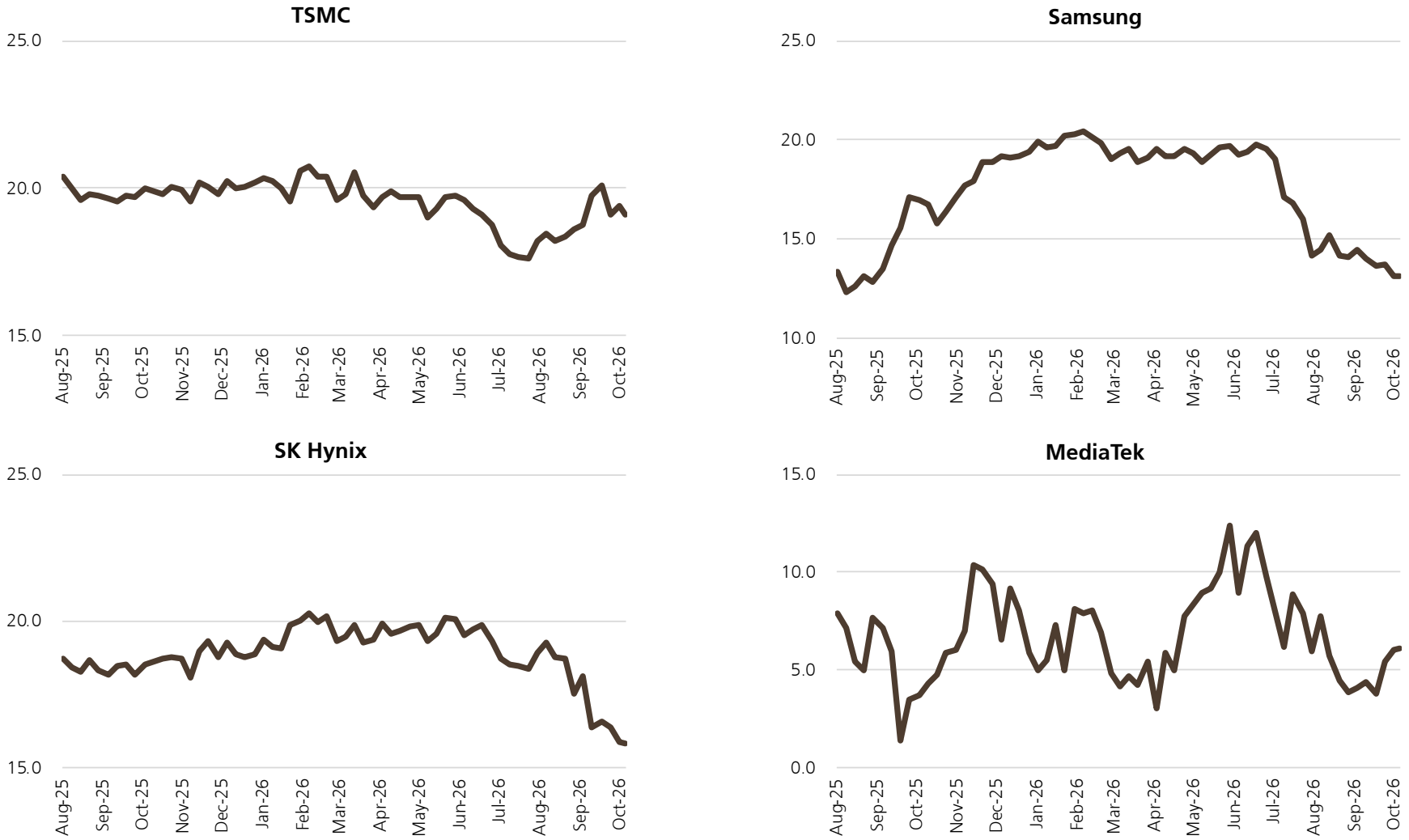
Japan	
Kioxia	16.0
Tokyo Electron	13.1
Advantest	7.6
Murata	8.3
Shin-Etsu	5.6
Renesas	5.4
TDK	2.3
Kyocera	2.7
Ibiden	-4.0
Lasertec	-4.0
SCREEN	0.3
Kokusai	4.4

Hong Kong / China	
Xiaomi	-12.9
NAURA	6.3
SMIC	-2.4
AMEC	10.7
Lenovo	-3.6
Avary	5.8
JCET	1.8
ACM Research	-0.8
Omnivision	-2.5
CR Micro	4.5

US	
NVDA	25.1
AAPL	11.9
AVGO	26.4
MU	26.2
AMD	22.1
ASML	21.7
INTC	26.0
LRCX	27.1
AMAT	27.3
TXN	27.1
MRVL	7.8
QCOM	10.7
ADI	27.7
MCHP	13.9

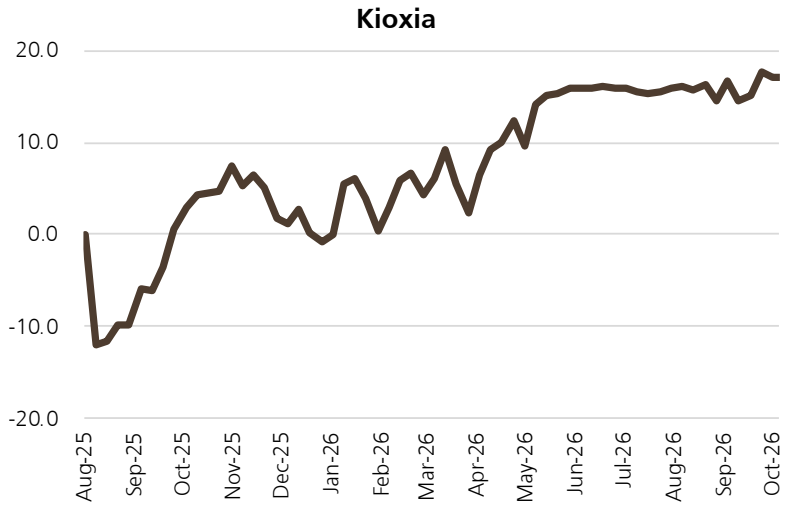
Source: UBS Quant Research (as of October 7th, 2026). Top 50/15 market cap stocks in APAC/US&EU

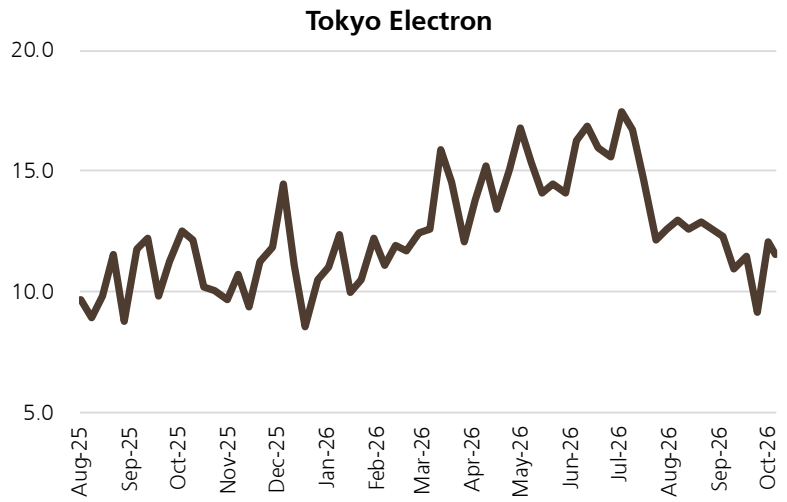
APAC Tech Crowding Scores

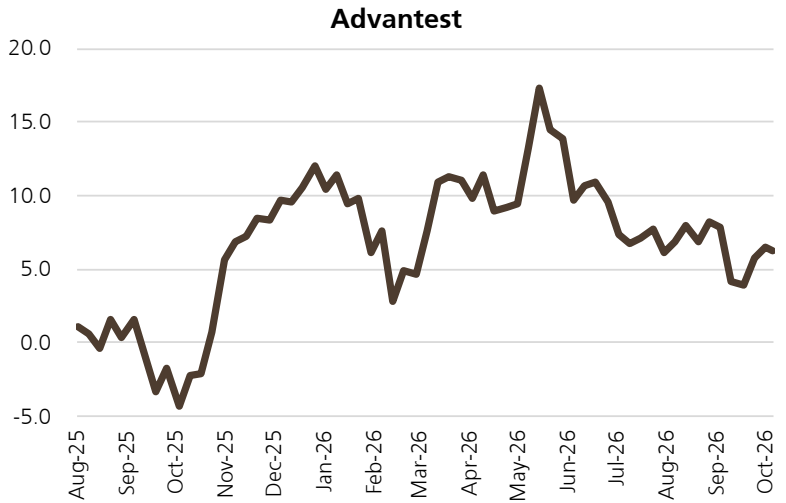


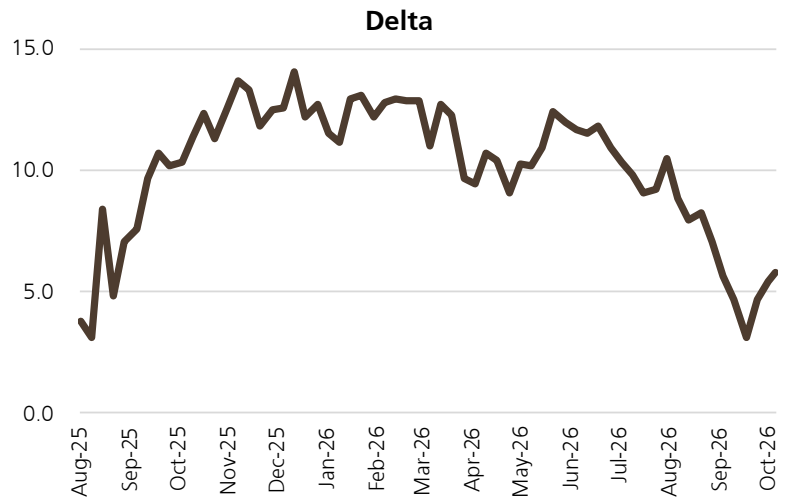
Source: UBS Quant Research (as of October 12th, 2026)

APAC Tech Crowding Scores



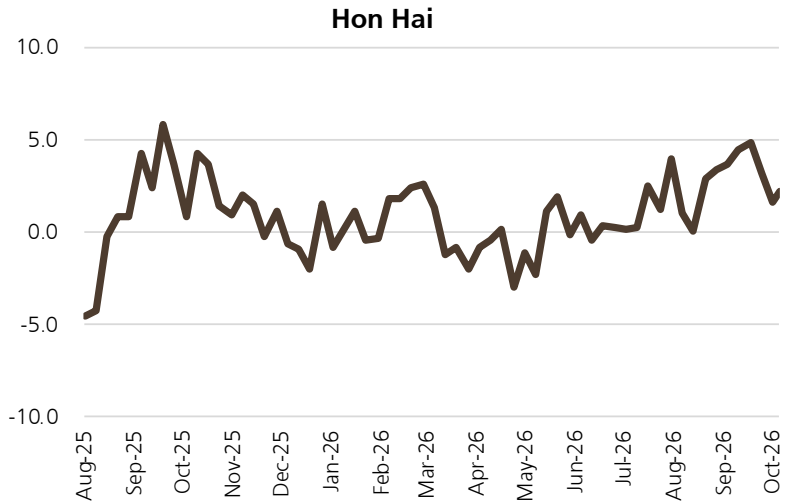


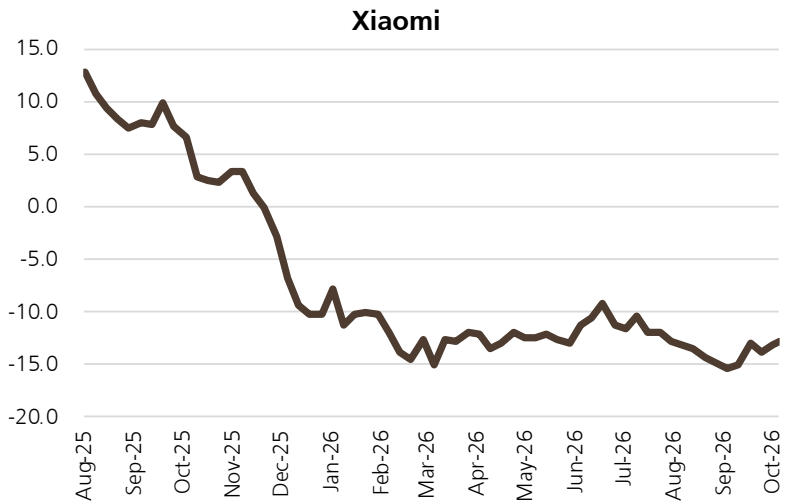


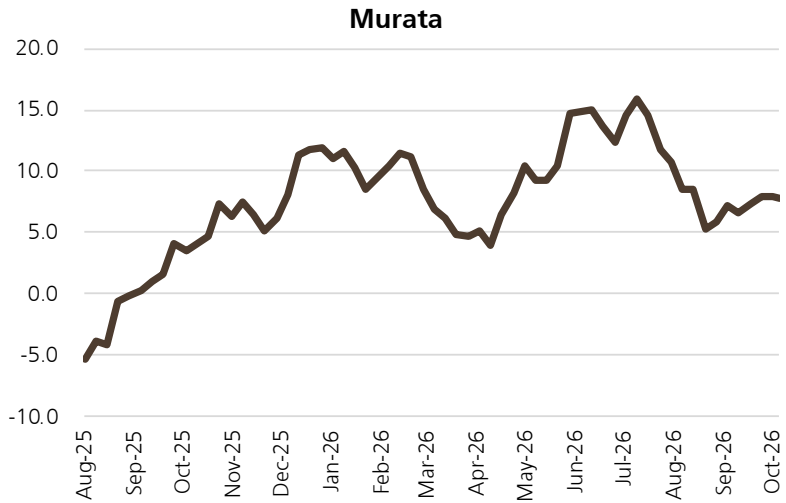


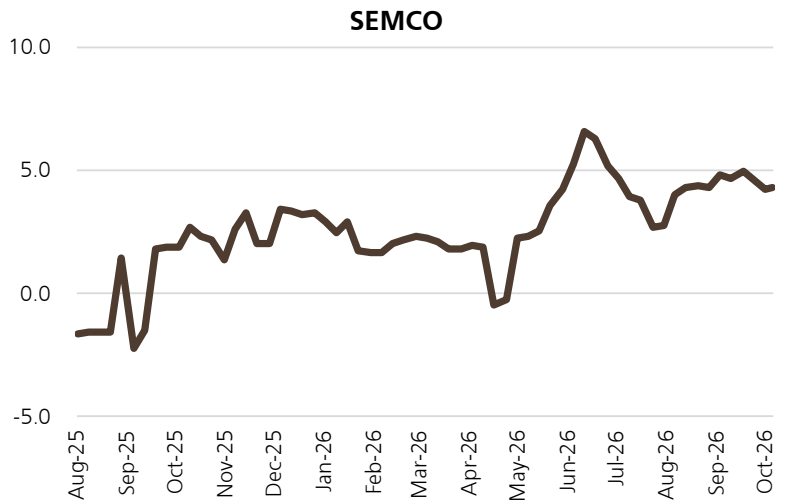
Source: UBS Quant Research (as of October 12th, 2026)

APAC Tech Crowding Scores





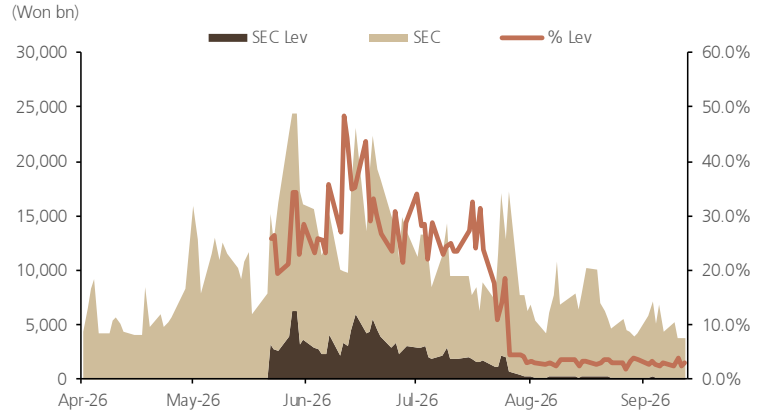




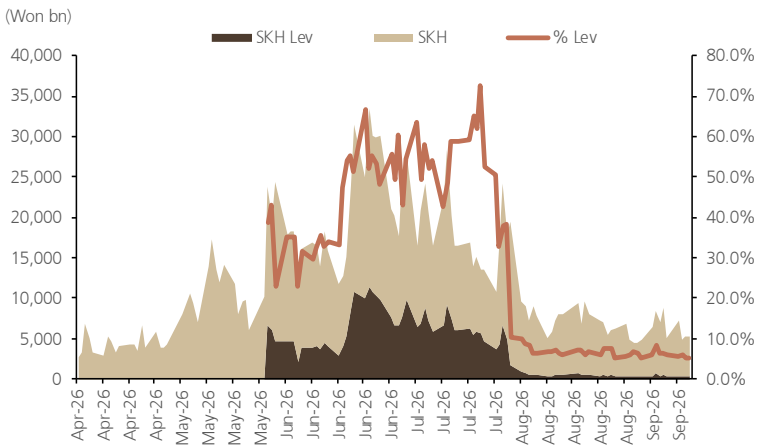
Source: UBS Quant Research (as of October 12th, 2026)

Investor Positioning

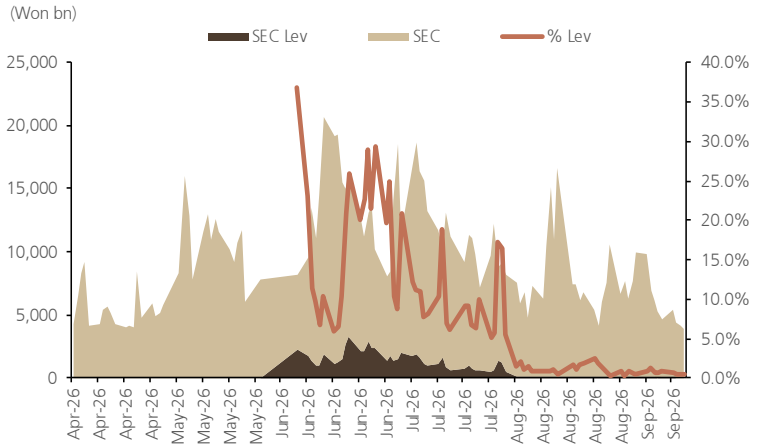
Samsung underlying vs leveraged ETF trading value (total)



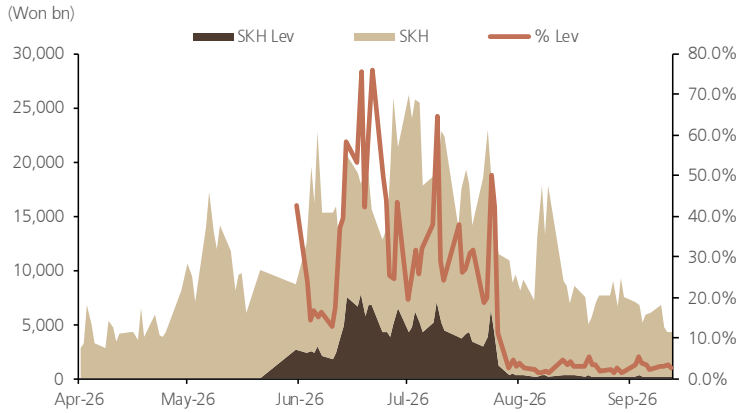
SK Hynix underlying vs leveraged ETF trading value (total)



Samsung underlying vs leverage ETF trading value (retail)



SK Hynix underlying vs leveraged ETF trading value (retail)



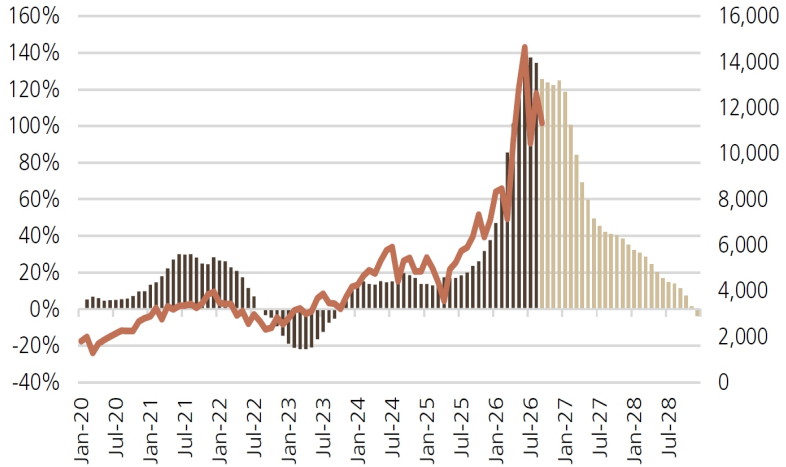
Source: UBS Quant Research (as of September 30th, 2026)

Semiconductor Cycle Key Lead Indicators – 1

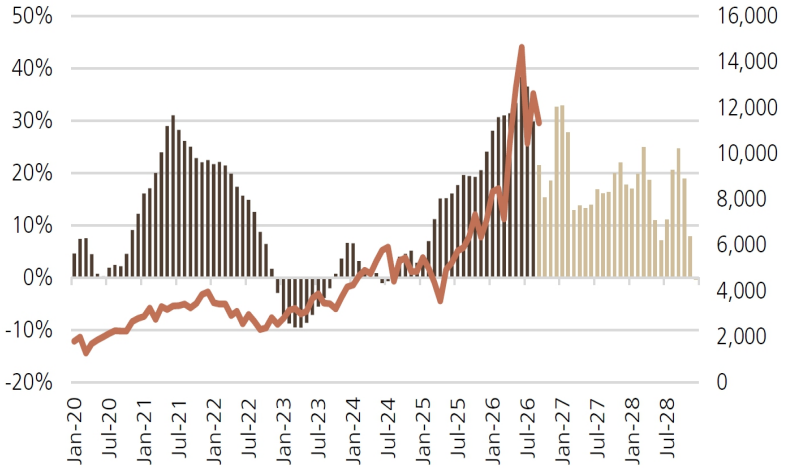
Summary of lead indicators and implications on stocks

Lead indicator	Lead Indication 'Color'	Recent Inflection (Peak/Trough)	Next Inflection (Peak / Trough)	Suggested Stock Peak / Trough
3MMA YoY semis revenue growth	Red	July '26 (Peak)	Dec '28 (Trough)	2CQ28E (Trough)
3MMA YoY semis ex-memory revenue growth	Green	June '26 (Peak)	March '27 (Trough)	3CQ26 (Trough)
Foundry capacity utilisation rate	Green	3Q23 (Trough)	4CQ28E (Peak)	2CQ28E (Peak)
YoY assembly equipment revenue growth	Green	3Q25 (Trough)	2CQ27E (Peak)	2CQ27E (Peak)
Memory operating profits	Green	1Q25 (Trough)	2CQ28E (Peak)	4CQ27E (Peak)

Semis YoY 3MMA revenue growth vs SOX



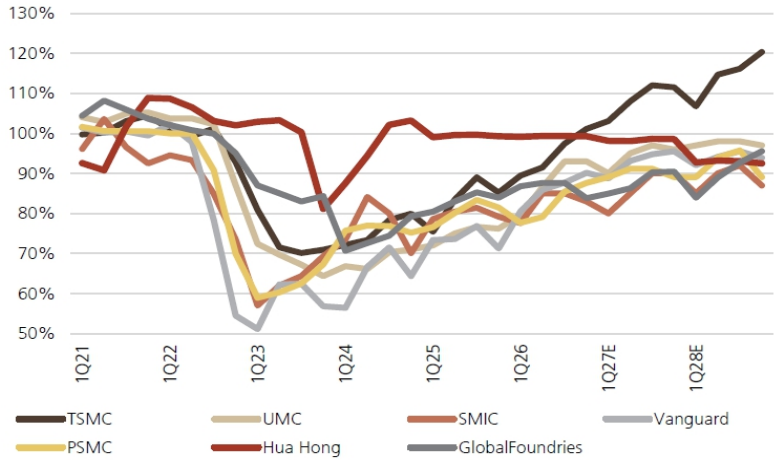
Semis ex-Memory YoY 3MMA revenue growth vs SOX



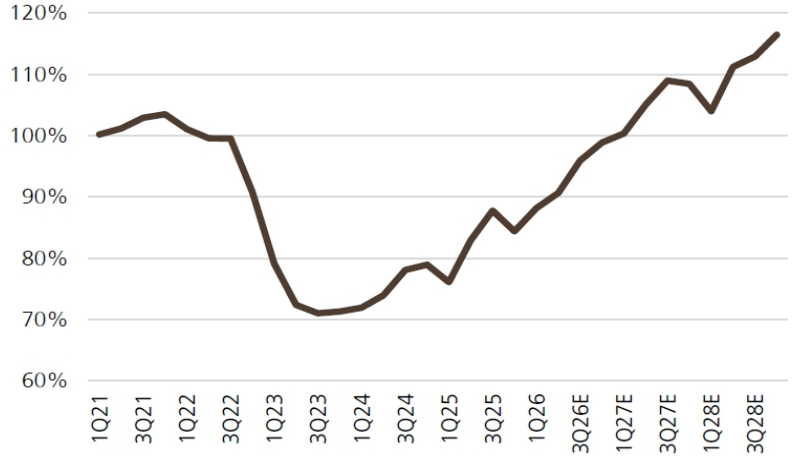
Source: UBS, Bloomberg

Semiconductor Cycle Key Lead Indicators – 1

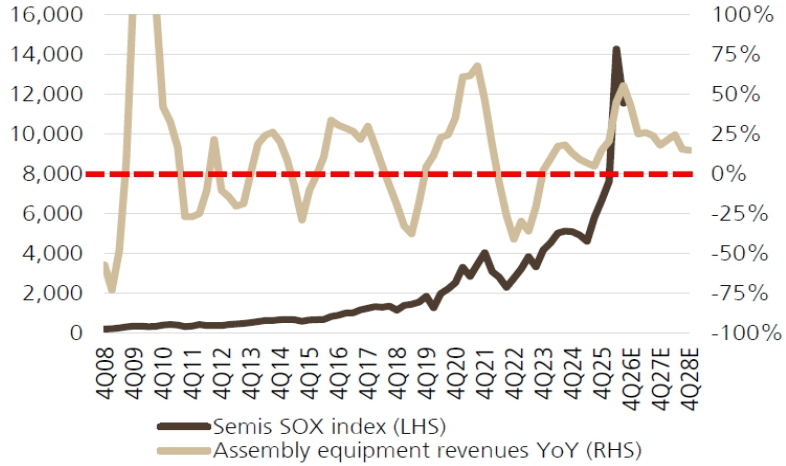
Foundry capacity utilization rates, 1Q11-4Q28E



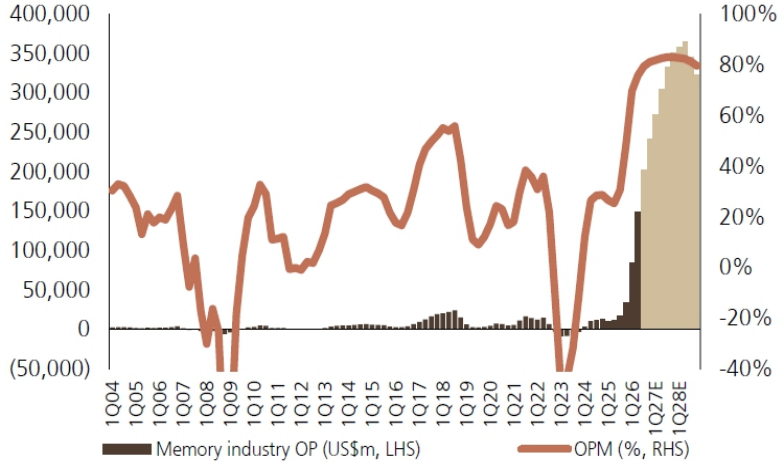
Weighted average of foundry capacity utilization rates, 1Q21-4Q28E



YoY assembly equipment revenue growth

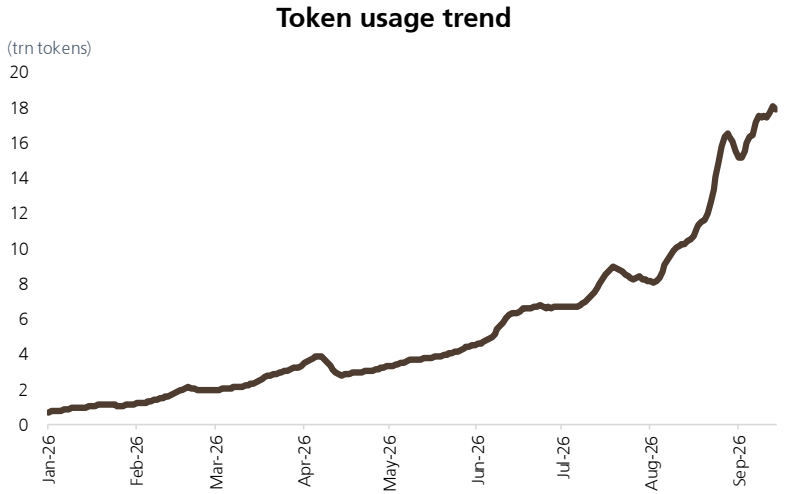


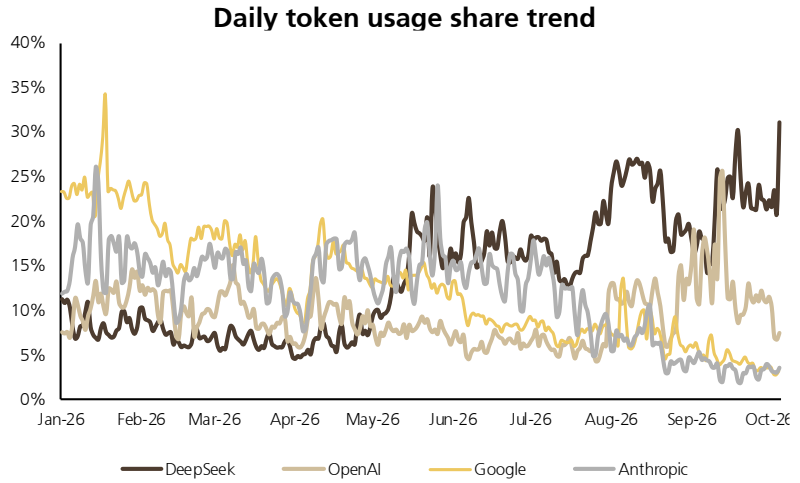
Memory industry OP (US$m) and OPM

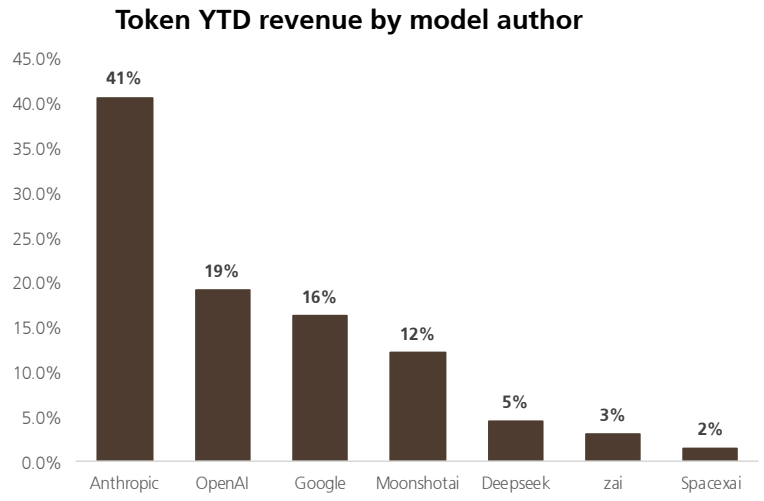


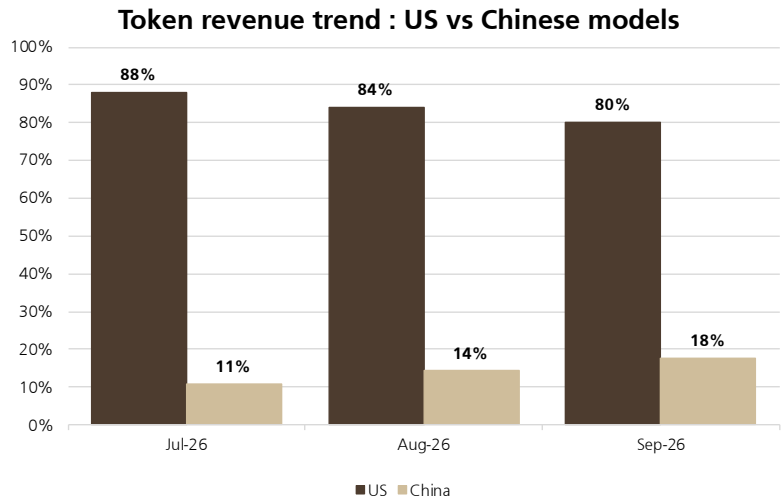
Source: UBS estimates, Company Data

AI Demand – Token Usage/Revenue Trends





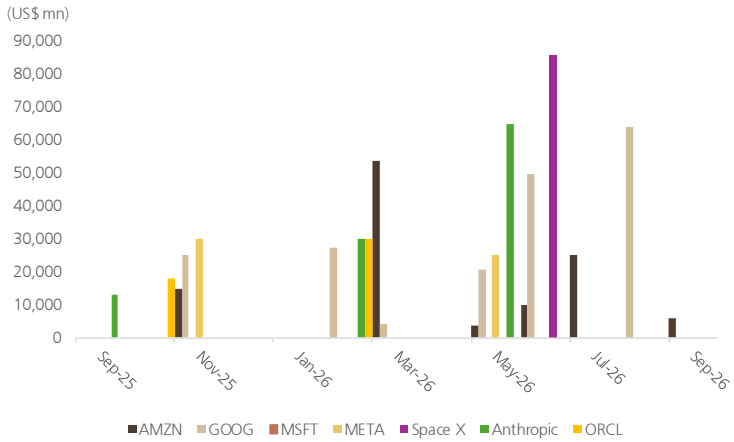




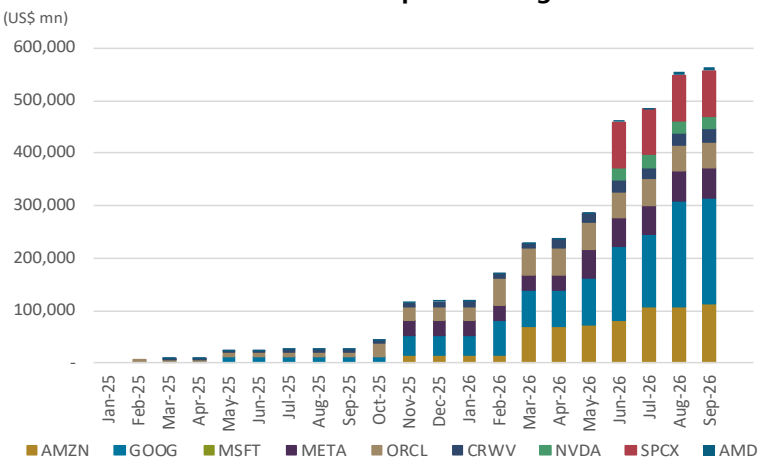
Source: OpenRouter, Vercel AI

Hyperscaler Capital Raising, Cash Flow, Credit Spreads

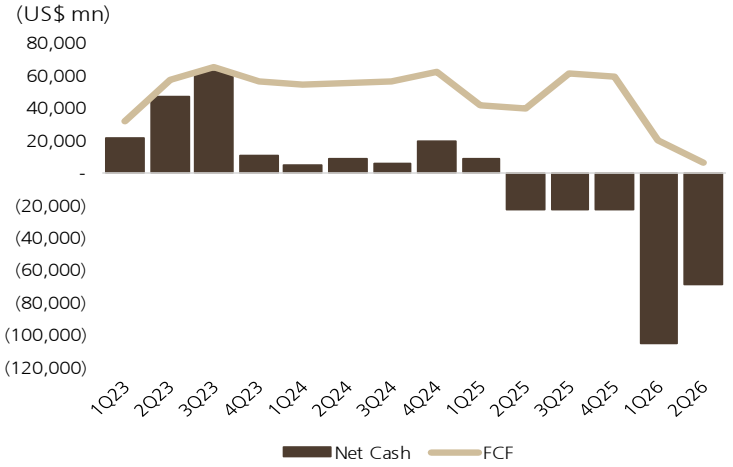
Hyperscaler monthly capital raising



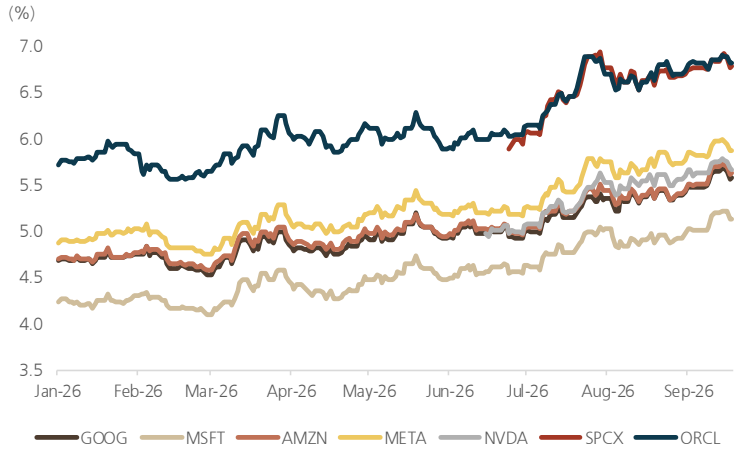
Cumulative capital raising



Hyperscaler net cash / free cash flow



Hyperscaler credit spreads



Source: SEC filings, Bloomberg, Company data (as of September 30th, 2026)

Power and AI - Visibility Through 2028, Constraints Beyond

1/ Sufficient power capacity through 2028, but execution remains tight

- Semiconductor-implied demand suggests sufficient powered-shell capacity through 2028 if ~40-50% of early-stage grid projects convert, though 76GW of demand vs. 50GW confirmed capacity increases reliance on behind-the-meter solutions.

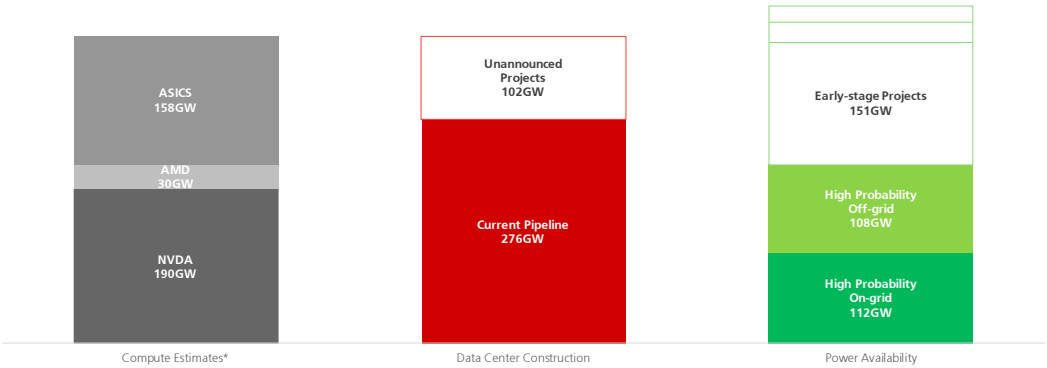
2/ Post-2028 demand step-up intensifies infrastructure risks

- AI data center requirements rise to ~260GW in 2028-30E, nearly 3x 2025-27E, with 2030 additions of ~105GW vs. market expectations of ~30-50GW. Grid connections take >4 years, transformers up to 210 weeks, and gas/backup power >3 years.

3/ Higher compute density drives power architecture and content

- 800VDC and 660kW-1.2MW power racks enable 227kW Rubin racks while increasing energy storage and HV distribution content. ASIC share rises from ~35% in 2026 to ~45% by 2030E, while AI-load stabilizers lift content/watt from $0.57 to $0.87.

Compute Expectations vs DC and Power Availability (2026-2030E Cumulatively)



17

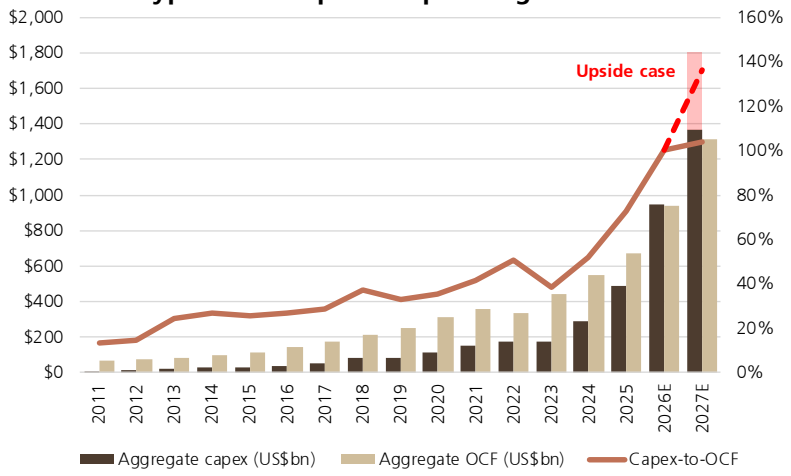
Source: UBS estimates, UBS Evidence Lab *Note: our compute estimates imply ~380GW of data center capacity when utilizing a PUE of 1.12.

Hyperscaler Capex

Top 12 hyperscaler capex summary

Top 12 Hyperscale Capex	1Q26	2Q26	3Q26	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2026E	2027E	2028E
Meta	$18,997	$30,116	$42,734	$52,833	$58,040	$55,681	$63,164	$68,048	$144,681	$244,932	$293,634
Google	$35,674	$44,924	$55,149	$69,880	$72,739	$87,218	$90,513	$102,617	$205,627	$353,087	$379,221
Amazon	$44,203	$54,208	$58,708	$61,995	$66,525	$67,070	$64,638	$66,648	$219,114	$264,882	$293,858
Microsoft (+Leases)	$31,900	$41,000	$50,504	$51,982	$54,193	$56,596	$58,848	$61,013	$175,387	$230,650	$266,426
Oracle	$18,635	$16,493	$28,499	$20,707	$22,756	$25,840	$23,421	$25,923	$84,334	$97,939	$100,269
Corew eave (+Leases)	$6,786	$9,352	$12,385	$10,180	$9,184	$13,121	$11,371	$10,059	$38,704	$43,735	$48,109
SpaceX (AI / Terafab)	$7,723	$15,828	$16,000	$16,000	$30,000	$30,480	$33,478	$30,000	$55,551	$123,958	$155,480
Nebius	$2,473	$5,657	$6,789	$7,851	$8,401	$7,981	$6,783	$7,575	$22,770	$30,740	$36,273
Lambda Labs	$420	$480	$520	$580	$525	$600	$650	$725	$2,000	$2,500	$3,000
Baidu	$858	$1,677	$696	$249	$733	$908	$838	$1,013	$3,480	$3,492	$3,443
Alibaba	$4,408	$5,037	$5,457	$6,087	$4,436	$5,070	$5,492	$6,126	$20,989	$21,125	$20,236
Tencent	$4,612	$7,756	$10,304	$13,364	$7,149	$7,453	$7,757	$8,061	$35,949	$30,420	$18,647
Top 12 Hyperscale capex (US$mn)	**$176,688**	**$232,530**	**$287,746**	**$311,709**	**$334,681**	**$358,017**	**$366,954**	**$387,808**	**$1,008,586**	**$1,447,460**	**$1,618,595**
YoY growth	**85.6%**	**100.5%**	**111.5%**	**92.6%**	**89.4%**	**54.0%**	**27.5%**	**24.4%**	**98.5%**	**43.5%**	**11.8%**
QoQ growth	**9.2%**	**31.6%**	**23.7%**	**8.3%**	**7.4%**	**7.0%**	**2.5%**	**5.7%**	**42.3%**	**49.3%**	**45.8%**

Hyperscaler capex to operating cash flow



Source: Company data, UBS estimates

Memory Spend as % of Total Hyperscaler Capex

Hyperscaler memory spend	2025	2026E	2027E
AI Server HBM Demand (mil Gb)	17,486	32,171	61,520
% Hyperscaler	*90%*	*90%*	*90%*
HBM Blended ASP (US$/Gb)	$1.52	$1.74	$3.14
AI Server HBM Capex (US$m)	**$23,924**	**$50,518**	**$173,583**
AI Server DDR Demand (mil Gb)	26,625	50,009	146,119
% Hyperscaler	*90%*	*90%*	*90%*
Server DDR Blended ASP (US$/Gb)	$0.43	$1.84	$2.51
AI Server DDR Capex (US$m)	**$10,299**	**$82,787**	**$329,813**
Conventional Server DDR Demand (mil Gb)	57,795	77,685	98,247
% Hyperscaler	*80%*	*80%*	*80%*
Server DDR Blended ASP (US$/Gb)	$0.43	$1.84	$2.51
Conventional Server DDR Capex (US$m)	**$19,871**	**$114,314**	**$197,118**
Server SSD NAND Demand (mil GB)	230,454	433,513	672,772
% Hyperscaler	*80%*	*80%*	*80%*
NAND Blended ASP (US$/GB)	$0.07	$0.25	$0.35
Server SSD NAND Capex (US$m)	**$12,104**	**$87,507**	**$187,343**
Storage SSD NAND Demand (mil GB)	88,546	182,763	265,294
% Hyperscaler	*80%*	*80%*	*80%*
NAND Blended ASP (US$/GB)	$0.07	$0.25	$0.35
Storage SSD NAND Capex (US$m)	**$4,651**	**$36,892**	**$73,875**

Summary	2025	2026E	2027E	2027E Upside
Top 12 hyperscaler capex	**$508,118**	**$1,008,586**	**$1,447,460**	**$1,800,000**
% YoY growth	*71%*	*98%*	*44%*	*78%*
Total HBM spend (US$m)	$23,924	$50,518	$173,583	$173,583
Total DDR spend (US$m)	$30,170	$197,101	$526,931	$526,931
Total NAND spend (US$m)	$16,755	$124,398	$261,218	$261,218
Total memory spend (US$m)	**$70,848**	**$372,017**	**$961,733**	**$961,733**
% YoY growth		*425%*	*159%*	159%
As % of total hyperscaler capex	***14%***	***37%***	***66%***	***53%***

Source: Company data, UBS estimates

Memory as % of VR200/VR300 BOM Costs

VR200

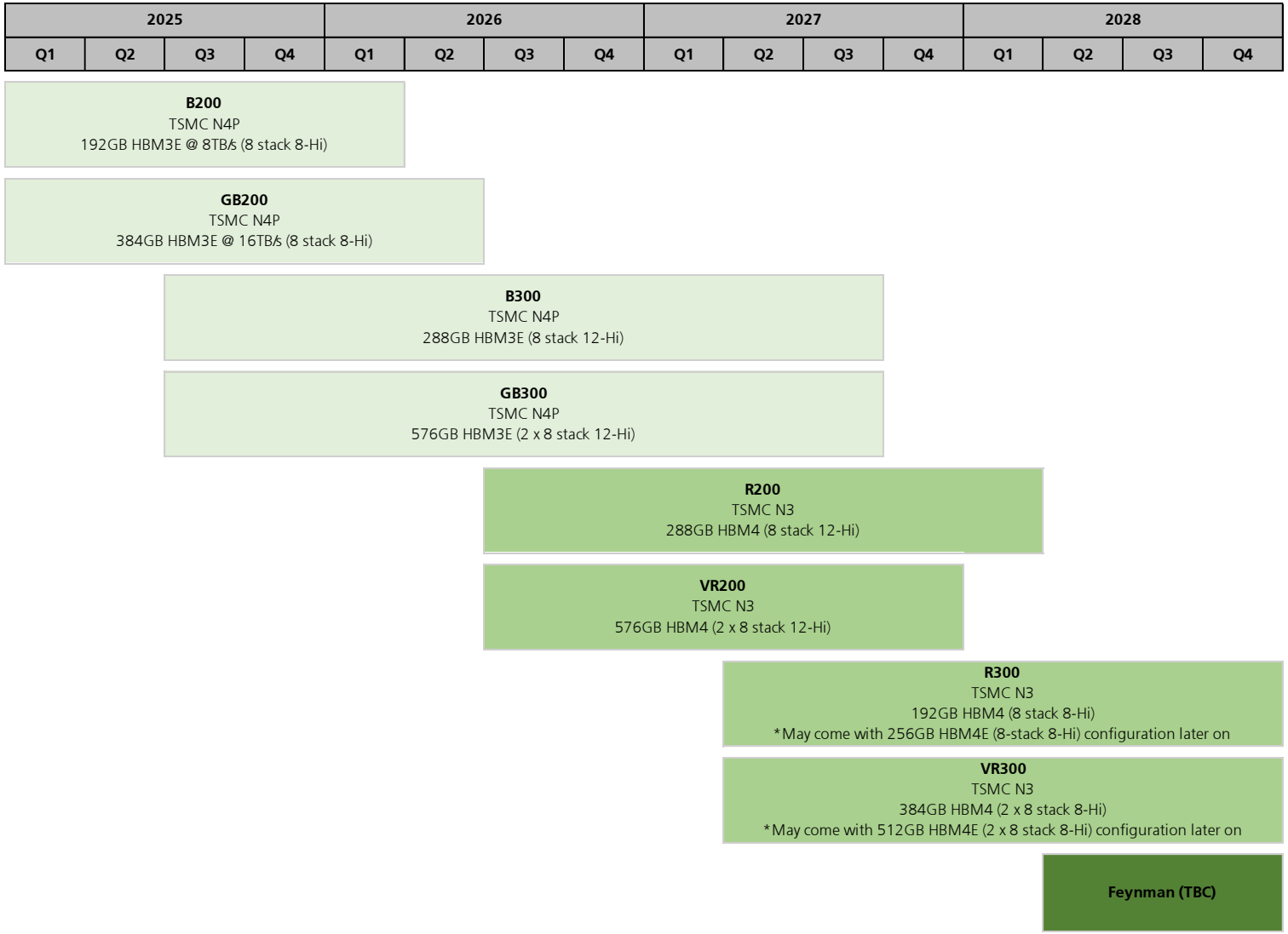
Subsystem	Assumptions	*Assuming 2026 pricing* Cost ($)	% of Total
2x GPU die		**$7,408**	
Total die logic cost for 2x GPU	*TSMC 3nm; 8 known good dies per 12" wafer (US$19,000)*	*$4,750*	
Interposer + CoWoS packaging	*CoWoS-L*	*$2,658*	
GPU memory	**288GB of HBM4 (12-Hi x 8 cubes) x 2**	**$9,976**	
Periphery components		$1,200	
(A) Total GPU cost		**$18,584**	**62%**
CPU die	TSMC 3nm; 27 known good dies per 12" wafer (US$19,000)	$704	
CPU memory	**768GB of SOCAMM2**	**$10,179**	
(B) Vera CPU		**$10,883**	**36%**
(C) Other board components		**$350**	**1%**
VR200 bill of material costs	*(A) Total GPU cost + (B) Vera CPU cost + (C) Components*	**$29,817**	**100%**
Total memory		**$20,155**	**68%**

VR300

Subsystem	Assumptions	*Assuming 2027 pricing* Cost ($)	
2x GPU die		**$12,724**	
Total die logic cost for 2x GPU	*TSMC 3nm; 8 known good dies per 12" wafer (US$19,000)*	*$4,750*	
Interposer + CoWoS packaging	*CoWoS-L*	*$7,974*	
GPU memory	**192GB of HBM4 (8-Hi x 8 cubes) x 2**	**$10,564**	
Periphery components		$1,200	
(A) Total GPU cost		**$24,488**	**61%**
CPU die	TSMC 3nm; 27 known good dies per 12" wafer (US$19,000)	$704	
CPU memory	**768GB of SOCAMM2**	**$14,299**	
Vera CPU		**$15,003**	**38%**
Other board components		**$500**	**1%**
VR300 bill of material costs	*(A) Total GPU cost + (B) Vera CPU cost + (C) Components*	**$39,991**	**100%**
Total memory		**$24,863**	**62%**

Source: Company data, UBS estimates

NVIDIA AI GPU Roadmap



Source: Company data

AI ASIC Roadmap

ASIC vendor	Products	IC design partner	Process technology	HBM generation	HBM content (GB)	Packages / CoWoS wafer	2025	2026E	2027E
Google	TPU v5e	Broadcom	5nm	HBM2E	32	60	134		
	TPU v5p	Broadcom	5nm	HBM2E	96	32	1,916		
	TPU v6e Trillium	Broadcom	4nm	HBM3E	32	50	426	640	
	TPU v7 Ironwood	Broadcom	3nm	HBM3E	192	17	90	2,790	1,500
	TPU v8i (Dual die)	Broadcom	3nm	HBM3E	288	14		250	3,500
	TPU v8t (Single die)	MediaTek	3nm	HBM3E	216	22		467	4,000
Google's total							**2,565**	**4,147**	**9,000**
Amazon	Trainium 1	Alchip	7nm	HBM2E	32	39	176		
	Trainium 2	Marvell	5nm	HBM3E	96	35	1,335	590	60
	Trainium 2.5	Marvell	3nm	HBM3	144	25		96	25
	Trainium 3	Alchip	3nm	HBM3E	144	25		1,780	2,800
	Trainium 4	Alchip	2nm	HBM4	256	15		0	280
Amazon's total							**1,511**	**2,466**	**3,165**
Meta	MTIA v1	Broadcom	7nm	LPDDR5	64	50	421	25	
	MTIA v2	Broadcom	5nm	LPDDR5	128	34		30	
	MTIA v3	Broadcom	3nm	HBM3E	192	15		110	105
	MTIA v4	Broadcom	2nm	HBM4	432	10			340
Meta's total							**421**	**165**	**445**
Intel / Habana	Gaudi 3	Alchip	5nm	HBM2 / 3E	288	36	188		
	Gaudi 4	Alchip	3nm	HBM3E	288	32			
Intel's total							**188**		
Microsoft	Maia 100	GUC	5nm	HBM3	120	34	34	4	
	Maia 200	GUC	3nm	HBM3E	288	14		100	35
	Maia 300	Marvell	2nm	HBM4	288	14			450
Microsoft's total							**34**	**104**	**485**
Tesla	Dojo	Broadcom	5nm	HBM3	64	36		64	63
Tesla's total								**64**	**63**
Others									
Total ASICs and others CoWoS wafer demand									

Source: Company data, UBS estimates

AI GPU/ASIC Roadmap and Supply Chain

Semiconductors	Google	Meta	Microsoft	Amazon	Oracle	NVIDIA
Foundry + CoWoS	TSMC	TSMC	TSMC	TSMC	N/A	TSMC
Assembly Outsourcing	ASE, Amkor	ASE	ASE	ASE	N/A	ASE, Amkor
Back-end Test	King Yuan (Broadcom, NVIDIA)	King Yuan (Broadcom, NVIDIA)	King Yuan	ASE, King Yuan	N/A	King Yuan, ASE (2nd source)
ASIC Design Service	Broadcom through v8t Mediatek v8i, GUC Axion	Broadcom	GUC Maia 100/200 & Cobalt, MRVL Maia 300	Tranium 2/3 MRVL Alchip T1/T3	N/A	N/A
IC Distribution	WT Micro (TPU)	WT Micro (Broadcom)	N/A	N/A	N/A	N/A
HBM	ASIC server: Hynix 67% / Samsung 33%	ASIC server: Micron 63% / Samsung 37%	NVDA server: (Hynix 60%+, Micron 24%, Samsung 15%	ASIC Server: Hynix 84%, Samsung 16%	N/A	Hynix 60%+, Micron 24%, Samsung 15%
BMC	Google internal	Aspeed	Aspeed	Aspeed	N/A	Aspeed
Test Platform	Advantest	Advantest	Advantest	Teradyne	N/A	Advantest FT, Chroma SLT, Teradyne
Probe Card	MPI (CHPT load board)	MPI	NVDA server: TPI	MPI	N/A	TPI

Hardware	Google	Meta	Microsoft	Amazon	Oracle	NVIDIA
ASIC Server	Celestica (TPU tray & L11), Hon Hai (CPU tray & L11, Inventec (CPU L6)	Quanta + Celestica	Hon Hai	Accton, Fabrinet L6 Wiwynn, Jabil L11	N/A	N/A
GPU Server	NVIDIA	NVIDIA, AMD	NVIDIA, AMD	NVIDIA	NVIDIA	NVIDIA
GPU Server Racks	Quanta, Hon Hai from VR	Hon Hai, Quanta	Hon Hai, Quanta	Quanta	Hon Hai, Quanta, Wistron, Dell, SMCI	Hon Hai, Quanta, Wistron, SMCI etc.
Switch/Networking	Celestica	Celestica, Nvidia, Cisco, Arista, Accton	Arista, Nvidia	Accton, Celestica	Mitac	Wistron ODM Spectrum, Hon Hai Switch Tray
Substrate	Unimicron, Nanya PCB (late 26)	Unimicron	Unimicron	Unimicron, Semco	N/A	Ibiden, Unimicron
PCB	Victory, ISU Petasys, Shennan, DSBJ	Isu Petasys (Switch), Shennan	ISU Petasys (Switch), Avary (Small share)	Shengyi Elect., Avary (small share)	N/A	HDI: Victory, Unimicron. MLB: Wus, TTM, ISU, Avary
CCL	Panasonic, Shengyi, Elite Material	Elite Material	N/A	Elite Material, Shengyi Tech	N/A	Panasonic, Isola, Shengyi, Elite, Doosan
Heatspreader/Stiffener	GPU lid: Jentech, TPU stiffener: Jentech	GPU lid: Jentech	GPU lid: Jentech	GPU lid: Jentech, AWS lid/stiffener: Jentech		Lid/stiffener main: Jentech, legacy: I-Chiun
Liquid Cooling (ASIC)	Boyd, Cooler Master, AVC	AVC, Auras	AVC	AVC, Auras		N/A
Liquid cooling GPU	Cooler master, Vertiv, AVC	AVC, Auras, Delta (L2A)	AVC, Delta (L2A)	AVC, Auras, Delta (L2A)		AVC, Auras, Jentech, CoolerMaster
Power Supply Unit	Delta	Delta	Delta, Lite-On	Delta, Lite-On (T3/T4 rack)		Delta, Lite-On, Megmeet

Source: Company data, UBS estimates

AI Server Demand

GB200/GB300 Rack Units	Q126	Q226	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
Quanta	4,000	5,200	5,800	5,100	**20,100**	2,700	1,200	705	700	**5,305**
Hon Hai	5,400	6,200	7,620	7,300	**26,520**	3,500	2,500	1,000	500	**7,500**
Wistron	3,905	4,250	5,100	3,900	**17,155**	2,700	1,400	900	500	**5,500**
Total (Top 3)	**13,305**	**15,650**	**18,520**	**16,300**	**63,775**	**8,900**	**5,100**	**2,605**	**1,700**	**18,305**
SMCI, Wiwynn, GigaByte, Inventec, Asus, Lenovo, Pega, etc	**2,861**	**3,130**	**3,704**	**3,260**	**12,955**	**1,335**	**765**	**261**	**170**	**2,531**
Total GB200/300 racks (market-wide)	**16,166**	**18,780**	**22,224**	**19,560**	**76,730**	**10,235**	**5,865**	**2,866**	**1,870**	**20,836**
Total GB200/300 GPUs into racks	**1,163,921**	**1,352,160**	**1,600,128**	**1,408,320**	**5,524,529**	**736,920**	**422,280**	**206,316**	**134,640**	**1,500,156**

VR200 NVL72 Rack Units	Q126	Q226	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
Quanta				1,300	**1,300**	4,000	6,400	5,100	4,300	**19,800**
Hon Hai				1,700	**1,700**	6,500	7,000	6,700	6,500	**26,700**
Wistron				200	**200**	2,600	4,400	3,900	3,500	**14,400**
Total (Top 3)				**3,200**	**3,200**	**13,100**	**17,800**	**15,700**	**14,300**	**60,900**
SMCI, Wiwynn, GigaByte, Inventec, Asus, Lenovo, Pega, etc				160	**160**	2,227	3,560	3,140	2,860	**11,787**
Others share %				5%	**5%**	17%	20%	20%	20%	**19%**
Total VR200 racks (market-wide)				**3,360**	**3,360**	**15,327**	**21,360**	**18,840**	**17,160**	**72,687**
Total VR200 GPUs into racks				**241,920**	**241,920**	**1,103,544**	**1,537,920**	**1,356,480**	**1,235,520**	**5,233,464**

VR300 Rack Units	Q126	Q226	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
Quanta									475	**475**
Hon Hai									900	**900**
Wistron									200	**200**
Total (Top 3)									**1,575**	**1,575**
SMCI, Wiwynn, GigaByte, Inventec, Asus, Lenovo, Pega, etc										
Total VR300 (market-wide)									**1,575**	**1,575**
Total VR300 GPUs into racks									**113,400**	**113,400**

Total NVIDIA Racks	Q126	Q226	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
GB200/GB300	16,166	18,780	22,224	19,560	**76,730**	10,235	5,865	2,866	1,870	**20,836**
VR200				3,360	**3,360**	15,327	21,360	18,840	17,160	**72,687**
VR300									1,575	**1,575**
Total Racks	**16,166**	**18,780**	**22,224**	**22,920**	**80,090**	**25,562**	**27,225**	**21,706**	**20,605**	**95,098**
Total GPUs into Racks	**1,163,921**	**1,352,160**	**1,600,128**	**1,650,240**	**5,766,449**	**1,840,464**	**1,960,200**	**1,562,796**	**1,483,560**	**6,847,020**

Source: Company data, UBS estimates

AI Server GPU/Accelerator Unit Forecasts ('000 units)

	HBM Gen.	HBM Content (GB)			2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E
NVIDIA		**2025E**	**2026E**	**2027E**											
L40S	GDDR6	48			38										
H200	HBM3E	141			593										
H20 / H20E	HBM3 / 3E	96			441										
B200 (8Hi)	HBM3E	192			1,441										
GB200 (2x GPU - 8Hi)	HBM3E	384	384		1,080	103				103					
B300 (12Hi)	HBM3E	288	288	288	587	267	300	200	100	867	50				50
GB300 (2x GPU - 12Hi)	HBM3E	576	576	576	778	658	670	580	450	2,358	360	270	180	90	900
R200 (12Hi)	HBM4		288	288			20	200	320	540	360	400	320	240	1,320
RV200 (2x GPU - 12Hi)	HBM4		576	576			40	250	450	740	600	720	690	630	2,640
R300 (8Hi)	HBM4			192									250		250
R300 (8Hi)	HBM4E			256										400	400
RV300 (2x GPU - 8Hi)	HBM4			384									250		250
RV300 (2x GPU - 8Hi)	HBM4E			512										550	550
*Other GPU/Accel	HBM3 / 3E	288	384	576	107	60	80	80	80	300	70	70	80	80	300
Total Units					**5,066**	**1,088**	**1,110**	**1,310**	**1,400**	**4,908**	**1,440**	**1,460**	**1,770**	**1,990**	**6,660**
% YoY					*3%*	*-17%*	*0%*	*3%*	*3%*	*-3%*	*32%*	*32%*	*35%*	*42%*	*36%*
% QoQ						*-20%*	*2%*	*18%*	*7%*		*3%*	*1%*	*21%*	*12%*	
Total SiPs					**6,924**	**1,849**	**1,820**	**2,140**	**2,300**	**8,109**	**2,400**	**2,450**	**2,890**	**3,260**	**11,000**
% YoY					*38%*	*27%*	*23%*	*15%*	*8%*	*17%*	*30%*	*35%*	*35%*	*42%*	*36%*
% QoQ						*-13%*	*-2%*	*18%*	*7%*		*4%*	*2%*	*18%*	*13%*	
AMD															
MI300	HBM3	192			71										
MI325X (12Hi)	HBM3E	256	256		224	18				18					
MI355X (12Hi)	HBM3E	288	288		221	170	150	80	70	470					
MI308X	HBM3	192	192		96	29	28	5		62					
MI450 (12Hi)	HBM4		432	432				45	230	275	350	450	470	380	1,650
MI5XX (16Hi)	HBM4E			1024									60	160	220
*Other GPU/Accel	HBM3	192	192	288	4	1	1	1	1	6	2	2	2	2	8
Total Units					**616**	**218**	**179**	**131**	**301**	**831**	**352**	**452**	**532**	**542**	**1,878**
% YoY					*41%*	*81%*	*86%*	*-26%*	*37%*	*35%*	*61%*	*152%*	*305%*	*80%*	*126%*
% QoQ						*-1%*	*-18%*	*-27%*	*129%*		*17%*	*28%*	*18%*	*2%*	
Google															
TPU v5e	HBM2E	32			134										
TPU v5p	HBM2E	96			1,916										
TPU v6e	HBM3E	32	32		426	205	224	141	70	640					
TPU v7	HBM3E	192	192	192	90	520	720	800	750	2,790	600	450	300	150	1,500
TPU v8i (Dual die AX)	HBM3E		288	288				25	225	250	700	1,000	1,300	1,000	4,000
TPU v8t (Single die X)	HBM3E		216	216				70	380	450	700	1,200	1,500	1,000	4,400
*Other TPU/Accel	HBM3	96	144	144	77	7	9	10	10	37	25	25	25	25	100
Total Units					**2,643**	**732**	**953**	**1,046**	**1,435**	**4,167**	**2,025**	**2,675**	**3,125**	**2,175**	**10,000**
% YoY					*19%*	*14%*	*66%*	*82%*	*68%*	*58%*	*177%*	*181%*	*199%*	*52%*	*140%*
% QoQ						*-14%*	*30%*	*10%*	*37%*		*41%*	*32%*	*17%*	*-30%*	
Amazon															
Inferentia 2	HBM2E	32			88										
Trainium 2 / 2.5	HBM3E	96	144	144	496	130	220	140	100	590	50	10			60
Inferentia 3	HBM3	96	96	96	162	30	28	20	18	96	15	10			25
Trainium 3	HBM3E		144	144			320	620	840	1,780	840	780	640	540	2,800
Trainium 4	HBM4			256									40	240	280
*Other Accel	HBM3	144	144	144	37	5	6	8	10	28	8	8	7	7	30
Total Units					**783**	**165**	**574**	**788**	**968**	**2,494**	**913**	**808**	**687**	**787**	**3,195**
% YoY					*-8%*	*-17%*	*180%*	*302%*	*424%*	*218%*	*454%*	*41%*	*-13%*	*-19%*	*28%*
% QoQ						*-11%*	*248%*	*37%*	*23%*		*-6%*	*-12%*	*-15%*	*15%*	

Source: Company data, UBS estimates

AI Server GPU/Accelerator Unit Forecasts ('000 units)

	HBM Gen.	HBM Content (GB)			2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E
Intel / Habana															
Gaudi 3	HBM2 / 3E	288	288		188	30	18	8		56					
Gaudi 4	HBM3E		288	288				10	25	35	25	20	15	10	70
*Others	HBM2 / 3E	192	192	192	25	6	4	4	5	18	4	4	4	4	16
Total Units					**213**	**36**	**22**	**22**	**30**	**109**	**29**	**24**	**19**	**14**	**86**
% YoY					*-32%*	*-20%*	*-60%*	*-63%*	*-46%*	*-49%*	*-19%*	*11%*	*-12%*	*-53%*	*-21%*
% QoQ						*-36%*	*-40%*	*0%*	*39%*		*-3%*	*-17%*	*-21%*	*-26%*	
Microsoft															
Maia 100	HBM3	120	120		34	3	1			4					
Maia 200	HBM3E		288	288		2	6	10	14	33	14	10	7	4	35
Maia 300	HBM4			288									8	14	22
*Others	HBM3	120	192	192	8	2	2	2	2	8	2	2	2	2	8
Total Units					**42**	**7**	**9**	**12**	**16**	**45**	**16**	**12**	**17**	**20**	**65**
% YoY					*22%*	*-27%*	*-19%*	*4%*	*84%*	*7%*	*119%*	*35%*	*36%*	*24%*	*45%*
% QoQ						*-17%*	*22%*	*40%*	*29%*		*0%*	*-25%*	*42%*	*18%*	
Meta															
MTIA v1	LPDDR5	64	64		421	20	5			25					
MTIA v2	LPDDR5	128	128		30	20	8	2		30					
MTIA v3	HBM3E	192	192	192	5	15	20	35	40	110	40	30	20	15	105
MTIA v4	HBM4			432							40	80	100	120	340
*Others	LPDDR5	64	128	128	28	3	1	1	1	6	5	5	5	5	20
Total Units					**484**	**58**	**34**	**38**	**41**	**171**	**85**	**115**	**125**	**140**	**465**
% YoY					*19%*	*-43%*	*-71%*	*-71%*	*-68%*	*-65%*	*47%*	*235%*	*228%*	*240%*	*171%*
% QoQ						*-56%*	*-41%*	*11%*	*8%*		*106%*	*35%*	*9%*	*12%*	
Tesla															
Dojo	HBM3	64	64	64	62	17	18	15	14	64	15	15	18	15	63
*Others	HBM3	64	64	64	44	7	7	6	6	26	6	6	5	5	22
Total Units					**106**	**24**	**25**	**21**	**20**	**90**	**21**	**21**	**23**	**20**	**85**
% YoY					*-39%*	*-12%*	*-9%*	*-24%*	*-16%*	*-15%*	*-12%*	*-17%*	*10%*	*2%*	*-5%*
% QoQ						*2%*	*6%*	*-17%*	*-7%*		*7%*	*0%*	*10%*	*-13%*	
Others															
Startups	HBM3	192	288	288	657	150	150	155	155	610	170	180	185	190	725
Others	HBM2E /3	192	288	288	361	30	30	39	39	138	60	63	65	67	254
Total Units					**1,019**	**180**	**180**	**194**	**194**	**748**	**230**	**243**	**250**	**257**	**979**
% YoY					*59%*	*-28%*	*-32%*	*-26%*	*-20%*	*-27%*	*28%*	*35%*	*29%*	*32%*	*31%*
% QoQ						*-26%*	*0%*	*8%*	*0%*		*18%*	*6%*	*3%*	*3%*	
Total GPU / Accelerator Units ('000)					**12,829**	**3,269**	**3,797**	**4,412**	**5,455**	**16,933**	**6,231**	**6,970**	**7,848**	**7,445**	**28,493**
% YoY					*27%*	*15%*	*34%*	*34%*	*42%*	*32%*	*91%*	*84%*	*78%*	*36%*	*68%*
% QoQ						*-15%*	*16%*	*16%*	*24%*		*14%*	*12%*	*13%*	*-5%*	

Source: Company data, UBS estimates

CoWoS Demand

CoWoS capacity & allocation

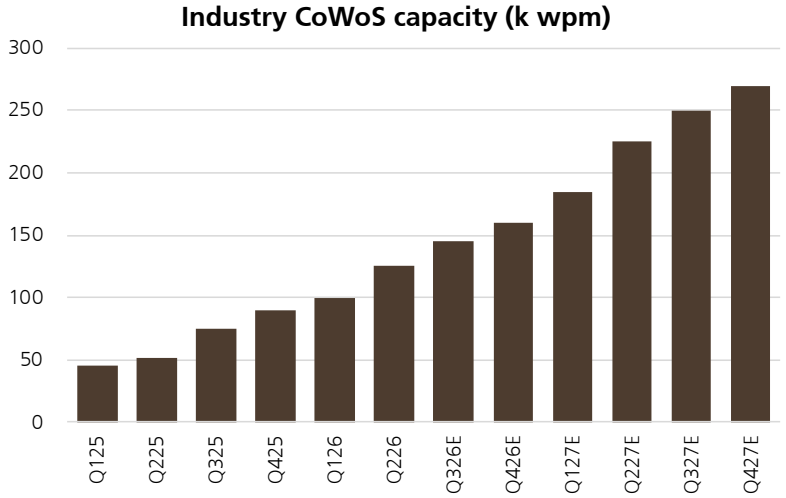
	Q125	Q225	Q325	Q425	Q126	Q226	Q326E	Q426E	Q127E	Q227E	Q327E	Q427E	2025	2026E	2027E
Base case															
TSMC CoWoS	40	45	65	70	80	105	120	130	140	160	170	180	55	109	163
Non-TSMC	5	7	10	20	20	20	25	30	45	65	80	90	11	24	70
Industry CoWoS capacity (kwpm)	45	52	75	90	100	125	145	160	185	225	250	270	66	133	233
TSMC CoWoS % of industry capacity	*89%*	*87%*	*87%*	*78%*	*80%*	*84%*	*83%*	*81%*	*76%*	*71%*	*68%*	*67%*	*84%*	*82%*	*70%*
Allocation taken by Nvidia	61%	68%	58%	46%	43%	41%	53%	60%	49%	44%	46%	46%	57%	50%	46%
Nvidia's volume estimates (kwpm)	27.5	35.2	43.6	41.8	42.7	51.3	76.9	95.4	91.3	99.9	114.1	124.7	37.0	66.6	107.5
Demand ex-NVIDIA (kwpm)															
AMD	2.6	1.8	3.7	4.5	3.6	3.8	9.9	25.7	30.2	40.5	41.4	46.1	3.1	10.7	39.6
ASICs and others	12.9	16.1	17.4	18.0	26.2	32.8	35.4	36.7	58.1	72.6	78.4	81.3	16.1	32.8	72.6
Google TPU - Broadcom													6.3	16.2	28.2
Google TPU - MediaTek													0.0	1.8	15.2
Amazon Trainium - Alchip													0.4	5.9	10.9
Amazon Trainium - Marvell													3.2	1.7	0.2
Others													6.2	7.1	18.1
Total demand (kwpm)	43.0	53.1	64.7	64.3	72.5	87.9	122.2	157.8	179.6	213.0	233.9	252.1	56.3	110.1	219.7
Industry UTR	*96%*	*102%*	*86%*	*71%*	*73%*	*70%*	*84%*	*99%*	*97%*	*95%*	*94%*	*93%*	*86%*	*83%*	*94%*

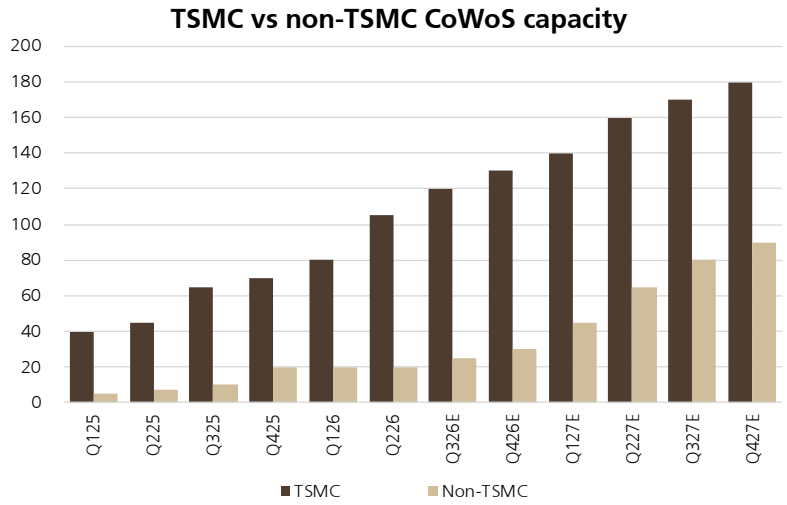
Select CPU supply chain build units

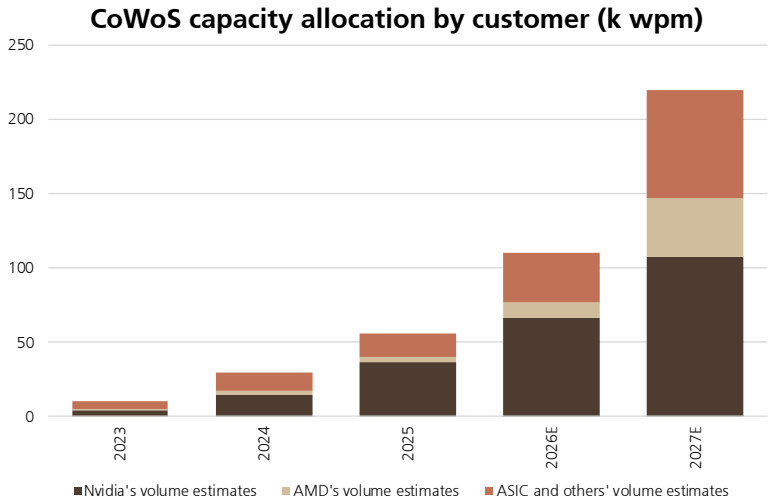
Supply Chain Build Units (k chips)	Q126	Q226	Q326E	Q426E	Q127E	Q227E	Q327E	Q427E	2026E	2027E
NVIDIA										
Vera CPU		100	428	1,054	1,025	1,180	1,543	1,754	1,582	5,502
AMD										
Venice CPU			400	900	1,000	1,300	1,300	1,400	1,300	5,000

Source: Company data, UBS estimates

CoWoS Capacity Forecasts





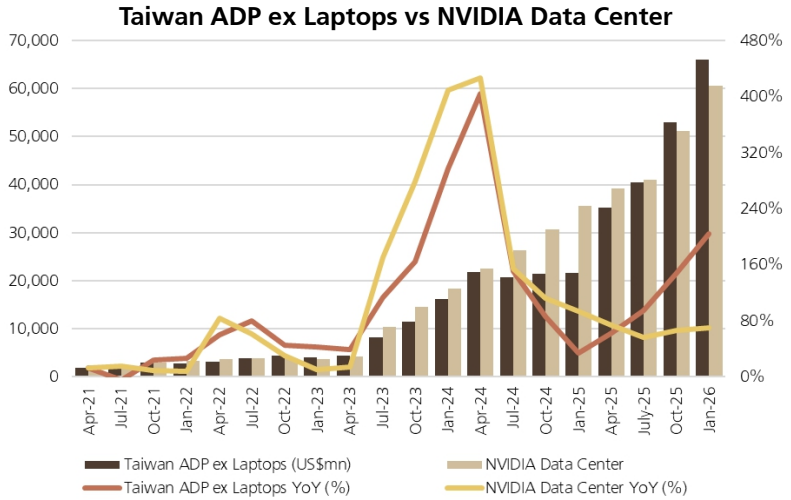


Source: Company data, UBS estimates

UBS Server Model

UBS global server model forecasts

	2023	2024	2025	2026E	2027E
Total Server shipments (K unit)	**14,087**	**16,025**	**20,078**	**23,874**	**27,844**
Yoy growth (%)	***-5.7%***	***13.8%***	***25.3%***	***18.9%***	***16.6%***
Out of which conventional servers	**13,304**	**14,435**	**18,835**	**22,728**	**26,085**
Yoy growth (%)	*-6.4%*	*8.5%*	*30.5%*	*20.7%*	*14.8%*
% of total	*94%*	*90%*	*94%*	*95%*	*94%*
Traditional Enterprise	**4,217**	**4,283**	**4,214**	**4,298**	**4,384**
Yoy growth (%)	*-16.1%*	*1.6%*	*-4.0%*	*-6.0%*	*-6.0%*
% of total	*30%*	*27%*	*21%*	*18%*	*16%*
Hyperscalers	**9,915**	**9,883**	**11,102**	**15,864**	**19,576**
Yoy growth (%)	*12.4%*	*-0.3%*	*12.3%*	*42.9%*	*23.4%*
% of total	*70%*	*62%*	*55%*	*66%*	*70%*
Standard servers	**8,873**	**8,047**	**9,600**	**14,452**	**17,543**
Yoy growth (%)	*13.4%*	*-9.3%*	*19.3%*	*50.5%*	*21.4%*
% of total hyperscalers	*89%*	*81%*	*86%*	*91%*	*90%*
% of total servers	*63%*	*50%*	*48%*	*61%*	*63%*
AI Servers	**783**	**1,591**	**1,243**	**1,146**	**1,759**
Yoy growth (%)	*7.3%*	*103.1%*	*-21.8%*	*-7.8%*	*53.5%*
% of total hyperscalers	*8%*	*16%*	*11%*	*7%*	*9%*
% of total servers	*6%*	*10%*	*6%*	*5%*	*6%*
Database Servers	**259**	**246**	**258**	**266**	**274**
Yoy growth (%)	*2.0%*	*4.0%*	*4.0%*	*5.0%*	*5.0%*
% of total hyperscalers	*3%*	*2%*	*2%*	*2%*	*1%*
% of total servers	*2%*	*2%*	*1%*	*1%*	*1%*



Source: Company data, Dell'Oro, Gartner, UBS estimates
*Note: Capex forecasts excluding ByteDance from Top 8

Hyperscale Server Demand by Key Customers

(K Units)	2022	2023	2024	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E
Quarterly														
Alphabet	1,039	1,085	1,247	1,845	530	600	580	640	2,350	720	720	760	750	2,950
AWS	2,237	2,131	2,276	3,087	940	960	1,020	1,000	3,920	1,150	1,180	1,250	1,240	4,820
Microsoft	1,056	1,274	1,468	2,457	680	760	830	820	3,090	850	920	1,000	970	3,740
Meta	2,192	2,027	2,188	3,409	950	1,080	1,100	1,050	4,180	1,200	1,280	1,250	1,260	4,990
Baidu	71	70	37	30	10	10	11	10	41	13	14	13	12	52
Alibaba	376	188	305	456	135	145	155	160	595	158	167	175	180	680
Tencent	118	149	303	268	75	82	88	82	327	82	92	100	95	369
ByteDance	358	550	770	1,306	360	420	430	440	1,650	450	480	520	490	1,940
Total Top 8	**7,448**	**7,473**	**8,594**	**12,858**	**3,680**	**4,057**	**4,214**	**4,202**	**16,153**	**4,623**	**4,853**	**5,068**	**4,997**	**19,541**
% YoY	*13.5%*	*0.3%*	*15.0%*	*49.6%*	*28.2%*	*31.7%*	*25.0%*	*18.8%*	*25.6%*	*25.6%*	*19.6%*	*20.3%*	*18.9%*	*21.0%*
% QoQ					*4.1%*	*10.2%*	*3.9%*	*-0.3%*		*10.0%*	*5.0%*	*4.4%*	*-1.4%*	
Others	2,468	2,410	2,901	1,879	787	839	889	908	3,423	922	999	1,021	977	3,919
Total hyperscale	**9,915**	**9,883**	**11,494**	**14,737**	**4,467**	**4,896**	**5,103**	**5,110**	**19,576**	**5,545**	**5,852**	**6,089**	**5,974**	**23,460**
% YoY	*12.4%*	*-0.3%*	*16.3%*	*28.2%*	*38.8%*	*51.2%*	*26.5%*	*20.3%*	*32.8%*	*24.1%*	*19.5%*	*19.3%*	*16.9%*	*19.8%*
% QoQ					*5.2%*	*9.6%*	*4.2%*	*0.1%*		*8.5%*	*5.5%*	*4.0%*	*-1.9%*	
YoY (%)														
Alphabet	*24.0%*	*4.4%*	*15.0%*	*48.0%*	*51.8%*	*6.8%*	*26.9%*	*34.1%*	*27.3%*	*35.8%*	*20.0%*	*31.0%*	*17.2%*	*25.5%*
AWS	*24.5%*	*-4.7%*	*6.8%*	*35.6%*	*40.0%*	*63.4%*	*20.0%*	*2.3%*	*27.0%*	*22.3%*	*22.9%*	*22.5%*	*24.0%*	*23.0%*
Microsoft	*-2.9%*	*20.6%*	*15.3%*	*67.4%*	*17.9%*	*29.3%*	*20.4%*	*35.9%*	*25.7%*	*25.0%*	*21.1%*	*20.5%*	*18.3%*	*21.0%*
Meta	*20.7%*	*-7.5%*	*8.0%*	*55.8%*	*23.0%*	*21.3%*	*23.9%*	*22.4%*	*22.6%*	*26.3%*	*18.5%*	*13.6%*	*20.0%*	*19.4%*
Baidu	*-34.0%*	*-1.6%*	*-47.9%*	*-16.5%*	*44.8%*	*42.5%*	*3.4%*	*68.6%*	*34.5%*	*30.0%*	*40.0%*	*18.2%*	*20.0%*	*26.8%*
Alibaba	*-10.6%*	*-50.2%*	*62.5%*	*49.6%*	*81.9%*	*10.3%*	*41.3%*	*13.8%*	*30.5%*	*17.0%*	*15.2%*	*12.9%*	*12.5%*	*14.3%*
Tencent	*-56.1%*	*25.9%*	*103.3%*	*-11.6%*	*2.3%*	*42.6%*	*103.2%*	*-12.2%*	*22.2%*	*9.3%*	*12.2%*	*13.6%*	*15.9%*	*12.8%*
ByteDance	*60.6%*	*53.8%*	*39.9%*	*69.7%*	*4.0%*	*63.3%*	*32.9%*	*16.0%*	*26.3%*	*25.0%*	*14.3%*	*20.9%*	*11.4%*	*17.6%*

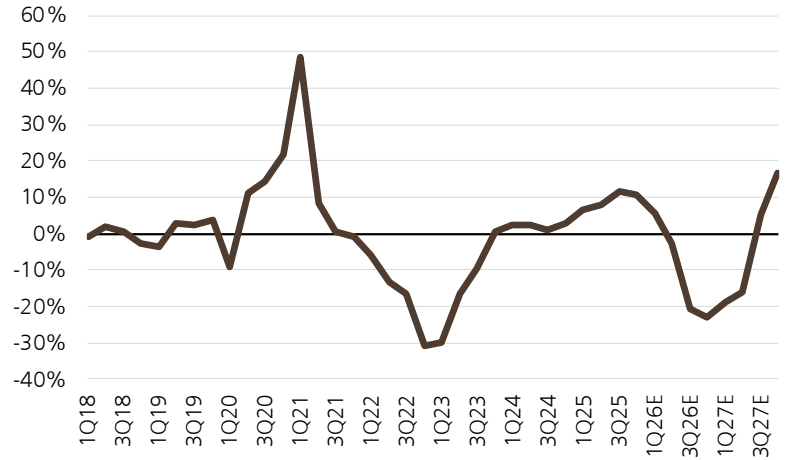
Source: Dell'Oro, UBS estimates

-11%/-4% YoY PC Unit Growth in '26E/'27E

UBS global PC model

(m units)		2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E
Desktop units		**108.0**	**102.3**	**99.8**	**100.3**	**79.8**	**85.3**	**76.1**	**62.4**	**61.1**	**68.3**	**60.5**	**58.8**
	% YoY	*-8.7%*	*-5.3%*	*-2.4%*	*0.5%*	*-20.4%*	*6.9%*	*-10.9%*	*-17.9%*	*-2.1%*	*11.8%*	*-11.4%*	*-2.7%*
Notebook units		**171.6**	**173.2**	**174.9**	**178.6**	**228.3**	**256.4**	**208.0**	**179.9**	**186.3**	**202.5**	**180.6**	**172.2**
	% YoY	*-2.7%*	*0.9%*	*1.0%*	*2.1%*	*27.8%*	*12.3%*	*-18.9%*	*-13.5%*	*3.6%*	*8.7%*	*-10.8%*	*-4.6%*
Chromebook		9.4	12.8	14.9	16.3	32.6	35.3	19.4	16.6	18.3	19.9	17.2	14.7
	% YoY	*38.2%*	*35.7%*	*16.8%*	*9.4%*	*100.0%*	*8.2%*	*-45.2%*	*-14.2%*	*10.2%*	*8.8%*	*-13.4%*	*-14.6%*
Consumer PC units		**123.3**	**111.1**	**105.9**	**98.4**	**117.5**	**130.2**	**105.3**	**88.4**	**86.9**	**93.8**	**82.5**	**77.3**
	% YoY	*-10.9%*	*-9.9%*	*-4.7%*	*-7.0%*	*19.4%*	*10.7%*	*-19.1%*	*-16.1%*	*-1.7%*	*7.9%*	*-12.1%*	*-6.3%*
Commercial PC units		**156.2**	**164.3**	**168.8**	**180.5**	**190.6**	**211.6**	**178.8**	**153.9**	**160.5**	**177.0**	**158.6**	**153.8**
	% YoY	*0.1%*	*5.2%*	*2.8%*	*6.9%*	*5.6%*	*11.0%*	*-15.5%*	*-13.9%*	*4.3%*	*10.3%*	*-10.4%*	*-3.0%*
Total units		**279.5**	**275.4**	**274.7**	**278.9**	**308.2**	**341.7**	**284.1**	**242.3**	**247.5**	**270.8**	**241.0**	**231.1**
	% YoY	***-5.1%***	***-1.5%***	***-0.3%***	***1.5%***	***10.5%***	***10.9%***	***-16.9%***	***-14.7%***	***2.1%***	***9.4%***	***-11.0%***	***-4.1%***

% YoY global PC unit growth

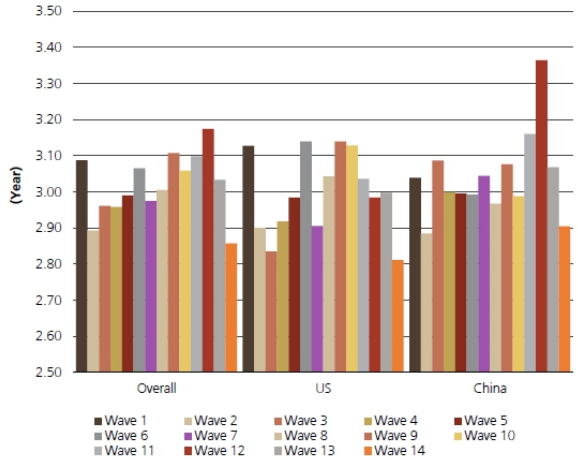


Source: Company data, Gartner, UBS estimates

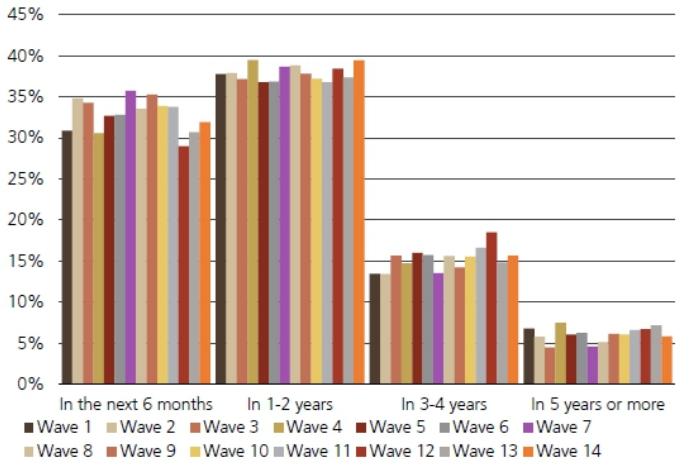
UBS Evidence Lab – Consumer PC Survey 3Q26

Powered by UBS Evidence Lab

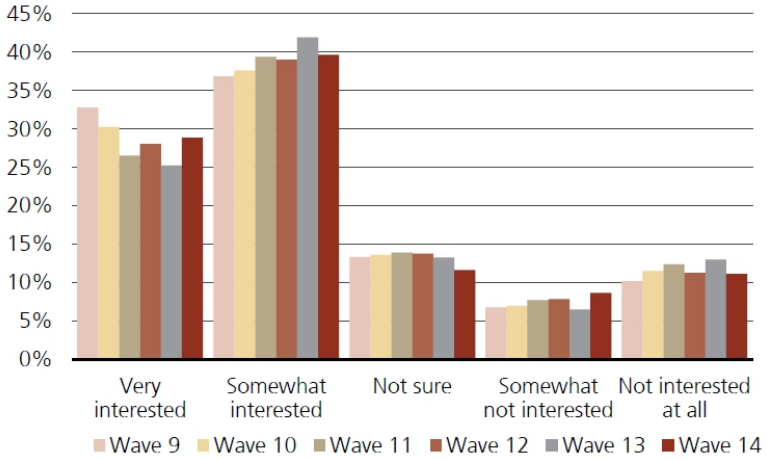
PC replacement cycle (years)



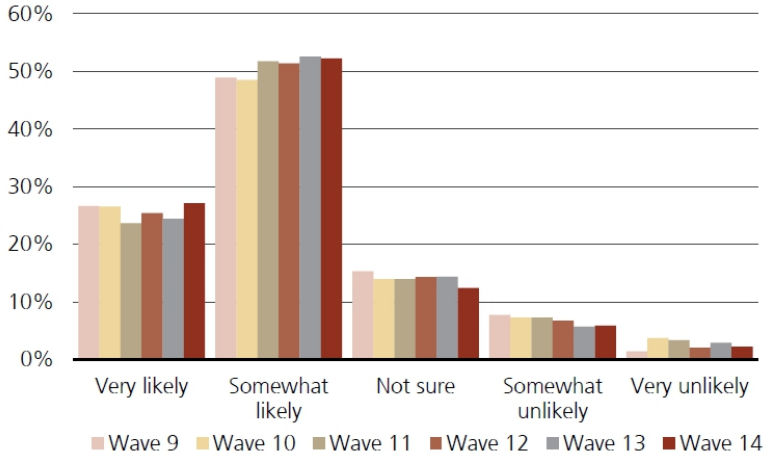
Forward purchase intent



Consumer interest in AI PC



Consumer purchase intent for AI PC



Source: UBS Evidence Lab

Notebook ODM Shipment Forecasts

('000 units)		2023	2024	1Q25	2Q25	3Q25	4Q25	2025	1Q26	2Q26	3Q26E	4Q26E	2026E
Quanta		**46,900**	**45,900**	**10,800**	**12,100**	**12,700**	**10,900**	**46,500**	**10,000**	**11,500**	**9,000**	**8,500**	**39,000**
	%QoQ			*-3%*	*12%*	*5%*	*-14%*		*-8%*	*15%*	*-22%*	*-6%*	
	%YoY	*-18.7%*	*-2.1%*	*3%*	*3%*	*1%*	*-2%*	*1.3%*	*-7%*	*-5%*	*-29%*	*-22%*	*-16.1%*
Compal		**33,900**	**32,300**	**7,000**	**7,100**	**7,100**	**6,800**	**28,000**	**5,900**	**6,200**	**5,200**	**5,700**	**23,000**
	%QoQ			*-10%*	*1%*	*0%*	*-4%*		*-13%*	*5%*	*-16%*	*10%*	
	%YoY	*-13.7%*	*-4.7%*	*-7%*	*-18%*	*-14%*	*-13%*	*-13.3%*	*-16%*	*-13%*	*-27%*	*-16%*	*-17.9%*
Pegatron		**7,925**	**8,055**	**1,885**	**2,375**	**2,400**	**2,410**	**9,070**	**1,800**	**1,750**	**1,425**	**1,750**	**6,725**
	%QoQ			*-3%*	*26%*	*1%*	*0%*		*-25%*	*-3%*	*-19%*	*23%*	
	%YoY	*-1.9%*	*1.6%*	*22%*	*16%*	*-4%*	*24%*	*12.6%*	*-5%*	*-26%*	*-41%*	*-27%*	*-25.9%*
Inventec		**18,700**	**20,000**	**5,000**	**5,600**	**5,400**	**5,300**	**21,300**	**5,400**	**5,400**	**4,800**	**4,400**	**20,000**
	%QoQ			*-7%*	*12%*	*-4%*	*-2%*		*2%*	*0%*	*-11%*	*-8%*	
	%YoY	*-5.6%*	*7.0%*	*11%*	*14%*	*4%*	*-2%*	*6.5%*	*8%*	*-4%*	*-11%*	*-17%*	*-6.1%*
Wistron		**19,000**	**20,200**	**4,900**	**5,900**	**6,400**	**6,900**	**24,100**	**6,100**	**5,500**	**4,650**	**4,750**	**21,000**
	%QoQ			*-8%*	*20%*	*8%*	*8%*		*-12%*	*-10%*	*-15%*	*2%*	
	%YoY	*-10.4%*	*6.3%*	*7%*	*16%*	*23%*	*30%*	*19.3%*	*24%*	*-7%*	*-27%*	*-31%*	*-12.9%*
Total		**126,425**	**126,455**	**29,585**	**33,075**	**34,000**	**32,310**	**128,970**	**29,200**	**30,350**	**25,075**	**25,100**	**109,725**
	%QoQ			*-6%*	*12%*	*3%*	*-5%*		*-10%*	*4%*	*-17%*	*0%*	
	%YoY	*-13%*	*0%*	*3%*	*2%*	*1%*	*2%*	*2%*	*-1%*	*-8%*	*-26%*	*-22%*	*-15%*

Source: Company data, UBS estimates

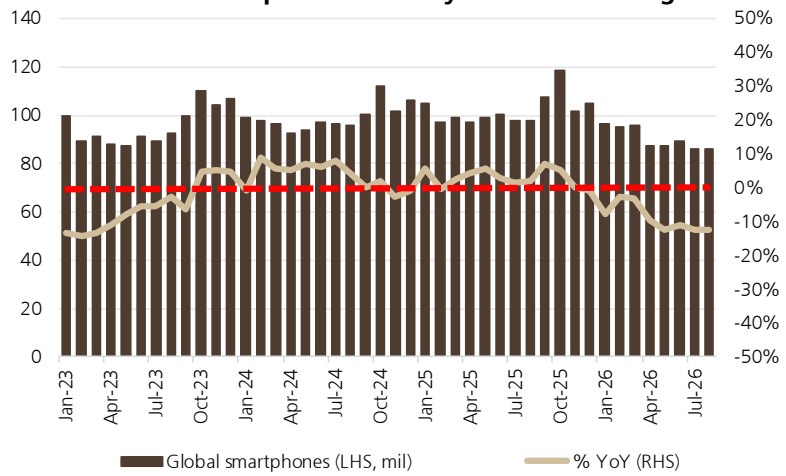
-10%/-3%/+2% Smartphone Unit Growth in ’26E/’27E/’28E

Smartphone volume (m)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	1Q28E	2Q28E	3Q28E	4Q28E	2028E
Samsung	**240**	**60**	**65**	**59**	**57**	**240**	**64**	**60**	**62**	**57**	**243**	**65**	**62**	**64**	**60**	**251**
% YoY	*7%*	*-2%*	*12%*	*-3%*	*-5%*	*0%*	*7%*	*-7%*	*5%*	*0%*	*1%*	*2%*	*3%*	*3%*	*5%*	*4%*
% of total	*19%*	*20%*	*23%*	*22%*	*19%*	*21%*	*23%*	*22%*	*23%*	*19%*	*22%*	*23%*	*23%*	*23%*	*20%*	*22%*
Apple	**241**	**60**	**56**	**55**	**90**	**261**	**63**	**59**	**58**	**94**	**273**	**66**	**61**	**60**	**96**	**283**
% YoY	*9%*	*17%*	*23%*	*3%*	*0%*	*8%*	*5%*	*5%*	*5%*	*4%*	*5%*	*4%*	*4%*	*4%*	*3%*	*4%*
% of total	*19%*	*20%*	*20%*	*20%*	*30%*	*23%*	*23%*	*22%*	*22%*	*32%*	*25%*	*24%*	*22%*	*22%*	*32%*	*25%*
Huawei	**48**	**14**	**16**	**14**	**13**	**56**	**14**	**16**	**14**	**13**	**57**	**14**	**16**	**14**	**14**	**58**
% YoY	*0%*	*3%*	*24%*	*27%*	*12%*	*16%*	*4%*	*3%*	*0%*	*0%*	*2%*	*0%*	*0%*	*0%*	*8%*	*2%*
% of total	*4%*	*5%*	*6%*	*5%*	*4%*	*5%*	*5%*	*6%*	*5%*	*4%*	*5%*	*5%*	*6%*	*5%*	*5%*	*5%*
OPPO*	**102**	**31**	**29**	**21**	**18**	**99**	**22**	**22**	**22**	**22**	**88**	**22**	**22**	**22**	**22**	**88**
% YoY	*-3%*	*31%*	*20%*	*-24%*	*-33%*	*-3%*	*-28%*	*-24%*	*5%*	*22%*	*-11%*	*0%*	*0%*	*0%*	*0%*	*0%*
% of total	*8%*	*10%*	*10%*	*8%*	*6%*	*9%*	*8%*	*8%*	*8%*	*7%*	*8%*	*8%*	*8%*	*8%*	*7%*	*8%*
Vivo	**106**	**21**	**21**	**20**	**21**	**83**	**19**	**19**	**19**	**19**	**76**	**19**	**19**	**19**	**19**	**76**
% YoY	*6%*	*-7%*	*-22%*	*-31%*	*-22%*	*-21%*	*-10%*	*-10%*	*-5%*	*-10%*	*-9%*	*0%*	*0%*	*0%*	*0%*	*0%*
% of total	*8%*	*7%*	*8%*	*7%*	*7%*	*7%*	*7%*	*7%*	*7%*	*6%*	*7%*	*7%*	*7%*	*7%*	*6%*	*7%*
Xiaomi	**165**	**34**	**31**	**29**	**29**	**123**	**29**	**28**	**27**	**27**	**111**	**31**	**30**	**29**	**29**	**119**
% YoY	*-2%*	*-19%*	*-26%*	*-32%*	*-24%*	*-26%*	*-14%*	*-10%*	*-8%*	*-7%*	*-10%*	*7%*	*7%*	*7%*	*10%*	*8%*
% of total	*13%*	*12%*	*11%*	*11%*	*10%*	*11%*	*11%*	*10%*	*10%*	*9%*	*10%*	*11%*	*11%*	*11%*	*10%*	*11%*
Transsion	**96**	**19**	**20**	**18**	**17**	**74**	**19**	**20**	**19**	**18**	**75**	**19**	**21**	**20**	**21**	**81**
% YoY	*-9%*	*-10%*	*-16%*	*-37%*	*-25%*	*-23%*	*0%*	*-3%*	*1%*	*6%*	*1%*	*0%*	*8%*	*8%*	*18%*	*8%*
% of total	*7%*	*6%*	*7%*	*6%*	*5%*	*6%*	*7%*	*7%*	*7%*	*6%*	*6%*	*6%*	*7%*	*7%*	*7%*	*7%*
Lenovo	**55**	**13**	**13**	**13**	**13**	**51**	**12**	**12**	**14**	**13**	**50**	**12**	**12**	**13**	**13**	**51**
% YoY	*3%*	*6%*	*-8%*	*-8%*	*-15%*	*-7%*	*-10%*	*-6%*	*3%*	*3%*	*-3%*	*5%*	*6%*	*-2%*	*0%*	*2%*
% of total	*4%*	*4%*	*5%*	*5%*	*4%*	*5%*	*4%*	*4%*	*5%*	*4%*	*5%*	*4%*	*5%*	*5%*	*4%*	*5%*
Honor	**69**	**19**	**14**	**14**	**14**	**61**	**14**	**14**	**14**	**14**	**56**	**14**	**14**	**14**	**14**	**56**
% YoY	*8%*	*12%*	*-13%*	*-20%*	*-24%*	*-12%*	*-26%*	*0%*	*0%*	*0%*	*-8%*	*0%*	*0%*	*0%*	*0%*	*0%*
% of total	*5%*	*6%*	*5%*	*5%*	*5%*	*5%*	*5%*	*5%*	*5%*	*5%*	*5%*	*5%*	*5%*	*5%*	*5%*	*5%*
Other	**143**	**24**	**12**	**29**	**25**	**89**	**18**	**19**	**20**	**20**	**77**	**15**	**16**	**17**	**16**	**64**
% YoY	*8%*	*-42%*	*-65%*	*-22%*	*-22%*	*-38%*	*-26%*	*60%*	*-30%*	*-17%*	*-13%*	*-17%*	*-16%*	*-15%*	*-23%*	*-18%*
% of total	*11%*	*8%*	*4%*	*10%*	*8%*	*8%*	*7%*	*7%*	*7%*	*7%*	*7%*	*5%*	*6%*	*6%*	*5%*	*6%*
Total	**1,265**	**294**	**276**	**273**	**295**	**1,138**	**273**	**268**	**268**	**296**	**1,105**	**277**	**274**	**272**	**304**	**1,127**
% YoY	*3%*	*-4%*	*-7%*	*-16%*	*-13%*	*-10%*	*-7%*	*-3%*	*-2%*	*0%*	*-3%*	*1%*	*2%*	*2%*	*3%*	*2%*

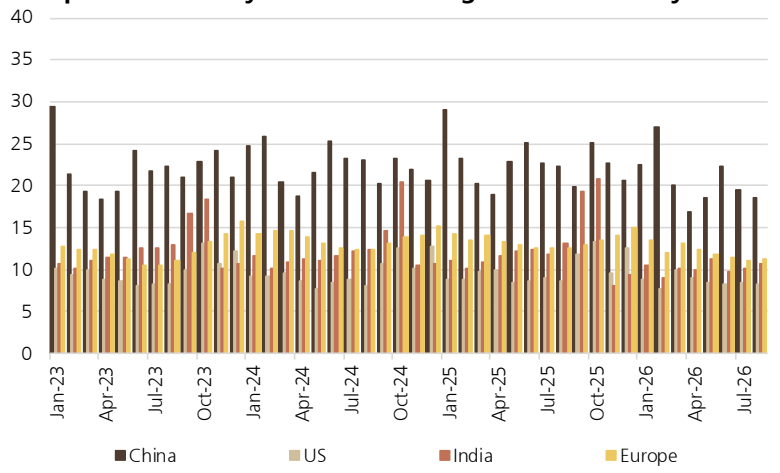
Source: Company data, Gartner, UBS estimates. *OPPO numbers include OnePlus but excludes Realme.

Smartphone Monthly Sell-Through – 1

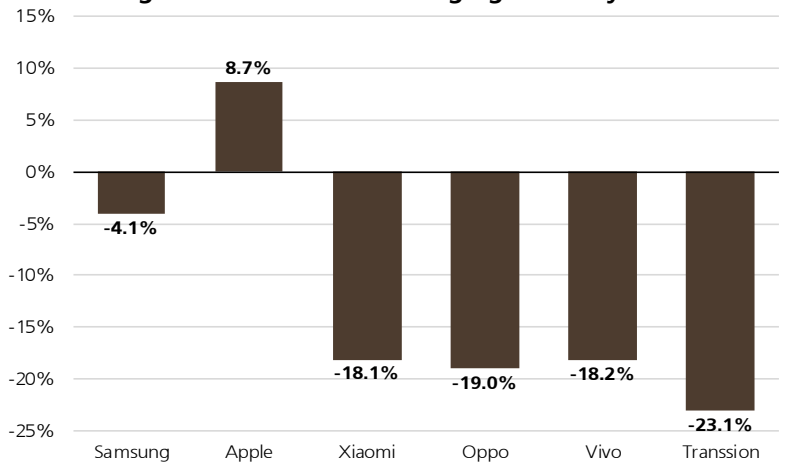
Global smartphone monthly units sell-through



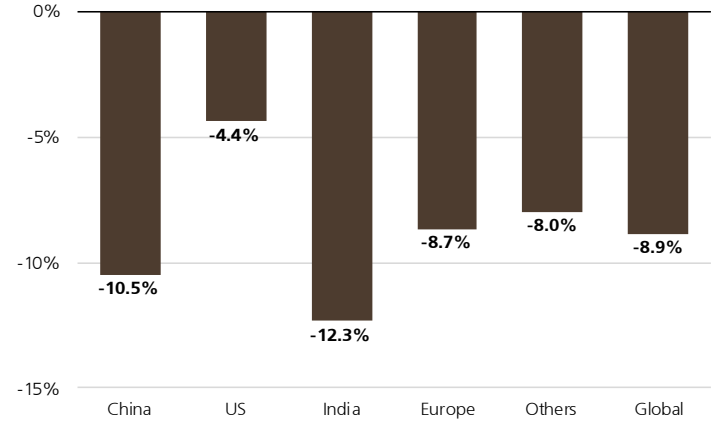
Smartphone monthly units sell-through breakdown by markets



Aug '26 YTD YoY sell-through growth by OEM

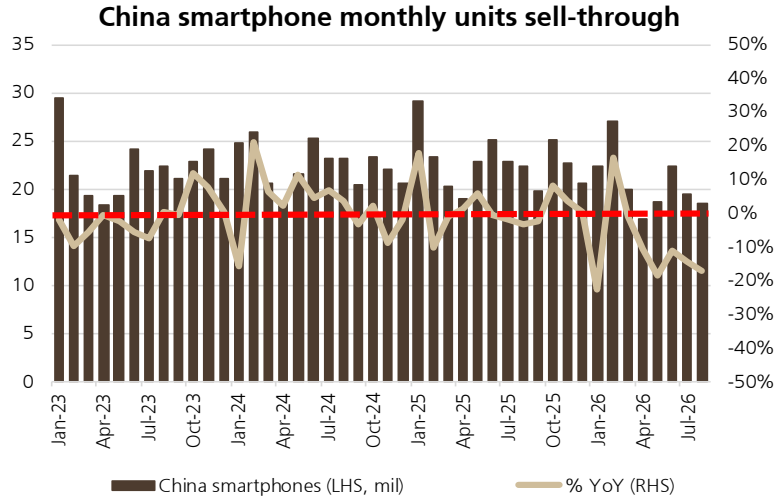


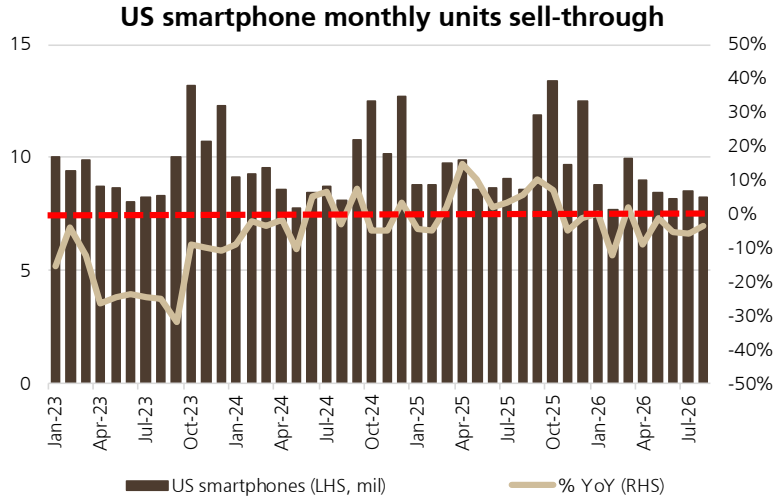
Aug '26 YTD YoY sell-through growth by region

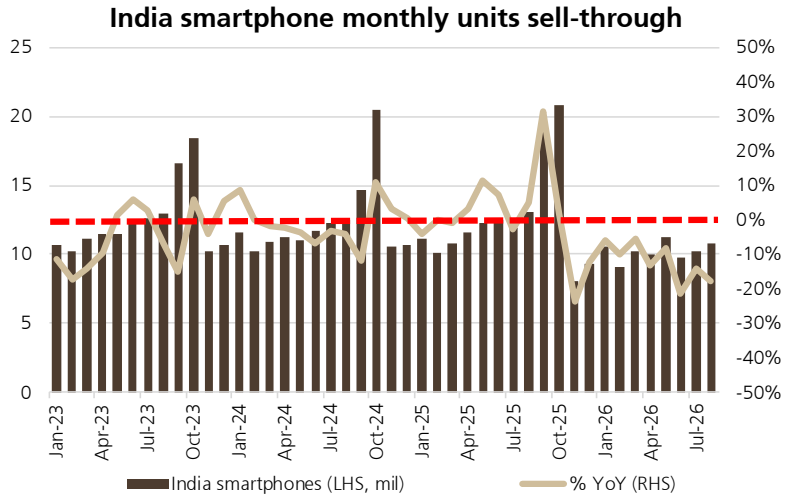


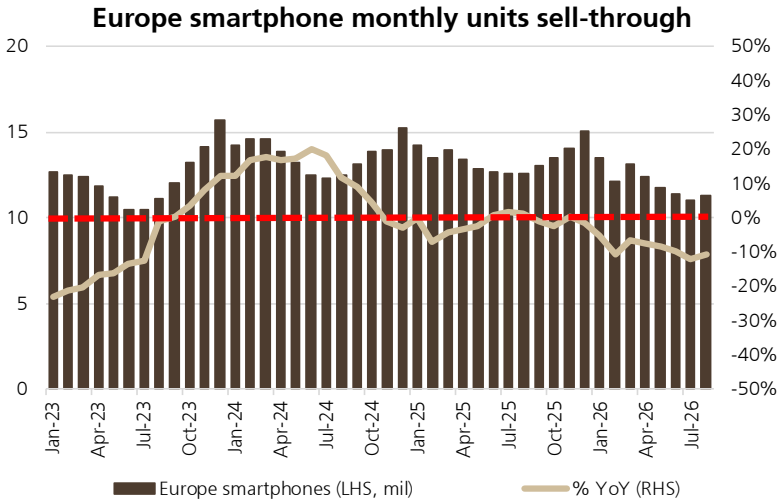
Source: Counterpoint

Smartphone Monthly Sell-Through – 2







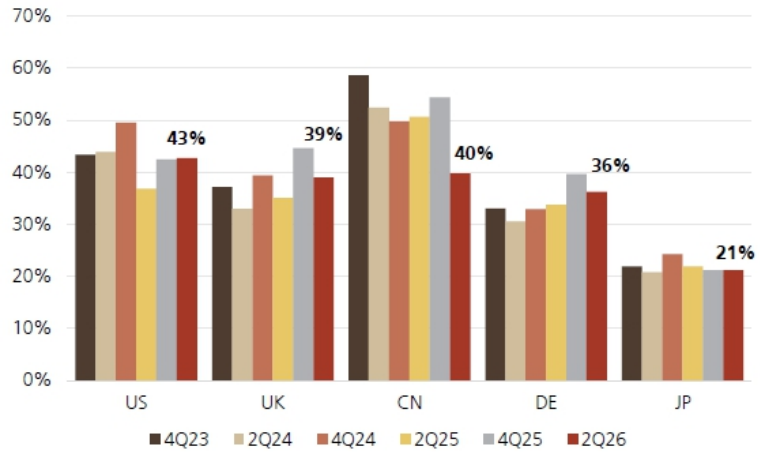


Source: Counterpoint

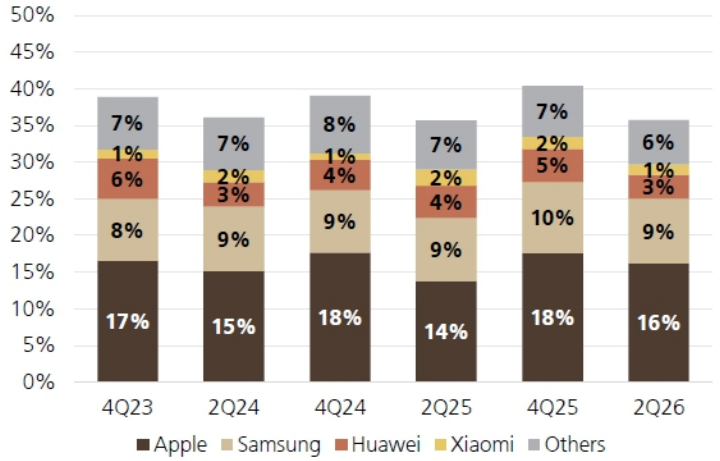
UBS Evidence Lab – Smartphones Survey 2Q26 – 1

Powered by UBS Evidence Lab

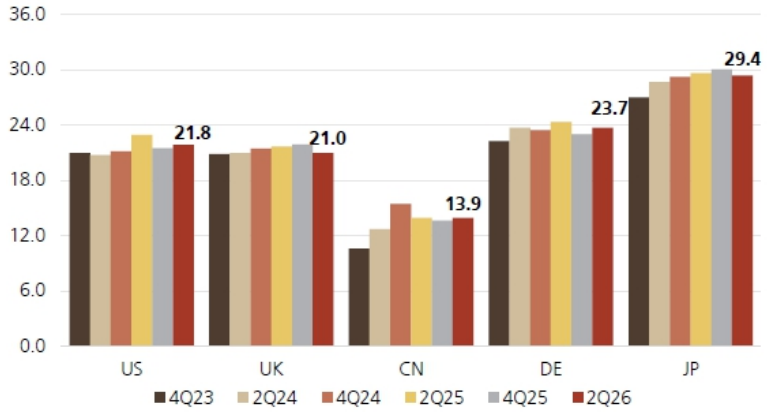
12-month forward purchasing intent (HoH)



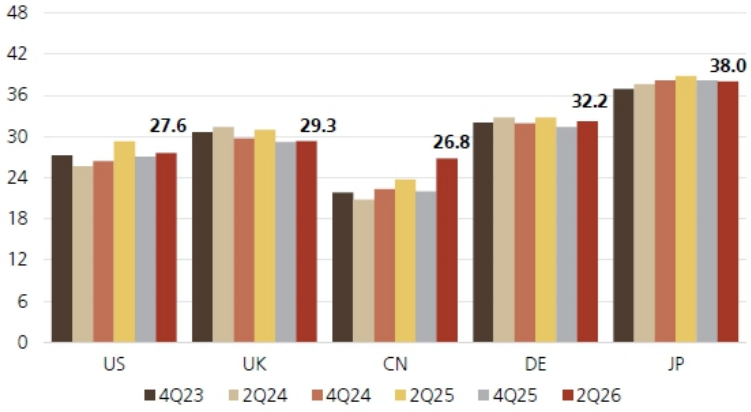
12-month forward purchase by OEM



Average age of installed base by region (months)



Expected pace of smartphone replacement (months)

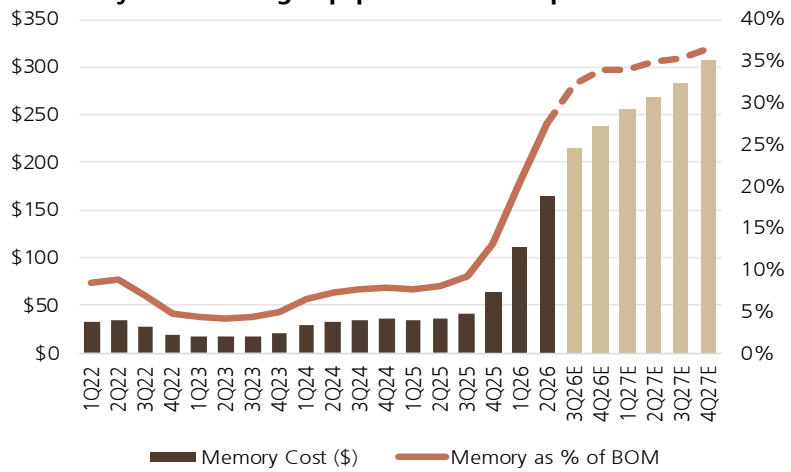


Source: UBS Evidence Lab

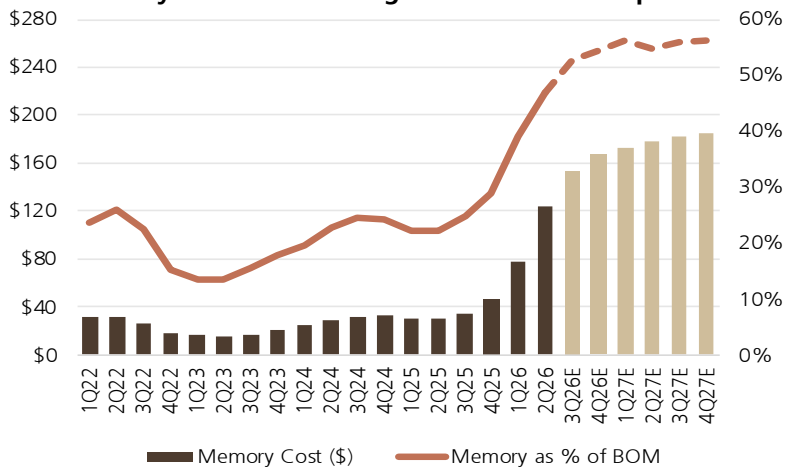
UBS Evidence Lab – Smartphones Survey 2Q26 – 2

Powered by UBS Evidence Lab

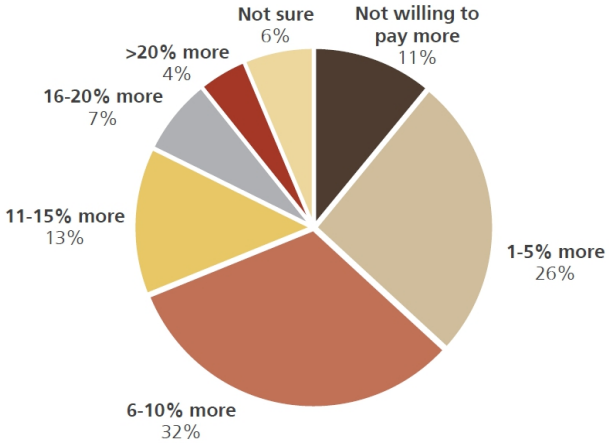
Memory as % of flagship/premium smartphone BOM costs



Memory as % of mid-range/lower-end smartphones



Consumer willingness to accommodate ASP hike



Source: UBS Evidence Lab

Global Semis Revs by Product Segment (US$bn, Sell-In)

Revenues (US$bn)	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
MPU	**48**	**52**	**58**	**51**	**45**	**54**	**60**	**81**	**110**	**137**
% YoY	*2%*	*8%*	*11%*	*-12%*	*-11%*	*19%*	*11%*	*35%*	*35%*	*25%*
MCU	**16**	**15**	**20**	**25**	**28**	**22**	**22**	**25**	**28**	**31**
% YoY	*-7%*	*-2%*	*27%*	*27%*	*11%*	*-19%*	*-4%*	*16%*	*13%*	*8%*
Analog	**54**	**56**	**74**	**89**	**81**	**80**	**87**	**103**	**123**	**135**
% YoY	*-8%*	*3%*	*33%*	*20%*	*-9%*	*-2%*	*9%*	*19%*	*19%*	*10%*
CMOS Logic	**109**	**121**	**158**	**178**	**176**	**190**	**253**	**365**	**415**	**473**
% YoY	*-3%*	*11%*	*31%*	*13%*	*-1%*	*8%*	*33%*	*44%*	*14%*	*14%*
Memory	**106**	**117**	**154**	**130**	**92**	**165**	**230**	**972**	**1,702**	**1,909**
% YoY	*-33%*	*10%*	*31%*	*-16%*	*-29%*	*79%*	*39%*	*322%*	*75%*	*12%*
Total IC	**333**	**361**	**463**	**473**	**423**	**511**	**651**	**1,546**	**2,378**	**2,685**
% YoY	*-15%*	*8%*	*28%*	*2%*	*-10%*	*21%*	*27%*	*137%*	*54%*	*13%*
Discrete, Opto & Passives	**79**	**79**	**93**	**100**	**98**	**91**	**95**	**95**	**98**	**100**
% YoY	*5%*	*0%*	*17%*	*7%*	*-1%*	*-7%*	*4%*	*0%*	*3%*	*3%*
Total Semiconductors	**412**	**440**	**556**	**572**	**521**	**602**	**746**	**1,641**	**2,475**	**2,785**
% YoY	*-12%*	*7%*	*26%*	*3%*	*-9%*	*15%*	*24%*	*120%*	*51%*	*13%*
Semis ex memory	**306**	**323**	**402**	**443**	**429**	**437**	**516**	**669**	**773**	**876**
% YoY	*-2%*	*6%*	*25%*	*10%*	*-3%*	*2%*	*18%*	*30%*	*16%*	*13%*
Semis ex memory ex CPU	**258**	**271**	**344**	**392**	**384**	**383**	**456**	**588**	**663**	**739**
% YoY	*-2%*	*5%*	*27%*	*14%*	*-2%*	*0%*	*19%*	*29%*	*13%*	*11%*

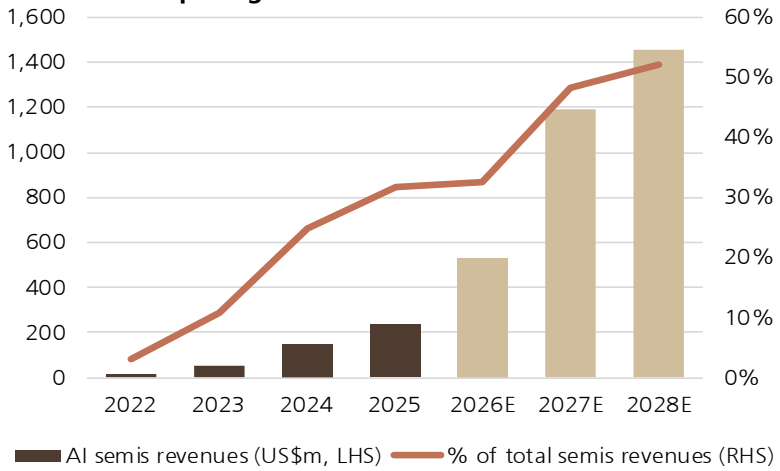
Source: Company data, UBS estimates

AI Cloud Computing Semis Revenues

Semis industry revenues breakdown

	2023	2024	2025	2026E	2027E	2028E
Total semis industry revenues (US$ bn)	**521**	**602**	**746**	**1,641**	**2,475**	**2,785**
YoY growth (%)	*-100%*	*15%*	*24%*	*120%*	*51%*	*13%*
AI cloud computing semis revenues (US$ bn)	**56**	**150**	**237**	**532**	**1,192**	**1,453**
YoY growth (%)	*-100%*	*168%*	*59%*	*124%*	*124%*	*22%*
As % of total semis revenues	***11%***	***25%***	***32%***	***32%***	***48%***	***52%***
Semis industry revenues ex-AI cloud computing (US$ bn)	**465**	**452**	**508**	**1,109**	**1,284**	**1,332**
YoY growth (%)	*-100%*	*-3%*	*12%*	*118%*	*16%*	*4%*
As % of total semis revenues	***89%***	***75%***	***68%***	***68%***	***52%***	***48%***

AI cloud computing semis revs and as % of total semis revs



Source: Company data, UBS estimates

Foundry Capacity Forecasts

Leading-edge foundry capacity forecasts

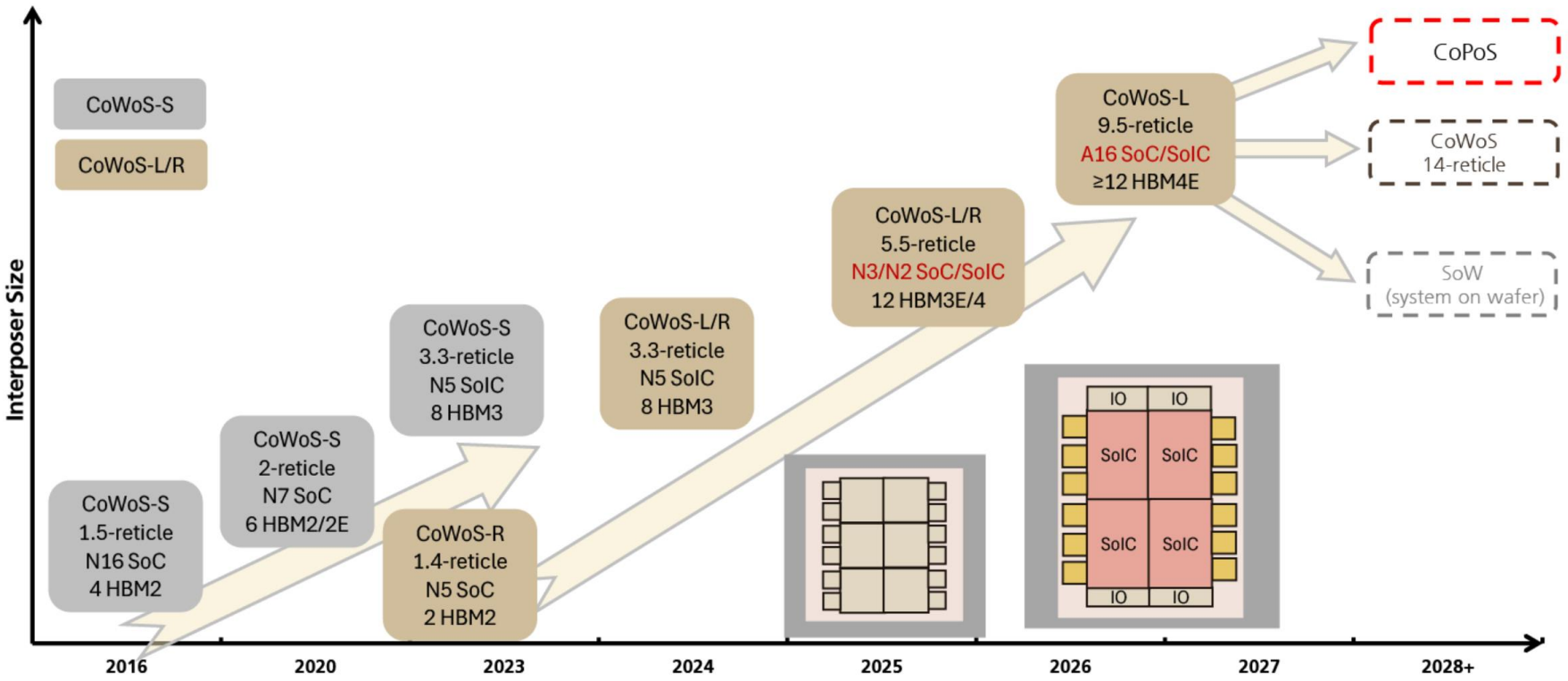
Period-end capacity (kwpm)	2019	2020	2021	2022	2023	2024	2025	2026E	2027E
FinFET foundry capacity (kwpm)									
TSMC	235.0	280.0	335.0	405.0	445.0	470.0	570.0	650.0	740.0
Samsung Foundry	80.0	120.0	160.0	175.0	180.0	190.0	193.0	215.0	255.0
Intel	81.3	111.0	151.3	100.3	175.3	199.5	225.6	232.5	205.9
UMC	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	13.0
GlobalFoundries	22.0	22.0	22.0	22.0	22.0	22.0	22.0	18.0	16.0
China	3.0	11.0	15.0	15.0	15.0	20.0	21.3	36.3	66.3
Total FinFET supply	**424.3**	**547.0**	**686.3**	**720.3**	**840.3**	**904.5**	**1,035.0**	**1,154.8**	**1,296.2**
YoY	*39%*	*29%*	*25%*	*5%*	*17%*	*8%*	*14%*	*12%*	*12%*

N2 foundry capacity forecasts

Period-end capacity (kwpm)	2025	2026E	2027E	2028E
2nm				
TSMC	40.0	90.0	140.0	210.0
Samsung Foundry	3.0	10.0	25.0	40.0
Intel	4.0	11.0	35.0	61.7
Total wafer supply	**47.0**	**111.0**	**200.0**	**311.7**
YoY		*136%*	*80%*	*56%*

Source: Company data, UBS estimates

TSMC's Tech Roadmap for Advanced Packaging



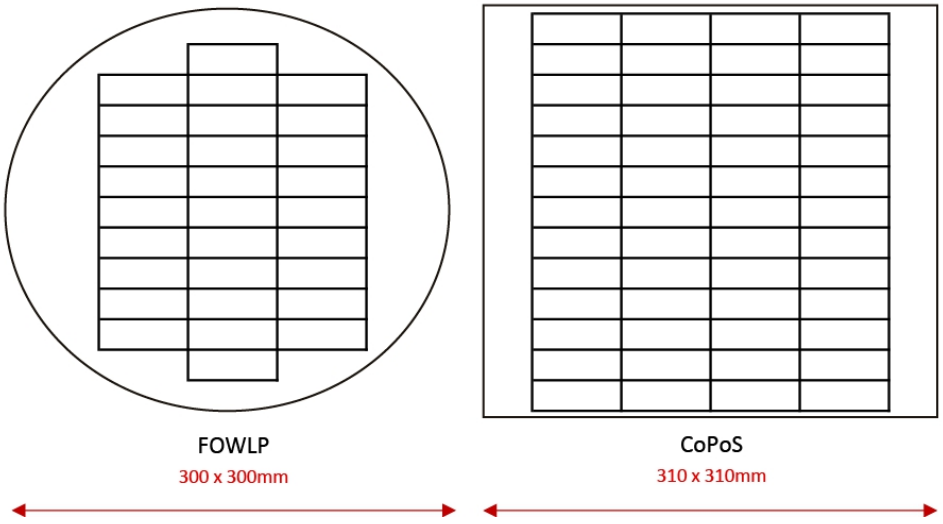
Source: Company data, UBS estimates

CoPoS vs EMIB-T

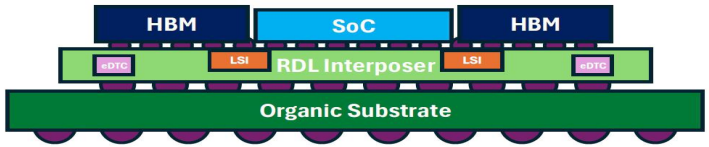
Advanced packaging is key to scaling Cloud AI performance while improving power efficiency.

- **CoPoS** (Chip-on-Panel-on-Substrate) is TSMC's next-generation 2.5D packaging technology that uses panel-level integration to support larger package sizes, higher interconnect density, and improved manufacturing scalability.
- **EMIB-T** is Intel's advanced 2.5D packaging technology that replaces large silicon interposers with embedded silicon bridges and adds TSV-based vertical power delivery to support higher-power AI packages.

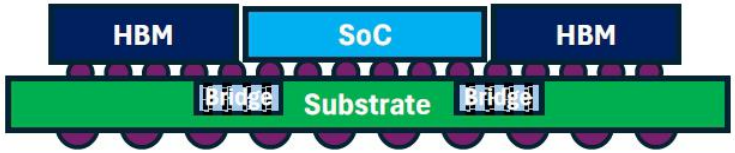
CoPoS (or panel-level packaging) has better area utilisation and production efficiency than CoWoS



CoPoS structure: interposer-based, embedded eDTC, LSI inside/underneath the interposer



EMIB-T with silicon bridge and TSV for vertical power delivery



Source: Company data, UBS

CoPoS vs EMIB-T

CoPoS

- **Architecture**

– TSMC's next gen 2.5D platform

– Migrates from wafer to square-panel format

– Die attached on interposer/panel before substrate integration

- **Timeline/Customer**

– Mass production targeted in FY28 with first-gen on 310x310mm² panel, further enhancing to 510 x 515mm²

– NVIDIA Feynman expected as first major customer, followed by AMD and ASICs in 2029-2030

CoPoS: Supply Chain

TSMC CoPoS supply chain	Suppliers
Deposition and re-distribution layer	Applied Materials, SCREEN, Tokyo Electron, Canon
Baking and thermal treatment	APT, C Sun, GMM, Kokusai Electric
Wet process	GFTC, Scientech
Debonding	SUSS MicroTec, Tokyo Electron, EVG, Scientech, E&R engineering, Daxin
CMP	Applied Materials, Disco, Ebara, LAM research, Tokyo Electron, GPM
Molding equipment	Apic Yamada, Towa
Pod	Gudeng, Chung King
Underfill dispenser	All Ring, Asymtek
Die attach	Kulicke & Soffa, ASMPT, Shibaura
Metrology, inspection, and automation	KLA, Chroma, Camtek, Favite, Ta Liang Technology, V5 Technologies
OSAT partner	ASE/SPIL
Substrate	Ibiden, Unimicron, Toppan, Shinko, Nanya PCB, Kinsus

EMIB-T

- **Architecture**

– Compute dies/HBMs attached directly to substrate, elimination the silicon interposer

– Incorporates TSVs (through silicon vias) for vertical power delivery to HBM & compute dies.

- **Timeline/Customer**

– Introduced in late 2024 as Intel's next-gen EMIB platform

– Targets H227 commercial ramp; Google TPU v9 expected as first major external product

EMIB-T: Supply Chain

Intel EMIB-T supply chain	Suppliers
Substrate	Ibiden, Unimicron
OSAT partner	Amkor
Die attach	Kulicke & Soffa, ASMPT

Source: Company data, UBS estimates

Co-Packaged Optics

- **Co-packaged optics (CPO)**

– CPO integrates silicon-photonics optical engines directly with switch ASICs or AI accelerators, reducing copper trace lengths to improve power efficiency and increase bandwidth density.

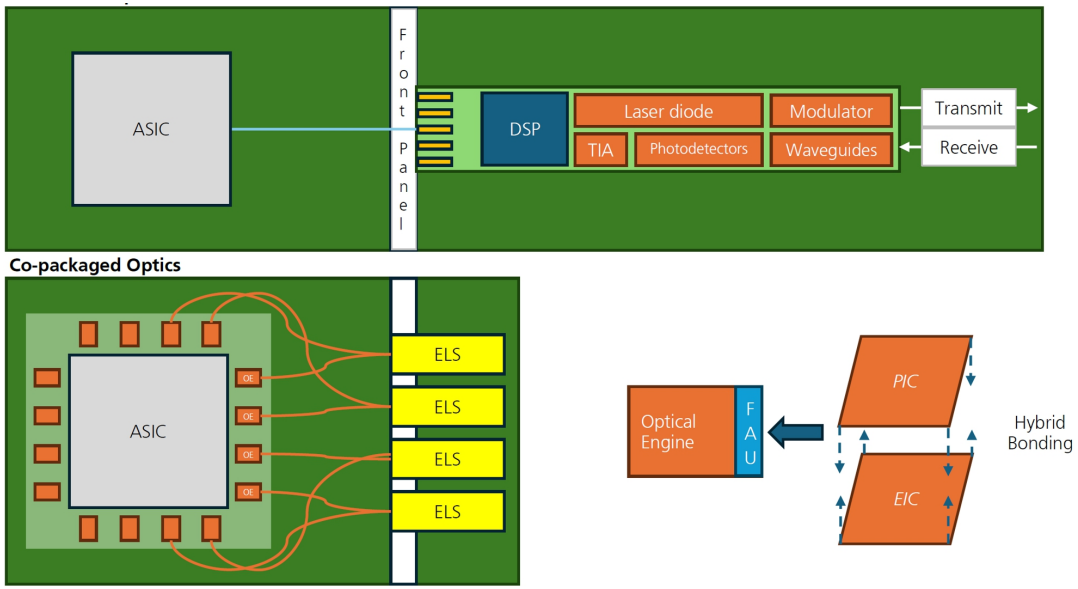
- **Key Components**

– **Optical engine (OE),** which combines a photonic IC (PIC) and electronic IC (EIC), is co-packaged with the ASIC to enable electrical-to-optical and optical-to-electrical signal conversion, allowing direct communication with fiber-optic links.

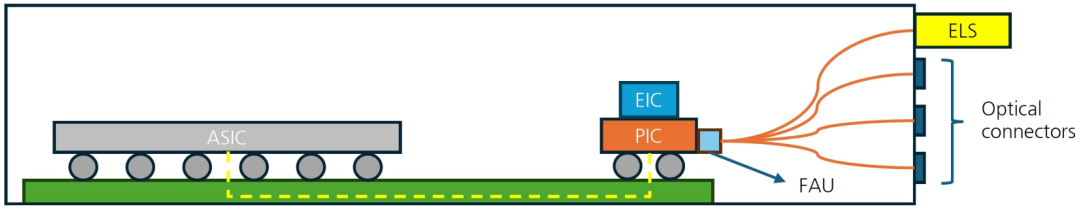
– **PIC** integrates optical functions such as waveguides, modulators, and photodiodes on a single chip, typically combining multiple material platforms, including silicon for optical routing and modulation and SiGe for photodetection.

– **Fibre array unit (FAU)** couple optical fibers to the optical engine (OE), routing incoming light to the PIC for signal reception and external-laser light to the modulators for data transmission, while requiring high-temperature-resistant materials for CPO assembly.

Traditional optical transceivers vs CPO structure



Simplified side view of the CPO system in the switch box



Source: Company data, UBS

Co-Packaged Optics – Comparison&Timeline

Comparison—fibre optics vs copper cable

	Fiber Optics	Copper
Bandwidth	More suitable for 800G /1.6T and above for longer reach interconnect.	In 800G /1.6T, the use should be limited to short-reach networking.
Transmission distance	Up to 100km. Single mode fiber could transmit data for longer distance than multi-mode fiber.	In general, <100m. For 800G /1.6T, the transmission distance could be <10m.
Environmental factor tolerance	Less susceptible to temperature /moisture	Susceptible to temperature and moisture
EMI	Immune	Susceptible to EM interference.
Useful life	30-50 years	5 years
Handling	Thinner, lighter, stronger pulling strength	Thicker, heavier, more prone to damage and breakage
Security	Better	Weaker - possibility to be tapped or causing fire when damaged

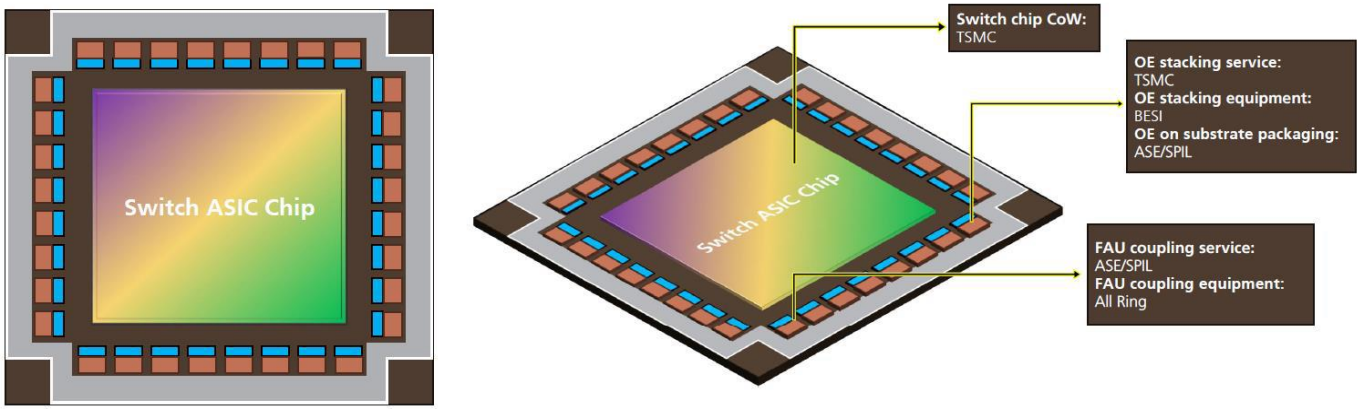
When will CPO inflection occur?

- For **scale out** (interconnect between racks), Nvidia may start to push CPO switch (Spectrum X) from H226E for Rubin racks
- For **scale up** (interconnect within a rack), Nvidia may adopt a hybrid approach to first upgrade the NVLINK switches to CPO from Kyber architecture (Rubin Ultra, Feynman) while GPU compute trays continue to use copper
- For the **CPO scale up for GPUs**, the technology may mature for adoption in 2029E, if not earlier.

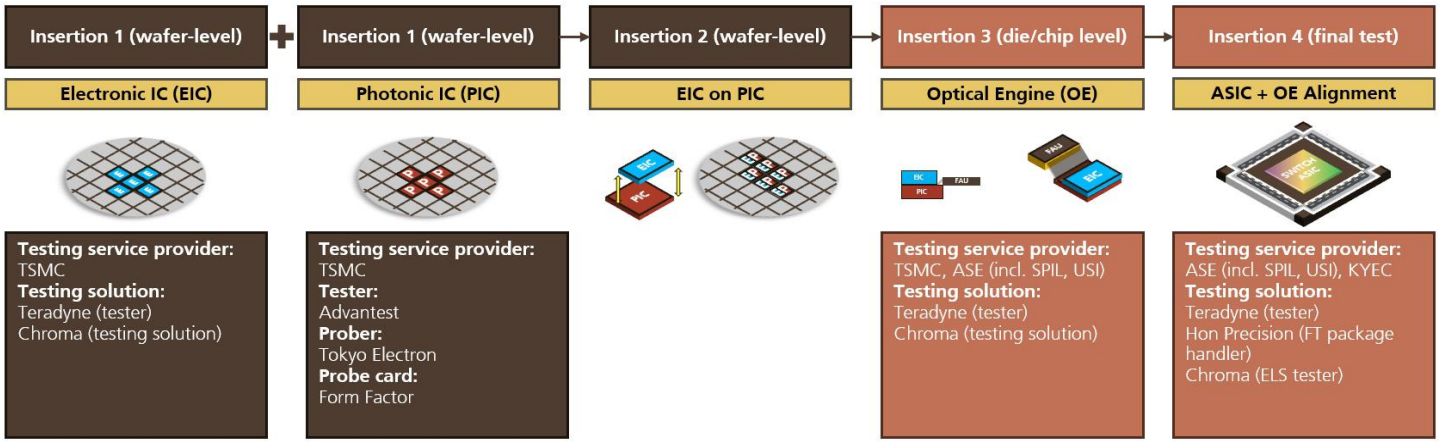
Source: Company data, UBS

Co-Packaged Optics – Process Flow

Nvidia's CPO packaging layout and notable suppliers



Nvidia's CPO test flow chart and notable suppliers



Source: Company data, UBS

Co-Packaged Optics – Revenue Opportunities

Optical transceiver SAM

	2025E	2026E	2027E	2028E	2029E	2030E	25-30E CAGR
Base case							
Market size (US$bn)	**12.0**	**15.6**	**19.5**	**23.4**	**28.1**	**32.3**	
Growth		*30%*	*25%*	*20%*	*20%*	*15%*	*22%*
o/w pluggable transceivers	**12.0**	**15.6**	**19.1**	**22.2**	**25.3**	**25.8**	
Share in optical transceivers	100%	100%	98%	95%	90%	80%	
o/w CPO	**0.0**	**0.0**	**0.4**	**1.2**	**2.8**	**6.5**	
Share in optical transceivers	0%	0%	2%	5%	10%	20%	
Upside scenario							
Market size (US$bn)	**15.0**	**20.3**	**26.7**	**34.7**	**44.1**	**55.2**	
Growth		*35%*	*32%*	*30%*	*27%*	*25%*	*30%*
o/w pluggable transceivers	**15.0**	**20.3**	**26.2**	**32.3**	**38.4**	**41.4**	
Share in optical transceivers	100%	100%	98%	93%	87%	75%	
o/w CPO	**0.0**	**0.0**	**0.5**	**2.4**	**5.7**	**13.8**	
Share in optical transceivers	0%	0%	2%	7%	13%	25%	

Notable CPO Suppliers

CPO Value Chain	Notable Suppliers
EpiWafers	VPEC, LandMark, IQE, Coherent, Lumentum
Fibers/cables	Corning, BizLink
FAUs	FOCI, Browave, TFC Communication, Senko, Sumitomo
Assembly, packaging, and testing	ASE/SPIL, FOCI, Amkor, ShunSin, Fabrinet, Foxconn, FIT
ELS lasers	Coherent, Lumentum, Sumitomo, Furukawa, Broadcom, LandMark, GCS
Optical engines	Nvidia, Broadcom, Marvell, Lightmatter, Ayar Labs
Switch ASICs/SerDes IP	Broadcom, Marvell, Nvidia, MediaTek
Transimpedance Amplifier (TIA)	Airoha, Broadcom, Marvell
Chip fabrication	TSMC, GlobalFoundries, Intel, Tower, UMC
Equipment vendors	BESI, ASMPT, K&S, All Ring, GPTC, Chroma, Hon Precision, Teradyne, Advantest, Tokyo Electron

Source: Company data, UBS estimates

Memory Semis: Key Debates

Q #1: Any softening in the memory upcycle near term?

UBS View: No. Undersupply could deepen into 2028

- Upside to near term pricing: OEMs still have to take on price increases to secure supply – e.g. Mobile DRAM contract pricing up c. 40% QoQ in 3Q26E. LTAs still in negotiations for many of them
- DRAM blended contract ASPs looking up 25% QoQ in 3Q26, +9% in 4Q (with upside); NAND +20% / +8% respectively
- DRAM overall fulfilment ratio only 60% currently
- Server DDR5 and server SSD demand forecasts for 2027 keep being revised higher. We forecast server DDR bit demand up 81% YoY in 2027, server SSD +50% (unconstrained demand x2)
- Key incremental driver for server SSD demand is KV cache, effectively boosted by HBM de-specing

Q #2: How will HBM demand and pricing shape up in 2027?

UBS View: De-specing occurred, but no impact on demand

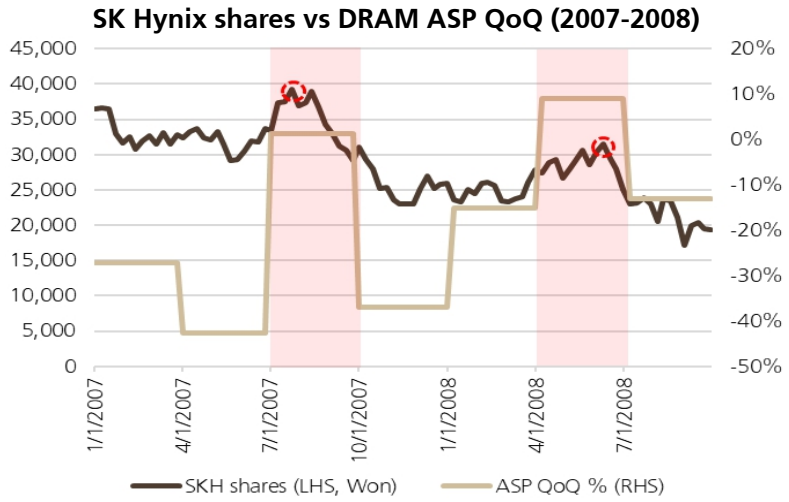
- Nvidia de-speced VR300 to HBM4 8-Hi from HBM4E 12-Hi. Yet, we estimate that their 2027 HBM bit demand in 2027 actually went slightly up. No evidence of moving to 4-Hi. But density increase to 12-Hi for Feynman TBC
- Upside in DDR demand is leading DRAM makers to re-locate supply from HBM. Hence we now forecast HBM bit supply in 2027 to only grow 47% - well below demand
- This means further HBM de-specing could occur for ASICs
- HBM pricing for SK Hynix could reflect volume discounts and passing on over time part of yield improvement benefits – but will not be materially lower than peers

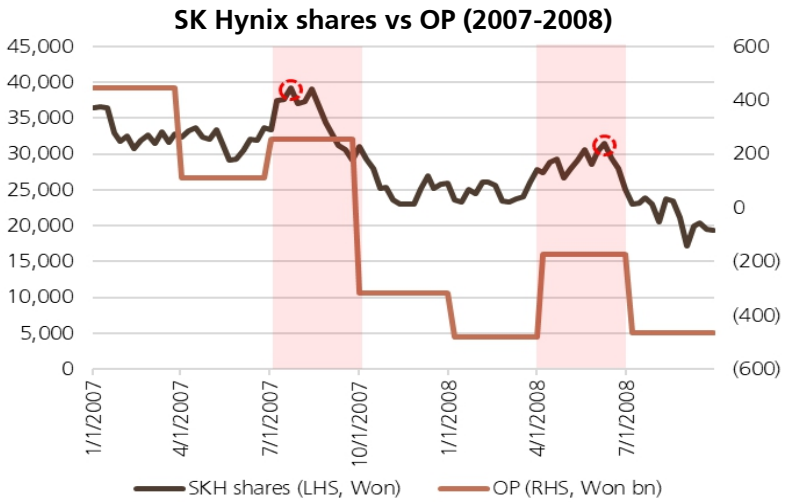
Q #3: Are LTAs ROE-accretive?

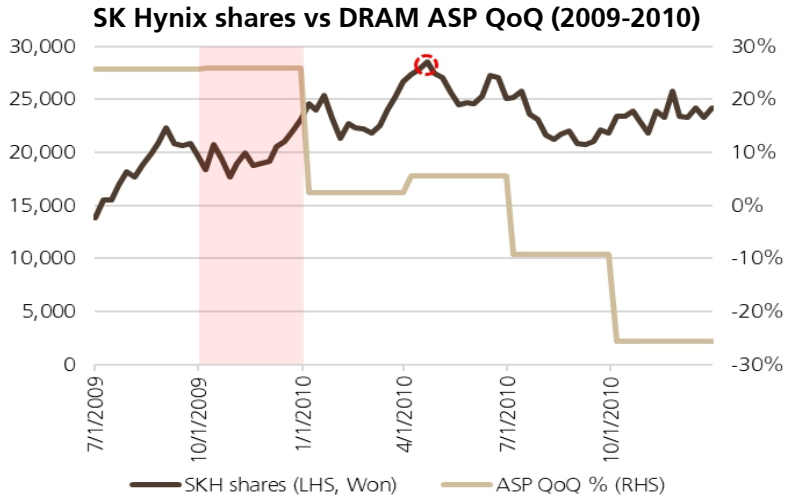
UBS View: Through-cycle they add over 2 pct points of ROE

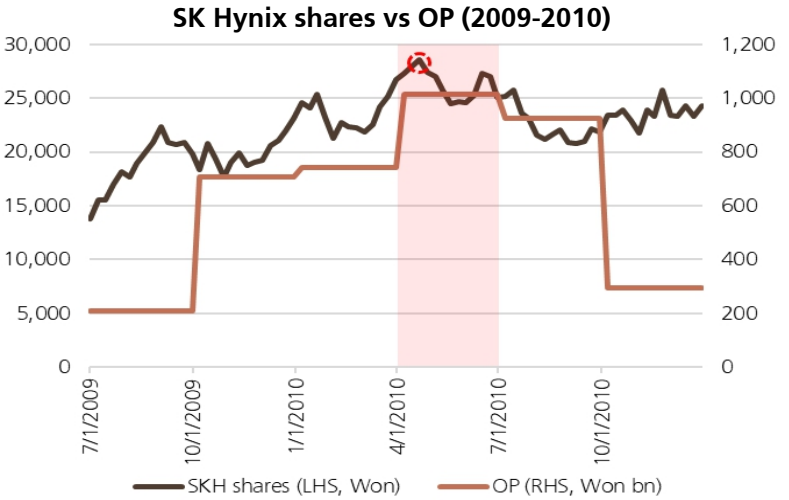
- LTA coverage for the industry could reach 50% of total bits by YE, and of that, 40% on fixed volume/pricing
- Memory makers confident that cash prepayment / callable sufficient to secure LTAs in a downcycle. Also, the opportunity cost of walking away from them for AI customers if AI demand continues to strengthen is high
- LTAs will not remove cyclicality, but narrow price bands over time and increase through-cycle margins

Memory Stocks – Which Lead Indicator Could (Still) Work? – 1



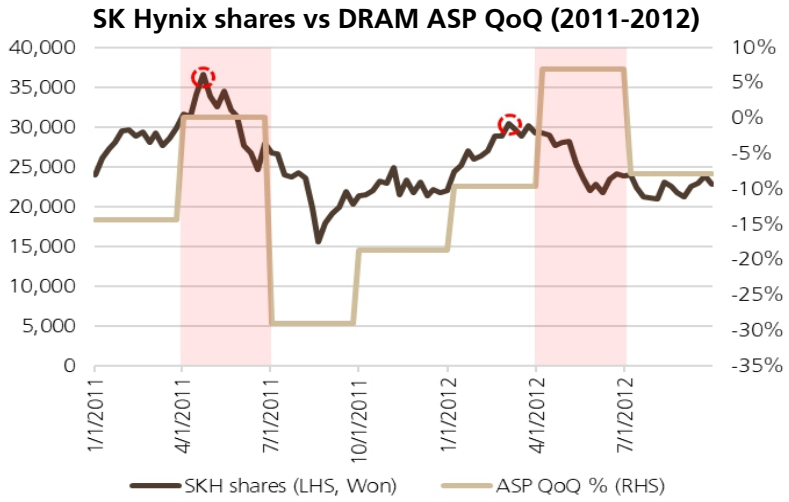


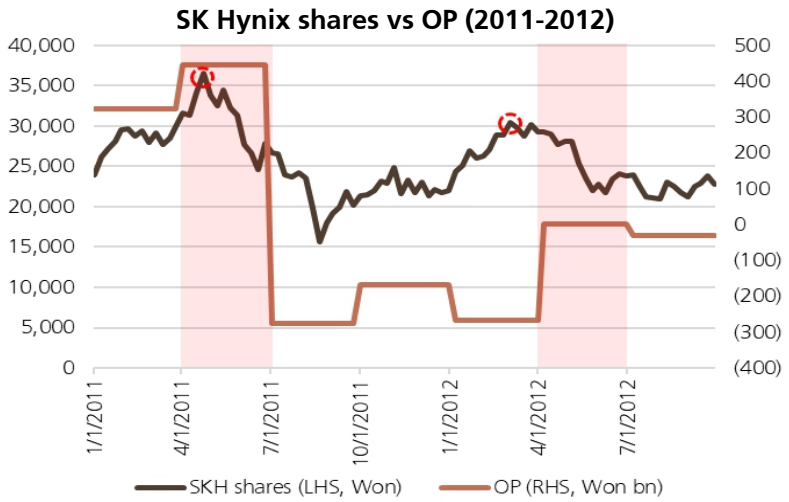


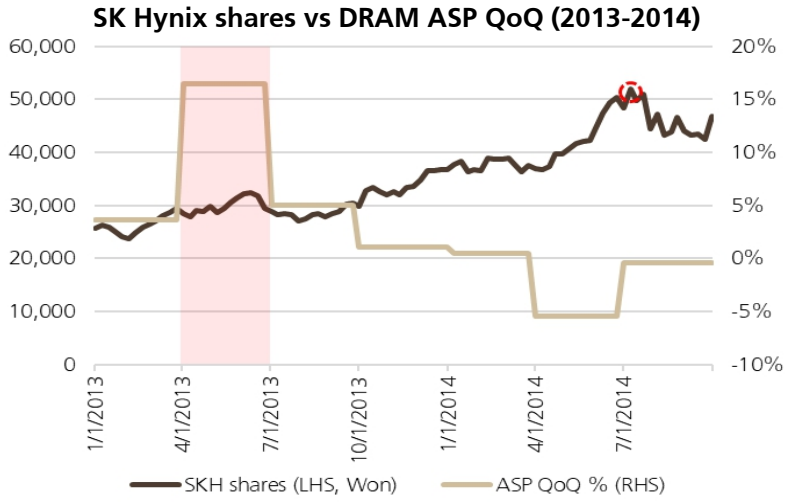


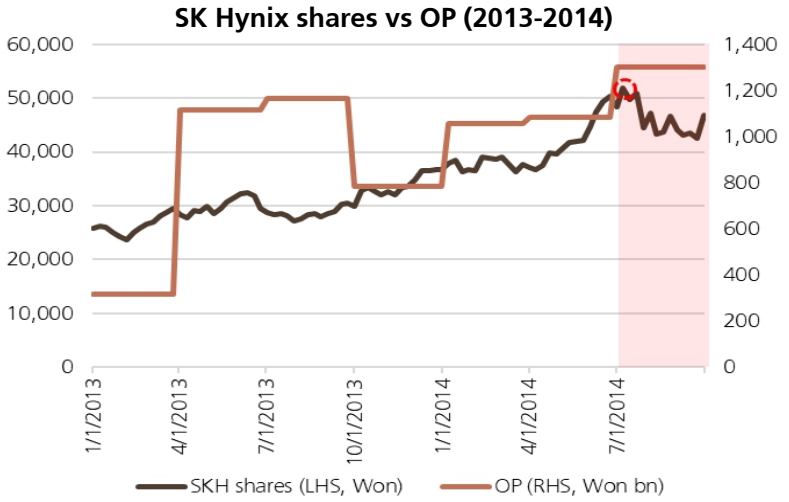
Source: Company data

Memory Stocks – Which Lead Indicator Could (Still) Work? – 2



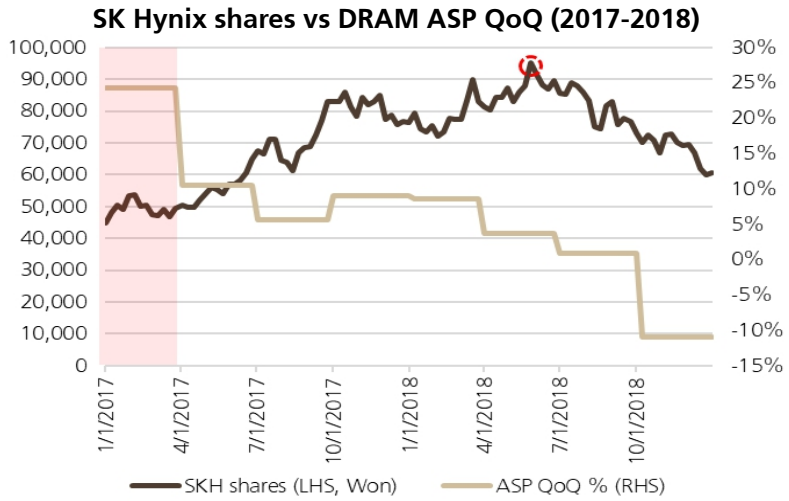


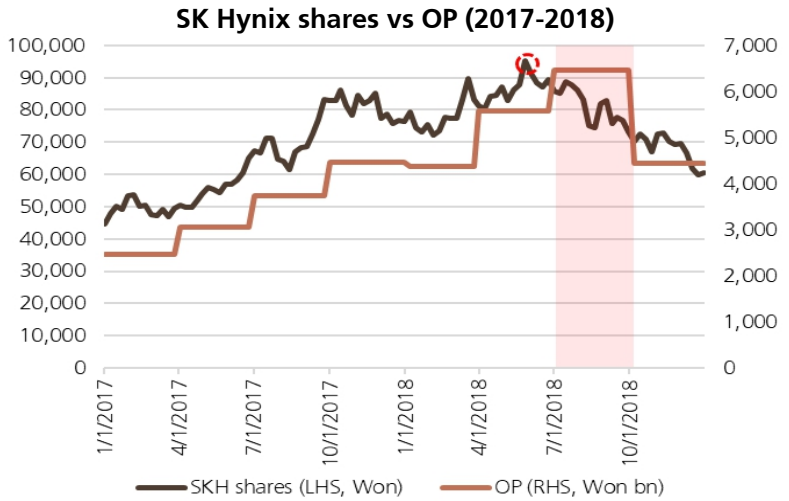


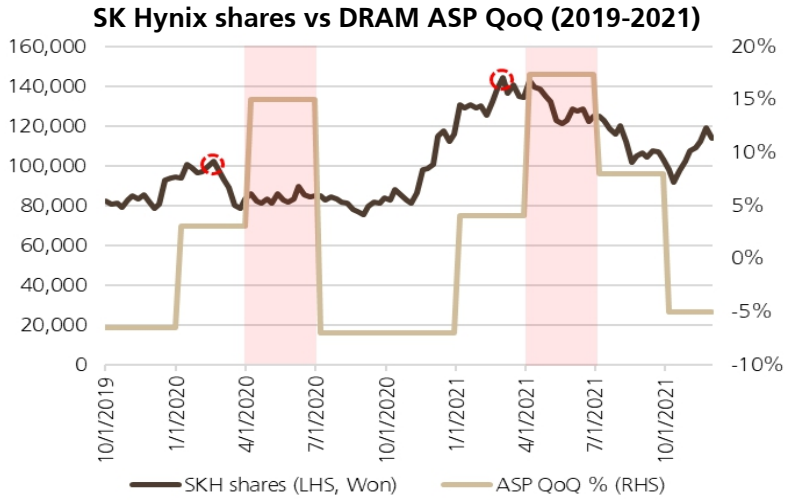


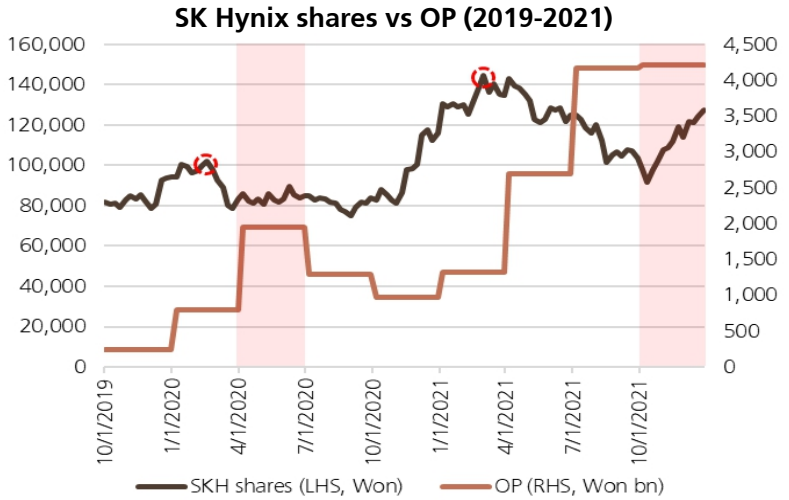
Source: Company data

Memory Stocks – Which Lead Indicator Could (Still) Work? – 3





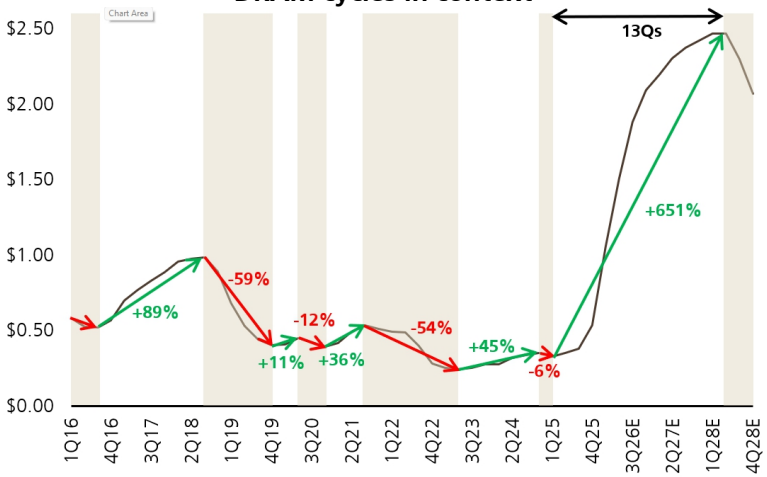




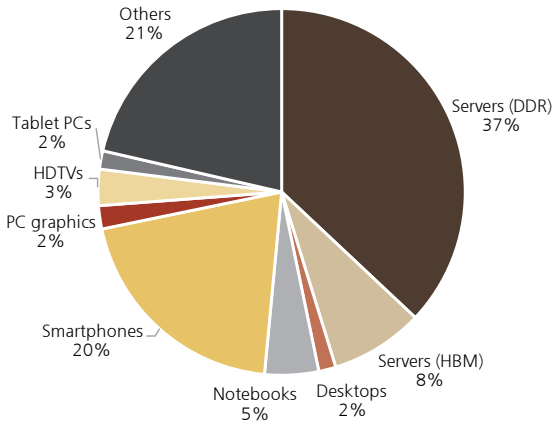
Source: Company data

DRAM Industry Key Charts

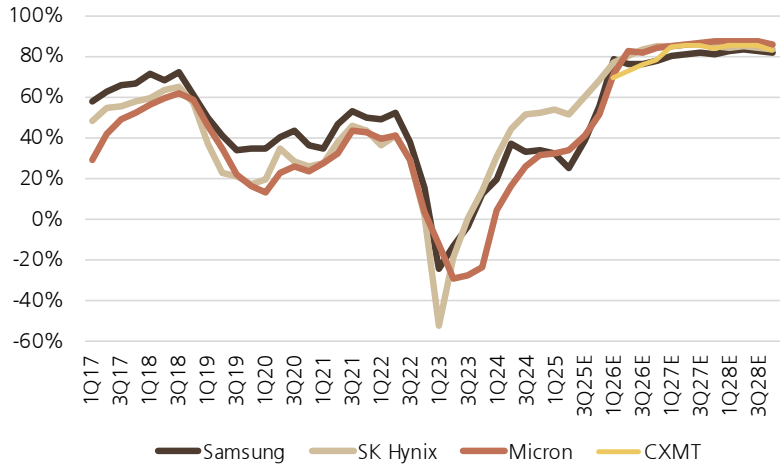
DRAM cycles in context



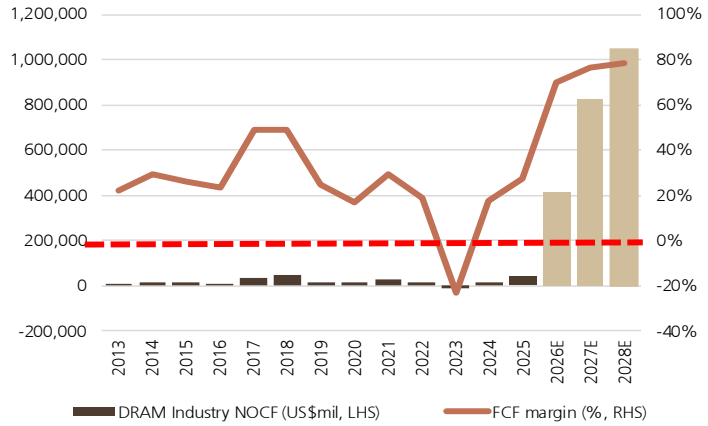
DRAM bit demand by applications (2027E)



DRAM operating profit margins



DRAM industry NOCF and FCF margin



Source: Company data, UBS estimates

DRAM Supply / Demand Outlook

	Q125	Q225	Q325	Q425	2025	Q126	Q226	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
Revenues ($ mil)	**26,415**	**31,776**	**39,234**	**52,958**	**150,384**	**98,124**	**146,329**	**193,867**	**227,347**	**665,667**	**258,641**	**288,546**	**316,883**	**337,869**	**1,201,939**
% QoQ	*-7.3%*	*20.3%*	*23.5%*	*35.0%*		*85.3%*	*49.1%*	*32.5%*	*17.3%*		*13.8%*	*11.6%*	*9.8%*	*6.6%*	
% YoY	*46.4%*	*39.0%*	*54.3%*	*85.8%*	*58.6%*	*271.5%*	*360.5%*	*394.1%*	*329.3%*	*342.6%*	*163.6%*	*97.2%*	*63.5%*	*48.6%*	*80.6%*
Supply (Shipments Based) (M 1Gb)	**64,317**	**75,938**	**85,576**	**87,400**	**313,229**	**89,789**	**96,748**	**103,476**	**109,808**	**399,821**	**114,091**	**121,028**	**128,428**	**133,830**	**497,376**
% QoQ	*-2.6%*	*18.1%*	*12.7%*	*2.1%*		*2.7%*	*7.8%*	*7.0%*	*6.1%*		*3.9%*	*6.1%*	*6.1%*	*4.2%*	
% YoY	*3.4%*	*14.2%*	*28.9%*	*32.3%*	*19.9%*	*39.6%*	*27.4%*	*20.9%*	*25.6%*	*27.6%*	*27.1%*	*25.1%*	*24.1%*	*21.9%*	*24.4%*
Demand (End Consumption) (M 1Gb)	**69,879**	**73,066**	**87,229**	**94,120**	**324,295**	**89,600**	**91,657**	**100,836**	**109,293**	**391,387**	**111,919**	**124,474**	**143,094**	**164,161**	**543,648**
% QoQ	*0.2%*	*4.6%*	*19.4%*	*7.9%*		*-4.8%*	*2.3%*	*10.0%*	*8.4%*		*2.4%*	*11.2%*	*15.0%*	*14.7%*	
% YoY	*17.5%*	*21.0%*	*35.1%*	*34.9%*	*27.6%*	*28.2%*	*25.4%*	*15.6%*	*16.1%*	*20.7%*	*24.9%*	*35.8%*	*41.9%*	*50.2%*	*38.9%*
DDR Demand (M 1Gb)					**306,809**					**359,216**					**482,128**
% YoY					*24.9%*					*17.1%*					*34.2%*
Inventory adjustment	-6,150	3,250	200	-700	-3,400	17,300	13,200	6,700	4,000	41,200	8,500	3,500	-7,500	-24,500	-20,000
Inventory Adj. Demand (M 1Gb)	**63,729**	**76,316**	**87,429**	**93,420**	**320,895**	**106,900**	**104,857**	**107,536**	**113,293**	**432,587**	**120,419**	**127,974**	**135,594**	**139,661**	**523,648**
% QoQ	*-5.8%*	*19.8%*	*14.6%*	*6.9%*		*14.4%*	*-1.9%*	*2.6%*	*5.4%*		*6.3%*	*6.3%*	*6.0%*	*3.0%*	
% YoY	*1.6%*	*10.5%*	*28.7%*	*38.1%*	*20.0%*	*67.7%*	*37.4%*	*23.0%*	*21.3%*	*34.8%*	*12.6%*	*22.0%*	*26.1%*	*23.3%*	*21.1%*
Wafer out (k wpm, 12" equi.)	**1,716**	**1,764**	**1,804**	**1,854**	**1,784**	**1,889**	**1,907**	**1,929**	**1,964**	**1,922**	**2,017**	**2,080**	**2,159**	**2,254**	**2,127**
% QoQ	*2.5%*	*2.8%*	*2.3%*	*2.8%*		*1.9%*	*1.0%*	*1.2%*	*1.8%*		*2.7%*	*3.1%*	*3.8%*	*4.4%*	
% YoY	*17.9%*	*16.5%*	*13.4%*	*10.8%*	*14.5%*	*10.1%*	*8.1%*	*7.0%*	*5.9%*	*7.7%*	*6.8%*	*9.1%*	*11.9%*	*14.8%*	*10.7%*
% Sufficiency Ratio	***0.9%***	***-0.5%***	***-2.1%***	***-6.4%***	***-2.4%***	***-16.0%***	***-7.7%***	***-3.8%***	***-3.1%***	***-7.6%***	***-5.3%***	***-5.4%***	***-5.3%***	***-4.2%***	***-5.0%***
Blended ASP incl. HBM ($/Gb)	**0.41**	**0.42**	**0.46**	**0.61**	**0.48**	**1.09**	**1.51**	**1.87**	**2.07**	**1.66**	**2.27**	**2.38**	**2.47**	**2.52**	**2.42**
% QoQ	*-4.8%*	*1.9%*	*9.6%*	*32.2%*		*80.4%*	*38.4%*	*23.9%*	*10.5%*		*9.5%*	*5.2%*	*3.5%*	*2.3%*	
% YoY	*41.5%*	*21.8%*	*19.7%*	*40.4%*	*32.2%*	*166.1%*	*261.4%*	*308.6%*	*241.7%*	*246.8%*	*107.4%*	*57.6%*	*31.7%*	*21.9%*	*45.1%*
DDR ASP ($/Gb)	**0.33**	**0.35**	**0.38**	**0.53**	**0.40**	**1.05**	**1.51**	**1.88**	**2.09**	**1.66**	**2.19**	**2.30**	**2.37**	**2.42**	**2.33**
% QoQ	*-6.2%*	*7.0%*	*7.0%*	*42.0%*		*97.0%*	*43.0%*	*25.0%*	*11.0%*		*5.0%*	*5.0%*	*3.0%*	*2.0%*	
% YoY	*20.3%*	*10.6%*	*11.8%*	*52.5%*	*26.4%*	*220.3%*	*328.0%*	*400.0%*	*290.9%*	*309.4%*	*108.3%*	*53.0%*	*26.0%*	*15.8%*	*40.5%*
HBM ASP ($/Gb)	**1.42**	**1.55**	**1.56**	**1.52**	**1.52**	**1.57**	**1.59**	**1.79**	**1.91**	**1.75**	**2.89**	**3.06**	**3.23**	**3.32**	**3.14**
% QoQ	*2.7%*	*9.4%*	*0.4%*	*-2.5%*		*3.0%*	*1.4%*	*12.7%*	*7.0%*		*50.8%*	*6.0%*	*5.7%*	*2.6%*	
% YoY	*25.6%*	*28.3%*	*16.9%*	*10.0%*	*15.6%*	*10.3%*	*2.3%*	*14.8%*	*26.0%*	*15.0%*	*84.3%*	*92.7%*	*80.7%*	*73.2%*	*79.8%*

Source: Company data, UBS estimates

DRAM demand: details by application

DRAM bit demand growth by applications

	Q125 QoQ	Q225 QoQ	Q325 QoQ	Q425 QoQ	2025 YoY	Q126 QoQ	Q226 QoQ	Q326E QoQ	Q426E QoQ	2026E YoY	Q127E QoQ	Q227E QoQ	Q327E QoQ	Q427E QoQ	2027E YoY
Total PC DRAM	**-6.1%**	**13.8%**	**18.5%**	**3.0%**	**20.2%**	**-15.2%**	**-2.0%**	**1.2%**	**-1.4%**	**-4.6%**	**-13.5%**	**9.3%**	**15.2%**	**5.5%**	**0.3%**
Desktops	0.4%	10.8%	9.3%	5.6%	22.8%	-12.9%	-5.4%	-6.1%	0.2%	-9.7%	-6.5%	9.3%	11.2%	10.6%	4.0%
Notebooks	-8.1%	14.6%	20.7%	2.2%	24.3%	-15.3%	-0.2%	2.5%	-1.1%	-2.3%	-13.9%	10.1%	17.0%	5.7%	2.6%
Upgrade Modules	-8.1%	17.3%	27.5%	1.9%	-10.9%	-20.0%	-8.3%	12.1%	-8.3%	-8.7%	-30.0%	0.0%	10.0%	-18.6%	-32.2%
Servers	**5.4%**	**7.2%**	**35.8%**	**11.1%**	**57.3%**	**0.9%**	**7.1%**	**18.6%**	**9.1%**	**49.9%**	**16.8%**	**19.7%**	**21.3%**	**16.8%**	**83.1%**
Graphics	13.2%	13.4%	17.5%	-1.0%	26.8%	-9.7%	5.3%	13.4%	0.0%	10.5%	-8.4%	4.9%	13.1%	-0.2%	8.8%
Game Consoles	22.7%	4.2%	4.0%	1.9%	18.3%	0.3%	2.3%	4.2%	-8.5%	6.5%	2.3%	2.7%	4.3%	-3.4%	1.5%
HDTVs	-17.5%	1.8%	21.2%	5.8%	9.7%	-21.5%	1.3%	16.2%	14.5%	2.0%	-21.1%	3.4%	15.7%	12.2%	7.2%
Total Mobile DRAM	**-5.6%**	**-1.0%**	**14.1%**	**10.2%**	**15.6%**	**-13.2%**	**-5.1%**	**2.6%**	**12.0%**	**-0.7%**	**-8.6%**	**-2.9%**	**3.5%**	**18.0%**	**3.4%**
Smartphones	**-4.8%**	**-1.4%**	**14.6%**	**10.1%**	**16.7%**	**-13.2%**	**-5.8%**	**2.4%**	**12.1%**	**-1.3%**	**-8.0%**	**-3.5%**	**3.1%**	**18.8%**	**3.4%**
Apple	-31.8%	-5.6%	29.4%	80.5%	31.3%	-32.6%	-6.7%	5.4%	70.0%	26.9%	-28.7%	-6.7%	1.7%	82.1%	19.8%
Samsung	26.2%	-7.0%	11.5%	-2.8%	17.6%	8.3%	4.5%	-3.1%	-5.2%	10.4%	16.8%	-10.0%	10.0%	-9.8%	-18.6%
Non-Apple/Samsung	-3.0%	1.7%	11.8%	-6.1%	12.0%	-9.7%	-9.6%	3.6%	-5.5%	-14.9%	-2.6%	1.8%	0.5%	-0.5%	4.6%
Tablet PCs	-16.0%	5.0%	8.9%	10.2%	3.2%	-13.1%	4.1%	5.1%	10.2%	6.8%	-16.2%	5.3%	8.4%	8.9%	3.3%
Others	4.3%	4.0%	2.8%	2.9%	12.2%	3.0%	2.6%	1.8%	7.3%	12.8%	-11.3%	2.6%	3.7%	8.8%	0.7%
Total Bit Demand	**0.2%**	**4.6%**	**19.4%**	**7.9%**	**27.6%**	**-4.8%**	**2.3%**	**10.0%**	**8.4%**	**20.7%**	**2.4%**	**11.2%**	**15.0%**	**14.7%**	**38.9%**

DRAM content assumptions (MB/Unit) by applications

	Q125	Q225	Q325	Q425	2025	Q126	Q226	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
Total PC	**12,038**	**12,569**	**12,982**	**13,302**	**12,766**	**13,312**	**13,548**	**13,745**	**13,965**	**13,637**	**13,996**	**14,201**	**14,686**	**15,214**	**14,560**
% YoY	*9%*	*12%*	*15%*	*16%*	*13%*	*11%*	*8%*	*6%*	*5%*	*7%*	*5%*	*5%*	*7%*	*9%*	*7%*
Desktops	11,860	12,370	12,422	12,422	12,283	12,422	12,622	12,822	13,022	12,708	13,022	13,222	13,722	14,222	13,582
Notebooks	12,401	13,001	13,601	14,081	13,328	14,081	14,281	14,481	14,681	14,379	14,681	14,881	15,381	15,881	15,247
Upgrade Modules	9,900	10,100	10,300	10,500	10,232	10,500	10,700	10,900	11,100	10,794	11,100	11,100	11,100	11,300	11,144
Servers	**667,847**	**707,847**	**770,462**	**836,694**	**752,827**	**871,694**	**916,694**	**966,694**	**1,026,434**	**949,280**	**1,176,434**	**1,376,434**	**1,576,434**	**1,792,029**	**1,490,002**
% YoY	*17%*	*20%*	*26%*	*34%*	*26%*	*31%*	*30%*	*25%*	*23%*	*26%*	*35%*	*50%*	*63%*	*75%*	*57%*
Smartphones	**7,739**	**7,829**	**8,255**	**8,611**	**8,126**	**8,664**	**8,680**	**8,977**	**9,317**	**8,912**	**9,277**	**9,123**	**9,406**	**10,098**	**9,491**
% YoY	*11%*	*11%*	*13%*	*15%*	*13%*	*12%*	*11%*	*9%*	*8%*	*10%*	*7%*	*5%*	*5%*	*8%*	*6%*
Apple	7,600	7,700	8,940	9,590	8,652	9,690	9,690	10,400	10,800	10,222	11,000	11,000	11,400	12,800	11,702
Samsung	8,619	8,359	8,866	8,764	8,655	9,573	9,202	9,746	9,626	9,528	10,054	9,578	10,195	10,005	7,682
Non-Apple/Samsung	7,500	7,700	7,900	8,100	7,801	8,000	8,100	8,200	8,300	8,144	8,200	8,200	8,300	8,400	8,275

Source: Company data, Gartner, UBS estimates

DRAM supply by vendor

DRAM vendors market share by bit shipments

	Q125	Q225	Q325	Q425	2025	Q126	Q226	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
Samsung	39.7%	37.5%	38.3%	38.2%	38.4%	37.6%	38.7%	38.0%	35.3%	37.3%	35.7%	36.4%	36.5%	36.5%	36.3%
SK Hynix	29.1%	30.9%	29.6%	29.2%	29.7%	28.3%	28.3%	29.2%	32.0%	29.6%	31.1%	30.6%	30.3%	29.8%	30.4%
Micron	23.0%	23.4%	23.9%	23.5%	23.5%	24.1%	23.0%	22.9%	22.8%	23.2%	23.1%	22.9%	22.9%	23.0%	23.0%
Nanya Tech	1.0%	1.4%	1.6%	1.8%	1.5%	1.6%	1.5%	1.3%	1.2%	1.4%	1.1%	1.1%	1.1%	1.2%	1.1%
CXMT	6.8%	6.5%	6.4%	7.0%	6.7%	8.2%	8.3%	8.3%	8.6%	8.3%	8.8%	8.9%	9.0%	9.4%	9.0%
Others	0.3%	0.3%	0.2%	0.2%	0.3%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%
Total Supply	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**

DRAM vendors bit shipments QoQ & YoY growth

	Q125 QoQ	Q225 QoQ	Q325 QoQ	Q425 QoQ	2025 YoY	Q126 QoQ	Q226 QoQ	Q326E QoQ	Q426E QoQ	2026E YoY	Q127E QoQ	Q227E QoQ	Q327E QoQ	Q427E QoQ	2027E YoY
Samsung	1.5%	11.5%	15.0%	2.0%	9.7%	1.0%	11.0%	5.0%	-1.6%	24.1%	5.3%	8.0%	6.6%	4.2%	21.0%
SK Hynix	-7.0%	25.0%	8.0%	0.8%	24.3%	-0.3%	7.5%	10.3%	16.4%	27.0%	0.9%	4.4%	5.1%	2.6%	28.0%
Micron	-8.0%	20.3%	15.0%	0.5%	21.8%	5.1%	3.0%	6.7%	5.5%	25.9%	5.0%	5.5%	6.0%	4.5%	23.3%
Nanya Tech	8.4%	68.0%	26.4%	11.9%	54.3%	-6.2%	-0.2%	-5.1%	-5.7%	19.1%	0.0%	0.5%	7.0%	12.6%	-0.2%
CXMT	14.0%	12.5%	10.8%	12.8%	67.5%	19.2%	9.4%	7.6%	9.1%	59.7%	7.2%	6.8%	7.9%	7.9%	34.8%
Others	-4.8%	2.5%	2.4%	2.4%	3.1%	-2.3%	0.0%	2.4%	0.0%	2.4%	-2.3%	0.0%	2.4%	0.0%	0.0%
Bit supply Growth	**-2.6%**	**18.1%**	**12.7%**	**2.1%**	**19.9%**	**2.7%**	**7.8%**	**7.0%**	**6.1%**	**27.6%**	**3.9%**	**6.1%**	**6.1%**	**4.2%**	**24.4%**

DRAM vendors wafer out (K wpm, 12” equivalent)

	Q125	Q225	Q325	Q425	2025	Q126	Q226	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
Samsung	663	667	668	674	668	676	678	685	691	682	703	724	752	790	742
SK Hynix	475	499	519	544	509	564	572	583	600	580	618	636	654	672	645
Micron	315	313	311	309	312	307	307	307	307	307	317	327	337	347	332
Nanya Tech	49	53	59	60	57	60	60	60	60	60	60	60	63	70	65
CXMT	164	182	198	217	190	232	240	244	256	243	269	284	304	325	295
Other	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50
Total	**1,716**	**1,764**	**1,804**	**1,854**	**1,784**	**1,889**	**1,907**	**1,929**	**1,964**	**1,922**	**2,017**	**2,080**	**2,159**	**2,254**	**2,127**
% YoY	*17.9%*	*16.5%*	*13.4%*	*10.8%*	*14.5%*	*10.1%*	*8.1%*	*7.0%*	*5.9%*	*7.7%*	*6.8%*	*9.1%*	*11.9%*	*14.8%*	*10.7%*

Source: Company data, Gartner, UBS estimates

Server DRAM: UBS Bottom-Up Model

	2020	2021	2022	2023	2024	2025	2026E	2027E
Avg. DRAM content (GB)	**392**	**457**	**461**	**524**	**576**	**735**	**927**	**1,455**
% YoY	***32.5%***	***16.7%***	***0.9%***	***13.5%***	***9.9%***	***27.7%***	***26.1%***	***57.0%***
Traditional enterprise	**356**	**417**	**420**	**464**	**470**	**480**	**500**	**520**
% YoY	*27.4%*	*17.3%*	*0.7%*	*10.3%*	*1.4%*	*2.1%*	*4.2%*	*4.0%*
Hyperscalers Blended	**416**	**480**	**482**	**549**	**650**	**803**	**1,021**	**1,630**
% YoY	*27.3%*	*15.5%*	*0.4%*	*13.8%*	*18.4%*	*23.5%*	*27.1%*	*59.7%*
Standard servers	**385**	**430**	**432**	**459**	**464**	**480**	**515**	**555**
% YoY	*28.3%*	*11.7%*	*0.5%*	*6.3%*	*1.1%*	*3.4%*	*7.3%*	*7.8%*
AI servers	**600**	**840**	**860**	**1,348**	**1,647**	**4,435**	**8,961**	**14,754**
% YoY	*7.1%*	*40.0%*	*2.4%*	*56.8%*	*22.1%*	*169.3%*	*102.1%*	*64.7%*
Database Servers	**900**	**1,100**	**1,120**	**1,200**	**1,210**	**1,280**	**1,340**	**1,420**
% YoY	*36.4%*	*22.2%*	*1.8%*	*7.1%*	*0.8%*	*5.8%*	*4.7%*	*6.0%*
Total DRAM Demand (mn Gb)	**42,491**	**51,095**	**55,158**	**59,039**	**73,831**	**118,088**	**177,057**	**324,124**
% YoY	***42.3%***	***20.2%***	***8.0%***	***7.0%***	***25.1%***	***59.9%***	***49.9%***	***83.1%***
Total DDR Demand (mn Gb)	**42,491**	**51,095**	**55,158**	**56,355**	**65,224**	**100,602**	**144,886**	**262,604**
% YoY	***42.3%***	***20.2%***	***8.0%***	***2.2%***	***15.7%***	***54.2%***	***44.0%***	***81.2%***
Traditional enterprise	**15,382**	**17,191**	**16,908**	**15,641**	**16,106**	**16,181**	**17,193**	**18,238**
% YoY	*19.8%*	*11.8%*	*-1.6%*	*-7.5%*	*3.0%*	*0.5%*	*6.3%*	*6.1%*
Hyperscalers	**27,110**	**33,904**	**38,250**	**43,398**	**57,725**	**101,907**	**159,864**	**305,886**
% YoY	*59.3%*	*25.1%*	*12.8%*	*13.5%*	*33.0%*	*76.5%*	*56.9%*	*91.3%*
Standard servers	**22,652**	**27,225**	**30,813**	**32,465**	**34,393**	**55,152**	**74,835**	**95,136**
% YoY	*58.9%*	*20.2%*	*13.2%*	*5.4%*	*5.9%*	*60.4%*	*35.7%*	*27.1%*
AI servers	**2,496**	**4,234**	**5,022**	**8,448**	**20,952**	**44,111**	**82,179**	**207,639**
% YoY	*85.7%*	*69.6%*	*18.6%*	*68.2%*	*148.0%*	*110.5%*	*86.3%*	*152.7%*
HBM DRAM Demand				***2,684***	***8,607***	***17,486***	***32,171***	***61,520***
% YoY					*220.7%*	*103.2%*	*84.0%*	*91.2%*
DDR DRAM Demand				***6,081***	***12,345***	***26,625***	***50,009***	***146,119***
% YoY					*103.0%*	*115.7%*	*87.8%*	*192.2%*
LPDDR / SOCAMM Demand					***468***	***7,613***	***14,626***	***38,001***
% YoY						*1527.9%*	*92.1%*	*159.8%*
Database Servers	**1,962**	**2,445**	**2,415**	**2,484**	**2,380**	**2,643**	**2,850**	**3,111**
% YoY	*39.1%*	*24.7%*	*-1.2%*	*2.9%*	*-4.2%*	*11.1%*	*7.8%*	*9.1%*

Source: Gartner, UBS estimates

HBM Demand Forecasts by GPU/Accelerator – 1

	HBM Gen.	HBM Content (GB)			2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E
NVIDIA		**2025E**	**2026E**	**2027E**											
H200	HBM3E	141			83,571										
H20 / H20E	HBM3 / 3E	96			42,336										
B200 (8Hi)	HBM3E	192			276,649										
GB200 (2x GPU - 8Hi)	HBM3E	384	384		414,872	39,504				39,504					
B300 (12Hi)	HBM3E	288	288	288	169,176	76,896	86,400	57,600	28,800	249,696	14,400				14,400
GB300 (2x GPU - 12Hi)	HBM3E	576	576	576	448,259	379,008	385,920	334,080	259,200	1,358,208	207,360	155,520	103,680	51,840	518,400
R200 (12Hi)	HBM4		288	288			5,760	57,600	92,160	155,520	103,680	115,200	92,160	69,120	380,160
RV200 (2x GPU - 12Hi)	HBM4		576	576			23,040	144,000	259,200	426,240	345,600	414,720	397,440	362,880	1,520,640
R300 (12Hi)	HBM4			192									48,000		48,000
R300 (12Hi)	HBM4E			256										102,400	102,400
RV300 (4x GPU - 12Hi)	HBM4			384									96,000		96,000
RV300 (4x GPU - 12Hi)	HBM4E			512										281,600	281,600
*Other GPU/Accel	HBM3 / 3E	288	384	576	30,787	23,040	30,720	30,720	30,720	115,200	40,320	40,320	46,080	46,080	172,800
Total ('000 GB)					**1,465,650**	**518,448**	**531,840**	**624,000**	**670,080**	**2,344,368**	**711,360**	**725,760**	**783,360**	**913,920**	**3,134,400**
Total (bn Gb)					**11.73**	**4.15**	**4.25**	**4.99**	**5.36**	**18.75**	**5.69**	**5.81**	**6.27**	**7.31**	**25.08**
AMD															
MI300	HBM3	192			13,685										
MI325X (12Hi)	HBM3E	256	256		57,294	4,608				4,608					
MI355X (12Hi)	HBM3E	288	288		63,569	48,960	43,200	23,040	20,160	135,360					
MI308X	HBM3	192	192		18,394	5,568	5,376	960		11,904					
MI450 (12Hi)	HBM4		432	432				19,440	99,360	118,800	151,200	194,400	203,040	164,160	712,800
MI5XX (16Hi)	HBM4E			1,024									61,440	163,840	225,280
*Other GPU/Accel	HBM3	192	192	288	812	274	285	282	281	1,123	576	576	576	576	2,304
Total ('000 GB)					**153,754**	**59,410**	**48,861**	**43,722**	**119,801**	**271,795**	**151,776**	**194,976**	**265,056**	**328,576**	**940,384**
Total (bn Gb)					**1.23**	**0.48**	**0.39**	**0.35**	**0.96**	**2.17**	**1.21**	**1.56**	**2.12**	**2.63**	**7.52**
Google															
TPU v5e	HBM2E	32			4,282										
TPU v5p	HBM2E	96			183,913										
TPU v6e	HBM3E	32	32		13,632	6,554	7,168	4,506	2,253	20,480					
TPU v7	HBM3E	192	192	192	17,280	99,840	138,240	153,600	144,000	535,680	115,200	86,400	57,600	28,800	288,000
TPU v8i (Dual die AX)	HBM3E		288	288				7,200	64,800	72,000	201,600	288,000	374,400	288,000	1,152,000
TPU v8t (Single die X)	HBM3E		216	216				15,120	82,080	97,200	151,200	259,200	324,000	216,000	950,400
*Other TPU/Accel	HBM3	96	144	144	7,389	1,044	1,359	1,492	1,440	5,335	3,600	3,600	3,600	3,600	14,400
Total ('000 GB)					**226,495**	**107,437**	**146,767**	**181,917**	**294,573**	**730,695**	**471,600**	**637,200**	**759,600**	**536,400**	**2,404,800**
Total (bn Gb)					**1.81**	**0.86**	**1.17**	**1.46**	**2.36**	**5.85**	**3.77**	**5.10**	**6.08**	**4.29**	**19.24**
Amazon															
Inferentia 2	HBM2E	32			2,823										
Trainium 2 / 2.5	HBM3E	96	144	144	47,574	18,720	31,680	20,160	14,400	84,960	7,200	1,440			8,640
Inferentia 3	HBM3	96	96	96	15,563	2,880	2,688	1,920	1,728	9,216	1,440	960			2,400
Trainium 3	HBM3E		144	144			46,080	89,280	120,960	256,320	120,960	112,320	92,160	77,760	403,200
Trainium 4	HBM4			256									10,240	61,440	71,680
*Other Accel	HBM3	144	144	144	5,375	691	818	1,123	1,380	4,012	1,152	1,152	1,008	1,008	4,320
Total ('000 GB)					**71,336**	**22,291**	**81,266**	**112,483**	**138,468**	**354,508**	**130,752**	**115,872**	**103,408**	**140,208**	**490,240**
Total (bn Gb)					**0.57**	**0.18**	**0.65**	**0.90**	**1.11**	**2.84**	**1.05**	**0.93**	**0.83**	**1.12**	**3.92**

Source: Company data, UBS estimates

HBM Demand Forecasts by GPU/Accelerator – 2

	HBM Gen.	HBM Content (GB)			2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E
Intel															
Gaudi 3	HBM2 / 3E	288	288		54,212	8,640	5,184	2,304		16,128					
Gaudi 4	HBM3E		288	288				2,880	7,200	10,080	7,200	5,760	4,320	2,880	20,160
*Others	HBM2 / 3E	192	192	192	4,789	1,152	691	691	960	3,494	768	768	768	768	3,072
Total ('000 GB)					**59,001**	**9,792**	**5,875**	**5,875**	**8,160**	**29,702**	**7,968**	**6,528**	**5,088**	**3,648**	**23,232**
Total (bn Gb)					**0.47**	**0.08**	**0.05**	**0.05**	**0.07**	**0.24**	**0.06**	**0.05**	**0.04**	**0.03**	**0.19**
Microsoft															
Maia 100	HBM3	120	120		4,050	360	120			480					
Maia 200	HBM3E		288	288		660	1,696	3,016	4,052	9,424	4,032	2,880	2,016	1,152	10,080
Maia 300	HBM4			288									2,304	4,032	6,336
*Others	HBM3	120	192	192	960	384	384	384	384	1,536	384	384	384	384	1,536
Total ('000 GB)					**5,010**	**1,404**	**2,200**	**3,400**	**4,436**	**11,440**	**4,416**	**3,264**	**4,704**	**5,568**	**17,952**
Total (bn Gb)					**0.04**	**0.01**	**0.02**	**0.03**	**0.04**	**0.09**	**0.04**	**0.03**	**0.04**	**0.04**	**0.14**
Meta															
MTIA v3	HBM3E	192	192	192	960	2,880	3,840	6,720	7,680	21,120	7,680	5,760	3,840	2,880	20,160
MTIA v4	HBM4			432							17,280	34,560	43,200	51,840	146,880
Total ('000 GB)					**960**	**2,880**	**3,840**	**6,720**	**7,680**	**21,120**	**24,960**	**40,320**	**47,040**	**54,720**	**167,040**
Total (bn Gb)					**0.01**	**0.02**	**0.03**	**0.05**	**0.06**	**0.17**	**0.20**	**0.32**	**0.38**	**0.44**	**1.34**
Others															
Tesla	HBM3	64	64		6,780	1,523	1,613	1,344	1,255	5,735	1,344	1,344	1,472	1,280	5,440
Startups / Etc.	HBM2E / 3	192	288		195,559	51,840	51,840	55,800	55,800	215,280	66,096	69,984	71,928	73,872	281,880
Total ('000 GB)					**202,339**	**53,363**	**53,453**	**57,144**	**57,055**	**221,015**	**67,440**	**71,328**	**73,400**	**75,152**	**287,320**
Total (bn Gb)					**1.62**	**0.43**	**0.43**	**0.46**	**0.46**	**1.77**	**0.54**	**0.57**	**0.59**	**0.60**	**2.30**
Total HBM Demand ('000 GB)					**2,184,545**	**775,025**	**874,103**	**1,039,582**	**1,332,654**	**4,021,364**	**1,604,832**	**1,831,968**	**2,098,536**	**2,154,672**	**7,690,008**
Total HBM Demand (bn Gb)					**17.48**	**6.20**	**6.99**	**8.32**	**10.66**	**32.17**	**12.84**	**14.66**	**16.79**	**17.24**	**61.52**
% YoY					*103%*	*97%*	*103%*	*72%*	*77%*	*84%*	*107%*	*110%*	*102%*	*62%*	*91%*
% QoQ						*3%*	*13%*	*19%*	*28%*		*20%*	*14%*	*15%*	*3%*	

Source: Company data, UBS estimates

HBM Supply Forecasts by Vendor

(M 1Gb)		2025	Q126	Q226	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
SK Hynix		**12,628**	**3,803**	**3,804**	**4,559**	**5,273**	**17,438**	**4,611**	**4,812**	**5,058**	**5,587**	**20,068**
	% YoY	*86%*	*31%*	*20%*	*41%*	*59%*	*38%*	*21%*	*26%*	*11%*	*6%*	*15%*
	% QoQ		*15%*	*0%*	*20%*	*16%*		*-13%*	*4%*	*5%*	*10%*	
Samsung		**4,327**	**1,519**	**2,061**	**3,384**	**4,452**	**11,416**	**4,891**	**5,281**	**5,864**	**6,356**	**22,392**
	% YoY	*-15%*	*161%*	*170%*	*140%*	*183%*	*164%*	*222%*	*156%*	*73%*	*43%*	*96%*
	% QoQ		*-3%*	*36%*	*64%*	*32%*		*10%*	*8%*	*11%*	*8%*	
Micron		**4,584**	**1,612**	**1,821**	**1,985**	**2,243**	**7,662**	**2,468**	**2,714**	**3,067**	**3,466**	**11,716**
	% YoY	*417%*	*106%*	*60%*	*57%*	*60%*	*67%*	*53%*	*49%*	*55%*	*55%*	*53%*
	% QoQ		*15%*	*13%*	*9%*	*13%*		*10%*	*10%*	*13%*	*13%*	
CXMT		**53**	**21**	**26**	**31**	**44**	**122**	**53**	**74**	**111**	**165**	**403**
	% YoY		*168%*	*142%*	*93%*	*139%*	*130%*	*149%*	*191%*	*264%*	*272%*	*232%*
	% QoQ		*15%*	*20%*	*20%*	*44%*		*20%*	*40%*	*50%*	*48%*	
Total Supply		**21,591**	**6,955**	**7,712**	**9,959**	**12,012**	**36,638**	**12,022**	**12,882**	**14,100**	**15,574**	**54,579**
	% YoY	*69%*	*63%*	*52%*	*68%*	*90%*	*70%*	*73%*	*67%*	*42%*	*30%*	*49%*
	% QoQ		*10%*	*11%*	*29%*	*21%*		*0%*	*7%*	*9%*	*10%*	

Source: Company data, UBS estimates

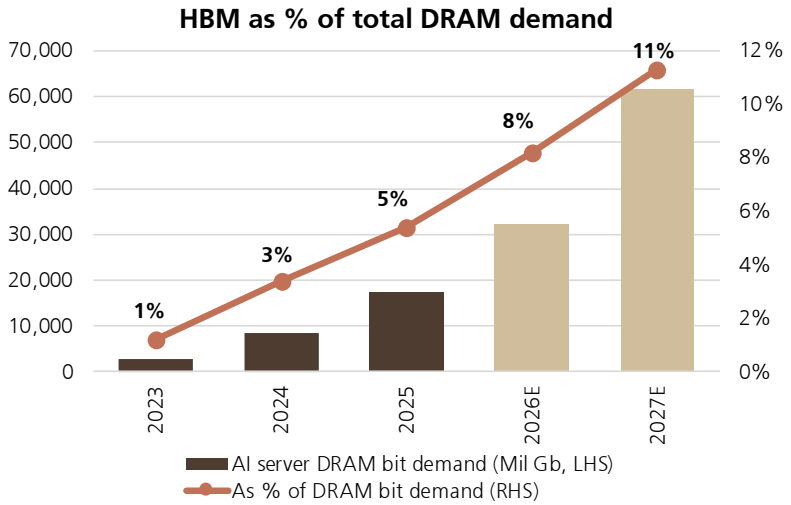
HBM Vendor Market Share by Product

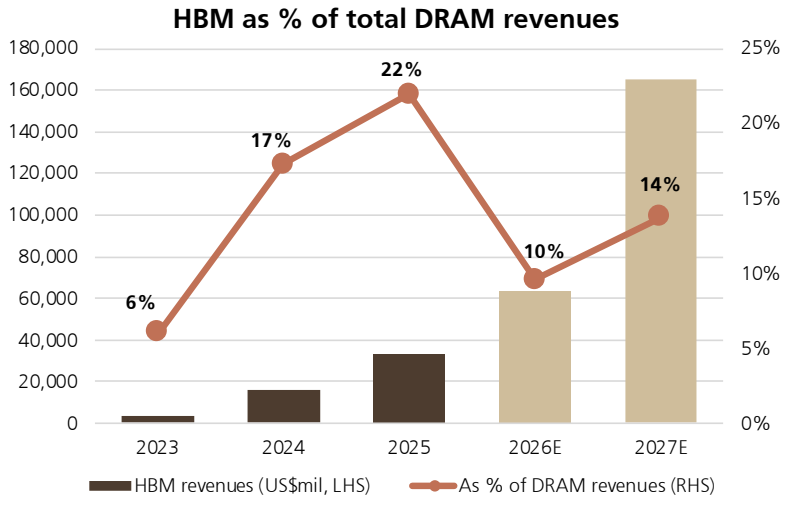
Customer	Product	Configuration	Supplier	2026E	2027E
Nvidia	B300	HBM3E	SK Hynix	56%	45%
		12-Hi x 8 cubes	Samsung	16%	25%
		288GB	Micron	28%	30%
	GB300	HBM3E	SK Hynix	55%	45%
		12-Hi x 8 cubes x 2	Samsung	17%	25%
		576GB	Micron	29%	30%
	R200	HBM4	SK Hynix	60%	55%
		12-Hi x 8 cubes	Samsung	30%	35%
		288GB	Micron	10%	10%
	VR200	HBM4	SK Hynix	60%	55%
		12-Hi x 8 cubes x 2	Samsung	30%	35%
		576GB	Micron	10%	10%
	VR300*	HBM4	SK Hynix		50%
		8-Hi x 8 cubes x 2	Samsung		35%
		384GB	Micron		15%
AMD	MI325X	HBM3E	SK Hynix	0%	
		12-Hi x 8 cubes	Samsung	70%	
		256GB	Micron	30%	
	MI355X	HBM3E	SK Hynix	2%	
		12-Hi x 8 cubes	Samsung	68%	
		288GB	Micron	30%	
	MI450	HBM4	SK Hynix	10%	10%
		12-Hi x 12 cubes	Samsung	60%	60%
		432GB	Micron	30%	30%
	MI5XX	HBM4E	SK Hynix		15%
		16-Hi x 16 cubes	Samsung		50%
		1,024GB	Micron		35%
Meta	MTIA v3	HBM3E	SK Hynix	0%	0%
		8-Hi x 8 cubes	Samsung	37%	40%
		192GB	Micron	63%	60%
	MTIA v4	HBM4 12-Hi	SK Hynix		0%
		12-Hi x 12 cubes	Samsung		40%
		432GB	Micron		60%

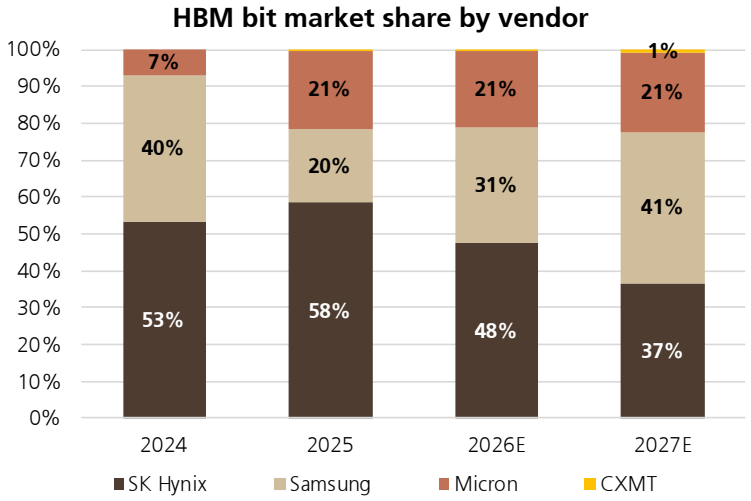
Customer	Product	Configuration	Supplier	2026E	2027E
Google	TPU v6e	HBM3E	SK Hynix	60%	
			Samsung	40%	
		32GB	Micron	0%	
	TPU v7	HBM3E	SK Hynix	70%	60%
		8-Hi x 8 cubes	Samsung	30%	40%
		192GB	Micron	0%	0%
	TPU v8i	HBM3E	SK Hynix	50%	35%
			Samsung	40%	50%
		288GB	Micron	10%	15%
	TPU v8t	HBM3E	SK Hynix	70%	70%
			Samsung	30%	30%
		216GB	Micron	0%	0%
	TPU v9	HBM4E	Base case: 12-Hi; could be 8-Hi due to potential de-spec.		
		12-Hi x 8 cubes			
		384GB			
Amazon	Trainium 2/2.5	HBM3E			90%
			Samsung	10%	10%
		144GB	Micron	0%	0%
	Inferentia 3	HBM3	SK Hynix	80%	80%
			Samsung	20%	20%
		96GB	Micron	0%	0%
	Trainium 3	HBM3E	SK Hynix	70%	60%
			Samsung	30%	40%
		144GB	Micron	0%	40%
	Trainium 4	HBM4 12-Hi	SK Hynix		40%
		12-Hi x 8 cubes	Samsung		40%
		256GB	Micron		20%

Source: Company data, UBS estimates. *VR300 may later come with a HBM4E 8-Hi configuration

HBM Industry Key Charts – 1



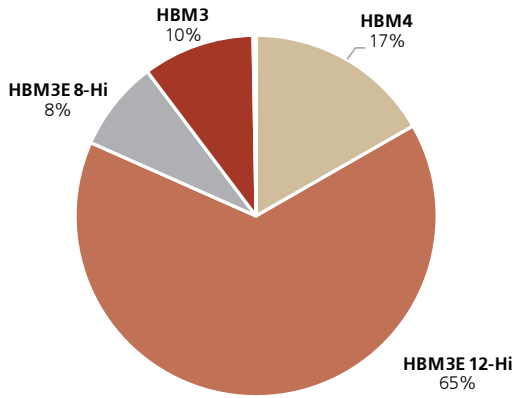




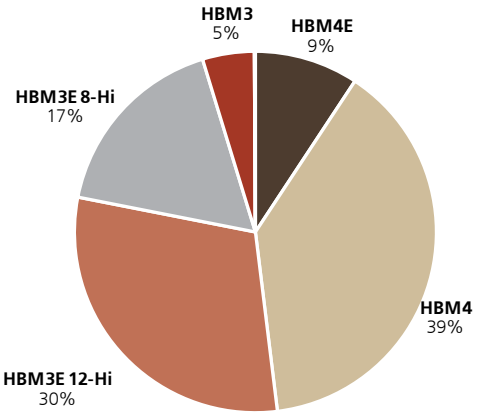
Source: Company data, UBS estimates

HBM Industry Key Charts – 2

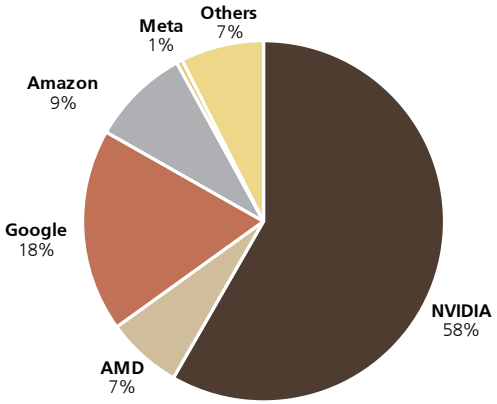
HBM demand by product (2026E)



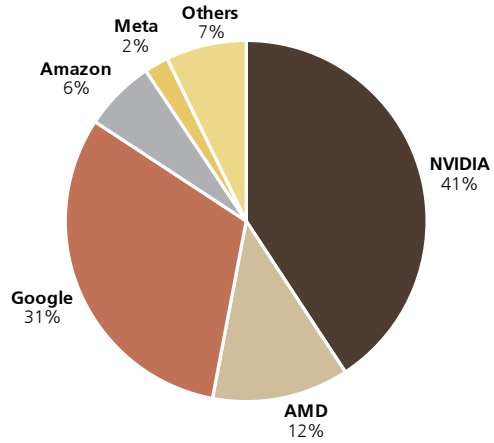
HBM demand by product (2027E)



HBM demand by customer (2026E)

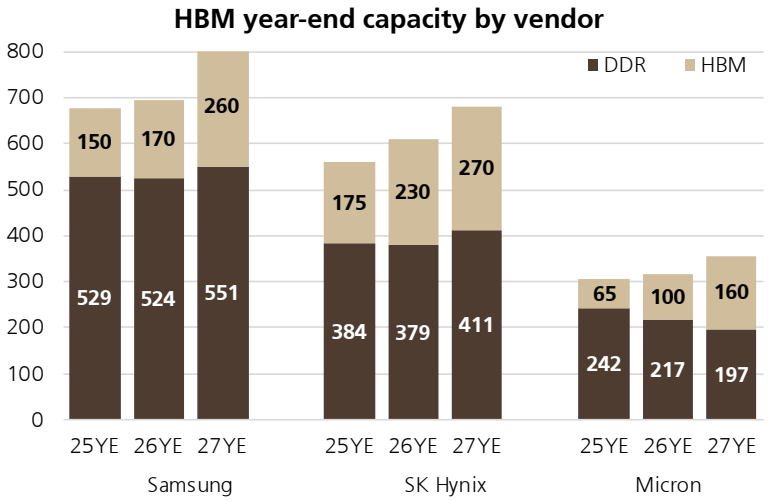


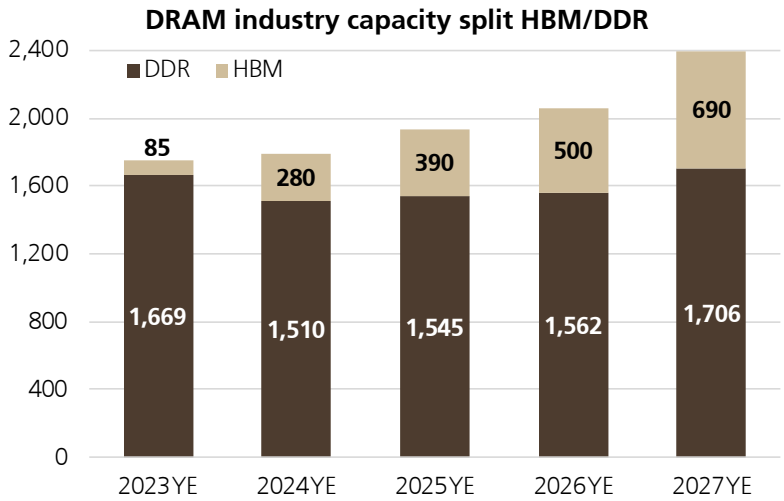
HBM demand by customer (2027E)

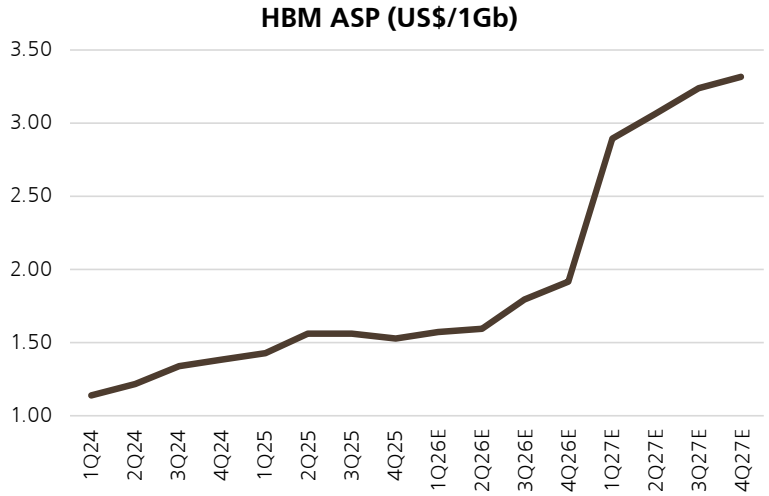


Source: Company data, UBS estimates

HBM Industry Key Charts – 3







Source: Company data, UBS estimates

HBM Industry Key Charts – 4

Summary of HBM inventory accumulation ahead of consumption

HBM vendors	>	CoWoS	>	Compute boards	>	ODMs

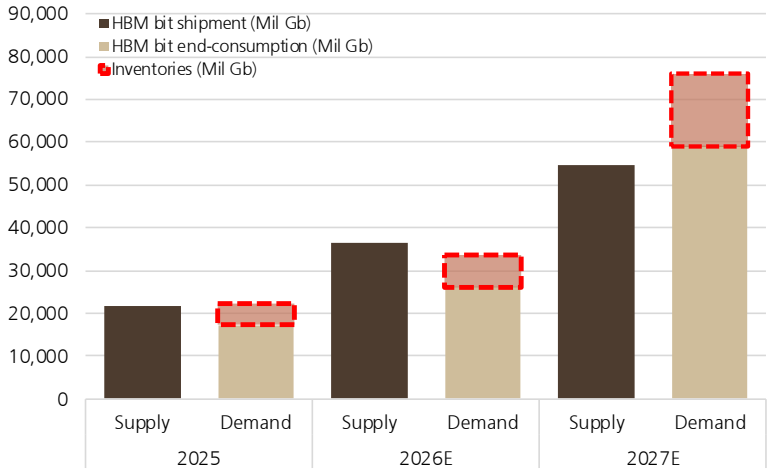
HBM vendors recognise revenues upon shipment
HBM solutions sit in inventory hubs on site at TSMC, awaiting CoWoS process completion

Assembled and tested AI compute solutions (AI accelerators and HBM) are shipped to another inventory hub with compute board makers

AI accelerator vendors recognise revenues as compute boards are shipped to ODMs for rack assembly
***This is the point at which we measure end consumption*

3 months for newer productions, up to 4.5 months for more legacy solutions

Adjusted HBM supply vs demand



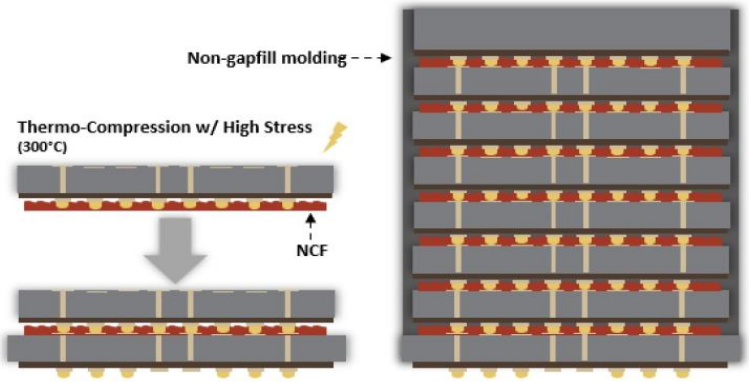
Source: Company data, UBS estimates

HBM Industry Key Charts – 5

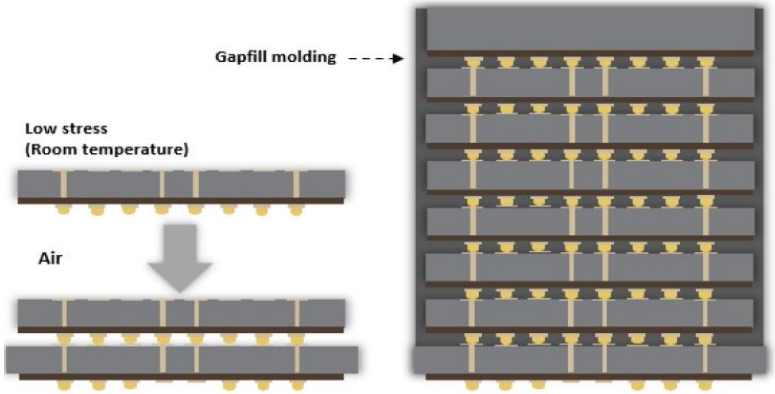
DRAM / Logic base die technology by HBM generation and supplier

	HBM3E	HBM4	HBM4E
DRAM process			
SK Hynix	1b nm	1b nm	1c nm
Micron	1ß nm	1ß nm	1γ nm
Samsung	1a nm	1c nm	1c nm
Logic base die process			
SK Hynix	DRAM	TSMC (12 nm)	TSMC (12 nm)
Micron	DRAM	DRAM (2z nm)	TSMC
Samsung	DRAM	Samsung Foundry (4nm)	Samsung Foundry or TSMC

TC-NCF Process



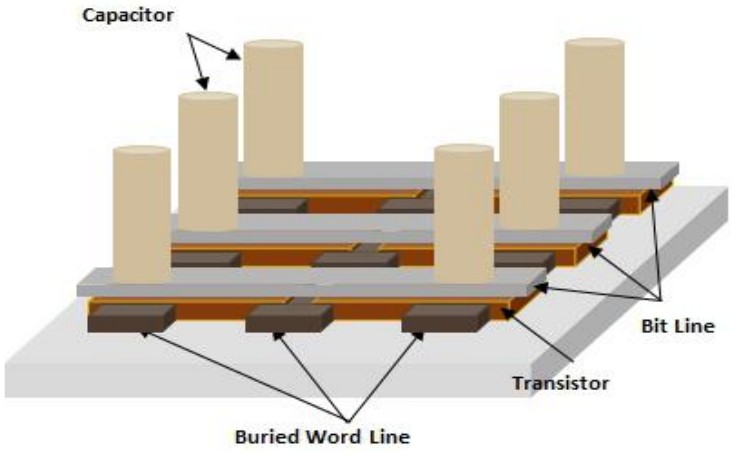
MR-MUF Process



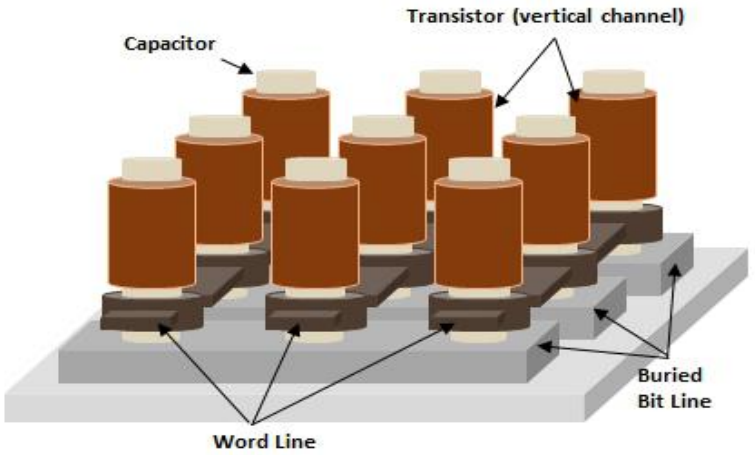
Source: UBS

4F² VCAT DRAM Key Charts – 1

Cross section of 6F² DRAM



Cross section of 4F² VCAT DRAM



DRAM tech migration and expected 4F² DRAM adoption roadmap for mass production

	2022				2023				2024				2025				2026E				2027E				2028E				2029E				2030E				2031E			
	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Samsung	1a								1b								1c				1d								0a								0b			
SK Hynix	1a								1b								1c				1d								0a								0b			
Micron	1α				1β (Beta)												1γ (Gamma)				1δ (Delta)								1ε (Epsilon)								0a			
CXMT	G1				G2				G3				G4 (≈1z)				G5	• 4F²	CXMT product launch — not mass production																							

Further process transition is subject to DUV/EUV availability

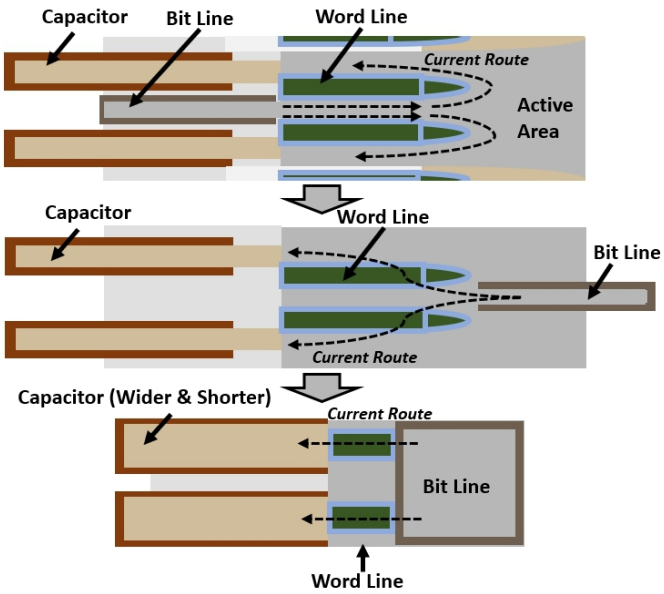
4F² mass production adoption

4F² adoption point (CXMT)

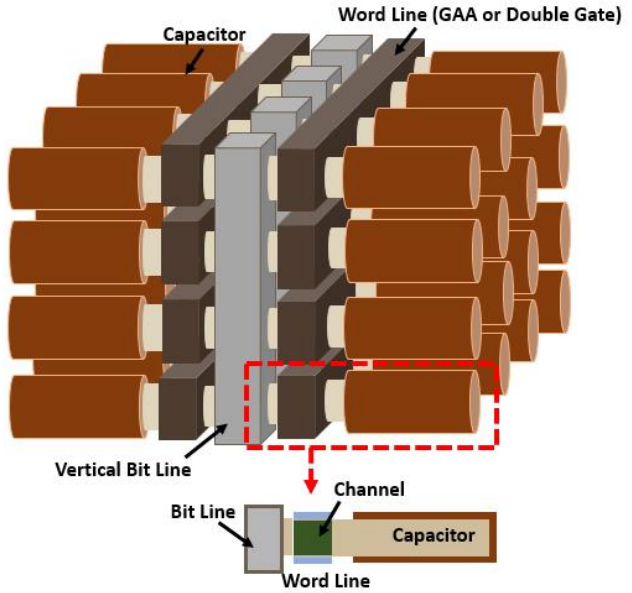
Source: Company data, UBS estimates

4F² VCAT DRAM Key Charts – 2

Possible architecture change from 2D to 3D stacked DRAM



Cross section of 3D stacked DRAM



Source: Company data, UBS estimates

$4F^2$ VCAT DRAM Key Charts – 3

Samsung DRAM technology roadmap base case

	1a nm	1b nm	1c nm	1d nm	0a nm	0b nm
Half pitch	13.8 nm	12.5 nm	11 nm	10 nm	9.x nm	9.x/8.x nm
Mass production	3Q21	3Q23	4Q25	2Q27E	2H28E	2H30E
EUV litho	Yes	Yes	Yes	Yes	Yes	Yes
Number of EUV layers	5 layers	5 layers	8 layers	10-12 layers	< 12 layers	TBD
Numerical aperture (NA)	0.33 NA	0.33 NA	0.33 NA	0.33 NA	0.33 NA	0.55 NA
Channel transistor	6F2	6F2	6F2	6F2	4F2	4F2

SK Hynix DRAM technology roadmap base case

	1a nm	1b nm	1c nm	1d nm	0a nm	0b nm
Half pitch	14.7 nm	12.7 nm	11 nm	10 nm	9.x nm	9.x/8.x nm
Mass production	3Q21	4Q23	3Q25	2Q27E	1H29E	2H30E
EUV litho	Yes	Yes	Yes	Yes	Yes	Yes
Number of EUV layers	1 layer	4 layers	6 layers	c. 10 layers	TBD	TBD
Numerical aperture (NA)	0.33 NA	0.33 NA	0.33 NA	0.33 NA	0.33 NA	0.33/0.55 NA
Channel transistor	6F2	6F2	6F2	6F2	6F2	4F2

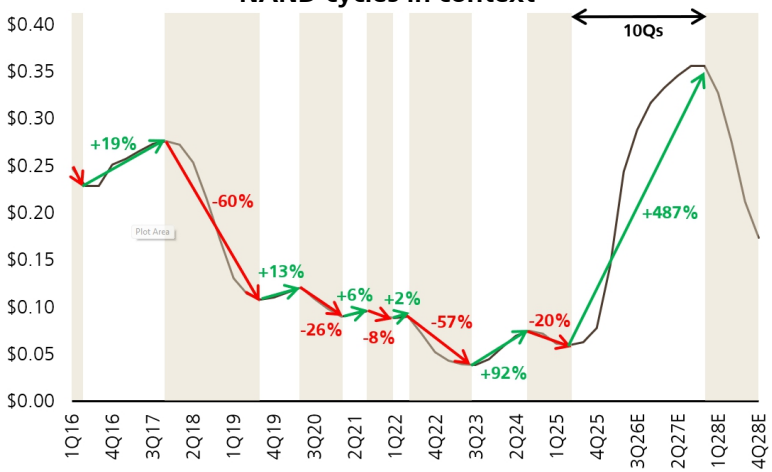
Micron DRAM technology roadmap base case

	1α nm	1β nm	1γ nm	1δ nm	1ε nm	0a nm
Half pitch	14.6 nm	13.0 nm	12 nm	11 nm	10 nm	9.x nm
Mass production	4Q20	4Q22	3Q25	2Q27E	1H29E	2H30E
EUV litho	No	No	Yes	Yes	Yes	Yes
Number of EUV layers	-	-	1 layer	1 layer	c. 3 layers	TBD
Numerical aperture (NA)	-	-	0.33 NA	0.33 NA	0.33 NA	0.33/0.55 NA
Channel transistor	6F2	6F2	6F2	6F2	6F2	4F2

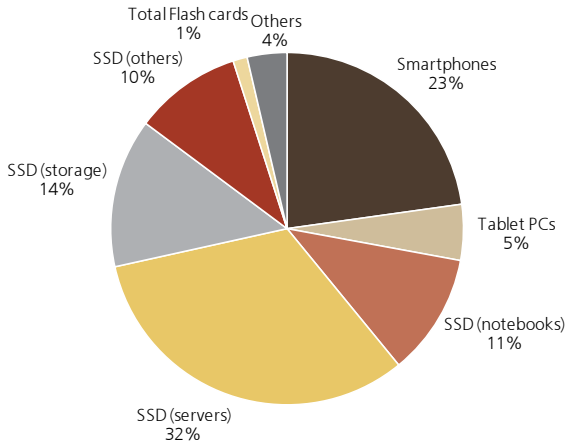
Source: Company data, UBS estimates

NAND Industry Key Charts

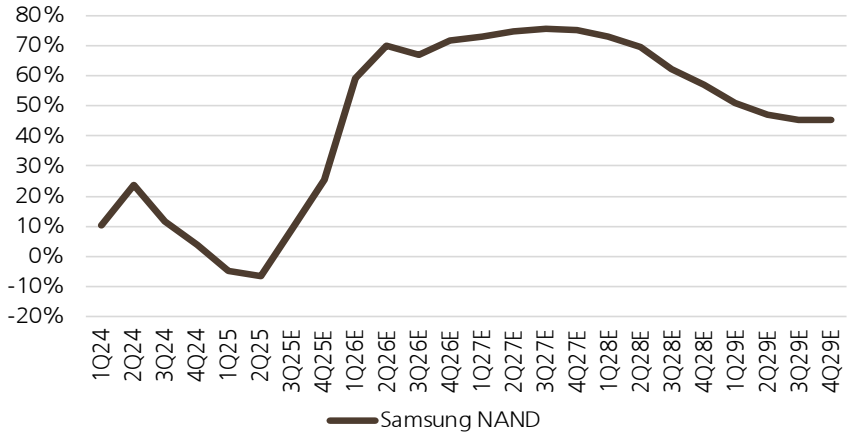
NAND cycles in context



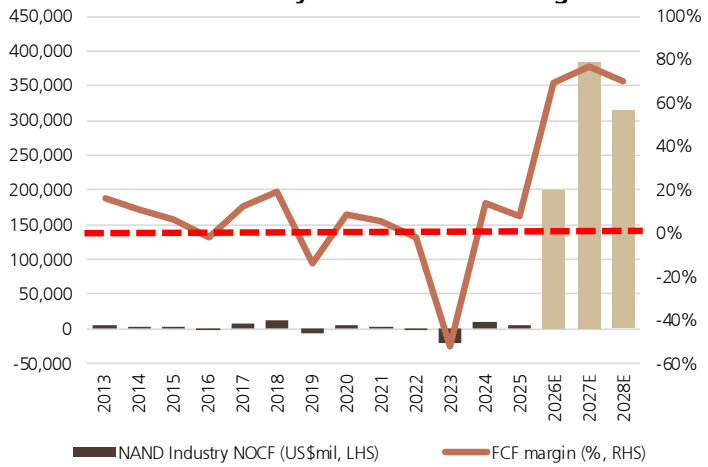
NAND bit demand by applications (2027E)



NAND operating profit margin



NAND industry NOCF and FCF margin



Source: Company data, Gartner, UBS estimates

NAND flash supply / demand outlook

	Q125	Q225	Q325	Q425	2025	Q126	Q226	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
Revenues ($ mil)	**12,302**	**14,726**	**16,969**	**21,320**	**65,317**	**39,006**	**70,496**	**88,144**	**102,417**	**300,063**	**111,872**	**121,870**	**134,230**	**140,439**	**508,411**
% QoQ	*-20.1%*	*19.7%*	*15.2%*	*25.6%*		*83.0%*	*80.7%*	*25.0%*	*16.2%*		*9.2%*	*8.9%*	*10.1%*	*4.6%*	
% YoY	*-1.0%*	*-1.1%*	*4.9%*	*38.5%*	*10.9%*	*217.1%*	*378.7%*	*419.4%*	*380.4%*	*359.4%*	*186.8%*	*72.9%*	*52.3%*	*37.1%*	*69.4%*
Supply (Shipments Based) (M 1GB)	**201,828**	**247,795**	**271,936**	**273,324**	**994,882**	**270,305**	**289,071**	**306,300**	**323,547**	**1,189,224**	**336,586**	**352,566**	**377,010**	**394,451**	**1,460,613**
% QoQ	*-6.0%*	*22.8%*	*9.7%*	*0.5%*		*-1.1%*	*6.9%*	*6.0%*	*5.6%*		*4.0%*	*4.7%*	*6.9%*	*4.6%*	
% YoY	*-6.3%*	*15.1%*	*25.6%*	*27.3%*	*15.4%*	*33.9%*	*16.7%*	*12.6%*	*18.4%*	*19.5%*	*24.5%*	*22.0%*	*23.1%*	*21.9%*	*22.8%*
Demand (End Consumption) (M 1GB)	**233,782**	**243,225**	**283,072**	**320,542**	**1,080,621**	**288,920**	**305,787**	**341,820**	**400,068**	**1,336,595**	**374,129**	**389,468**	**421,432**	**468,022**	**1,653,052**
% QoQ	*-8.8%*	*4.0%*	*16.4%*	*13.2%*		*-9.9%*	*5.8%*	*11.8%*	*17.0%*		*-6.5%*	*4.1%*	*8.2%*	*11.1%*	
% YoY	*10.5%*	*13.2%*	*21.0%*	*25.1%*	*17.9%*	*23.6%*	*25.7%*	*20.8%*	*24.8%*	*23.7%*	*29.5%*	*27.4%*	*23.3%*	*17.0%*	*23.7%*
Inventory adjustment	-38,000	3,700	-8,000	-32,900	-75,200	37,000	30,000	-24,000	-67,000	-24,000	-31,000	-29,700	-42,000	-73,500	-176,200
Inventory Adj. Demand (M 1GB)	**195,782**	**246,925**	**275,072**	**287,642**	**1,005,421**	**325,920**	**335,787**	**317,820**	**333,068**	**1,312,595**	**343,129**	**359,768**	**379,432**	**394,522**	**1,476,852**
% QoQ	*-8.1%*	*26.1%*	*11.4%*	*4.6%*		*13.3%*	*3.0%*	*-5.4%*	*4.8%*		*3.0%*	*4.8%*	*5.5%*	*4.0%*	
% YoY	*-14.4%*	*10.1%*	*24.8%*	*35.0%*	*13.4%*	*66.5%*	*36.0%*	*15.5%*	*15.8%*	*30.6%*	*5.3%*	*7.1%*	*19.4%*	*18.5%*	*12.5%*
Wafer out (k wpm, 12" equi.)	**1,418**	**1,288**	**1,268**	**1,352**	**1,331**	**1,366**	**1,333**	**1,338**	**1,351**	**1,347**	**1,331**	**1,307**	**1,303**	**1,299**	**1,310**
% QoQ	*-0.4%*	*-9.2%*	*-1.5%*	*6.6%*		*1.1%*	*-2.5%*	*0.4%*	*1.0%*		*-1.5%*	*-1.8%*	*-0.3%*	*-0.3%*	
% YoY	*23.7%*	*2.5%*	*-5.8%*	*-5.1%*	*2.9%*	*-3.6%*	*3.5%*	*5.5%*	*-0.1%*	*1.2%*	*-2.6%*	*-2.0%*	*-2.6%*	*-3.9%*	*-2.8%*
% Sufficiency Ratio	***3.1%***	***0.4%***	***-1.1%***	***-5.0%***	***-1.0%***	***-17.1%***	***-13.9%***	***-3.6%***	***-2.9%***	***-9.4%***	***-1.9%***	***-2.0%***	***-0.6%***	***0.0%***	***-1.1%***
NAND ASP ($/1GB)	**0.06**	**0.06**	**0.06**	**0.08**	**0.07**	**0.14**	**0.24**	**0.29**	**0.32**	**0.25**	**0.33**	**0.35**	**0.36**	**0.36**	**0.35**
% QoQ	*-15.0%*	*-2.5%*	*5.0%*	*25.0%*		*85.0%*	*69.0%*	*18.0%*	*10.0%*		*5.0%*	*4.0%*	*3.0%*	*0.0%*	
% YoY	*5.8%*	*-14.1%*	*-16.5%*	*8.8%*	*-3.9%*	*136.7%*	*310.4%*	*361.2%*	*305.8%*	*284.3%*	*130.3%*	*41.7%*	*23.7%*	*12.5%*	*38.0%*

Source: Company data, Gartner, UBS estimates

NAND demand: details by application

NAND flash demand growth by application, QoQ and YoY%

	Q125 QoQ	Q225 QoQ	Q325 QoQ	Q425 QoQ	2025 YoY	Q126 QoQ	Q226 QoQ	Q326E QoQ	Q426E QoQ	2026E YoY	Q127E QoQ	Q227E QoQ	Q327E QoQ	Q427E QoQ	2027E YoY
Total Embedded	**-8.8%**	**4.1%**	**16.8%**	**13.6%**	**18.7%**	**-9.7%**	**6.0%**	**12.0%**	**17.3%**	**24.9%**	**-6.4%**	**4.2%**	**8.3%**	**11.2%**	**24.2%**
Smartphones	**-7.9%**	**-3.2%**	**19.0%**	**26.3%**	**25.4%**	**-21.5%**	**-7.6%**	**-1.6%**	**24.8%**	**-1.0%**	**-14.8%**	**-6.6%**	**1.1%**	**30.6%**	**0.2%**
iPhones	-29.0%	-7.7%	24.8%	69.7%	28.1%	-34.5%	-5.8%	-1.0%	65.2%	12.4%	-29.4%	-5.9%	2.0%	68.3%	11.3%
Samsung	48.8%	-12.2%	13.4%	-3.6%	24.9%	12.6%	-0.9%	-3.8%	-7.5%	7.8%	26.2%	-13.9%	6.3%	-11.7%	4.2%
Non- iPhone/Samsung	1.7%	6.1%	16.3%	-1.5%	22.7%	-15.0%	-13.5%	-0.9%	-9.8%	-20.0%	-4.8%	-1.8%	-4.3%	-5.4%	-19.8%
Tablet PCs	-21.6%	10.0%	9.5%	17.2%	6.4%	-20.2%	12.2%	4.1%	17.1%	11.2%	-24.4%	13.1%	10.6%	14.3%	6.8%
SSD-Total	**-6.1%**	**7.6%**	**18.6%**	**8.3%**	**21.8%**	**2.6%**	**11.7%**	**18.1%**	**15.8%**	**48.8%**	**-1.7%**	**6.8%**	**10.1%**	**6.9%**	**36.1%**
SSD (notebooks)	-7.2%	11.8%	18.0%	0.9%	21.6%	-15.3%	-1.0%	1.7%	-1.9%	-5.8%	-13.8%	9.7%	14.3%	4.5%	-0.5%
SSD (servers)	-10.9%	7.2%	35.4%	16.7%	29.9%	7.0%	16.5%	27.4%	19.1%	88.1%	5.2%	6.9%	10.6%	7.2%	55.2%
SSD (storage)	6.2%	4.1%	3.9%	4.3%	33.4%	33.0%	29.3%	20.9%	27.4%	106.4%	-3.6%	6.9%	6.6%	6.4%	45.2%
SSD (others)	-4.8%	5.8%	4.0%	4.6%	3.7%	-4.7%	-2.7%	6.5%	7.2%	4.9%	-13.4%	2.6%	9.5%	9.1%	2.2%
Total Flash cards	**-7.7%**	**0.0%**	**0.8%**	**-4.5%**	**-9.7%**	**-21.1%**	**-5.4%**	**-2.9%**	**-6.3%**	**-28.8%**	**-10.3%**	**-2.2%**	**0.2%**	**-0.8%**	**-18.6%**
Mobile Phones - Flash cards	-6.3%	-1.5%	5.3%	-1.6%	-0.8%	-3.8%	-4.9%	-0.9%	-7.2%	-8.5%	3.2%	0.3%	0.0%	-4.3%	-5.0%
Total Bit Demand	**-8.8%**	**4.0%**	**16.4%**	**13.2%**	**17.9%**	**-9.9%**	**5.8%**	**11.8%**	**17.0%**	**23.7%**	**-6.5%**	**4.1%**	**8.2%**	**11.1%**	**23.7%**

NAND flash content assumptions (MB/Unit) by application

		Q125	Q225	Q325	Q425	2025	Q126	Q226	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
Smartphones - Blended		**206**	**207**	**235**	**273**	**231**	**242**	**236**	**235**	**275**	**247**	**251**	**238**	**241**	**288**	**255**
	% YoY	*18%*	*22%*	*20%*	*32%*	*24%*	*18%*	*14%*	*0%*	*1%*	*7%*	*4%*	*1%*	*3%*	*5%*	*3%*
iPhones		517	512	573	579	538	568	573	579	584	563	589	594	617	640	599
Samsung		235	216	232	228	228	258	235	248	239	245	269	245	252	242	253
Non- iPhone/Samsung		137	146	156	168	152	156	151	146	142	149	137	132	127	122	129
SSD - Notebook PCs		**692**	**704**	**758**	**778**	**735**	**763**	**751**	**779**	**789**	**770**	**779**	**766**	**794**	**812**	**789**
	% YoY	*10%*	*13%*	*15%*	*16%*	*14%*	*10%*	*7%*	*3%*	*1%*	*5%*	*2%*	*2%*	*2%*	*3%*	*2%*
SSD - Servers		**3,189**	**3,433**	**3,726**	**4,117**	**3,656**	**4,409**	**4,898**	**5,386**	**6,070**	**5,252**	**6,265**	**6,460**	**6,656**	**6,851**	**6,570**
	% YoY	*4%*	*12%*	*21%*	*33%*	*19%*	*38%*	*43%*	*45%*	*47%*	*44%*	*42%*	*32%*	*24%*	*13%*	*25%*
SSD - Storage		**11,895**	**12,383**	**12,871**	**13,420**	**12,642**	**14,885**	**16,838**	**18,792**	**20,745**	**18,095**	**21,428**	**22,112**	**22,795**	**23,479**	**22,483**
	% YoY	*51%*	*37%*	*26%*	*18%*	*31%*	*25%*	*36%*	*46%*	*55%*	*43%*	*44%*	*31%*	*21%*	*13%*	*24%*

Source: Company data, Gartner, UBS estimates

Semis Capex by Segment

	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Capex US$m										
Memory Total	41,189	47,010	61,035	63,414	46,097	62,173	74,454	131,051	187,669	204,251
DRAM	17,816	23,790	31,252	38,061	34,633	47,461	59,126	105,143	151,781	159,724
NAND/non-volatile	23,373	23,221	29,783	25,353	11,463	14,712	15,328	25,908	35,889	44,526
Large non Memory IDMs	18,960	16,384	22,312	33,940	35,754	30,778	23,655	26,772	38,260	48,169
Tier-one/two foundries	25,340	35,125	51,103	60,782	58,076	52,678	63,073	91,220	126,053	145,693
Other & SAT	17,044	19,049	28,713	32,446	31,969	30,228	36,899	41,122	22,809	21,972
Total	**102,533**	**117,569**	**163,163**	**190,581**	**171,896**	**175,857**	**198,081**	**290,164**	**374,791**	**420,084**
% YoY										
Memory Total	-25.4%	14.1%	29.8%	3.9%	-27.3%	34.9%	19.8%	76.0%	43.2%	8.8%
DRAM	-28.9%	33.5%	31.4%	21.8%	-9.0%	37.0%	24.6%	77.8%	44.4%	5.2%
NAND/non-volatile	-22.5%	-0.7%	28.3%	-14.9%	-54.8%	28.3%	4.2%	69.0%	38.5%	24.1%
Large non Memory IDMs	29.1%	-13.6%	36.2%	52.1%	5.3%	-13.9%	-23.1%	13.2%	42.9%	25.9%
Tier-one/two foundries	25.8%	38.6%	45.5%	18.9%	-4.5%	-9.3%	19.7%	44.6%	38.2%	15.6%
Total	**-4.0%**	**14.7%**	**38.8%**	**16.8%**	**-9.8%**	**2.3%**	**12.6%**	**46.5%**	**29.2%**	**12.1%**
% Capital intensity										
Memory Total	38.7%	40.0%	39.7%	48.9%	49.9%	37.7%	32.4%	13.5%	11.3%	
DRAM	28.5%	37.0%	33.6%	48.9%	66.7%	50.0%	38.3%	16.4%	13.6%	
NAND	58.2%	47.0%	53.2%	53.9%	31.8%	22.3%	21.6%	7.9%	6.7%	
Non-memory	20.1%	21.9%	25.4%	28.7%	29.3%	26.0%	24.0%	24.0%	25.3%	
Total	**24.9%**	**26.7%**	**29.4%**	**33.3%**	**33.0%**	**29.2%**	**26.6%**	**17.7%**	**15.6%**	

Source: Company data, UBS estimates

WFE by Segment

	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
WFE US$m										
Memory Total	22,850	29,450	44,650	39,000	28,377	42,066	52,237	77,899	118,104	130,988
DRAM	10,250	13,400	23,100	22,500	19,988	33,156	41,080	59,330	94,204	102,052
NAND	12,600	16,050	21,550	16,500	8,389	8,910	11,157	18,569	23,900	28,936
Non-memory Total	28,050	30,550	45,850	61,550	62,173	60,434	64,263	80,101	107,896	143,513
Total	**51,500**	**61,500**	**90,500**	**100,550**	**90,550**	**102,500**	**116,500**	**158,000**	**226,000**	**274,500**
China WFE	7,000	14,000	17,000	22,792	31,998	40,000	45,000	49,000	63,000	70,000
% of total	*13.6%*	*22.8%*	*18.8%*	*22.7%*	*35.3%*	*39.0%*	*38.6%*	*31.0%*	*27.9%*	*25.5%*
Domestic China (excl. multinational)	3,500	9,100	12,750	18,502	29,498	38,800	44,000	47,800	61,000	68,000
% YoY										
Memory Total	-33.5%	28.9%	51.6%	-12.7%	-27.2%	48.2%	24.2%	49.1%	51.6%	10.9%
DRAM	-34.1%	30.7%	72.4%	-2.6%	-11.2%	65.9%	23.9%	44.4%	58.8%	8.3%
NAND	-33.0%	27.4%	34.3%	-23.4%	-49.2%	6.2%	25.2%	66.4%	28.7%	21.1%
Non-memory Total	43.8%	8.9%	50.1%	34.2%	1.0%	-2.8%	6.3%	24.6%	34.7%	33.0%
Total	**-4.4%**	**19.4%**	**47.2%**	**11.1%**	**-9.9%**	**13.2%**	**13.7%**	**35.6%**	**43.0%**	**21.5%**
China WFE	16.7%	100.0%	21.4%	34.1%	40.4%	25.0%	12.5%	8.9%	28.6%	11.1%
Domestic China (excl. multinational)	29.6%	160.0%	40.1%	45.1%	59.4%	31.5%	13.4%	8.6%	27.6%	11.5%
Segment % of total WFE										
Memory Total	44.4%	47.9%	49.3%	38.8%	31.3%	41.0%	44.8%	49.3%	52.3%	47.7%
DRAM	19.9%	21.8%	25.5%	22.4%	22.1%	32.3%	35.3%	37.6%	41.7%	37.2%
NAND	24.5%	26.1%	23.8%	16.4%	9.3%	8.7%	9.6%	11.8%	10.6%	10.5%
Non-memory Total	54.5%	49.7%	50.7%	61.2%	68.7%	59.0%	55.2%	50.7%	47.7%	52.3%
Total	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**

Source: Company data, UBS estimates

Bottom-Up Semis Capex Model (US$m)

	2024	% YoY	2025	% YoY	2026E	% YoY	2027E	% YoY	2028E	% YoY
North America										
Intel	23,944	*-7%*	17,672	*-26%*	20,215	*14%*	28,500	*41%*	40,000	*40%*
SpaceX							6,000		20,000	*233%*
Micron	9,796	*40%*	18,040	*84%*	37,213	*106%*	52,000	*40%*	47,000	*-10%*
Texas Instruments	4,820	*-5%*	4,550	*-6%*	2,590	*-43%*	2,900	*12%*	3,100	*7%*
Global Foundries	625	*-91%*	722	*16%*	1,702	*136%*	2,339	*37%*	1,883	*-19%*
SanDisk	601	*-89%*	1,197	*99%*	1,750	*46%*	2,750	*57%*	2,750	*0%*
Europe										
ST Micro	2,642	*-48%*	1,843	*-30%*	2,192	*19%*	2,514	*15%*	2,413	*-4%*
Infineon	2,892	*60%*	2,390	*-17%*	3,059	*28%*	5,946	*94%*	4,456	*-25%*
NXP	876	*-10%*	537	*-39%*	913	*70%*	1,275	*40%*	1,214	*-5%*
Korea										
Samsung Memory	22,805	*-10%*	24,520	*8%*	38,736	*58%*	49,315	*27%*	62,083	*26%*
Samsung Foundry / LSI	13,987	*20%*	9,428	*-33%*	10,232	*9%*	11,320	*11%*	13,395	*18%*
SK Hynix	6,283	*-57%*	11,644	*85%*	33,290	*186%*	46,138	*39%*	52,138	*13%*
Japan										
Kioxia	1,463	*-96%*	1,478	*1%*	3,100	*110%*	3,100	*0%*	3,100	*0%*
Sony	1,300	*-94%*	1,750	*35%*	1,306	*-25%*	1,300	*0%*	1,300	*0%*
Rapidus			3,500	*NA*	3,500	*0%*	1,120	*-68%*	1,120	*0%*
Taiwan										
TSMC	29,760	*113%*	40,739	*37%*	63,000	*55%*	90,000	*43%*	105,000	*17%*
UMC	2,701	*-58%*	1,536	*-43%*	2,052	*34%*	3,227	*57%*	2,581	*-20%*
Nanya	509	*-67%*	425	*-17%*	1,644	*287%*	6,000	*265%*	5,250	*-13%*

Source: Company data, UBS estimates

Bottom-Up Semis Capex Model (US$m)

	2024	% YoY	2025	% YoY	2026E	% YoY	2027E	% YoY	2028E	% YoY
China										
SMIC	7,664	*0%*	8,400	*10%*	8,400	*0%*	10,000	*19%*	12,000	*20%*
Hua Hong	2,739	*5%*	1,814	*-34%*	2,333	*29%*	3,056	*31%*	2,911	*-5%*
Shanghai Huali	2,500	*100%*	4,500	*80%*	5,833	*30%*	9,167	*57%*	10,833	*18%*
YMTC	3,500	*250%*	4,000	*14%*	7,143	*79%*	12,857	*80%*	17,143	*33%*
CXMT	10,340	*41%*	4,604	*-55%*	10,648	*131%*	17,859	*68%*	17,137	*-4%*
SMEC	494	*-66%*	466	*-6%*	513	*10%*	564	*10%*	621	*10%*
CR Micro	990	*-87%*	1,271	*28%*	1,321	*4%*	1,604	*21%*	1,078	*-33%*
Silan	383	*-58%*	553	*44%*	403	*-27%*	830	*106%*	818	*-1%*
Yandong Micro	496	*-60%*	1,925	*288%*	1,500	*-22%*	1,600	*7%*	1,600	*0%*
Sanan	376	*-62%*	200	*-47%*	221	*11%*	221	*0%*	221	*0%*
Yangjie	120	*-88%*	142	*18%*	142	*0%*	142	*0%*	142	*0%*
GalaxyCore	228	*38%*	328	*44%*	361	*10%*	397	*10%*	437	*10%*
Total	**175,857**	***2%***	**198,081**	***13%***	**290,164**	***46%***	**374,791**	***29%***	**420,084**	***12%***
WFE	**102,500**	***13%***	**116,500**	***14%***	**158,000**	***36%***	**226,000**	***43%***	**274,500**	***21%***

Source: Company data, UBS estimates

WFE Spending Summary (US$m)

		2024	% YoY	2025	% YoY	2026E	% YoY	2027E	% YoY	2028E	% YoY
Taiwan											
TSMC	Foundry	15,624	*-12%*	20,370	*30%*	28,980	*42%*	49,500	*71%*	65,625	*33%*
UMC	Foundry	1,351	*-9%*	768	*-43%*	1,026	*34%*	1,613	*57%*	1,419	*-12%*
Nanya	DRAM	255	*6%*	265	*4%*	484	*83%*	4,000	*726%*	2,625	*-34%*
China											
SMIC	Foundry	6,100	*16%*	7,000	*15%*	7,000	*0%*	8,500	*21%*	10,000	*18%*
Hua Hong	Foundry	2,050	*242%*	1,700	*-17%*	2,000	*18%*	2,500	*25%*	2,500	*0%*
Shanghai Huali	Foundry	1,500	*100%*	3,000	*100%*	3,500	*17%*	5,500	*57%*	6,500	*18%*
YMTC	NAND	3,000	*50%*	3,500	*17%*	5,000	*43%*	9,000	*80%*	12,000	*33%*
CXMT	DRAM	8,000	*60%*	5,500	*-31%*	9,000	*64%*	13,500	*50%*	14,000	*4%*
SMEC	Foundry	371	*-58%*	350	*-6%*	385	*10%*	423	*10%*	466	*10%*
Nexchip	Foundry	1,286	*104%*	950	*-26%*	1,173	*23%*	1,349	*15%*	1,551	*15%*
CR Micro	IDM	644	*-8%*	890	*38%*	951	*7%*	1,187	*25%*	819	*-31%*
Silan	IDM	268	*45%*	387	*44%*	282	*-27%*	581	*106%*	573	*-1%*
Yandong Micro	IDM	347	*72%*	1,443	*316%*	1,125	*-22%*	1,200	*7%*	1,200	*0%*
Sanan	IDM	263	*24%*	140	*-47%*	154	*10%*	154	*0%*	154	*0%*
Yangjie	IDM	84	*116%*	100	*19%*	100	*0%*	100	*0%*	100	*0%*
GalaxyCore	IDM	159	*60%*	230	*45%*	253	*10%*	278	*10%*	306	*10%*
Other											
Others (mostly China)		13,728		16,756		15,565		-1,180		-7,996	
Total		**102,500**	***-10%***	**116,500**	***13%***	**158,000**	***36%***	**226,000**	***43%***	**274,500**	***21%***
Upside Scenario								**239,000**	***51%***	**296,000**	***24%***

Source: Company data, UBS estimates. Upside scenario assumes $16bn/$17bn in Others in '27E/'28E

Update On Backend Equipment - Testing & Packaging

Testing

1/ Upside to SoC test demand driven incrementally by ASICs/CPU both on units but also expected pick up in test intensity/insertions

- SoC test TAM for ASICs potentially on course to overtake GPUs in next few years
- Test intensity of CPUs, with advanced packaging adoption, set to potentially come up on par with GPUs mid-LT

2/ Impact of HBM de-spec on SoC test TAM likely net neutral to slightly positive

- RV300 HBM de-spec to HBM4 8Hi (vs HBM4E 12Hi) likely slight negative to test intensity on like for like basis
- Better HBM yields and in turn higher volumes / SiP in turn to offset decline in test time
- Custom HBM4E could introduce another layer of test complexity; incremental positive to test TAM

3/ Memory test TAM also seeing upside but also with ongoing supplier diversification (localization)

- Upside to HBM but also conventional memory (DDR/NAND) test demand on test intensity (HBM – BIT) and capacity increase
- Advantest expects to cede memory test M/S from c.61% in CY25 (UBSe competition from KR peers DI and YC Corp and Teradyne

Advanced Packaging

1/ Logic order momentum continuing to remain strong on AI related CoWoS capa expansion into 2027E

Driven incrementally by ASICs and CPU both on units but also expected pick up in test intensity/insertions

- More meaningful transition of CoW TCB orders from FCB at TSMC expected into 2028/29E
- Transition to CoPoS, EMIB-T as well as growing CPO demand potentially offering upside to logic TCB TAM

2/ Seeing longer use case of HBM TCB tools; hybrid bonding adopt potential push-out

- HBM suppliers dual-track assessing both TCB as well as hybrid bonder use for HBM5 (ASMPT JDP w/ Samsung)
- Cost (higher clean room req + BE tools) but also recent HBM de-spec (12Hi > 8Hi) potential factors
- Beyond stack height, memory suppliers likely to weigh thermal (Cu-Cu w/ less resistivity / better thermal) and B/W improvement in assessment for HB adoption

Stocks – Core Coverage

Samsung Electronics (Nicolas Gaudois) – APAC Key Call Buy, PT Won535,000

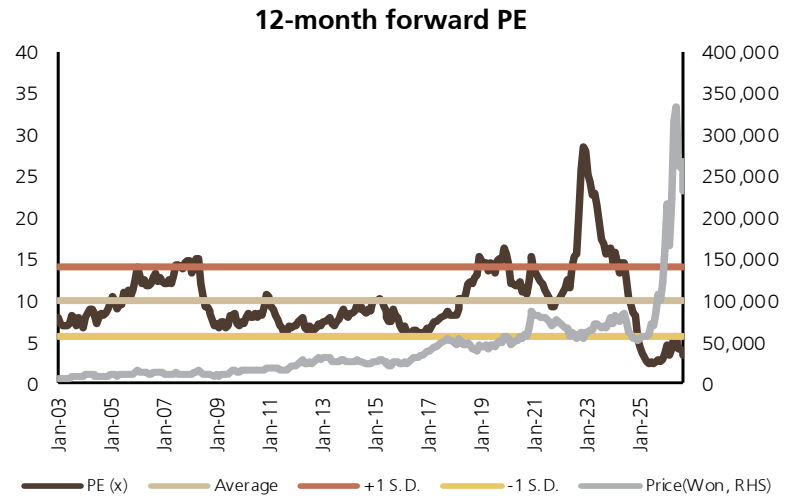
- **Our View**
  - We believe fundamentals remain strong with (1) agentic AI driving a significant uplift in memory demand, (2) improving HBM execution, and (3) enhanced LTAs likely to benefit Samsung more than peers. At current valuation, we do not believe Samsung shares fully reflect structurally higher memory profitability, stronger FCF generation, and significantly enhanced returns to shareholders likely to come through.
- **Will memory undersupply continue into 2028?**
  - Yes, in our view. Agentic AI is boosting further memory demand on multiple fronts beyond HBM, including DDR5/LPDDR5 for conventional servers and CPU head nodes in AI servers, as well as NAND flash for KV Cache and Storage. We now forecast an acceleration of memory demand growth into 2027 with DRAM bit end-consumption growing 39% YoY vs 21% in 2026E, and NAND flash 24%. As almost all of the incremental wafer capacity addition for DRAM is going to HBM, and no new capacity ex China is being added to NAND, supply is not catching up with demand. The main risk remains affordability as memory industry revenues are approaching US$1.64tn in 2027E.
- **Will HBM execution improve?**
  - To some extent. Samsung has improved yields and performance for HBM4. We forecast Samsung to surpass SK Hynix in terms of HBM bit share in 2027 at 41% (SK Hynix 37%, Micron 22%). It also retains the advantage of being able to deploy more new wafer capacity than peers thanks to still significant floor space available in its P4 fab, and the pull-in of P5 to first clean room readiness by 2Q27.
- **Will shareholder returns grow?**
  - We expect Samsung to 1/ consider buybacks/dividend incrementally for the reminder of the 2024-2026 policy. 2/ To reset its policy for 2027 onwards in January. We model returning 50% of FCF ex M&A to shareholders, but believe upside to this is likely. Considering our FCF forecasts of Won253tn in 2026 and Won448tn in 2027, this could imply additional (on top of existing quarterly dividend) shareholder returns of Won110tn and Won214tn for those years respectively.

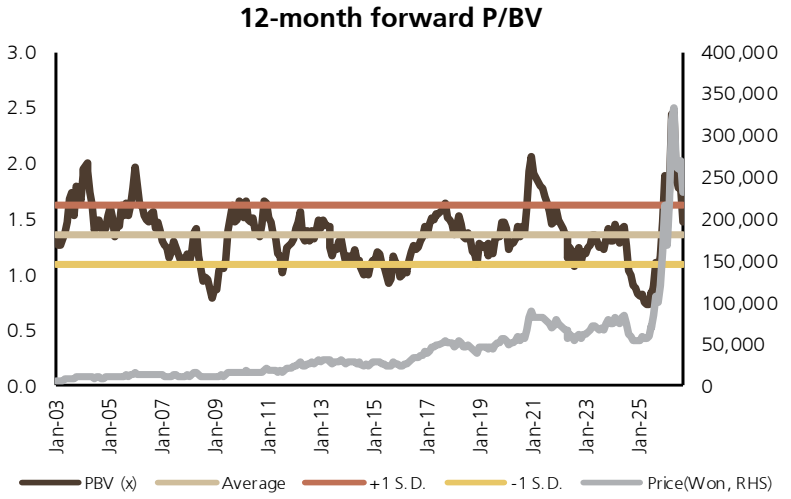
- **PT Won535k –** We value Samsung shares at 3.39x NTM P/BV based on our LT ROE estimate of 29.5% and CoE of 10.0%.

Samsung Electronics (Nicolas Gaudois) – Valuation

UBS price target (Won)	**535,000**
Current stock price (Won)	232,000
Number of shares (bn, ex-treasury shrs)	5.764
Treasury shares	0.082
Market cap. (Won bn)	1,356,337
Preference share price (Won)	194,000
Number of shares (bn, ex-treasury shrs)	0.8
Treasury shares	0.0
Adjusted market cap (Won bn)	1,514,636
Debt (Won bn)	22,409
Cash & equivalents (Won bn)	189,999
Associates (Won bn)	36,568
Adjusted EV (Won bn, ex-treasury shrs)	1,310,477

	2025	2026E	2027E	2028E	2029E	2030E
Current valuation (x)						
EV/Revenues	3.93	1.85	1.24	1.08	1.20	1.06
P/Book	3.76	2.19	1.36	1.01	0.88	0.73
P/E (ex-treasury shares)	35.80	5.49	3.07	2.65	3.31	2.83
P/E (incl. treasury shares)	36.36	5.57	3.12	2.69	3.36	2.87
P/E Stub (ex-treasury shares)	35.46	5.44	3.04	2.63	3.28	2.80
P/E Stub (incl. treasury shares)	36.03	5.52	3.09	2.66	3.32	2.84
EV/EBITDA	15.03	3.08	1.75	1.51	1.84	1.58
UBS valuation (x)						
EV/Revenues	9.10	4.28	2.87	2.51	2.79	2.45
P/Book	8.17	4.76	2.94	2.20	1.91	1.59
P/E (ex-treasury shares)	82.55	12.66	7.09	6.11	7.63	6.52
P/E (incl. treasury shares)	83.85	12.84	7.19	6.20	7.74	6.61
P/E Stub (ex-treasury shares)	82.21	12.61	7.06	6.09	7.60	6.49
P/E Stub (incl. treasury shares)	83.51	12.79	7.16	6.18	7.71	6.59
EV/EBITDA	35.34	7.24	4.12	3.56	4.32	3.72
Other						
EPS (ex-treasury shares)	6,481	42,247	75,498	87,512	70,134	82,044
EPS (incl. treasury shares)	6,381	41,653	74,438	86,283	69,149	80,892
% ROE	***10.6%***	***50.4%***	***55.6%***	***45.0%***	***29.5%***	***29.3%***
Weighted av tangible book per share (Won	60,401	103,659	167,610	224,272	258,312	310,088
Cash per share (Won)	18,684	48,504	101,726	145,231	172,404	211,719
Free cash flow per share (Won)	3,762	43,142	76,907	88,329	75,132	82,541
% Free cash flow yield	*1.6%*	*18.6%*	*33.1%*	*38.1%*	*32.4%*	*35.6%*





Source: Company data, UBS estimates

Samsung Electronics – Key Charts

UBS forecasts vs Consensus

	4Q26E			1Q27E			2026E			2027E			2028E			2029E		
(Won bn)	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**
Sales	208,872	214,763	-2.7%	234,209	226,402	3.4%	709,246	725,607	-2.3%	1,058,367	992,918	6.6%	1,210,897	1,125,824	7.6%	1,089,687	1,373,859	-20.7%
Operating Profit	**124,172**	**121,711**	**2.0%**	**146,185**	**128,473**	**13.8%**	**378,297**	**378,038**	**0.1%**	**696,268**	**574,230**	**21.3%**	**807,408**	**640,558**	**26.0%**	**648,146**	**806,616**	**-19.6%**
% OP margin	*59.4%*	*56.7%*	*278bps*	*62.4%*	*56.7%*	*567bps*	*53.3%*	*52.1%*	*124bps*	*65.8%*	*57.8%*	*795bps*	*66.7%*	*56.9%*	*978bps*	*59.5%*	*58.7%*	*77bps*
EPS	**12,521**	**14,593**	**-14.2%**	**16,834**	**15,298**	**10.0%**	**42,247**	**45,140**	**-6.4%**	**75,498**	**68,564**	**10.1%**	**87,512**	**73,053**	**19.8%**	**70,134**	**61,189**	**14.6%**

Value drivers	DRAM ASP % YoY '27E/'28E	NAND ASP % YoY '27E/'28E	Operating Profit (Won) '27E/'28E
Won600k upside	+70%/+10%	+50%/-15%	750tn/850tn
Won535k base	+53%/+3%	+37%/-28%	696tn/807tn
Won160k downside	+35%/-5%	+15%/-35%	620tn/750tn

Source: Company data, Visible Alpha, UBS estimates

Samsung Electronics – Divisional Forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues														
Device Solutions (Semis)	129,990	81,700	127,500	158,083	175,305	542,588	196,823	221,337	241,872	254,836	914,868	1,046,821	902,144	1,034,092
Displays (SDC)	29,961	6,700	7,500	8,007	8,930	31,137	6,361	6,369	7,359	7,784	27,872	26,550	25,197	23,924
MX & Networks	129,599	38,100	33,200	36,081	31,711	139,092	39,926	31,759	38,645	33,880	144,209	157,521	165,091	171,769
Consumer Electronics	57,299	14,300	14,500	13,565	12,555	54,921	11,759	12,578	12,809	13,747	50,893	52,524	52,770	53,019
Harman	15,800	3,800	4,600	4,830	5,072	18,302	3,990	4,738	5,072	5,325	19,125	20,081	21,085	22,139
Others	-30,143	-11,027	-16,101	-25,865	-25,000	-77,993	-25,000	-25,000	-25,000	-25,000	-100,000	-94,000	-78,000	-66,000
Total	**333,606**	**133,873**	**171,499**	**195,000**	**208,872**	**709,246**	**234,209**	**252,130**	**281,106**	**290,922**	**1,058,367**	**1,210,897**	**1,089,687**	**1,240,343**
Op Profit														
Device Solutions (Semis)	24,900	53,700	89,200	109,184	125,805	377,890	146,340	168,158	185,707	195,755	695,961	799,919	635,042	738,614
Displays (SDC)	4,535	400	700	309	287	1,696	206	198	496	663	1,563	2,233	2,167	2,065
MX & Networks	12,880	2,800	-700	-960	-1,462	-322	-267	-1,044	62	-545	-1,793	2,779	7,560	12,048
Consumer Electronics	-204	201	0	-329	-440	-569	-134	-142	-35	40	-271	1,148	1,446	1,453
Harman	1,500	200	400	435	482	1,516	239	379	456	533	1,607	1,730	1,931	2,128
Total	**43,601**	**57,233**	**89,492**	**107,400**	**124,172**	**378,297**	**146,185**	**167,349**	**186,487**	**196,246**	**696,268**	**807,408**	**648,146**	**756,309**
% OP margin														
Device Solutions (Semis)	19.2%	65.7%	70.0%	69.1%	71.8%	69.6%	74.4%	76.0%	76.8%	76.8%	76.1%	76.4%	70.4%	71.4%
Displays (SDC)	15.1%	6.0%	9.3%	3.9%	3.2%	5.4%	3.2%	3.1%	6.7%	8.5%	5.6%	8.4%	8.6%	8.6%
MX & Networks	9.9%	7.3%	-2.1%	-2.7%	-4.6%	-0.2%	-0.7%	-3.3%	0.2%	-1.6%	-1.2%	1.8%	4.6%	7.0%
Consumer Electronics	-0.4%	1.4%	0.0%	-2.4%	-3.5%	-1.0%	-1.1%	-1.1%	-0.3%	0.3%	-0.5%	2.2%	2.7%	2.7%
Harman	9.5%	5.3%	8.7%	9.0%	9.5%	8.3%	6.0%	8.0%	9.0%	10.0%	8.4%	8.6%	9.2%	9.6%
Total	**13.1%**	**42.8%**	**52.2%**	**55.1%**	**59.4%**	**53.3%**	**62.4%**	**66.4%**	**66.3%**	**67.5%**	**65.8%**	**66.7%**	**59.5%**	**61.0%**

Source: Company data, UBS estimates

Samsung Electronics – DRAM Forecasts

	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
DRAM revenues (Won bn)	**70,045**	**53,048**	**84,233**	**107,186**	**114,870**	**359,337**	**134,904**	**153,331**	**169,262**	**181,107**	**638,605**	**808,803**	**718,871**	**822,882**
% QoQ / % YoY	*38.7%*	*100.2%*	*58.8%*	*27.2%*	*7.2%*	*413.0%*	*17.4%*	*13.7%*	*10.4%*	*7.0%*	*77.7%*	*26.7%*	*-11.1%*	*14.5%*
DRAM revenues (US$m)	**50,178**	**36,188**	**58,237**	**75,568**	**85,233**	**255,225**	**100,098**	**113,771**	**125,592**	**134,380**	**473,842**	**600,128**	**533,399**	**610,575**
% QoQ / % YoY	*35.2%*	*93.8%*	*60.9%*	*29.8%*	*12.8%*	*408.6%*	*17.4%*	*13.7%*	*10.4%*	*7.0%*	*85.7%*	*26.7%*	*-11.1%*	*14.5%*
DRAM unit shipment (1Gb eq)	120,251	33,761	37,467	39,349	38,712	149,289	40,756	44,008	46,911	48,896	180,571	222,832	265,367	322,022
% QoQ / % YoY	*9.7%*	*1.0%*	*11.0%*	*5.0%*	*-1.6%*	*24.1%*	*5.3%*	*8.0%*	*6.6%*	*4.2%*	*21.0%*	*23.4%*	*19.1%*	*21.3%*
Avg. wafer out per month (K 12" eq)	668	676	678	685	691	682	703	724	752	790	742	844	893	957
% QoQ / % YoY	*8.1%*	*0.4%*	*0.2%*	*1.1%*	*0.9%*	*2.2%*	*1.7%*	*2.9%*	*3.9%*	*5.1%*	*8.7%*	*13.7%*	*5.9%*	*7.1%*
DRAM ASP - US$ (1Gb eq)	0.42	1.07	1.55	1.92	2.20	1.71	2.46	2.59	2.68	2.75	2.62	2.69	2.01	1.90
% QoQ / % YoY	*23.2%*	*91.9%*	*45.0%*	*23.6%*	*14.6%*	*309.7%*	*11.6%*	*5.3%*	*3.6%*	*2.7%*	*53.5%*	*2.6%*	*-25.4%*	*-5.7%*
Operating Profit (Won bn)	**28,915**	**41,532**	**64,403**	**81,826**	**89,500**	**277,262**	**108,027**	**124,452**	**137,978**	**147,489**	**517,947**	**667,482**	**567,246**	**656,322**
% OP margin	*41.1%*	*78.3%*	*76.5%*	*76.3%*	*77.9%*	*77.2%*	*80.1%*	*81.2%*	*81.5%*	*81.4%*	*81.1%*	*82.5%*	*78.9%*	*79.8%*
Fully Loaded Cost per unit (1Gb US$)	0.24	0.23	0.37	0.45	0.49	0.39	0.49	0.49	0.49	0.51	0.50	0.47	0.42	0.38
% QoQ / % YoY	*-0.1%*	*-6.4%*	*57.3%*	*24.2%*	*7.0%*	*63.0%*	*0.6%*	*-0.5%*	*1.6%*	*3.1%*	*27.0%*	*-5.1%*	*-9.9%*	*-9.5%*
Cash Cost per Unit (Unit 1Gb US$)	0.13	0.12	0.26	0.35	0.37	0.28	0.37	0.37	0.38	0.40	0.38	0.36	0.32	0.29
% QoQ / % YoY	*-8.2%*	*-19.1%*	*116.0%*	*32.1%*	*5.9%*	*111.1%*	*1.0%*	*0.4%*	*2.8%*	*4.0%*	*36.3%*	*-5.1%*	*-11.8%*	*-10.3%*
DDR revenues (US$m)	**44,725**	**33,984**	**55,146**	**69,984**	**76,987**	**236,101**	**85,961**	**97,204**	**106,174**	**112,279**	**401,618**	**479,347**	**349,448**	**344,108**
% QoQ / % YoY	*42.4%*	*104.2%*	*62.3%*	*26.9%*	*10.0%*	*427.9%*	*11.7%*	*13.1%*	*9.2%*	*5.8%*	*70.1%*	*19.4%*	*-27.1%*	*-1.5%*
DDR unit shipment (1Gb eq)	115,924	32,242	35,407	35,965	34,260	137,873	35,865	38,727	41,047	42,539	158,179	190,469	221,522	265,576
% QoQ / % YoY	*10.9%*	*1.2%*	*9.8%*	*1.6%*	*-4.7%*	*18.9%*	*4.7%*	*8.0%*	*6.0%*	*3.6%*	*14.7%*	*20.4%*	*16.3%*	*19.9%*
DDR ASP - US$ (1Gb eq)	0.39	1.05	1.56	1.95	2.25	1.71	2.40	2.51	2.59	2.64	2.54	2.52	1.58	1.30
% QoQ / % YoY	*28.4%*	*101.7%*	*47.8%*	*24.9%*	*15.5%*	*343.9%*	*6.7%*	*4.7%*	*3.1%*	*2.0%*	*48.3%*	*-0.9%*	*-37.3%*	*-17.9%*
HBM revenues (US$m)	**5,453**	**2,203**	**3,091**	**5,584**	**8,246**	**19,124**	**14,137**	**16,568**	**19,418**	**22,101**	**72,224**	**120,781**	**183,951**	**266,467**
% QoQ / % YoY	*-4.4%*	*8.7%*	*40.3%*	*80.6%*	*47.7%*	*250.7%*	*71.4%*	*17.2%*	*17.2%*	*13.8%*	*277.7%*	*67.2%*	*52.3%*	*44.9%*
HBM unit shipment (1Gb eq)	4,327	1,519	2,061	3,384	4,452	11,416	4,891	5,281	5,864	6,356	22,392	32,363	43,845	56,446
% QoQ / % YoY	*-14.6%*	*-3.3%*	*35.6%*	*64.2%*	*31.6%*	*163.9%*	*9.9%*	*8.0%*	*11.0%*	*8.4%*	*96.1%*	*44.5%*	*35.5%*	*28.7%*
HBM ASP - US$ (1Gb eq)	1.26	1.45	1.50	1.65	1.85	1.68	2.89	3.14	3.31	3.48	3.23	3.73	4.20	4.72
% QoQ / % YoY	*12.0%*	*12.4%*	*3.4%*	*10.0%*	*12.3%*	*32.9%*	*56.1%*	*8.5%*	*5.6%*	*5.0%*	*92.5%*	*15.7%*	*12.4%*	*12.5%*

Source: Company data, UBS estimates

Samsung Electronics – NAND Forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues	**33,944**	**21,752**	**36,567**	**43,669**	**52,195**	**154,184**	**54,255**	**59,576**	**63,758**	**64,877**	**242,467**	**200,934**	**143,441**	**168,696**
% YoY	*-0.2%*	*222.0%*	*375.3%*	*390.8%*	*392.6%*	*354.2%*	*149.4%*	*62.9%*	*46.0%*	*24.3%*	*57.3%*	*-17.1%*	*-28.6%*	*17.6%*
% QoQ		*105.3%*	*68.1%*	*19.4%*	*19.5%*		*3.9%*	*9.8%*	*7.0%*	*1.8%*				
Revenues (US$m)	**24,248**	**14,838**	**25,282**	**30,788**	**38,729**	**109,636**	**40,257**	**44,205**	**47,308**	**48,139**	**179,909**	**149,092**	**106,432**	**125,172**
% YoY	*-3.1%*	*216.7%*	*346.4%*	*370.7%*	*426.3%*	*352.1%*	*171.3%*	*74.8%*	*53.7%*	*24.3%*	*64.1%*	*-17.1%*	*-28.6%*	*17.6%*
% QoQ		*101.6%*	*70.4%*	*21.8%*	*25.8%*		*3.9%*	*9.8%*	*7.0%*	*1.8%*				
Unit shipments (1 GB eq m)	**305,670**	**83,786**	**85,481**	**89,760**	**101,618**	**360,645**	**100,819**	**106,731**	**111,698**	**113,659**	**432,907**	**497,979**	**582,831**	**685,561**
% YoY	*3.0%*	*34.1%*	*8.6%*	*3.7%*	*30.4%*	*18.0%*	*20.3%*	*24.9%*	*24.4%*	*11.8%*	*20.0%*	*15.0%*	*17.0%*	*17.6%*
% QoQ		*7.5%*	*2.0%*	*5.0%*	*13.2%*		*-0.8%*	*5.9%*	*4.7%*	*1.8%*				
Avg. wafer out per month (K 12" eq)	**438**	**412**	**405**	**400**	**400**	**404**	**398**	**393**	**388**	**383**	**390**	**370**	**356**	**350**
% YoY	*2.6%*	*-12.5%*	*-9.8%*	*-5.5%*	*-2.2%*	*-7.7%*	*-3.4%*	*-3.1%*	*-3.1%*	*-4.4%*	*-3.5%*	*-5.1%*	*-3.9%*	*-1.6%*
% QoQ		*-6.0%*	*-1.6%*	*-1.2%*	*0.0%*		*-1.6%*	*-1.3%*	*-1.3%*	*-1.3%*				
ASP (US$ 1 GB eq)	**0.08**	**0.18**	**0.30**	**0.34**	**0.38**	**0.30**	**0.40**	**0.41**	**0.42**	**0.42**	**0.42**	**0.30**	**0.18**	**0.18**
% YoY	*-6.0%*	*136.3%*	*311.0%*	*353.9%*	*303.5%*	*283.2%*	*125.5%*	*40.0%*	*23.5%*	*11.1%*	*36.7%*	*-28.0%*	*-39.0%*	*0.0%*
% QoQ		*87.5%*	*67.0%*	*16.0%*	*11.1%*		*4.8%*	*3.7%*	*2.3%*	*0.0%*				
Op Profit	**2,640**	**12,901**	**25,574**	**29,344**	**37,505**	**105,324**	**39,687**	**44,543**	**48,270**	**48,807**	**181,307**	**133,765**	**67,750**	**80,905**
% OP margin	***7.6%***	***59.3%***	***69.9%***	***67.2%***	***71.9%***	***68.4%***	***73.1%***	***74.8%***	***75.7%***	***75.2%***	***74.8%***	***66.6%***	***47.2%***	***48.0%***
Fully Loaded Cost per unit(1GB US$)	**0.07**	**0.07**	**0.09**	**0.11**	**0.11**	**0.10**	**0.11**	**0.10**	**0.10**	**0.10**	**0.10**	**0.10**	**0.10**	**0.10**
% YoY	*-7.9%*	*-8.4%*	*16.0%*	*63.3%*	*52.3%*	*31.0%*	*48.8%*	*17.5%*	*-8.6%*	*-2.2%*	*9.1%*	*-4.5%*	*-3.7%*	*-1.4%*
% QoQ		*2.3%*	*23.4%*	*26.6%*	*-4.7%*		*0.0%*	*-2.5%*	*-1.6%*	*2.0%*				
Cash Cost per unit (1 GB US$)	**0.05**	**0.06**	**0.07**	**0.10**	**0.10**	**0.08**	**0.10**	**0.09**	**0.09**	**0.09**	**0.09**	**0.09**	**0.09**	**0.09**
% YoY	*-7.2%*	*2.1%*	*26.6%*	*92.1%*	*85.8%*	*51.9%*	*66.9%*	*25.8%*	*-7.0%*	*-1.2%*	*13.8%*	*-3.7%*	*-3.4%*	*-0.8%*
% QoQ		*12.1%*	*28.8%*	*33.6%*	*-3.6%*		*0.6%*	*-2.9%*	*-1.2%*	*2.4%*				

Source: Company data, UBS estimates

Samsung Electronics – DS Division Forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues														
Memory	**103,990**	**74,800**	**120,800**	**150,855**	**167,065**	**513,520**	**189,160**	**212,907**	**233,021**	**245,984**	**881,071**	**1,009,737**	**862,312**	**991,578**
% YoY	*23.1%*	*292.4%*	*471.7%*	*465.0%*	*350.3%*	*393.8%*	*152.9%*	*76.2%*	*54.5%*	*47.2%*	*71.6%*	*14.6%*	*-14.6%*	*15.0%*
% QoQ		*101.6%*	*61.5%*	*24.9%*	*10.7%*		*13.2%*	*12.6%*	*9.4%*	*5.6%*				
System LSI & Foundry	**26,000**	**6,900**	**6,700**	**7,227**	**8,241**	**29,068**	**7,664**	**8,430**	**8,852**	**8,852**	**33,797**	**37,084**	**39,832**	**42,514**
% YoY	*-2.1%*	*15.0%*	*0.0%*	*12.9%*	*19.4%*	*11.8%*	*11.1%*	*25.8%*	*22.5%*	*7.4%*	*16.3%*	*9.7%*	*7.4%*	*6.7%*
% QoQ		*0.0%*	*-2.9%*	*7.9%*	*14.0%*		*-7.0%*	*10.0%*	*5.0%*	*0.0%*				
Total Semis	**129,990**	**81,700**	**127,500**	**158,083**	**175,305**	**542,588**	**196,823**	**221,337**	**241,872**	**254,836**	**914,868**	**1,046,821**	**902,144**	**1,034,092**
% YoY	*17.0%*	*226.0%*	*358.2%*	*377.6%*	*298.4%*	*317.4%*	*140.9%*	*73.6%*	*53.0%*	*45.4%*	*68.6%*	*14.4%*	*-13.8%*	*14.6%*
% QoQ		*85.7%*	*56.1%*	*24.0%*	*10.9%*		*12.3%*	*12.5%*	*9.3%*	*5.4%*				
Total Semis (US $m)	**93,007**	**55,733**	**88,151**	**111,451**	**130,076**	**385,411**	**146,042**	**164,231**	**179,468**	**189,087**	**678,828**	**776,737**	**669,387**	**767,292**
% YoY	*13.8%*	*220.8%*	*330.3%*	*358.1%*	*322.1%*	*314.4%*	*162.0%*	*86.3%*	*61.0%*	*45.4%*	*76.1%*	*14.4%*	*-13.8%*	*14.6%*
% QoQ		*80.8%*	*58.2%*	*26.4%*	*16.7%*		*12.3%*	*12.5%*	*9.3%*	*5.4%*				
Operating Profit														
Memory	**31,555**	**54,433**	**89,977**	**111,170**	**127,005**	**382,586**	**147,714**	**168,995**	**186,249**	**196,297**	**699,254**	**801,247**	**634,996**	**737,227**
% OP margin	*30.3%*	*72.8%*	*74.5%*	*73.7%*	*76.0%*	*74.5%*	*78.1%*	*79.4%*	*79.9%*	*79.8%*	*79.4%*	*79.4%*	*73.6%*	*74.3%*
System LSI & Foundry	**-6,655**	**-733**	**-777**	**-1,986**	**-1,200**	**-4,696**	**-1,373**	**-837**	**-542**	**-542**	**-3,293**	**-1,328**	**46**	**1,387**
% OP margin	*-25.6%*	*-10.6%*	*-11.6%*	*-27.5%*	*-14.6%*	*-16.2%*	*-17.9%*	*-9.9%*	*-6.1%*	*-6.1%*	*-9.7%*	*-3.6%*	*0.1%*	*3.3%*
Total Device Solutions (Semis)	**24,900**	**53,700**	**89,200**	**109,184**	**125,805**	**377,890**	**146,340**	**168,158**	**185,707**	**195,755**	**695,961**	**799,919**	**635,042**	**738,614**
% OP margin	*19.2%*	*65.7%*	*70.0%*	*69.1%*	*71.8%*	*69.6%*	*74.4%*	*76.0%*	*76.8%*	*76.8%*	*76.1%*	*76.4%*	*70.4%*	*71.4%*

Source: Company data, UBS estimates

Samsung Electronics – Samsung Display Forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Large Panel OLED revenues	**1,431**	**296**	**369**	**374**	**434**	**1,482**	**302**	**310**	**333**	**406**	**1,350**	**1,208**	**1,081**	**967**
% YoY	*19.9%*	*-14.7%*	*-11.7%*	*20.5%*	*22.4%*	*3.5%*	*1.8%*	*-16.1%*	*-11.1%*	*-6.4%*	*-8.9%*	*-10.5%*	*-10.5%*	*-10.5%*
% QoQ		*-16.4%*	*24.4%*	*1.5%*	*16.0%*		*-30.5%*	*2.6%*	*7.5%*	*22.1%*				
Large Panel OLED area ('000 sq.m.)	**1,600**	**357**	**416**	**461**	**580**	**1,814**	**416**	**440**	**487**	**613**	**1,957**	**1,977**	**1,998**	**2,020**
% YoY	*20.3%*	*-4.1%*	*-4.9%*	*24.1%*	*38.6%*	*13.4%*	*16.7%*	*5.6%*	*5.7%*	*5.7%*	*7.8%*	*1.1%*	*1.1%*	*1.1%*
% QoQ		*-14.9%*	*16.8%*	*10.7%*	*25.9%*		*-28.3%*	*5.7%*	*10.8%*	*25.9%*				
Large Panel OLED OP	**-560**	**-102**	**-87**	**-100**	**-71**	**-360**	**-100**	**-88**	**-85**	**-46**	**-319**	**-297**	**-272**	**-390**
% OP margin	*NM*	*NM*	*NM*	*NM*	*NM*	*NM*	*NM*	*NM*	*NM*	*NM*	*NM*	*NM*	*NM*	*NM*
Smartphone OLED panel unit (mn)	**376**	**82**	**89**	**94**	**117**	**383**	**92**	**92**	**101**	**106**	**390**	**387**	**384**	**381**
% YoY	*1.3%*	*-3.3%*	*5.4%*	*-7.0%*	*11.6%*	*1.9%*	*11.6%*	*2.4%*	*7.3%*	*-9.9%*	*1.8%*	*-0.6%*	*-0.8%*	*-0.8%*
% QoQ		*-22.0%*	*9.0%*	*5.0%*	*25.0%*		*-22.0%*	*0.0%*	*10.0%*	*5.0%*				
Mobile OLED panel area (mn inches2)	**9,499**	**2,101**	**2,280**	**2,351**	**2,824**	**9,556**	**2,270**	**2,277**	**2,536**	**2,672**	**9,755**	**10,032**	**10,112**	**10,169**
% YoY	*0.9%*	*7.3%*	*4.4%*	*-10.6%*	*3.6%*	*0.6%*	*8.0%*	*-0.1%*	*7.9%*	*-5.4%*	*2.1%*	*2.8%*	*0.8%*	*0.6%*
% QoQ		*-22.9%*	*8.5%*	*3.1%*	*20.1%*		*-19.6%*	*0.3%*	*11.4%*	*5.4%*				
Mobile OLED revenues	**28,535**	**6,403**	**7,131**	**7,633**	**8,496**	**29,818**	**6,059**	**6,059**	**7,026**	**7,378**	**26,522**	**25,342**	**24,116**	**22,957**
% YoY	*2.0%*	*15.3%*	*19.2%*	*-2.0%*	*-7.7%*	*4.5%*	*-5.4%*	*-15.0%*	*-7.9%*	*-13.2%*	*-11.1%*	*-4.4%*	*-4.8%*	*-4.8%*
% QoQ		*-30.4%*	*11.4%*	*7.0%*	*11.3%*		*-28.7%*	*0.0%*	*16.0%*	*5.0%*				
Mobile OLED OP	**5,095**	**501**	**786**	**410**	**358**	**2,055**	**306**	**286**	**581**	**710**	**1,882**	**2,531**	**2,440**	**2,455**
% OP margin	*17.9%*	*7.8%*	*11.0%*	*5.4%*	*4.2%*	*6.9%*	*5.0%*	*4.7%*	*8.3%*	*9.6%*	*7.1%*	*10.0%*	*10.1%*	*10.7%*
Total panel revenues	**29,961**	**6,700**	**7,500**	**8,007**	**8,930**	**31,137**	**6,361**	**6,369**	**7,359**	**7,784**	**27,872**	**26,550**	**25,197**	**23,924**
% YoY	*2.0%*	*13.6%*	*17.2%*	*-1.2%*	*-6.6%*	*3.9%*	*-5.1%*	*-15.1%*	*-8.1%*	*-12.8%*	*-10.5%*	*-4.7%*	*-5.1%*	*-5.1%*
% QoQ		*-29.9%*	*11.9%*	*6.8%*	*11.5%*		*-28.8%*	*0.1%*	*15.5%*	*5.8%*				
Op Profit	**4,535**	**400**	**700**	**309**	**287**	**1,695**	**206**	**198**	**496**	**663**	**1,563**	**2,233**	**2,167**	**2,065**
% OP margin	*15.1%*	*6.0%*	*9.3%*	*3.9%*	*3.2%*	*5.4%*	*3.2%*	*3.1%*	*6.7%*	*8.5%*	*5.6%*	*8.4%*	*8.6%*	*8.6%*

Source: Company data, UBS estimates

Samsung Electronics – MX & Networks Forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Mobile Devices revenues	**126,500**	**37,500**	**32,300**	**35,135**	**30,766**	**135,701**	**39,075**	**30,908**	**37,667**	**32,804**	**140,454**	**153,578**	**161,001**	**167,527**
% YoY	*10.6%*	*3.6%*	*13.3%*	*4.9%*	*8.7%*	*7.3%*	*4.2%*	*-4.3%*	*7.2%*	*6.6%*	*3.5%*	*9.3%*	*4.8%*	*4.1%*
% QoQ		*32.5%*	*-13.9%*	*18.3%*	*-16.5%*		*26.5%*	*-19.2%*	*21.5%*	*-13.9%*				
Op Profit - mobile devices	**12,605**	**2,747**	**-718**	**-960**	**-1,414**	**-345**	**-309**	**-1,069**	**62**	**-512**	**-1,829**	**2,730**	**7,497**	**11,971**
% YoY	*19.8%*	*-6.5%*	*13.7%*	*-0.8%*	*18.7%*	*-102.7%*	*3.4%*	*2.8%*	*2.9%*	*-1.8%*	*430.8%*	*-249.3%*	*174.6%*	*59.7%*
% QoQ		*-2.3%*	*0.5%*	*10.0%*	*10.0%*		*-15.0%*	*0.0%*	*10.0%*	*5.0%*				
% OP margin	*10.0%*	*7.3%*	*-2.2%*	*-2.7%*	*-4.6%*	*-0.3%*	*-0.8%*	*-3.5%*	*0.2%*	*-1.6%*	*-1.3%*	*1.8%*	*4.7%*	*7.1%*
Total mobile devices (m units)	**266.5**	**65.3**	**69.8**	**64.4**	**62.3**	**261.8**	**69.0**	**65.0**	**67.5**	**63.0**	**264.5**	**274.0**	**277.0**	**279.0**
% YoY	*5.8%*	*-3.3%*	*7.4%*	*-5.3%*	*-5.6%*	*-1.8%*	*5.7%*	*-6.9%*	*4.8%*	*1.1%*	*1.0%*	*3.6%*	*1.1%*	*0.7%*
% QoQ		*-1.1%*	*6.9%*	*-7.7%*	*-3.3%*		*10.8%*	*-5.8%*	*3.8%*	*-6.7%*				
Smartphone unit shipments (mil)	**239.5**	**59.5**	**64.7**	**59.2**	**56.8**	**240.2**	**63.5**	**60.0**	**62.0**	**57.0**	**242.5**	**251.0**	**254.0**	**256.0**
% YoY	*6.9%*	*-1.7%*	*11.6%*	*-3.0%*	*-5.3%*	*0.3%*	*6.7%*	*-7.3%*	*4.7%*	*0.4%*	*1.0%*	*3.5%*	*1.2%*	*0.8%*
% QoQ		*-0.8%*	*8.7%*	*-8.5%*	*-4.1%*		*11.8%*	*-5.5%*	*3.3%*	*-8.1%*				
Smartphones ASP (US$)	**286**	**350**	**260**	**329**	**295**	**308**	**372**	**295**	**358**	**320**	**337**	**361**	**377**	**391**
% YoY	*-1.8%*	*7.4%*	*-3.7%*	*8.1%*	*20.9%*	*7.4%*	*6.2%*	*13.4%*	*8.8%*	*8.3%*	*9.5%*	*7.0%*	*4.5%*	*3.9%*
% QoQ		*43.5%*	*-25.7%*	*26.4%*	*-10.2%*		*26.0%*	*-20.7%*	*21.3%*	*-10.7%*				
Op Profit - mobile phones/tablet	**11,742**	**2,747**	**-668**	**-793**	**-1,170**	**116**	**-205**	**-966**	**119**	**-452**	**-1,505**	**2,554**	**6,894**	**11,096**
% OP margin	*11.1%*	*8.5%*	*-2.4%*	*-2.7%*	*-4.7%*	*0.1%*	*-0.6%*	*-3.8%*	*0.4%*	*-1.7%*	*-1.3%*	*2.0%*	*5.0%*	*7.8%*
Network revenues	**3,100**	**600**	**900**	**945**	**945**	**3,390**	**851**	**851**	**978**	**1,076**	**3,755**	**3,943**	**4,090**	**4,243**
% YoY	*9.9%*	*-25.0%*	*28.6%*	*57.6%*	*-5.5%*	*9.4%*	*41.8%*	*-5.5%*	*3.5%*	*13.9%*	*10.8%*	*5.0%*	*3.7%*	*3.7%*
% QoQ		*-40.0%*	*50.0%*	*5.0%*	*0.0%*		*-10.0%*	*0.0%*	*15.0%*	*10.0%*				
OP Profit - networks	**275**	**52**	**18**	**0**	**-47**	**23**	**43**	**26**	**0**	**-32**	**36**	**49**	**62**	**77**
% OP margin	*8.9%*	*8.7%*	*2.0%*	*0.0%*	*-5.0%*	*0.7%*	*5.0%*	*3.0%*	*0.0%*	*-3.0%*	*1.0%*	*1.2%*	*1.5%*	*1.8%*
Total revenues	**129,599**	**38,100**	**33,200**	**36,081**	**31,711**	**139,092**	**39,926**	**31,759**	**38,645**	**33,880**	**144,209**	**157,521**	**165,091**	**171,769**
% YoY	*10.6%*	*3.0%*	*13.7%*	*5.8%*	*8.2%*	*7.3%*	*4.8%*	*-4.3%*	*7.1%*	*6.8%*	*3.7%*	*9.2%*	*4.8%*	*4.0%*
% QoQ		*30.0%*	*-12.9%*	*8.7%*	*-12.1%*		*25.9%*	*-20.5%*	*21.7%*	*-12.3%*				
Op Profit - MX & Networks	**12,880**	**2,800**	**-700**	**-960**	**-1,462**	**-322**	**-267**	**-1,044**	**62**	**-545**	**-1,793**	**2,779**	**7,560**	**12,048**
% OP margin	*9.9%*	*7.3%*	*-2.1%*	*-2.7%*	*-4.6%*	*-0.2%*	*-0.7%*	*-3.3%*	*0.2%*	*-1.6%*	*-1.2%*	*1.8%*	*4.6%*	*7.0%*

Source: Company data, UBS estimates

Samsung Electronics – Smartphone Unit Forecasts

(mil units)		2023	2024	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E
Galaxy S22		2.5												
Galaxy S23		31.0	2.5											
Galaxy S24			35.5	2.5										
Galaxy S25				34.1	1.5	1.0	0.7	0.5	3.7					
Galaxy S26					10.8	8.2	7.0	6.0	32.0	1.0	0.5	0.5	0.5	2.5
Galaxy S27										13.0	9.0	7.5	6.0	35.5
Galaxy foldable		8.6	6.7	7.2	0.5	0.5	4.5	3.3	8.8	0.5	0.5	5.0	3.5	9.5
Other high end		15.0	10.6	9.3	2.7	4.0	3.0	4.0	13.7	3.0	3.0	3.0	4.0	13.0
> US$400		**57.1**	**55.3**	**53.1**	**15.5**	**13.7**	**15.2**	**13.8**	**58.2**	**17.5**	**13.0**	**16.0**	**14.0**	**60.5**
	% YoY	*-17.7%*	*-3.2%*	*-4.0%*	*-6.6%*	*24.5%*	*8.6%*	*20.0%*	*9.6%*	*12.9%*	*-5.1%*	*5.3%*	*1.4%*	*4.0%*
	% QoQ				*34.8%*	*-11.6%*	*10.9%*	*-9.2%*		*26.8%*	*-25.7%*	*23.1%*	*-12.5%*	
Galaxy S+Note		**33.5**	**38.0**	**36.6**	**12.3**	**9.2**	**7.7**	**6.5**	**35.7**	**14.0**	**9.5**	**8.0**	**6.5**	**38.0**
	% YoY	*-17.7%*	*-3.2%*	*-3.7%*	*-12.8%*	*8.2%*	*2.7%*	*0.0%*	*-2.5%*	*13.8%*	*3.3%*	*3.9%*	*0.0%*	*6.4%*
	% QoQ				*89.2%*	*-25.2%*	*-16.3%*	*-15.6%*		*115.4%*	*-32.1%*	*-15.8%*	*-18.8%*	
US$200 to US$300		**24.5**	**26.5**	**31.0**	**8.3**	**8.0**	**7.0**	**7.0**	**30.3**	**8.0**	**7.0**	**8.0**	**7.0**	**30.0**
	% YoY	*-15.5%*	*8.2%*	*17.0%*	*18.6%*	*14.3%*	*-17.6%*	*-17.6%*	*-2.3%*	*-3.6%*	*-12.5%*	*14.3%*	*0.0%*	*-1.0%*
	% QoQ				*-2.4%*	*-3.6%*	*-12.5%*	*0.0%*		*14.3%*	*-12.5%*	*14.3%*	*-12.5%*	
< US$200		**143.4**	**142.2**	**155.4**	**35.7**	**43.0**	**37.0**	**36.0**	**151.7**	**38.0**	**40.0**	**38.0**	**36.0**	**152.0**
	% YoY	*-10.2%*	*-0.8%*	*9.3%*	*-3.3%*	*7.5%*	*-3.9%*	*-10.0%*	*-2.4%*	*6.4%*	*-7.0%*	*2.7%*	*0.0%*	*0.2%*
	% QoQ				*-10.8%*	*20.4%*	*-14.0%*	*-2.7%*		*5.6%*	*5.3%*	*-5.0%*	*-5.3%*	
Total smartphone units		**225.0**	**224.0**	**239.5**	**59.5**	**64.7**	**59.2**	**56.8**	**240.2**	**63.5**	**60.0**	**62.0**	**57.0**	**242.5**
	% YoY	*-12.8%*	*-0.4%*	*6.9%*	*-1.7%*	*11.6%*	*-3.0%*	*-5.3%*	*0.3%*	*6.7%*	*-7.3%*	*4.7%*	*0.4%*	*1.0%*
	% QoQ				*-0.8%*	*8.7%*	*-8.5%*	*-4.1%*		*11.8%*	*-5.5%*	*3.3%*	*-8.1%*	

Source: Company data, UBS estimates

Samsung Electronics – Consumer Electronics Forecasts

(Won bn)		2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Visual Display revenues		**30,899**	**7,700**	**7,700**	**7,265**	**7,725**	**30,391**	**6,353**	**6,655**	**6,874**	**8,073**	**27,954**	**28,332**	**28,260**	**28,186**
	% YoY	*-0.2%*	*-1.3%*	*10.0%*	*-0.5%*	*-12.2%*	*-1.6%*	*-17.5%*	*-13.6%*	*-5.4%*	*4.5%*	*-8.0%*	*1.4%*	*-0.3%*	*-0.3%*
	% QoQ		*-12.5%*	*0.0%*	*-5.6%*	*6.3%*		*-17.8%*	*4.8%*	*3.3%*	*17.4%*				
TV revenues		**24,565**	**5,776**	**5,505**	**4,851**	**5,311**	**21,443**	**4,301**	**4,602**	**4,658**	**5,590**	**19,151**	**19,290**	**19,248**	**19,204**
	% YoY	*-10.0%*	*-10.8%*	*13.0%*	*-22.8%*	*-23.4%*	*-12.7%*	*-25.5%*	*-16.4%*	*-4.0%*	*5.3%*	*-10.7%*	*0.7%*	*-0.2%*	*-0.2%*
	% QoQ		*-16.7%*	*-4.7%*	*-11.9%*	*9.5%*		*-19.0%*	*7.0%*	*1.2%*	*20.0%*				
LCD TV revenues		**20,359**	**4,718**	**4,428**	**3,774**	**4,006**	**16,927**	**3,392**	**3,561**	**3,586**	**4,218**	**14,757**	**14,713**	**14,418**	**14,129**
	% YoY	*-14.2%*	*-14.5%*	*8.0%*	*-27.6%*	*-27.5%*	*-16.9%*	*-28.1%*	*-19.6%*	*-5.0%*	*5.3%*	*-12.8%*	*-0.3%*	*-2.0%*	*-2.0%*
	% QoQ		*-14.6%*	*-6.2%*	*-14.8%*	*6.2%*		*-15.3%*	*5.0%*	*0.7%*	*17.6%*				
OLED TV revenues		**4,206**	**1,057**	**1,077**	**1,077**	**1,304**	**4,516**	**909**	**1,041**	**1,071**	**1,373**	**4,394**	**4,578**	**4,830**	**5,075**
	% YoY	*18.1%*	*10.4%*	*39.1%*	*0.6%*	*-7.1%*	*7.4%*	*-14.1%*	*-3.4%*	*-0.5%*	*5.2%*	*-2.7%*	*4.2%*	*5.5%*	*5.1%*
	% QoQ		*-24.6%*	*1.9%*	*-0.1%*	*21.1%*		*-30.3%*	*14.5%*	*2.9%*	*28.2%*				
TV units		**36,924**	**9,332**	**8,730**	**8,371**	**9,609**	**36,042**	**7,903**	**7,953**	**8,444**	**10,203**	**34,502**	**34,695**	**34,348**	**34,009**
	% YoY	*0.6%*	*2.4%*	*9.5%*	*-9.9%*	*-8.9%*	*-2.4%*	*-15.3%*	*-8.9%*	*0.9%*	*6.2%*	*-4.3%*	*0.6%*	*-1.0%*	*-1.0%*
	% QoQ		*-11.5%*	*-6.5%*	*-4.1%*	*14.8%*		*-17.8%*	*0.6%*	*6.2%*	*20.8%*				
LCD TV units ('000)		**34,934**	**8,796**	**8,180**	**7,771**	**8,859**	**33,605**	**7,353**	**7,353**	**7,794**	**9,353**	**31,852**	**31,845**	**31,298**	**30,759**
	% YoY	*-0.9%*	*1.1%*	*7.9%*	*-11.7%*	*-10.1%*	*-3.8%*	*-16.4%*	*-10.1%*	*0.3%*	*5.6%*	*-5.2%*	*0.0%*	*-1.7%*	*-1.7%*
	% QoQ		*-10.7%*	*-7.0%*	*-5.0%*	*14.0%*		*-17.0%*	*0.0%*	*6.0%*	*20.0%*				
OLED TV units ('000)		**1,990**	**537**	**550**	**600**	**750**	**2,437**	**550**	**600**	**650**	**850**	**2,650**	**2,850**	**3,050**	**3,250**
	% YoY	*38.2%*	*28.7%*	*39.9%*	*22.8%*	*8.5%*	*22.5%*	*2.5%*	*9.1%*	*8.3%*	*13.3%*	*8.8%*	*7.5%*	*7.0%*	*6.6%*
	% QoQ		*-22.4%*	*2.5%*	*9.1%*	*25.0%*		*-26.7%*	*9.1%*	*8.3%*	*30.8%*				
Visual Display OP		**596**	**267**	**269**	**46**	**26**	**608**	**136**	**97**	**148**	**161**	**543**	**791**	**957**	**956**
	% OP margin	*1.9%*	*3.5%*	*3.5%*	*0.6%*	*0.3%*	*2.0%*	*2.1%*	*1.5%*	*2.2%*	*2.0%*	*1.9%*	*2.8%*	*3.4%*	*3.4%*
Appliances Revenues		**25,904**	**6,600**	**6,500**	**6,000**	**4,500**	**23,600**	**5,175**	**5,692**	**5,692**	**5,408**	**21,967**	**23,228**	**23,552**	**23,881**
	% YoY	*4.4%*	*1.0%*	*-6.3%*	*-6.7%*	*-25.0%*	*-8.9%*	*-21.6%*	*-12.4%*	*-5.1%*	*20.2%*	*-6.9%*	*5.7%*	*1.4%*	*1.4%*
	% QoQ		*10.0%*	*-1.5%*	*-7.7%*	*-25.0%*		*15.0%*	*10.0%*	*0.0%*	*-5.0%*				
Appliances OP		**-793**	**-66**	**-254**	**-360**	**-450**	**-1,130**	**-259**	**-228**	**-171**	**-108**	**-765**	**415**	**537**	**545**
	% OP margin	*-3.1%*	*-1.0%*	*-3.9%*	*-6.0%*	*-10.0%*	*-4.8%*	*-5.0%*	*-4.0%*	*-3.0%*	*-2.0%*	*-3.5%*	*1.8%*	*2.3%*	*2.3%*
Revenues		**57,299**	**14,300**	**14,500**	**13,565**	**12,555**	**54,921**	**11,759**	**12,578**	**12,809**	**13,747**	**50,893**	**52,524**	**52,770**	**53,019**
	% YoY	*1.5%*	*-1.4%*	*2.8%*	*-2.4%*	*-15.2%*	*-4.2%*	*-17.8%*	*-13.3%*	*-5.6%*	*9.5%*	*-7.3%*	*3.2%*	*0.5%*	*0.5%*
	% QoQ		*-3.4%*	*1.4%*	*-6.4%*	*-7.4%*		*-6.3%*	*7.0%*	*1.8%*	*7.3%*				
Operating profit		**-204**	**201**	**0**	**-329**	**-440**	**-569**	**-134**	**-142**	**-35**	**40**	**-271**	**1,148**	**1,446**	**1,453**
	% OP margin	*-0.4%*	*1.4%*	*0.0%*	*-2.4%*	*-3.5%*	*-1.0%*	*-1.1%*	*-1.1%*	*-0.3%*	*0.3%*	*-0.5%*	*2.2%*	*2.7%*	*2.7%*

Source: Company data, UBS estimates

Samsung Electronics – Capex Forecasts

		2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
(Won bn)													
Total		**26,501**	**38,930**	**47,858**	**52,232**	**53,100**	**53,601**	**52,700**	**76,484**	**96,460**	**113,160**	**125,160**	**134,160**
	% YoY	*-9.9%*	*46.9%*	*22.9%*	*9.1%*	*1.7%*	*0.9%*	*-1.7%*	*45.1%*	*26.1%*	*17.3%*	*10.6%*	*7.2%*
Semis total		**22,201**	**33,330**	**43,258**	**47,532**	**48,400**	**46,301**	**47,500**	**71,784**	**91,060**	**107,060**	**119,060**	**128,060**
	% YoY	*-5.8%*	*50.1%*	*29.8%*	*9.9%*	*1.8%*	*-4.3%*	*2.6%*	*51.1%*	*26.9%*	*17.6%*	*11.2%*	*7.6%*
Display total		**2,200**	**3,900**	**2,600**	**2,500**	**2,400**	**4,800**	**2,800**	**2,000**	**2,500**	**3,000**	**3,000**	**3,000**
	% YoY	*-24.1%*	*77.3%*	*-33.3%*	*-3.8%*	*-4.0%*	*100.0%*	*-41.7%*	*-28.6%*	*25.0%*	*20.0%*	*0.0%*	*0.0%*
Others		**2,100**	**1,700**	**2,000**	**2,200**	**2,300**	**2,500**	**2,400**	**2,700**	**2,900**	**3,100**	**3,100**	**3,100**
	% YoY	*-29.1%*	*-19.0%*	*17.6%*	*10.0%*	*4.5%*	*8.7%*	*-4.0%*	*12.5%*	*7.4%*	*6.9%*	*0.0%*	*0.0%*
(US$ mil)													
Semis total		**19,046**	**28,244**	**38,830**	**36,922**	**36,793**	**33,948**	**33,408**	**48,968**	**60,635**	**75,479**	**88,342**	**89,336**
	% YoY	*-11.0%*	*48.3%*	*37.5%*	*-4.9%*	*-0.4%*	*-7.7%*	*-1.6%*	*46.6%*	*23.8%*	*24.5%*	*17.0%*	*1.1%*
DRAM		**5,713**	**9,558**	**14,940**	**14,638**	**16,469**	**18,352**	**21,287**	**32,780**	**41,472**	**52,370**	**61,053**	**62,284**
	% YoY	*-41.1%*	*67.3%*	*56.3%*	*-2.0%*	*12.5%*	*11.4%*	*16.0%*	*54.0%*	*26.5%*	*26.3%*	*16.6%*	*2.0%*
NAND		**7,034**	**9,280**	**11,683**	**10,632**	**6,337**	**6,169**	**4,945**	**5,956**	**7,843**	**9,713**	**11,707**	**11,007**
	% YoY	*-15.7%*	*31.9%*	*25.9%*	*-9.0%*	*-40.4%*	*-2.7%*	*-19.8%*	*20.4%*	*31.7%*	*23.9%*	*20.5%*	*-6.0%*
Memory		**12,747**	**18,838**	**26,623**	**25,270**	**22,805**	**24,520**	**26,232**	**38,736**	**49,315**	**62,083**	**72,760**	**73,291**
	% YoY	*-29.4%*	*47.8%*	*41.3%*	*-5.1%*	*-9.8%*	*7.5%*	*7.0%*	*47.7%*	*27.3%*	*25.9%*	*17.2%*	*0.7%*
Foundry/LSI		**6,299**	**9,406**	**12,207**	**11,652**	**13,987**	**9,428**	**7,176**	**10,232**	**11,320**	**13,395**	**15,582**	**16,045**
	% YoY	*87.5%*	*49.3%*	*29.8%*	*-4.5%*	*20.0%*	*-32.6%*	*-23.9%*	*42.6%*	*10.6%*	*18.3%*	*16.3%*	*3.0%*
Display total		**1,887**	**3,305**	**2,334**	**1,942**	**1,824**	**3,519**	**1,969**	**1,364**	**1,665**	**2,115**	**2,226**	**2,093**
	% YoY	*-28.4%*	*75.1%*	*-29.4%*	*-16.8%*	*-6.1%*	*92.9%*	*-44.0%*	*-30.7%*	*22.0%*	*27.1%*	*5.2%*	*-6.0%*
LCD		**172**	**85**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
	% YoY	*-73.0%*	*-50.6%*	*NA*	*NA*	*NA*	*NA*	*NA*	*NA*	*NA*	*NA*	*NA*	*NA*
Mobile OLED		**1,716**	**3,220**	**2,334**	**1,942**	**1,824**	**3,519**	**1,969**	**1,364**	**1,665**	**2,115**	**2,226**	**2,093**
	% YoY	*-14.2%*	*87.7%*	*-27.5%*	*-16.8%*	*-6.1%*	*92.9%*	*-44.0%*	*-30.7%*	*22.0%*	*27.1%*	*5.2%*	*-6.0%*

Source: Company data, UBS estimates

Samsung Electronics – WFE Forecasts

		2020	2021	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
(US$ mil)												
Semis WFE total		**10,671**	**19,663**	**28,558**	**22,589**	**18,537**	**18,388**	**19,270**	**29,868**	**41,534**	**47,820**	**56,084**	**57,849**
	% YoY	*-26.9%*	*84.3%*	*45.2%*	*-20.9%*	*-17.9%*	*-0.8%*	*4.8%*	*55.0%*	*39.1%*	*15.1%*	*17.3%*	*3.1%*
DRAM		**1,822**	**6,679**	**10,901**	**8,485**	**8,243**	**9,953**	**12,847**	**20,228**	**28,776**	**32,993**	**38,463**	**40,485**
	% YoY	*-74.4%*	*266.6%*	*63.2%*	*-22.2%*	*-2.9%*	*20.7%*	*29.1%*	*57.5%*	*42.3%*	*14.7%*	*16.6%*	*5.3%*
NAND		**4,440**	**6,400**	**8,990**	**6,530**	**3,300**	**3,250**	**2,835**	**3,500**	**5,400**	**6,119**	**7,492**	**6,934**
	% YoY	*-13.1%*	*44.1%*	*40.5%*	*-27.4%*	*-49.5%*	*-1.5%*	*-12.8%*	*23.5%*	*54.3%*	*13.3%*	*22.4%*	*-7.5%*
Memory		**6,262**	**13,079**	**19,891**	**15,015**	**11,543**	**13,203**	**15,682**	**23,728**	**34,176**	**39,113**	**45,956**	**47,419**
	% YoY	*-48.8%*	*108.9%*	*52.1%*	*-24.5%*	*-23.1%*	*14.4%*	*18.8%*	*51.3%*	*44.0%*	*14.4%*	*17.5%*	*3.2%*
Foundry/LSI		**4,409**	**6,584**	**8,667**	**7,574**	**6,994**	**5,185**	**3,588**	**6,139**	**7,358**	**8,707**	**10,128**	**10,429**
	% YoY	*87.5%*	*49.3%*	*31.6%*	*-12.6%*	*-7.7%*	*-25.9%*	*-30.8%*	*71.1%*	*19.8%*	*18.3%*	*16.3%*	*3.0%*

Source: Company data, UBS estimates

Samsung Electronics – Financial statement forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues	**333,606**	**133,873**	**171,499**	**195,000**	**208,872**	**709,246**	**234,209**	**252,130**	**281,106**	**290,922**	**1,058,367**	**1,210,897**	**1,089,687**	**1,240,343**
Gross profit	131,370	81,913	119,301	137,058	153,830	492,102	175,443	196,358	215,346	225,104	812,251	920,192	757,729	862,692
% GP margin	*39.4%*	*61.2%*	*69.6%*	*70.3%*	*73.6%*	*69.4%*	*74.9%*	*77.9%*	*76.6%*	*77.4%*	*76.7%*	*76.0%*	*69.5%*	*69.6%*
Operating profit	**43,601**	**57,233**	**89,492**	**107,400**	**124,172**	**378,297**	**146,185**	**167,349**	**186,487**	**196,246**	**696,268**	**807,408**	**648,146**	**756,309**
% OP margin	*13.1%*	*42.8%*	*52.2%*	*55.1%*	*59.4%*	*53.3%*	*62.4%*	*66.4%*	*66.3%*	*67.5%*	*65.8%*	*66.7%*	*59.5%*	*61.0%*
NI post MI & pref div	37,771	41,103	62,266	67,978	72,171	243,517	97,032	111,039	123,426	103,688	435,185	504,436	404,265	472,919
EPS	**6,481**	**7,053**	**10,802**	**11,793**	**12,521**	**42,247**	**16,834**	**19,264**	**21,413**	**17,988**	**75,498**	**87,512**	**70,134**	**82,044**
% EPS growth YoY	*35.8%*	*515.7%*	*1471.2%*	*573.3%*	*336.5%*	*551.8%*	*138.7%*	*78.3%*	*81.6%*	*43.7%*	*78.7%*	*15.9%*	*-19.9%*	*17.0%*
DPS	1,668	372	374	4,569	13,000	18,315	400	400	400	32,900	34,100	39,100	33,300	36,600
% payout ratio	*25.7%*	*5.3%*	*3.5%*	*38.7%*	*103.8%*	*43.4%*	*2.4%*	*2.1%*	*1.9%*	*182.9%*	*45.2%*	*44.7%*	*47.5%*	*44.6%*
Cash and cash equiv	**125,847**	**147,378**	**189,999**	**277,668**	**323,147**	**323,147**	**415,287**	**432,752**	**542,493**	**677,723**	**677,723**	**967,564**	**1,148,597**	**1,410,523**
Total assets	**566,942**	**633,340**	**759,481**	**837,593**	**898,467**	**898,467**	**1,010,527**	**1,047,536**	**1,187,756**	**1,330,000**	**1,330,000**	**1,713,630**	**1,948,355**	**2,299,542**
Total equity	436,320	486,636	579,310	657,482	720,421	720,421	829,917	867,167	1,003,712	1,147,380	1,147,380	1,525,983	1,753,972	2,100,162
Net cash (debt)	**-100,608**	**-119,239**	**-167,590**	**-255,259**	**-300,738**	**-300,738**	**-392,879**	**-410,344**	**-520,084**	**-655,315**	**-655,315**	**-945,156**	**-1,126,189**	**-1,388,114**
% ROE	*10.6%*	*40.8%*	*53.5%*	*52.1%*	*53.9%*	*50.4%*	*56.4%*	*59.0%*	*59.5%*	*54.4%*	*55.6%*	*45.0%*	*29.5%*	*29.3%*
Depreciation	**43,606**	**11,480**	**12,113**	**11,681**	**11,881**	**47,156**	**12,181**	**12,681**	**13,081**	**13,481**	**51,424**	**57,924**	**64,924**	**71,924**
CF from Operations	**85,315**	**40,274**	**105,081**	**112,749**	**98,106**	**356,211**	**116,255**	**129,573**	**136,482**	**161,972**	**544,282**	**626,921**	**562,946**	**614,753**
Capex	-47,522	-17,127	-14,108	-22,625	-22,625	-76,484	-24,115	-24,115	-24,115	-24,115	-96,460	-113,160	-125,160	-134,160
Free cash flow	**25,847**	**24,899**	**62,590**	**90,125**	**75,482**	**253,095**	**92,140**	**105,458**	**112,367**	**137,857**	**447,822**	**513,762**	**437,786**	**480,593**
CF from Investments	**-59,468**	**-15,375**	**-42,491**	**-22,625**	**-22,625**	**-103,116**	**-24,115**	**-24,115**	**-24,115**	**-24,115**	**-96,460**	**-113,160**	**-125,160**	**-134,160**
Dividend	-9,897	-14	-6,209	-2,456	-30,003	-38,681	0	-87,993	-2,627	-2,627	-93,246	-223,921	-256,753	-218,667
Share buyback	-8,189	-7,615	-5,634	0	0	-17,118	0	0	0	0	0	0	0	0
CF from Finance	**-13,478**	**-5,455**	**-17,885**	**-2,456**	**-30,003**	**-55,799**	**0**	**-87,993**	**-2,627**	**-2,627**	**-93,246**	**-223,921**	**-256,753**	**-218,667**
Net changes of cash	**13,195**	**21,531**	**42,621**	**87,669**	**45,479**	**197,300**	**92,140**	**17,465**	**109,740**	**135,231**	**354,576**	**289,841**	**181,033**	**261,926**

Source: Company data, UBS estimates

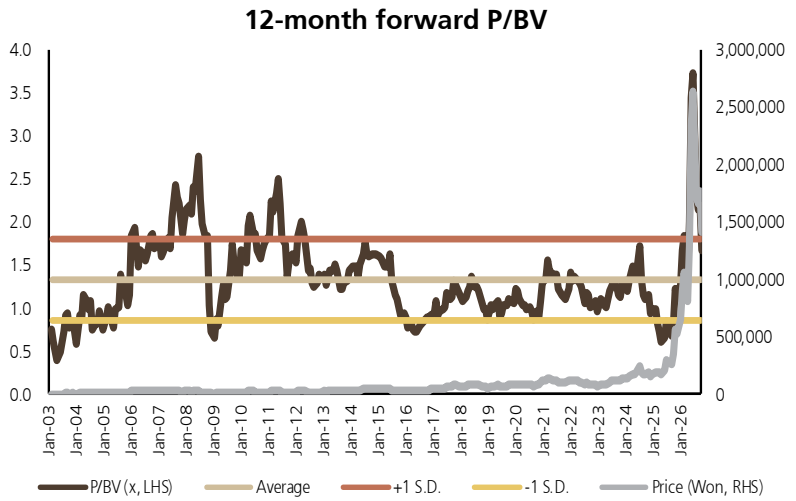
SK Hynix (Nicolas Gaudois) – Buy, PT Won3,000,000

- **UBS View**
  - Agentic continues to drive significant uplift in memory demand. Concerns on HBM pricing are overdone. Another wave of shareholders returns measures should be announced in October. FCF returns to shareholders will likely exceed 50%.
- **Will memory undersupply continue into 2028 across the industry?**
  - Yes in our view. Agentic AI is boosting further memory demand on multiple fronts beyond HBM, including DDR5/LPDDR5 for conventional servers and CPU head nodes in AI servers, as well as NAND flash for KV Cache and Storage. We now forecast an acceleration of memory demand growth into 2027 with DRAM bit end-consumption growing 39% YoY vs 21% in 2026E, and NAND flash 24%. As almost all of the incremental wafer capacity addition for DRAM is going to HBM, and no new capacity ex China is being added to NAND, supply is not catching up with demand. The main risk remains affordability as memory industry revenues are approaching US$1.64tn in 2027.
- **Will SK Hynix' lead in HBM change in 2027?**
  - To some extent. We now forecast SK Hynix HBM shipments to reach 20.1bn in 2027 (+15% YoY), undergrowing the market. This will lead SK Hynix to move below Samsung for HBM bit share in 2027 with 37% of industry share (Samsung 41%; Micron 22%). This comes from moving wafer allocation to DDR, in part due to continuing requests for upside from customers. Hence, we do not see SK Hynix chasing share through attractive pricing. Rather, their pricing to Nvidia will reflect, as before, volume discounts, and yield improvement benefits being shared with their customer.
- **Will SK Hynix engage in material shareholder returns?**
  - Yes. In spite of continuous upward revision of capex forecasts, we expect SK Hynix to generate FCF of Won186tn/Won315tn/Won363tn in 2026E/2027E/2028E. SK Hynix recently announced (1) a Won40tn share buyback/cancellation program (to be completed by November '26) and (2) plans to return at least 50% of cumulative FCF in the current policy period (2025-2027E) via buybacks and dividends with more details to be provided around 3Q26 results. Over the long run, we remain of the view that SK Hynix will continue to return at least 50% of FCF.
  - **PT Won3.0m** – We value SK Hynix shares at 4.25x NTM P/BV based on our LT ROE of 50.0% and CoE 11.8%.

SK Hynix (Nicolas Gaudois) – Valuation

UBS price target	**3,000,000**
Current stock price (Won)	1,419,000
Number of shares (bn) - ex treasury shares	0.711
Treasury shares (bn)	0.002
Market cap. (Won bn)	1,011,325
Debt (Won bn)	18,586
Cash & equivalents (Won bn)	87,958
Adjusted EV (Won bn)	941,953

	2025	2026E	2027E	2028E	2029E	2030E
Current valuation						
EV/Revenues	9.70	2.64	1.57	1.41	1.68	1.55
P/Book	8.38	2.85	1.73	1.23	1.02	0.86
P/E (ex-treasury shares)	22.96	3.59	2.33	1.90	2.18	1.79
P/E (including treasury shares)	24.05	3.59	2.33	1.90	2.18	1.79
EV/EBITDA	15.42	3.16	1.80	1.65	2.08	1.95
UBS valuation						
EV/Revenues	21.29	5.80	3.44	3.10	3.69	3.41
P/Book	17.72	6.02	3.67	2.60	2.17	1.82
P/E (ex-treasury shares)	48.54	7.58	4.92	4.02	4.62	3.78
P/E (including treasury shares)	50.85	7.59	4.92	4.02	4.62	3.78
EV/EBITDA	33.78	6.92	3.95	3.61	4.56	4.27
Other						
% ROE	***44.1%***	***116.5%***	***83.2%***	***60.1%***	***35.8%***	***31.6%***
Tangible book per share (Won)	164,001	492,740	812,382	1,146,234	1,377,959	1,643,425
Cash per share (Won)	49,140	209,570	422,505	672,895	871,540	1,077,138
Free cash flow per share (Won)	36,359	261,762	442,289	510,398	421,964	421,157
% Free cash flow yield	*2.6%*	*18.4%*	*31.2%*	*36.0%*	*29.7%*	*29.7%*



Source: Company data, UBS estimates

SK Hynix – Key Charts

UBS forecasts vs Consensus

	3Q26E			4Q26E			2026E			2027E			2028E			2029E		
(Won bn)	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**
Revenues	103,684	99,212	*4.5%*	121,134	113,021	*7.2%*	356,713	344,291	*3.6%*	600,701	541,060	*11.0%*	668,206	634,216	*5.4%*	561,085	706,270	*-20.6%*
Operating profit	**83,541**	**77,871**	**7.3%**	**99,546**	**89,986**	**10.6%**	**281,240**	**265,377**	**6.0%**	**499,907**	**424,405**	**17.8%**	**540,798**	**479,660**	**12.7%**	**414,233**	**532,938**	**-22.3%**
% OP margin	*80.6%*	*78.5%*	*208bps*	*82.2%*	*79.6%*	*256bps*	*78.8%*	*77.1%*	*176bps*	*83.2%*	*78.4%*	*478bps*	*80.9%*	*75.6%*	*530bps*	*73.8%*	*75.5%*	*-163bps*
EPS	93,096	87,982	5.8%	113,563	102,214	11.1%	395,053	360,659	9.5%	610,284	474,958	28.5%	745,367	531,165	40.3%	649,593	588,077	10.5%

Source: Company data, Visible Alpha, UBS estimates

SK Hynix – DRAM forecasts

	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
DRAM revenues (Won bn)	**75,349**	**41,008**	**58,473**	**78,181**	**93,041**	**270,702**	**102,263**	**111,195**	**121,248**	**127,028**	**461,735**	**556,898**	**494,727**	**526,959**
% QoQ / % YoY	*66.9%*	*63.6%*	*42.6%*	*33.7%*	*19.0%*	*259.3%*	*9.9%*	*8.7%*	*9.0%*	*4.8%*	*70.6%*	*20.6%*	*-11.2%*	*6.5%*
DRAM revenues (US$m)	**53,456**	**28,857**	**40,345**	**54,774**	**68,020**	**191,995**	**74,762**	**81,292**	**88,642**	**92,867**	**337,563**	**407,134**	**361,682**	**385,246**
% QoQ / % YoY	*62.8%*	*64.7%*	*39.8%*	*35.8%*	*24.2%*	*259.2%*	*9.9%*	*8.7%*	*9.0%*	*4.8%*	*75.8%*	*20.6%*	*-11.2%*	*6.5%*
DRAM unit shipment (1Gb eq)	93,014	25,453	27,368	30,191	35,150	118,162	35,467	37,017	38,906	39,907	151,298	178,609	207,974	227,632
% QoQ / % YoY	*24.3%*	*-0.3%*	*7.5%*	*10.3%*	*16.4%*	*27.0%*	*0.9%*	*4.4%*	*5.1%*	*2.6%*	*28.0%*	*18.1%*	*16.4%*	*9.5%*
Avg. wafer out per month (k 12"eq)	6,110	564	572	583	600	6,957	618	636	654	672	7,740	8,526	9,084	9,852
% QoQ / % YoY	*25.7%*	*3.7%*	*1.4%*	*1.9%*	*2.9%*	*13.9%*	*3.0%*	*2.9%*	*2.8%*	*2.8%*	*11.3%*	*10.2%*	*6.5%*	*8.5%*
DRAM ASP (US$ 1Gb eq)	0.57	1.13	1.47	1.81	1.94	1.62	2.11	2.20	2.28	2.33	2.23	2.28	1.74	1.69
% QoQ / % YoY	*31.0%*	*65.1%*	*30.0%*	*23.1%*	*6.7%*	*182.7%*	*8.9%*	*4.2%*	*3.7%*	*2.1%*	*37.3%*	*2.2%*	*-23.7%*	*-2.7%*
Operating profit (Won bn)	**44,895**	**31,588**	**46,791**	**65,184**	**78,886**	**222,448**	**87,290**	**95,055**	**103,911**	**108,186**	**394,442**	**468,380**	**392,523**	**408,886**
% OP margin	***59.6%***	***77.0%***	***80.0%***	***83.4%***	***84.8%***	***82.2%***	***85.4%***	***85.5%***	***85.7%***	***85.2%***	***85.4%***	***84.1%***	***79.3%***	***77.6%***
Fully Loaded Cost/Unit (1Gb US$)	0.23	0.26	0.29	0.30	0.29	0.29	0.31	0.32	0.33	0.35	0.33	0.36	0.36	0.38
% QoQ / % YoY	*-3.0%*	*19.1%*	*13.1%*	*2.4%*	*-2.4%*	*23.9%*	*4.8%*	*3.3%*	*2.2%*	*6.0%*	*12.5%*	*11.4%*	*-0.8%*	*5.5%*
Cash Cost/Unit (1Gb US$)	0.15	0.18	0.22	0.22	0.22	0.21	0.22	0.23	0.24	0.25	0.24	0.26	0.25	0.26
% QoQ / % YoY	*-1.0%*	*27.6%*	*18.7%*	*2.2%*	*-2.0%*	*37.9%*	*3.0%*	*2.5%*	*2.7%*	*7.1%*	*11.7%*	*9.0%*	*-4.4%*	*4.2%*
DDR revenues (US$m)	**32,935**	**22,699**	**34,106**	**46,211**	**57,844**	**160,860**	**62,146**	**67,788**	**73,598**	**75,949**	**279,480**	**317,719**	**231,058**	**207,047**
% QoQ / % YoY	*40.4%*	*88.3%*	*50.3%*	*35.5%*	*25.2%*	*388.4%*	*7.4%*	*9.1%*	*8.6%*	*3.2%*	*73.7%*	*13.7%*	*-27.3%*	*-10.4%*
DDR unit shipment (1Gb eq)	80,386	21,650	23,564	25,632	29,878	100,724	30,856	32,205	33,848	34,320	131,230	150,772	170,539	179,829
% QoQ / % YoY	*18.1%*	*-2.5%*	*8.8%*	*8.8%*	*16.6%*	*25.3%*	*3.3%*	*4.4%*	*5.1%*	*1.4%*	*30.3%*	*14.9%*	*13.1%*	*5.4%*
DDR ASP (US$ 1Gb eq)	0.41	1.05	1.45	1.80	1.94	1.60	2.01	2.10	2.17	2.21	2.13	2.11	1.35	1.15
% QoQ / % YoY	*18.8%*	*93.1%*	*38.1%*	*24.6%*	*7.4%*	*289.8%*	*4.0%*	*4.5%*	*3.3%*	*1.8%*	*33.4%*	*-1.1%*	*-35.7%*	*-15.0%*
HBM revenues (US$m)	**20,521**	**6,158**	**6,239**	**8,563**	**10,176**	**31,135**	**12,616**	**13,504**	**15,044**	**16,918**	**58,083**	**89,415**	**130,624**	**178,199**
% QoQ / % YoY	*118.9%*	*12.6%*	*1.3%*	*37.2%*	*18.8%*	*51.7%*	*24.0%*	*7.0%*	*11.4%*	*12.5%*	*86.6%*	*53.9%*	*46.1%*	*36.4%*
HBM unit shipment (1Gb eq)	12,628	3,803	3,804	4,559	5,273	17,438	4,611	4,812	5,058	5,587	20,068	27,836	37,435	47,803
% QoQ / % YoY	*86.0%*	*14.6%*	*0.0%*	*19.8%*	*15.7%*	*38.1%*	*-12.6%*	*4.4%*	*5.1%*	*10.5%*	*15.1%*	*38.7%*	*34.5%*	*27.7%*
HBM ASP (US$ 1Gb eq)	1.63	1.62	1.64	1.88	1.93	1.79	2.74	2.81	2.97	3.03	2.89	3.21	3.49	3.73
% QoQ / % YoY	*17.7%*	*-1.7%*	*1.3%*	*14.5%*	*2.8%*	*9.9%*	*41.8%*	*2.6%*	*6.0%*	*1.8%*	*62.1%*	*11.0%*	*8.6%*	*6.8%*

Source: Company data, UBS estimates

SK Hynix – Capex forecasts

	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Semis Capex (US$m)											
Semis Capex	**8,508**	**10,911**	**14,767**	**6,283**	**11,644**	**19,504**	**33,290**	**46,138**	**52,138**	**56,138**	**59,138**
% YoY	*-24.3%*	*28.2%*	*35.3%*	*-57.5%*	*85.3%*	*67.5%*	*70.7%*	*38.6%*	*13.0%*	*7.7%*	*5.3%*
Semis Capex (Won bn)	**10,069**	**12,487**	**19,010**	**8,325**	**15,920**	**27,519**	**47,153**	**63,110**	**71,317**	**76,788**	**80,892**
% YoY	*107.6%*	*24.0%*	*52.2%*	*-56.2%*	*91.2%*	*72.9%*	*71.3%*	*33.8%*	*13.0%*	*7.7%*	*5.3%*
Semis WFE (US$m)											
Semis WFE	**4,509**	**7,082**	**8,924**	**3,344**	**7,553**	**11,317**	**18,290**	**26,138**	**31,138**	**35,138**	**39,138**
% YoY	*-18.2%*	*57.1%*	*26.0%*	*-62.5%*	*125.9%*	*49.8%*	*61.6%*	*42.9%*	*19.1%*	*12.8%*	*11.4%*

Source: Company data, UBS estimates

SK Hynix – Financial Statement forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues	**97,147**	**52,576**	**79,319**	**103,684**	**121,134**	**356,713**	**132,496**	**144,933**	**158,186**	**165,086**	**600,701**	**668,206**	**561,085**	**606,045**
Gross profit	58,691	41,679	65,991	89,710	106,247	303,628	116,789	128,080	140,211	145,719	530,799	576,588	454,094	481,697
% GP margin	*60.4%*	*79.3%*	*83.2%*	*86.5%*	*87.7%*	*85.1%*	*88.1%*	*88.4%*	*88.6%*	*88.3%*	*88.4%*	*86.3%*	*80.9%*	*79.5%*
Operating profit	**47,206**	**37,610**	**60,543**	**83,541**	**99,546**	**281,240**	**109,753**	**120,589**	**132,265**	**137,299**	**499,907**	**540,798**	**414,233**	**437,729**
% OP margin	*48.6%*	*71.5%*	*76.3%*	*80.6%*	*82.2%*	*78.8%*	*82.8%*	*83.2%*	*83.6%*	*83.2%*	*83.2%*	*80.9%*	*73.8%*	*72.2%*
NI post MI	42,919	40,330	93,820	65,196	77,694	277,041	85,671	94,140	103,235	107,181	390,227	422,310	323,738	342,152
EPS	**61,810**	**56,940**	**131,941**	**93,096**	**113,563**	**395,541**	**128,251**	**144,421**	**163,091**	**174,521**	**610,284**	**745,367**	**649,593**	**793,637**
% EPS growth YoY	*115.2%*	*384.9%*	*1201.9%*	*410.3%*	*423.6%*	*539.9%*	*125.2%*	*9.5%*	*75.2%*	*53.7%*	*54.3%*	*22.1%*	*-12.8%*	*22.2%*
DPS	3,000	375	375	375	47,000	48,125	375	375	375	73,000	74,125	73,125	60,125	57,125
% payout ratio	*4.9%*	*0.7%*	*0.3%*	*0.4%*	*41.4%*	*12.2%*	*0.3%*	*0.3%*	*0.2%*	*41.8%*	*12.1%*	*9.8%*	*9.3%*	*7.2%*
Cash and cash equiv	**34,942**	**54,330**	**87,958**	**118,503**	**149,020**	**149,020**	**183,017**	**194,386**	**243,850**	**300,433**	**300,433**	**478,479**	**619,731**	**765,927**
Total assets	**176,108**	**222,829**	**348,862**	**393,864**	**440,777**	**440,777**	**496,364**	**529,009**	**596,949**	**668,522**	**668,522**	**907,363**	**1,073,029**	**1,263,821**
Total equity	120,667	164,380	262,693	307,622	355,054	355,054	410,725	443,219	511,210	583,153	583,153	821,358	986,899	1,176,372
Net cash (debt)	12,694	35,012	69,372	100,017	131,264	131,264	165,361	176,830	226,394	283,637	283,637	462,463	604,315	751,111
% ROE	*44.1%*	*113.2%*	*175.9%*	*91.5%*	*93.8%*	*116.5%*	*89.5%*	*88.2%*	*86.5%*	*78.4%*	*83.2%*	*60.1%*	*35.8%*	*31.6%*
Depreciation	**-13,890**	**-3,729**	**-4,030**	**-4,396**	**-4,637**	**-16,792**	**-5,088**	**-5,540**	**-5,854**	**-6,169**	**-22,652**	**-30,777**	**-38,437**	**-45,549**
CF from Operations	**53,373**	**26,330**	**65,710**	**65,323**	**75,922**	**233,285**	**79,874**	**88,893**	**100,586**	**108,257**	**377,611**	**434,248**	**376,837**	**380,367**
Capex	-27,519	-7,657	-10,671	-14,412	-14,412	-47,153	-15,777	-15,777	-15,777	-15,777	-63,110	-71,317	-76,788	-80,892
Free cash flow	**25,854**	**18,673**	**55,039**	**50,911**	**61,510**	**186,132**	**64,097**	**73,115**	**84,809**	**92,480**	**314,501**	**362,931**	**300,049**	**299,475**
CF from Investments	**-30,987**	**-7,128**	**-28,747**	**-14,412**	**-14,412**	**-64,699**	**-15,777**	**-15,777**	**-15,777**	**-15,777**	**-63,110**	**-71,317**	**-76,788**	**-80,892**
Dividend	-1,681	0	-1,594	-267	-263	-2,123	0	-31,646	-244	-237	-32,128	-44,105	-38,197	-27,679
CF from Finance	**-1,445**	**-2,951**	**-4,666**	**-20,366**	**-30,993**	**-58,976**	**-30,100**	**-61,746**	**-35,344**	**-35,897**	**-163,088**	**-184,885**	**-158,797**	**-153,279**
Net changes of cash	**20,786**	**19,388**	**33,628**	**30,545**	**30,517**	**114,078**	**33,997**	**11,369**	**49,465**	**56,582**	**151,413**	**178,046**	**141,252**	**146,196**

Source: Company data, UBS estimates

DI Corporation (Jimmy Yoon) – Buy, PT Won50,000

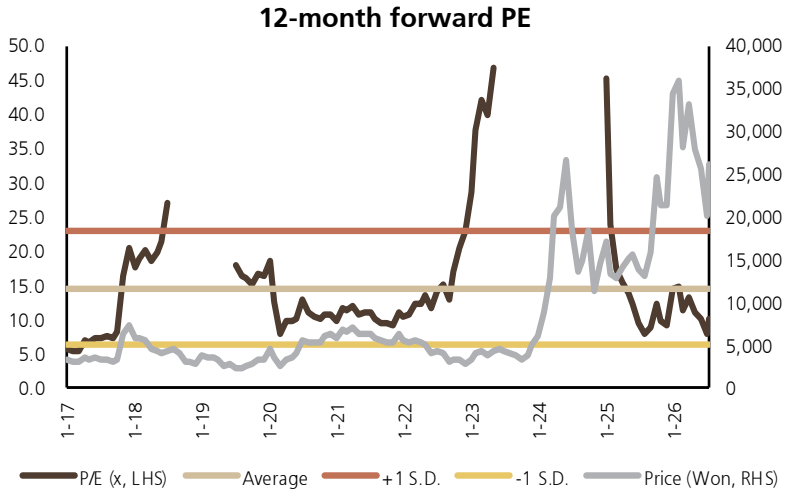
- **Our View**
  - We view DI as the best positioned among Korean memory test vendors to incrementally benefit from transition to HBM4. Revenue for HBM4 KGSD burn-in-testers from SK Hynix (c70% m/s) increased from Q126, along with recovery in conventional DRAM/NAND wafer/package test revenue to drive 205%/21% YoY earnings growth in '26/'27E. Based on higher revenue mix of HBM testers with higher ASP vs. conventional testers, we forecast DI's margins at c18%/18% in '26/'27E.
- **Will the addressable market for memory test tools grow YoY in '26E/'27E?**
  - Yes. We expect memory tester TAM to grow +20%/+18% YoY in '26E/'27E to US$2.5bn/$2.9bn. Transition to HBM4/HBM4E should meaningfully raise HBM test requirements (e.g., longer test times) and hence present upside for test tool units and ASP.
- **Can DI gain market share for HBM tools at key customers?**
  - Yes. We believe DI is well positioned to gain share in wafer tests at SK Hynix with the transition to HBM4, as we believe SK Hynix will diversify and de-risk its HBM test tool supplier base from tier-1 vendors (Advantest) to local vendors.
  - We expect DI to secure c. 70% market share of Known Good Stacked Die (KGSD) HBM wafer tests with SK Hynix as the primary vendor, starting with HBM4, which should help grow revenue/OP by 30%/144% YoY in '26E.
- **Will DI's operating margin expand to mid-teens % in the mid- to long-term?**
  - Yes. We believe DI should see operating margin expand to mid-teens % mid- to long-term.
  - We believe HBM4 KGSD burn-in-test tools command a higher ASP of Won1.9bn-2.0bn with an operating margin of >30% vs DDR wafer/package tester ASP of Won0.5bn-1.5bn with an operating margin of low-teens %.
  - As HBM-related wafer test revenues continue to outgrow conventional DDR test tools, we believe DI should expand margins >18% beyond 2027E (from c. 8% in 2025).

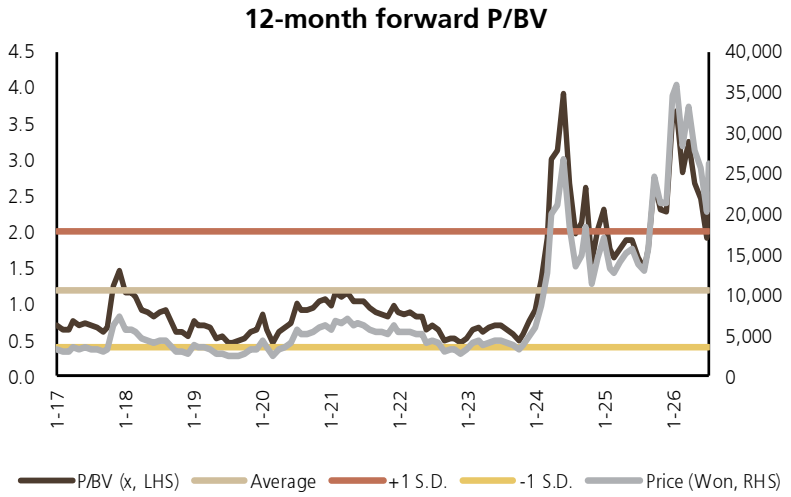
- **PT Won50k –** We value DI shares at 17.4x 2027E P/E, between historical average and +1StD P/E band of 14.7x-22.6x, based on our LT EPS CAGR of 14%.

DI Corporation (Jimmy Yoon) – Valuation

UBS price target	**50,000**
Current stock price (Won)	21,100
Number of shares (mil, ex-treasury shrs)	27.4
Treasury shares	0.9
Market cap. (Won bn)	578
Net debt (Won bn)	-72
EV (Won bn)	507

	2025	2026E	2027E	2028E	2029E	2030E
Current valuation (x)						
EV/Revenues	1.17	0.83	0.69	0.53	0.57	0.48
P/Book	2.95	2.24	1.77	1.38	1.19	0.99
P/E	107.73	8.94	7.35	5.56	6.62	4.92
EV/EBITDA	12.39	4.35	3.63	2.71	3.30	2.52
Target price valuation (x)						
EV/Revenues	3.00	2.14	1.76	1.35	1.47	1.23
P/Book	7.00	5.31	4.19	3.27	2.82	2.34
P/E	255.28	21.19	17.43	13.17	15.69	11.67
EV/EBITDA	31.77	11.17	9.31	6.94	8.46	6.46
Other						
EPS	196	2,359	2,869	3,795	3,186	4,286
% ROE	***2.9%***	***29.1%***	***26.9%***	***27.9%***	***19.3%***	***21.9%***
Book per share (Won)	7,145	9,409	11,925	15,289	17,716	21,365
Cash per share (Won)	4,117	6,349	8,949	11,415	14,227	16,977
Free cash flow per share (Won)	716	3,169	3,064	3,024	3,702	3,518
% Free cash flow yield	*3.4%*	*15.0%*	*14.5%*	*14.3%*	*17.5%*	*16.7%*



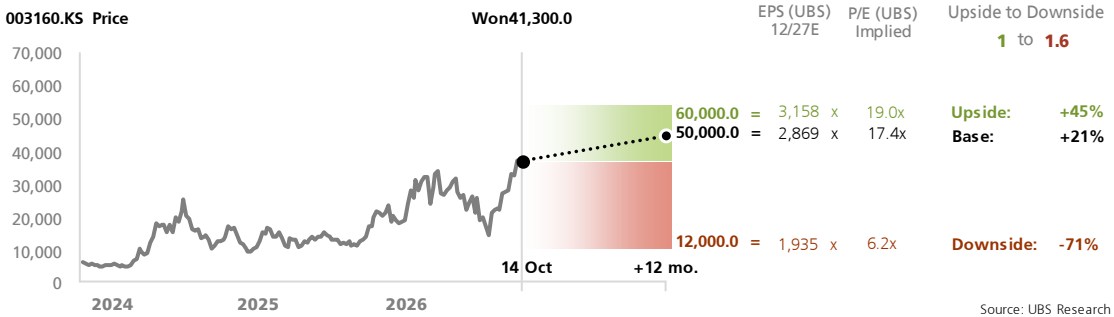


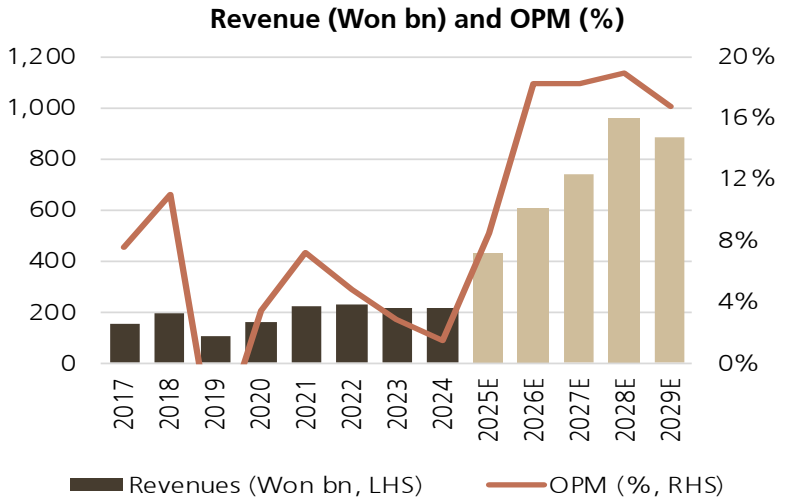
Source: Company data, UBS estimates

DI Corporation – Key Charts

UBS forecasts vs Consensus

	3Q26E			4Q26E			2026E			2027E			2028E		
(Won bn)	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**
Sales	191	175	9.3%	142	132	8.0%	608	579	5.1%	739	690	7.1%	959	746	28.5%
Operating profit	**35**	**29**	**23.2%**	**18**	**17**	**1.1%**	**111**	**96**	**16.2%**	**135**	**116**	**15.7%**	**182**	**136**	**34.0%**
% OP margin	18.4%	16.4%		12.4%	13.2%		18.3%	16.6%		18.2%	16.9%		19.0%	18.2%	
EPS (Won)	748	563	32.9%	338	197	71.8%	2,359	1,854	27.3%	2,869	2,344	22.4%	3,795	2,689	41.1%





Value drivers (2026/27E)	HBM tester market share	HBM tester units	Operating profit
Won60,000upside	85%/90%	150/ 220	Won128bn/150bn
Won50,000 base	70%/65%	130/179	Won111bn/135bn
Won12,000 downside	40%/35%	115/130	Won90bn/102bn

Source: Company data, UBS estimates

DI Corporation – Revenue Forecasts

	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Test Equipment	**340**	**82**	**144**	**168**	**120**	**514**	**103**	**154**	**187**	**165**	**609**	**777**	**678**	**819**
Burn-In Board	40	14	21	9	9	53	8	10	15	23	56	81	110	118
Parts & Maintenance	26	9	9	11	10	39	10	16	18	17	61	88	83	103
Others	27	1	(5)	8	8	26	8	8	8	8	33	33	33	34
Total Revenue (Won bn)	**432**	**106**	**168**	**197**	**147**	**632**	**129**	**189**	**229**	**213**	**759**	**980**	**904**	**1,074**
% YoY	*102%*	*-4%*	*41%*	*80%*	*57%*	*46%*	*22%*	*12%*	*16%*	*45%*	*20%*	*29%*	*-8%*	*19%*
% QoQ		*13%*	*59%*	*17%*	*-25%*		*-12%*	*46%*	*21%*	*-7%*				
Operating Profit (Won bn)	**37**	**19**	**40**	**32**	**15**	**105**	**12**	**34**	**46**	**31**	**123**	**170**	**137**	**184**
OP Margin (%)	*8%*	*18%*	*24%*	*16%*	*10%*	*17%*	*9%*	*18%*	*20%*	*15%*	*16%*	*17%*	*15%*	*17%*
Test Equipment Revenue														
HBM Wafer Test	153	48	84	66	40	239	64	92	99	73	329	495	368	458
DRAM Wafer	74	6	17	13	11	47	7	9	14	21	52	75	102	109
NAND Wafer	6	1	1	1	7	10	5	3	2	8	19	15	23	28
DRAM Package	99	24	38	74	50	186	23	42	56	52	172	167	136	167
NAND Package	8	2	4	13	12	32	3	8	17	10	38	26	50	56
Total Tester Revenue (Won bn)	**340**	**82**	**144**	**168**	**120**	**514**	**103**	**154**	**187**	**165**	**609**	**777**	**678**	**819**
Tester Shipments														
HBM Wafer Test	90	26	46	36	22	130	35	50	54	40	179	245	182	227
DRAM Wafer	80	7	18	14	12	51	8	10	15	23	56	81	110	118
NAND Wafer	10	2	1	2	10	15	8	5	3	12	28	22	34	42
DRAM Package	107	26	41	80	54	201	25	45	60	56	186	180	147	180
NAND Package	15	4	7	20	18	49	4	12	25	15	56	39	74	84
Total Shipments (Units)	**302**	**65**	**113**	**152**	**116**	**446**	**80**	**122**	**157**	**146**	**505**	**567**	**547**	**651**
Tool ASP														
HBM Wafer Test	1.7	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	2.0	2.0	2.0
DRAM Wafer	0.9	1.5	1.5	1.5	1.5	0.9	1.5	1.5	1.5	1.5	0.9	0.9	0.9	0.9
NAND Wafer	0.6	1.2	1.2	1.2	1.2	0.7	1.2	1.2	1.2	1.2	0.7	0.7	0.7	0.7
DRAM Package	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
NAND Package	0.6	0.6	0.6	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Blended ASP (Won bn)	**1.1**	**1.3**	**1.3**	**1.1**	**1.0**	**1.2**	**1.3**	**1.3**	**1.2**	**1.1**	**1.2**	**1.4**	**1.2**	**1.3**

Source: Company data, UBS estimates

DI Corporation – Financial Statement Forecasts

(Won bn)	2025E	1Q26E	2Q26E	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues	**432**	**106**	**168**	**191**	**142**	**608**	**124**	**184**	**224**	**208**	**739**	**959**	**883**	**1,052**
Gross profit	92	34	57	52	31	174	30	57	72	54	212	275	248	309
% GM	*21.3%*	*31.8%*	*33.8%*	*27.0%*	*22.0%*	*28.6%*	*24.0%*	*31.0%*	*32.0%*	*26.0%*	*28.7%*	*28.6%*	*28.1%*	*29.3%*
Operating profit	**37**	**19**	**40**	**32**	**15**	**105**	**12**	**34**	**46**	**31**	**123**	**170**	**137**	**184**
% OpM	*8.4%*	*17.9%*	*23.5%*	*16.9%*	*10.3%*	*17.3%*	*9.6%*	*18.4%*	*20.5%*	*15.0%*	*16.6%*	*17.8%*	*15.5%*	*17.5%*
Net profit post MI	**5**	**11**	**24**	**18**	**7**	**61**	**5**	**20**	**28**	**18**	**70**	**96**	**79**	**109**
EPS reported (Won)	**196**	**402**	**871**	**673**	**264**	**2,212**	**178**	**716**	**1,020**	**656**	**2,570**	**3,496**	**2,887**	**3,986**
Cash and cash equivalents	107	82	58	105	174	174	185	178	206	237	237	298	368	437
Total assets	**374**	**430**	**462**	**545**	**469**	**469**	**461**	**502**	**550**	**565**	**565**	**662**	**712**	**819**
Total liabilities	**188**	**220**	**224**	**289**	**205**	**205**	**193**	**223**	**243**	**239**	**239**	**252**	**242**	**255**
Shareholders' equity	**186**	**211**	**238**	**257**	**264**	**264**	**269**	**279**	**307**	**325**	**325**	**411**	**470**	**564**
Depreciation & amortization	4	1	1	1	1	5	1	1	1	1	5	5	5	5
Cash flow from operations	**49**	**(0)**	**(15)**	**47**	**69**	**100**	**17**	**5**	**30**	**34**	**86**	**85**	**103**	**98**
Capex	(36)	(1)	(0)	(0)	(0)	(3)	(3)	(3)	(3)	(3)	(10)	(10)	(10)	(10)
Cash flow from investments	**(30)**	**(11)**	**(1)**	**(0)**	**(0)**	**(13)**	**(3)**	**(3)**	**(3)**	**(3)**	**(10)**	**(10)**	**(10)**	**(10)**
Dividends paid	(3)	(4)	(4)	0	0	(8)	0	(9)	0	0	(9)	(11)	(19)	(16)
Cash flow from financing activities	**46**	**(14)**	**(7)**	**0**	**0**	**(21)**	**(3)**	**(9)**	**0**	**0**	**(12)**	**(14)**	**(23)**	**(19)**
Total cash flow	**64**	**(25)**	**(24)**	**47**	**69**	**66**	**11**	**(6)**	**28**	**31**	**64**	**61**	**71**	**69**

Source: Bloomberg, Company data, UBS estimates

Eugene Tech (Nicolas Gaudois) – Buy, PT Won182,000

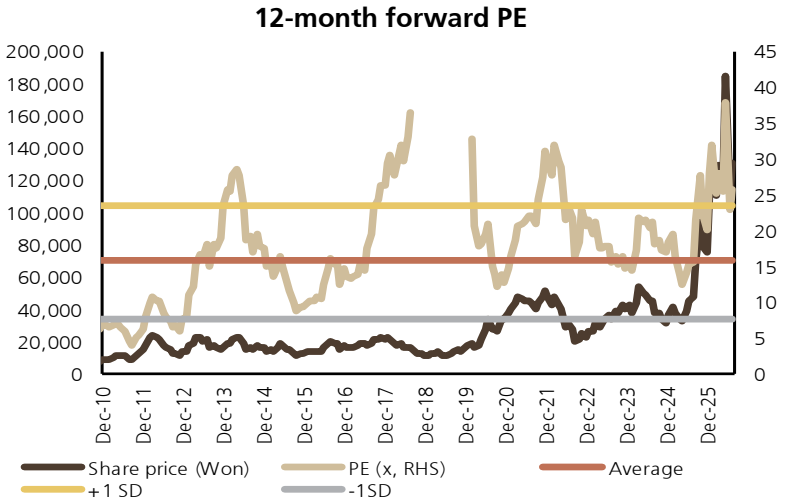
- **Our View**
  - We believe the overall set up for Eugene Tech is improving on the back of (1) growing DRAM WFE TAM, (2) increasing attach rates for etching tools to LPCVD and Plasma tools and (3) continuing share gains across multiple product segments including large batch PEALD and Plasma. We do not believe these positives are fully priced in. As the most concentrated play on a multi-year DRAM WFE uptrend, Eugene Tech is our top pick in Korea semiscap.
- **Will Samsung and SK Hynix' memory WFE spending recover meaningfully in the next few years?**
  - Yes. We now forecast combined Samsung and SK Hynix DRAM WFE to be up +54% YoY in 2026E, +18% YoY in 2027E, and +16% YoY again in 2028E.
  - We believe further upside could come from HBM demand, as well as the possible move to 4F2 by 2028 (initially Samsung, in our view) would also add to the TAM for both ALD and selective epitaxy.
- **Can Eugene Tech gain meaningful market share at Samsung and SK Hynix?**
  - Yes. We believe the focus is on qualification for plasma clustering tools for key memory customers, large batch PEALD tools for 1c nm / Logic and, later, epitaxy for 1c nm and PECVD tools for 1d nm.

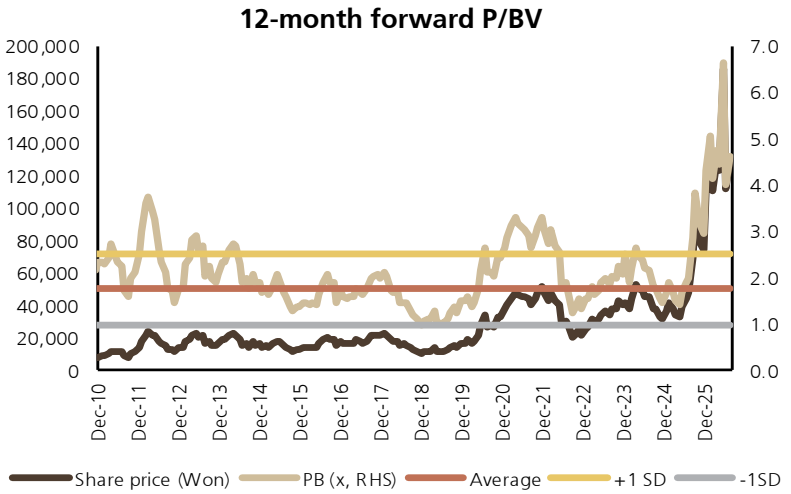
- **PT Won182k –** We value Eugene Tech shares at 21x 2028E P/E based on our long-term EPS CAGR of 23%.

Eugene Tech – Valuation

UBS price target	**182,000**
Current stock price (Won)	131,100
Number of shares (mil, ex-treasury shrs)	22.9
Treasury shares (mil)	0.3
Market cap. (Won bn)	3,004
Net debt (Won bn)	-90
EV (Won bn)	2,915

	2025	2026E	2027E	2028E	2029E	2030E
Current valuation (x)						
EV/Revenues	8.32	5.99	4.27	3.48	2.96	2.55
P/Book	6.03	4.75	3.89	3.12	2.51	2.02
P/E	70.76	22.78	20.09	15.23	12.32	10.14
EV/EBITDA	43.45	28.39	15.25	11.57	9.33	7.75
Target price valuation (x)						
EV/Revenues	11.75	8.46	6.03	4.92	4.18	3.60
P/Book	9.16	7.12	5.74	4.56	3.62	2.89
P/E	98.23	31.63	27.89	21.15	17.10	14.08
EV/EBITDA	61.37	40.10	21.54	16.35	13.18	10.94
Other						
EPS	1,853	5,753	6,523	8,603	10,637	12,924
% ROE	***9.2%***	***23.6%***	***21.6%***	***23.0%***	***22.9%***	***22.4%***
Tangible book per share (Won)	19,859	25,564	31,699	39,942	50,246	62,868
Cash per share (Won)	2,429	5,355	9,960	16,690	26,592	38,015
Free cash flow per share (Won)	505	3,136	5,206	7,330	10,502	12,023
% Free cash flow yield	*0.4%*	*2.4%*	*4.0%*	*5.6%*	*8.0%*	*9.2%*



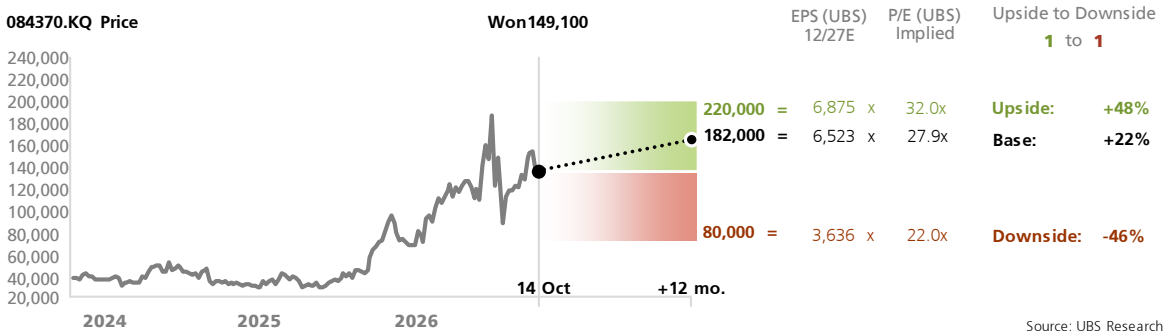


Source: Company data, UBS estimates

Eugene Tech – Key Charts

UBS forecasts vs Consensus

	3Q26E			4Q26E			2026E			2027E			2028E		
(Won bn)	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**
Sales	128	130	-2%	135	151	-10%	487	506	-4%	683	666	3%	837	806	4%
Operating Profit	**25**	**29**	**-14%**	**28**	**35**	**-21%**	**90**	**110**	**-18%**	**178**	**164**	**8%**	**236**	**212**	**12%**
% OP margin	*19.3%*	*22.1%*	*-278bps*	*20.6%*	*23.4%*	*-280bps*	*18.4%*	*21.7%*	*-325bps*	*26.1%*	*24.7%*	*141bps*	*28.2%*	*26.2%*	*194bps*
EPS	921	1,109	-17%	1,035	1,429	-28%	5,753	4,276	35%	6,523	5,992	9%	8,603	7,896	9%



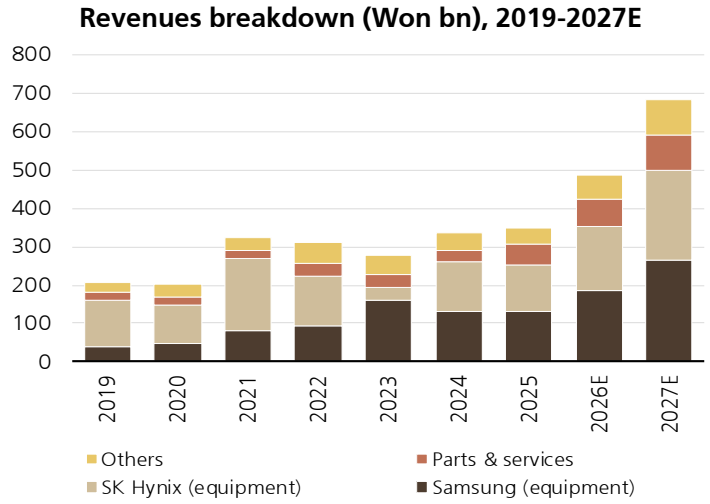
Source: UBS Research

Value drivers	'26E/'27E Revenues (Won)	'26E/'27E OP (Won)	'27E-'30E EPS CAGR
Won220,000 upside	540bn/740bn	110bn/215bn	30%
Won182,000 base	487bn/683bn	95bn/179bn	25%
Won80,000 downside	455bn/560bn	80bn/120bn	15%

Source: Company data, UBS estimates

Eugene Tech – Revenue breakdown

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Total revenues	**345**	**102**	**122**	**128**	**135**	**487**	**160**	**165**	**180**	**178**	**683**	**837**	**985**	**1,143**
Memory	**277**	**82**	**102**	**106**	**110**	**400**	**133**	**136**	**149**	**149**	**566**	**699**	**824**	**960**
Samsung	131	36	44	51	53	184	61	61	68	68	259	314	363	403
SK Hynix	121	36	45	42	45	167	55	58	61	61	234	283	328	391
Other	25	10	13	13	13	48	17	17	20	20	73	103	134	166
Non-Memory	**2**	**0**	**0**	**1**	**2**	**3**	**2**	**2**	**2**	**0**	**6**	**8**	**10**	**12**
Samsung Foundry	2	0	0	1	2	3	2	2	2	0	6	8	10	12
Parts and services	**52**	**16**	**16**	**17**	**19**	**68**	**21**	**23**	**25**	**25**	**94**	**110**	**126**	**142**
Eugene Materials	**15**	**4**	**4**	**4**	**4**	**16**	**4**	**4**	**4**	**5**	**17**	**20**	**24**	**29**



Source: Company data, UBS estimates

Eugene Tech – Financial Statement Forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues	**350**	**102**	**122**	**128**	**135**	**487**	**160**	**165**	**180**	**178**	**683**	**837**	**985**	**1,143**
% QoQ		*2.0%*	*19.9%*	*4.6%*	*5.8%*		*18.1%*	*3.5%*	*9.0%*	*-1.0%*				
% YoY	*3.6%*	*22.5%*	*16.4%*	*104.8%*	*35.2%*	*39.0%*	*56.7%*	*35.3%*	*41.0%*	*32.0%*	*40.3%*	*22.5%*	*17.6%*	*16.1%*
Gross profit	171	52	59	64	68	242	81	84	92	91	348	430	510	600
% GP margin	*48.8%*	*50.7%*	*48.6%*	*50.0%*	*50.0%*	*49.8%*	*51.0%*	*51.0%*	*51.0%*	*51.0%*	*51.0%*	*51.4%*	*51.8%*	*52.5%*
Operating profit	**52**	**19**	**18**	**25**	**28**	**90**	**43**	**42**	**49**	**44**	**178**	**236**	**292**	**356**
% OP margin	*14.7%*	*18.5%*	*14.9%*	*19.3%*	*20.6%*	*18.4%*	*27.2%*	*25.5%*	*27.1%*	*24.6%*	*26.1%*	*28.2%*	*29.7%*	*31.2%*
NI post MI	42	22	65	21	24	132	36	35	41	37	149	197	244	296
EPS	**1,853**	**949**	**2,847**	**921**	**1,035**	**5,754**	**1,588**	**1,547**	**1,784**	**1,606**	**6,526**	**8,607**	**10,641**	**12,929**
% EPS growth YoY	*-32.9%*	*188.7%*	*510.9%*	*167.8%*	*44.9%*	*210.5%*	*67.3%*	*-45.7%*	*93.7%*	*55.2%*	*13.4%*	*31.9%*	*23.6%*	*21.5%*
Cash and cash equiv	**55**	**71**	**90**	**92**	**121**	**121**	**132**	**139**	**170**	**225**	**225**	**377**	**601**	**860**
Total assets	**581**	**628**	**681**	**698**	**726**	**726**	**766**	**788**	**833**	**872**	**872**	**1,064**	**1,309**	**1,601**
Total equity	491	511	578	599	624	624	661	684	725	763	763	949	1,182	1,467
Net cash (debt)	-55	-71	-90	-92	-121	-121	-132	-139	-170	-225	-225	-377	-601	-860
% ROE	*9.2%*	*17.4%*	*48.8%*	*14.3%*	*15.5%*	*23.6%*	*22.7%*	*21.1%*	*23.2%*	*19.8%*	*21.6%*	*23.0%*	*22.9%*	*22.4%*
CF from Operations	**66**	**11**	**61**	**4**	**33**	**109**	**13**	**24**	**35**	**60**	**132**	**182**	**257**	**292**
Capex	-13	-4	-10	-3	-3	-20	-3	-3	-4	-4	-14	-16	-20	-20
Free cash flow	**11**	**15**	**25**	**1**	**30**	**71**	**10**	**21**	**31**	**56**	**118**	**166**	**237**	**272**
CF from Investments	**-55**	**5**	**-37**	**-3**	**-3**	**-38**	**-3**	**-3**	**-4**	**-4**	**-14**	**-16**	**-20**	**-20**
Dividend	-5	0	-5	0	0	-5	0	-14	0	0	-14	-14	-14	-14
CF from Finance	**14**	**0**	**-6**	**0**	**0**	**-6**	**0**	**-14**	**0**	**0**	**-14**	**-14**	**-14**	**-14**
Net changes of cash	**25**	**16**	**19**	**1**	**30**	**66**	**10**	**7**	**31**	**56**	**104**	**152**	**224**	**258**

Source: Company data, UBS estimates

Hanmi Semiconductor (Jimmy Yoon) – Sell, PT Won170,000

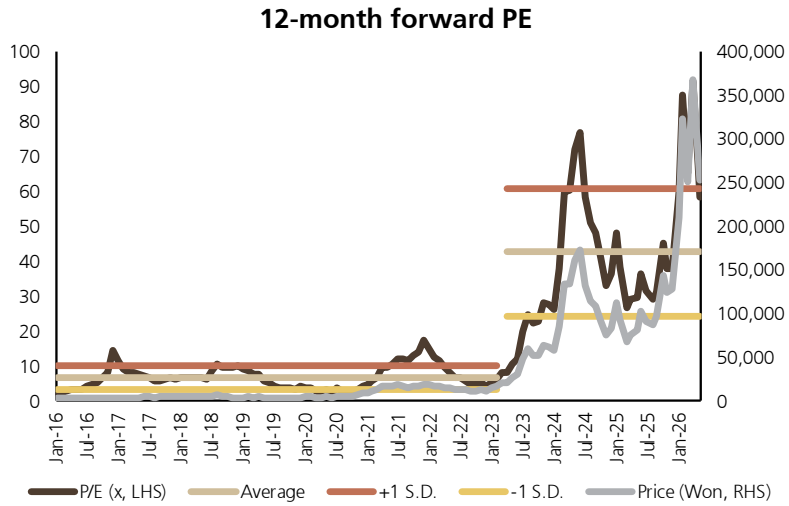
- **Our View**
  - We forecast Hanmi's revenue to grow 32%/60% YoY in 2026/27E on the back of HBM TCB unit shipments growing +15%/+71% YoY to c. 107/183 tools and blended HBM TCB tool ASP +6%/+7% YoY. That said, we expect a higher portion of HBM TCB spend to also be focused on upgrades (lower dollar opportunity) with SK Hynix having been able to meaningfully control TSV yields/process for HBM3E 12Hi, likely limiting required adjustments (less incremental tool intensity per k wpm) for HBM4 12Hi. Though with low visibility, we are not yet reflecting M/S gains into Samsung. While Hanmi should continue to play a key leading role for HBM customers' HBM TCB tool needs, we believe current valuation well reflects a potential upside scenario.
- **Will HBM back-end equipment spending and, in turn, Hanmi's TCB SAM grow YoY in 2026?**
  - We expect Hanmi's unit SAM to grow +10%/+82% YoY to 145/264 tools in 2026/2027, supported by SK Hynix and Micron adding net new HBM4 wafer capacity with flux TCB (1H26 loaded). While hard to fully assess, we see low probability of Hanmi securing HBM4 TCB orders from Samsung given Samsung's preference to in-source TCB tool needs from SEMES. We believe TCB demand from China should see a more meaningful YoY recovery into 2027E.
- **Can Hanmi maintain its HBM TCB tool market share at SK Hynix and Micron?**
  - While playing a leading role at both SK Hynix and Micron, we believe Hanmi may lose incremental market share to competitors over time given customer needs to diversify their TCB supplier base. We expect Hanmi's M/S at SK Hynix and Micron to come in at 50%/92% in 2026E and 46%/84% in 2027E on competition from ASMPT and, to a lesser extent, Hanwha Vision for flux TCB tools. Mid- to long-term we forecast Hanmi's TCB tool market share at SK Hynix and Micron to normalize at 45%/75% by 2030E.

- **PT Won170k –** We value Hanmi shares at 32x 2026E P/E based on our long-term EPS CAGR of 23%.

Hanmi Semiconductor – Valuation

UBS price target	**170,000**
Current stock price (Won)	258,000
Number of shares (bn)	94.9
Treasury shares	0.5
Market cap. (Won bn)	24,472
Net debt (Won bn)	-175
Adjusted EV (Won bn)	24,297

	2025	2026E	2027E	2028E	2029E	2030E
Current Valuation						
EV/Revenues	42.13	31.93	20.01	16.42	17.05	13.19
P/Book	35.45	28.43	20.59	16.18	14.33	11.87
P/E	114.35	76.54	48.60	40.69	44.07	35.60
EV/EBITDA	92.41	64.85	38.82	31.82	34.76	27.61
UBS valuation (x)						
EV/Revenues	27.66	20.96	13.14	10.78	11.20	8.66
P/Book	23.36	18.74	13.57	10.66	9.44	7.82
P/E	75.35	50.43	32.02	26.81	29.04	23.46
EV/EBITDA	60.66	42.57	25.48	20.89	22.82	18.13
Other data						
EPS	2,256	3,371	5,309	6,340	5,854	7,246
% ROE	***34.8%***	***41.2%***	***49.1%***	***44.5%***	***34.5%***	***36.5%***
Tangible book per share (Won)	7,278	9,074	12,529	15,949	17,174	21,733
Cash per share (Won)	2,912	4,633	6,883	9,925	11,654	14,660
FCF Per Share (Won)	2,604	3,251	4,105	5,962	5,689	6,363
% FCF yield	*1.0%*	*1.3%*	*1.6%*	*2.3%*	*2.2%*	*2.5%*

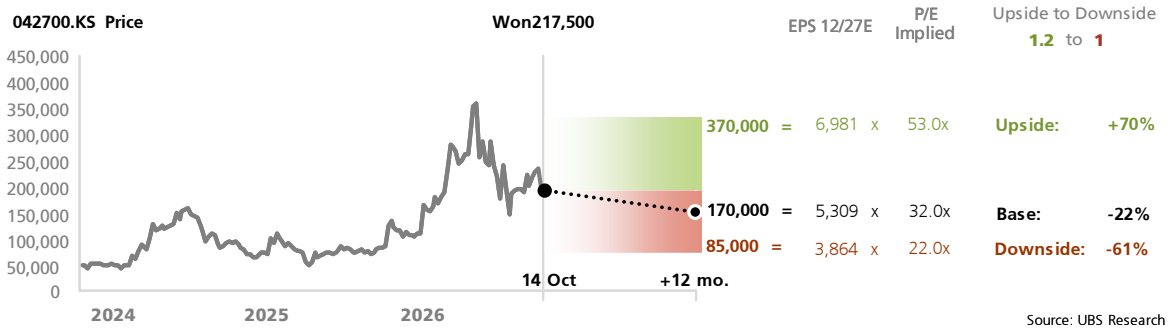


Source: Company data, UBS estimates

Hanmi Semiconductor – Key Charts

UBS forecasts vs Consensus

	3Q26E			4Q26E			2026E			2027E			2028E		
(Won bn)	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**
Sales	226	235	*-4.0%*	209	223	*-6.0%*	737	649	*13.6%*	1,028	1,154	*-10.9%*	1,490	1,342	*11.0%*
Op Profit	**112**	**113**	**-0.9%**	**99**	**114**	**-13.1%**	**350**	**279**	**25.2%**	**497**	**584**	**-15.0%**	**758**	**691**	**9.7%**
% OP margin	*49.5%*	*47.9%*	*156bps*	*47.3%*	*51.1%*	*-382bps*	*47.4%*	*43.0%*	*441bps*	*48.4%*	*50.7%*	*-229bps*	*50.9%*	*51.5%*	*-64bps*
EPS (Won)	953	983	-3.1%	850	929	-8.5%	3,146	2,522	24.7%	4,158	4,999	-16.8%	6,211	5,730	8.4%



Value drivers (2026E/2027E)	Qualification at Samsung (2027E)	TCB unit shipments (2026/27E)	TCB unit share (2026/27E)	Operating profit (2026/27E)
Won370,000 upside	Yes	125/235 tools	76%/78%	Won392/782bn
Won170,000 base	No	107/183 tools	74%/69%	Won363/614bn
Won85,000 downside	No	85/135 tools	72%/60%	Won318/422bn

33

Source: Company data, UBS estimates, Bloomberg

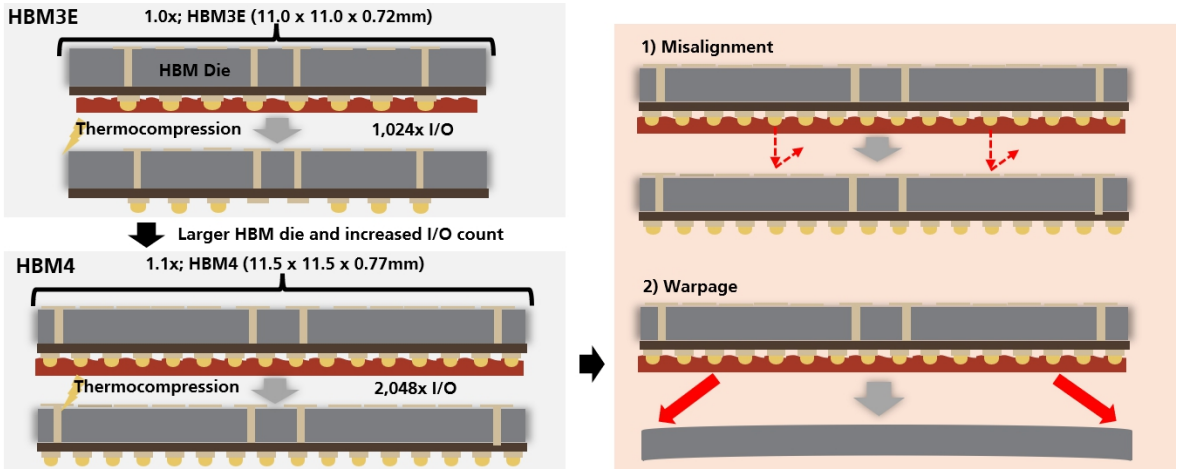
Hanmi Semiconductor – TCB TAM Analysis

	2025	1Q26	2Q26E	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
SK Hynix														
Total HBM Wafers Out (k wpm)	**148**	**175**	**185**	**145**	**175**	**174**	**210**	**240**	**240**	**250**	**235**	**318**	**355**	**410**
HBM + Logic Dies Out (mil)	**827**	**278**	**294**	**228**	**259**	**1,059**	**299**	**339**	**340**	**347**	**1,325**	**1,774**	**2,014**	**2,353**
Provision for Back-End Yields (%)		8%	8%	8%	8%		0%	0%	0%	0%				
# of TCB Tools Required		**240**	**250**	**190**	**216**		**268**	**309**	**315**	**326**				
Incremental # of TCB Tools	**54**	**27**	**10**	**0**	**26**	**63**	**52**	**41**	**7**	**12**	**112**	**147**	**85**	**180**
Micron														
Total HBM Wafers Out (k wpm)	**64**	**70**	**80**	**85**	**95**	**105**	**90**	**140**	**150**	**160**	**151**	**204**	**220**	**268**
HBM + Logic Dies Out (mil)	**333**	**108**	**125**	**135**	**144**	**512**	**133**	**194**	**209**	**222**	**757**	**1,040**	**1,229**	**1,506**
Provision for Back-End Yields (%)		8%	10%	12%	12%		12%	12%	12%	12%				
# of TCB Tools Required		**110**	**131**	**149**	**166**		**152**	**228**	**247**	**266**				
Incremental # of TCB Tools	**67**	**7**	**23**	**18**	**17**	**65**	**0**	**76**	**20**	**19**	**115**	**124**	**114**	**135**
Samsung														
Total HBM Wafers Out (k wpm)	**120**	**150**	**150**	**145**	**160**	**166**	**160**	**230**	**250**	**260**	**254**	**316**	**338**	**385**
HBM + Logic Dies Out (mil)	**492**	**158**	**178**	**191**	**223**	**750**	**227**	**313**	**342**	**347**	**1,231**	**1,591**	**1,839**	**2,128**
Provision for Back-End Yields (%)		10%	12%	12%	12%		12%	12%	10%	10%				
# of TCB Tools Required		**168**	**192**	**213**	**260**		**270**	**380**	**408**	**423**				
Incremental # of TCB Tools	**47**	**16**	**19**	**16**	**42**	**78**	**11**	**110**	**29**	**16**	**166**	**155**	**134**	**122**
Others (CXMT/Tongfu)														
Total HBM Wafers Out (k wpm)	**13**	**20**	**20**	**25**	**25**	**23**	**30**	**30**	**30**	**30**	**30**	**40**	**50**	**60**
HBM + Logic Dies Out (mil)	**21**	**6**	**7**	**11**	**12**	**36**	**17**	**20**	**24**	**25**	**86**	**152**	**222**	**307**
Provision for Back-End Yields (%)		30%	30%	30%	30%		20%	20%	20%	20%				
# of TCB Tools Required		**11**	**14**	**24**	**27**		**36**	**46**	**58**	**63**				
Incremental # of TCB Tools	**14**	**0**	**3**	**10**	**4**	**17**	**9**	**10**	**13**	**5**	**37**	**45**	**45**	**50**

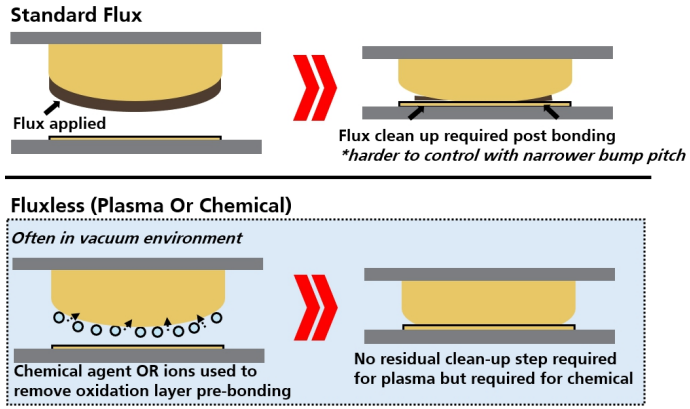
Source: Company data, UBS estimates

Summary of HBM bonding technology

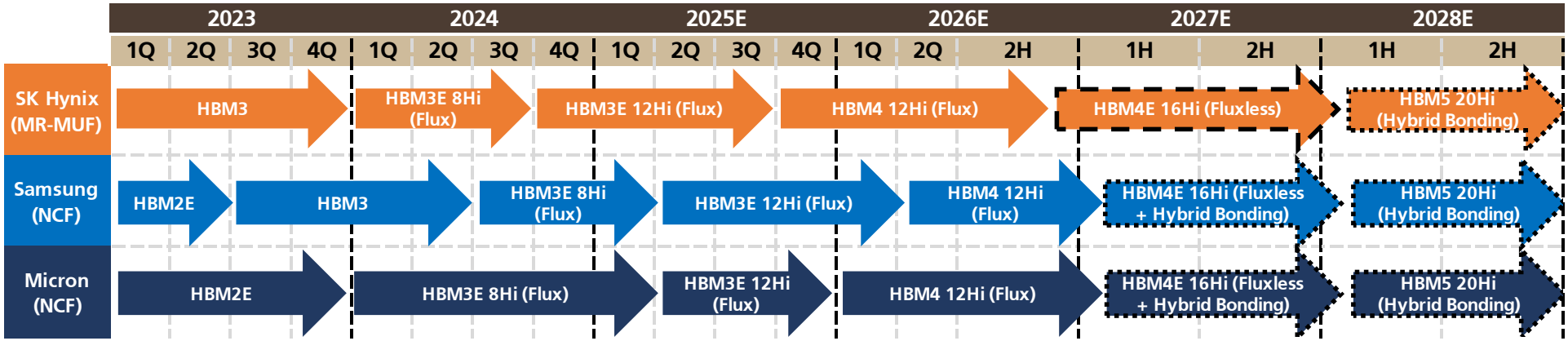
How would the transition from HBM3E to HBM4 impact HBM TCB tool TAM?



Flux vs Fluxless TCB



HBM production timeline and associated bonding technology



Source: Company data, UBS estimates

HBM TC/Hybrid boding equipment competitive landscape

Company	TC Bonder	*Hybrid Bonder
ASMPT	SK Hynix, Micron	SK Hynix, Micron
BE Semiconductor	Samsung, Micron	Samsung, SK Hynix, Micron
Hanmi Semiconductor	SK Hynix, Micron	SK Hynix, Micron
Hanwha Semitech	SK Hynix	SK Hynix
Kulicke and Soffa	Samsung, SK Hynix, Micron	-
SEMES	Samsung	Samsung
Shinkawa (Yamaha)	Samsung, Micron	-
Toray	Samsung	-

Source: Company data, UBS. *Reflects known engagements for HBM hybrid bonder development.

Hanmi Semiconductor – Revenue breakdown

(Won bn)	2025	1Q26	2Q26E	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E
Semis Equipment	**488**	**33**	**209**	**245**	**174**	**661**	**220**	**288**	**296**	**284**	**1,088**	**1,308**	**1,234**
% YoY	*-7%*	*-74%*	*28%*	*80%*	*194%*	*35%*	*558%*	*38%*	*21%*	*63%*	*65%*	*20%*	*-6%*
% QoQ		*-43%*	*525%*	*17%*	*-29%*		*26%*	*31%*	*3%*	*-4%*			
HBM TC Bonder	**369**	**4**	**170**	**188**	**115**	**477**	**173**	**237**	**231**	**222**	**863**	**1,056**	**965**
% of Revenue	*64%*	*8%*	*72%*	*68%*	*58%*	*63%*	*70%*	*74%*	*70%*	*70%*	*71%*	*71%*	*68%*
% YoY	*-17%*	*-96%*	*34%*	*88%*	*363%*	*29%*	*4105%*	*40%*	*23%*	*93%*	*81%*	*22%*	*-9%*
% QoQ		*-83%*	*4040%*	*11%*	*-39%*		*50%*	*37%*	*-3%*	*-4%*			
HBM TCB Shipments (Units)	**93**	**1**	**40**	**42**	**24**	**107**	**37**	**52**	**49**	**45**	**183**	**205**	**176**
% YoY	*-33%*	*-96%*	*21%*	*56%*	*300%*	*15%*	*3600%*	*30%*	*17%*	*88%*	*71%*	*12%*	*-14%*
SK Hynix	20	0	11	14	6	31	10	17	15	10	52	63	44
Micron	60	1	25	22	12	60	20	27	25	25	97	102	92
China	13	0	4	6	6	16	7	8	9	10	34	40	40
HBM TCB Revenue (Won bn)	**369**	**4**	**170**	**188**	**115**	**477**	**173**	**237**	**231**	**222**	**863**	**1,056**	**965**
% YoY	*-17%*	*-96%*	*34%*	*88%*	*363%*	*29%*	*4105%*	*40%*	*23%*	*93%*	*81%*	*22%*	*-9%*
SK Hynix	60	0	40	52	22	115	37	63	60	42	203	282	205
Micron	240	4	108	99	56	267	93	125	116	119	452	529	512
China	69	0	22	37	37	95	43	49	55	61	208	244	247
% of TCB Revenue													
SK Hynix	*16%*	*0%*	*24%*	*28%*	*19%*	*24%*	*22%*	*27%*	*26%*	*19%*	*24%*	*27%*	*21%*
Micron	*65%*	*100%*	*63%*	*53%*	*49%*	*56%*	*54%*	*53%*	*50%*	*54%*	*52%*	*50%*	*53%*
China	*19%*	*0%*	*13%*	*19%*	*32%*	*20%*	*25%*	*21%*	*24%*	*27%*	*24%*	*23%*	*26%*
MSVP	**93**	**20**	**27**	**32**	**26**	**105**	**28**	**30**	**33**	**30**	**121**	**131**	**127**
% YoY	*82%*	*140%*	*-8%*	*19%*	*-8%*	*13%*	*39%*	*11%*	*3%*	*15%*	*15%*	*8%*	*-3%*
% QoQ		*-29%*	*34%*	*19%*	*-19%*		*8%*	*7%*	*10%*	*-9%*			
Others (EMI Shield Line, etc.)	**19**	**7**	**9**	**10**	**11**	**37**	**10**	**11**	**12**	**10**	**43**	**48**	**56**
% YoY	*-12%*	*69%*	*82%*	*79%*	*164%*	*96%*	*35%*	*29%*	*20%*	*-9%*	*17%*	*12%*	*17%*
% QoQ		*77%*	*15%*	*18%*	*10%*		*-9%*	*10%*	*9%*	*-17%*			
Consumables (Parts)	**89**	**18**	**28**	**30**	**24**	**100**	**27**	**32**	**33**	**35**	**126**	**172**	**191**
% of Revenue	*15%*	*34%*	*12%*	*11%*	*12%*	*13%*	*11%*	*10%*	*10%*	*11%*	*10%*	*12%*	*13%*
% YoY	*140%*	*-1%*	*66%*	*1%*	*0%*	*13%*	*56%*	*12%*	*8%*	*44%*	*26%*	*36%*	*11%*
% QoQ		*-27%*	*62%*	*8%*	*-21%*		*14%*	*17%*	*3%*	*5%*			
Total	**577**	**51**	**237**	**275**	**198**	**761**	**247**	**320**	**329**	**319**	**1,214**	**1,480**	**1,425**
% YoY	*3%*	*-65%*	*32%*	*65%*	*138%*	*32%*	*385%*	*35%*	*19%*	*61%*	*60%*	*22%*	*-4%*
% QoQ		*-39%*	*366%*	*16%*	*-28%*		*25%*	*29%*	*3%*	*-3%*			

Source: Company data, UBS estimates

Hanmi Semiconductor – Financial Statement Forecasts

(Won bn)	2025	1Q26	2Q26E	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues	**577**	**51**	**237**	**275**	**198**	**761**	**247**	**320**	**329**	**319**	**1,214**	**1,480**	**1,425**	**1,842**
Gross profit	332	28	145	165	113	450	146	198	200	185	729	882	831	1,042
% GP margin	*57.6%*	*54.1%*	*61.0%*	*60.0%*	*57.0%*	*59.1%*	*59.0%*	*62.0%*	*61.0%*	*58.0%*	*60.1%*	*59.6%*	*58.4%*	*56.6%*
Operating profit	**251**	**8**	**120**	**142**	**93**	**363**	**120**	**167**	**171**	**157**	**614**	**753**	**690**	**872**
% OP margin	*43.6%*	*16.6%*	*50.6%*	*51.5%*	*46.9%*	*47.7%*	*48.5%*	*52.1%*	*52.0%*	*49.3%*	*50.6%*	*50.9%*	*48.4%*	*47.3%*
NI post MI	214	19	102	117	82	320	101	135	139	128	504	601	555	687
EPS	**2,256**	**201**	**1,071**	**1,237**	**863**	**3,371**	**1,069**	**1,426**	**1,460**	**1,354**	**5,309**	**6,340**	**5,854**	**7,246**
% EPS growth YoY	*41.6%*	*-65.2%*	*57.0%*	*78.5%*	*183.6%*	*49.4%*	*432.6%*	*33.2%*	*18.1%*	*56.9%*	*57.5%*	*19.4%*	*-7.7%*	*23.8%*
DPS	3,804	3,804	0	0	0	3,513	3,513	0	0	0	4,348	0	0	0
% payout ratio	*168.6%*	*1895.9%*	*0.0%*	*0.0%*	*0.0%*	*104.2%*	*328.7%*	*0.0%*	*0.0%*	*0.0%*	*81.9%*	*0.0%*	*0.0%*	*0.0%*
Cash and cash equiv	**276**	**175**	**221**	**304**	**439**	**439**	**510**	**401**	**532**	**653**	**653**	**941**	**1,120**	**1,391**
Total assets	**813**	**743**	**873**	**1,011**	**1,046**	**1,046**	**1,173**	**1,171**	**1,314**	**1,430**	**1,430**	**1,758**	**1,961**	**2,350**
Total equity	690	636	662	779	861	861	962	921	1,060	1,188	1,188	1,513	1,707	2,061
Net debt	-276	-175	-221	-304	-439	-439	-510	-401	-532	-653	-653	-941	-1,120	-1,391
% ROE	*34.8%*	*11.5%*	*62.6%*	*65.1%*	*39.9%*	*41.2%*	*44.5%*	*57.5%*	*55.9%*	*45.7%*	*49.1%*	*44.5%*	*34.5%*	*36.5%*
D&A	**12**	**3**	**3**	**3**	**3**	**12**	**3**	**3**	**3**	**3**	**12**	**10**	**9**	**8**
CF from Operations	**229**	**-38**	**127**	**89**	**141**	**318**	**75**	**71**	**135**	**126**	**407**	**579**	**549**	**609**
Capex	-75	-5	-5	-5	-5	-22	-4	-4	-4	-4	-18	-14	-10	-6
Free cash flow	**247**	**-32**	**122**	**84**	**135**	**308**	**71**	**67**	**130**	**121**	**389**	**566**	**540**	**604**
CF from Investments	**18**	**6**	**-5**	**-5**	**-5**	**-10**	**-4**	**-4**	**-4**	**-4**	**-18**	**-14**	**-10**	**-6**
Dividend	-68	-76	-76	0	0	-152	0	-176	0	0	-176	-277	-361	-333
CF from Finance	**-77**	**-76**	**-76**	**0**	**0**	**-152**	**0**	**-176**	**0**	**0**	**-176**	**-277**	**-361**	**-333**
Net changes of cash	**172**	**-102**	**46**	**84**	**135**	**163**	**71**	**-109**	**130**	**121**	**213**	**289**	**179**	**270**

Source: Company data, UBS estimates

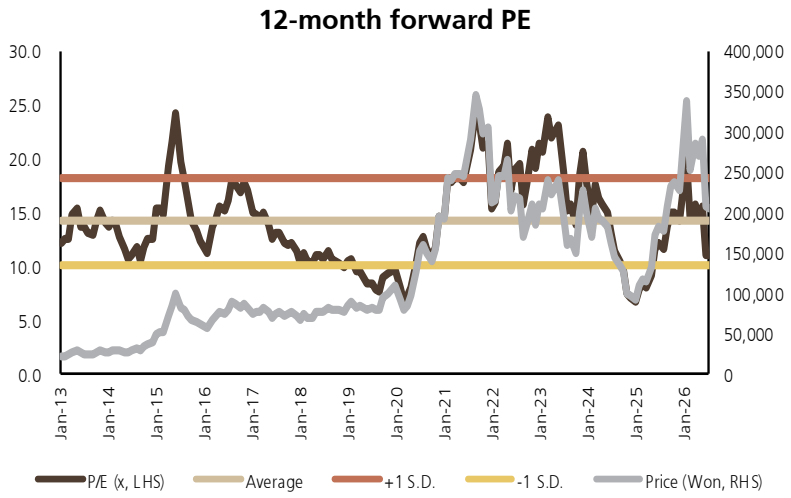
Hansol Chemical (Jimmy Yoon) – Buy, PT Won350,000

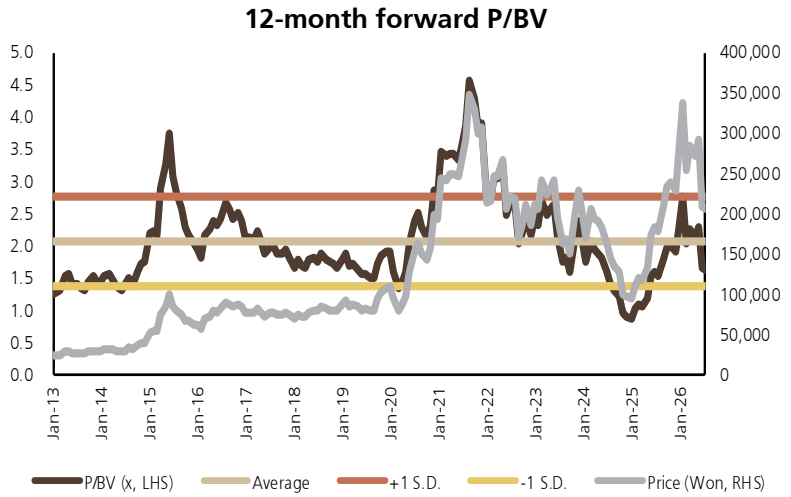
- **Our View**
  - Among Korean semis material vendors, we believe Hansol Chem is best positioned to benefit from the memory (in particular DRAM) upcycle and associated capacity addition via H^2O^2, precursor, specialty gas as well as ongoing market share gains for materials used in critical DRAM layers.
  - On the back of this, we forecast Hansol's semis material revenue to grow +26%/+25% YoY in 2026/27E. Medium to longer term, we expect ramping revenue contribution from margin-accretive non-semis material business (Battery/Display) to help lift margins above 23%.
- **Will Hansol be able to secure >15% YoY revenue growth in 2026E/2027E?**
  - Yes. We believe Hansol should be able to record sequential revenue growth of +19% YoY in 2027E on the back of resuming DRAM capacity additions at both Samsung and SK Hynix (P4 & M15X) leading to semis material revenues (H2O2, precursor, specialty gas) growing +25% YoY in 2027E.
  - Hansol is also continuing to gain share for precursors at key customers while continuous tech migration to 1c nm DRAM should also drive incremental demand for high-K materials.
- **Will Hansol be able to lift its OpM >23% in the medium- to long-term?**
  - Yes. We expect Hansol's OPM to see sequential YoY pick up into 2027E (c.21%) primarily spurred by the normalisation of margin as hydrogen peroxide (H2O2) shipments continue to recover on DRAM capacity addition/migration (high operating leverage) while mix also improves from higher margin precursor materials.
  - Beyond semis materials, we also expect steady YoY revenue growth from other margin-accretive non-semis material businesses including battery binders, SiC anodes and, to a lesser extent, QD materials to lift OpM >23% in the medium to long term.
- **PT Won350k –** We value Hansol shares at 17x 2027E P/E, slightly higher than global peers and between the historical average and +1StD of 14x-19x. We believe this is justified by higher-than-peers' CAGR of 14% in '27E-'30E.

Hansol Chemical (Jimmy Yoon) – Valuation

UBS price target (Won)	**350,000**
Current stock price (Won)	218,500
Number of shares (bn)	10.7
Treasury shares	0.1
Market cap. (Won bn)	2,344
Net debt (Won bn)	204
Adjusted EV (Won bn)	2,548

	2025	2026E	2027E	2028E	2029E	2030E
Current Valuation (x)						
EV/Revenues	2.88	2.63	2.20	1.86	1.65	1.49
P/Book	2.03	1.85	1.64	1.45	1.27	1.13
P/E	16.24	14.23	11.33	9.18	7.89	7.09
EV/EBITDA	11.89	10.85	8.28	6.89	6.02	5.47
UBS valuation (x)						
EV/Revenues	4.48	4.08	3.42	2.89	2.57	2.32
P/Book	3.26	2.97	2.63	2.32	2.04	1.80
P/E	26.02	22.79	18.15	14.70	12.64	11.35
EV/EBITDA	18.48	16.85	12.86	10.71	9.35	8.50
Other data						
EPS (Won)	13,452	15,355	19,284	23,805	27,683	30,834
% ROE	***13.1%***	***13.5%***	***15.4%***	***16.8%***	***17.2%***	***16.9%***
Tangible book per share (Won)	107,524	117,795	132,865	150,869	171,452	193,986
Cash per share (Won)	10,280	13,374	10,713	17,059	28,187	43,355
FCF Per Share (Won)	6,357	11,351	3,992	12,379	18,461	23,701
% FCF yield	*2.9%*	*5.2%*	*1.8%*	*5.7%*	*8.4%*	*10.8%*





Source: Company data, UBS estimates

Hansol Chemical (Jimmy Yoon) – Key Charts

UBS forecasts vs Consensus

	3Q26E			4Q26E			2026E			2027E			2028E		
(Won bn)	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**
Sales	245	253	*-3.2%*	246	258	*-4.5%*	971	1,000	*-2.9%*	1,159	1,158	*0.0%*	1,369	1,320	*3.7%*
Op Profit	**46**	**53**	**-11.9%**	**33**	**36**	**-8.2%**	**173**	**189**	**-8.5%**	**238**	**242**	**-1.8%**	**302**	**302**	**-0.1%**
% OP margin	*19.0%*	*20.8%*	*-187bps*	*13.5%*	*14.0%*	*-54bps*	*17.8%*	*18.9%*	*-108bps*	*20.5%*	*20.9%*	*-38bps*	*22.0%*	*22.9%*	*-85bps*
EPS (Won)	3,843	3,734	2.9%	2,924	2,817	3.8%	15,355	15,736	-2.4%	19,284	19,970	-3.4%	23,805	23,627	0.8%

Value drivers	H_2O_2 revenue CAGR ('26E-'30E)	Precursors as % of total revs ('26E-'30E)	Operating profit '26E/'27E
W420,000 upside	+19%	15%-26%	Won190bn/274bn
W350,000 base	+15%	13%-22%	Won173bn/238bn
W140,000 downside	+10%	12%-18%	Won159bn/214bn

41

Source: Company data, UBS estimates, Bloomberg

Hansol Chemical – Divisional Forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Fine Chemicals (incl. H2O2, SD, BPO)	**217.7**	**55.7**	**65.0**	**66.8**	**67.5**	**255.0**	**69.8**	**78.0**	**83.3**	**86.8**	**318.0**	**388.5**	**438.6**	**482.0**
% YoY	*8.9%*	*6.7%*	*17.7%*	*20.5%*	*23.1%*	*17.1%*	*25.3%*	*20.0%*	*24.8%*	*28.6%*	*24.7%*	*22.2%*	*12.9%*	*9.9%*
% QoQ		*1.5%*	*16.9%*	*2.7%*	*1.1%*		*3.3%*	*11.9%*	*6.8%*	*4.2%*				
Paper/Eco Materials (SB, NB latex)	**110.1**	**27.1**	**36.6**	**32.2**	**30.0**	**126.0**	**30.6**	**32.7**	**32.7**	**34.0**	**130.0**	**136.7**	**137.5**	**141.1**
% YoY	*1.8%*	*-2.4%*	*24.3%*	*25.5%*	*10.5%*	*14.4%*	*12.7%*	*-10.7%*	*1.5%*	*13.5%*	*3.2%*	*5.1%*	*0.6%*	*2.6%*
% QoQ		*0.0%*	*35.0%*	*-12.0%*	*-7.0%*		*2.0%*	*7.0%*	*0.0%*	*4.0%*				
Electronic & LiB Materials	**324.7**	**81.5**	**76.9**	**78.7**	**81.9**	**319.0**	**89.8**	**96.7**	**102.8**	**108.4**	**397.7**	**510.6**	**617.5**	**719.7**
% YoY	*37.1%*	*6.6%*	*-5.5%*	*-8.2%*	*0.9%*	*-1.8%*	*10.2%*	*25.7%*	*30.7%*	*32.4%*	*24.7%*	*28.4%*	*20.9%*	*16.6%*
% QoQ		*0.4%*	*-5.7%*	*2.3%*	*4.1%*		*9.7%*	*7.6%*	*6.3%*	*5.5%*				
Precursors (TSA, DIPAS etc.)	**130.8**	**42.0**	**41.2**	**38.7**	**41.1**	**163.0**	**45.2**	**51.9**	**57.1**	**60.6**	**214.8**	**288.1**	**363.7**	**447.9**
% YoY	*57.8%*	*55.6%*	*29.6%*	*7.6%*	*14.0%*	*24.6%*	*7.5%*	*26.0%*	*47.5%*	*47.5%*	*31.8%*	*34.2%*	*26.2%*	*23.1%*
% QoQ		*16.7%*	*-1.9%*	*-6.0%*	*6.0%*		*10.0%*	*15.0%*	*10.0%*	*6.0%*				
QD Material	**128.6**	**25.0**	**20.0**	**21.2**	**22.9**	**89.1**	**26.1**	**24.0**	**23.5**	**24.9**	**98.6**	**120.1**	**138.7**	**146.2**
% YoY	*12.0%*	*-30.0%*	*-36.5%*	*-31.6%*	*-24.7%*	*-30.7%*	*4.4%*	*20.1%*	*11.0%*	*8.9%*	*10.7%*	*21.8%*	*15.5%*	*5.4%*
% QoQ		*-17.8%*	*-20.0%*	*6.0%*	*8.0%*		*14.0%*	*-8.0%*	*-2.0%*	*6.0%*				
Battery Material (Binder etc.)	**48.4**	**11.5**	**15.0**	**15.8**	**15.0**	**57.2**	**15.6**	**17.7**	**19.2**	**19.9**	**72.4**	**90.4**	**103.1**	**113.6**
% YoY	*49.4%*	*-0.9%*	*11.1%*	*26.0%*	*38.5%*	*18.2%*	*35.3%*	*18.3%*	*21.6%*	*33.2%*	*26.5%*	*24.8%*	*14.1%*	*10.3%*
% QoQ	*0.0%*	*6.5%*	*30.4%*	*5.0%*	*-5.0%*	*0.0%*	*4.0%*	*14.0%*	*8.0%*	*4.0%*	*0.0%*	*0.0%*	*0.0%*	*0.0%*
Tapex	**141.1**	**38.1**	**40.0**	**38.1**	**36.7**	**152.9**	**39.6**	**42.7**	**41.8**	**42.8**	**166.8**	**186.6**	**201.8**	**215.9**
% YoY	*3.8%*	*23.7%*	*13.1%*	*2.4%*	*-2.9%*	*8.3%*	*4.0%*	*6.8%*	*9.5%*	*16.6%*	*9.1%*	*11.8%*	*8.2%*	*7.0%*
% QoQ		*0.7%*	*5.0%*	*-4.6%*	*-3.7%*		*7.8%*	*7.8%*	*-2.1%*	*2.5%*				
Elec Materials (Xi'an)	**39.1**	**10.8**	**11.9**	**12.2**	**12.1**	**47.1**	**12.5**	**12.8**	**12.8**	**12.7**	**50.8**	**51.0**	**50.6**	**50.6**
% YoY	*-3.4%*	*1.3%*	*27.0%*	*40.8%*	*17.6%*	*20.5%*	*15.6%*	*7.2%*	*5.2%*	*4.3%*	*7.9%*	*0.5%*	*-0.8%*	*0.0%*
% QoQ		*5.1%*	*10.1%*	*1.9%*	*-0.1%*		*3.2%*	*2.1%*	*0.0%*	*-1.0%*				
Others	51.3	18.7	17.1	17.0	18.0	70.7	23.9	23.9	23.9	23.9	95.5	124.1	149.0	174.3
Total	**884.0**	**231.9**	**247.5**	**245.0**	**246.3**	**970.7**	**266.1**	**286.8**	**297.3**	**308.6**	**1,158.8**	**1,368.9**	**1,541.6**	**1,704.9**
% YoY	*13.9%*	*10.6%*	*11.5%*	*6.5%*	*10.7%*	*9.8%*	*14.8%*	*15.8%*	*21.4%*	*25.3%*	*19.4%*	*18.1%*	*12.6%*	*10.6%*
% QoQ		*4.2%*	*6.8%*	*-1.0%*	*0.5%*		*8.1%*	*7.8%*	*3.7%*	*3.8%*				
% of Revenues														
Fine Chemicals	*24.6%*	*24.0%*	*26.3%*	*27.3%*	*27.4%*	*26.3%*	*26.2%*	*27.2%*	*28.0%*	*28.1%*	*27.4%*	*28.4%*	*28.4%*	*28.3%*
Paper/Eco Materials	*12.5%*	*11.7%*	*14.8%*	*13.2%*	*12.2%*	*13.0%*	*11.5%*	*11.4%*	*11.0%*	*11.0%*	*11.2%*	*10.0%*	*8.9%*	*8.3%*
Electronic & LiB Materials	*36.7%*	*35.2%*	*31.1%*	*32.1%*	*33.3%*	*32.9%*	*33.8%*	*33.7%*	*34.6%*	*35.1%*	*34.3%*	*37.3%*	*40.1%*	*42.2%*
Tapex	*16.0%*	*16.4%*	*16.1%*	*15.6%*	*14.9%*	*15.7%*	*14.9%*	*14.9%*	*14.0%*	*13.9%*	*14.4%*	*13.6%*	*13.1%*	*12.7%*
Other	*10.2%*	*12.7%*	*11.7%*	*11.9%*	*12.2%*	*12.1%*	*13.7%*	*12.8%*	*12.3%*	*11.8%*	*12.6%*	*10.7%*	*9.5%*	*8.6%*

Source: Company data, UBS estimates

Hansol Chemical – Financial Statement Forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues	**884**	**232**	**248**	**245**	**246**	**971**	**266**	**287**	**297**	**309**	**1,159**	**1,369**	**1,542**	**1,705**
Gross profit	250	67	74	73	63	277	81	91	92	83	346	430	508	567
% GP margin	*28.3%*	*28.7%*	*30.0%*	*30.0%*	*25.5%*	*28.5%*	*30.4%*	*31.7%*	*30.8%*	*26.9%*	*29.9%*	*31.4%*	*33.0%*	*33.3%*
Operating profit	**156**	**44**	**49**	**46**	**33**	**173**	**54**	**64**	**65**	**55**	**238**	**302**	**356**	**401**
% OP margin	*17.7%*	*19.2%*	*19.7%*	*19.0%*	*13.5%*	*17.8%*	*20.4%*	*22.2%*	*21.8%*	*17.9%*	*20.5%*	*22.0%*	*23.1%*	*23.5%*
NI post MI	147	41	51	41	31	165	47	55	56	49	207	255	297	331
EPS	**13,452**	**3,799**	**4,789**	**3,843**	**2,924**	**15,355**	**4,422**	**5,137**	**5,186**	**4,539**	**19,284**	**23,805**	**27,683**	**30,834**
% EPS growth YoY	*22.3%*	*15.5%*	*7.3%*	*-7.8%*	*92.5%*	*14.2%*	*16.4%*	*7.3%*	*35.0%*	*55.2%*	*25.6%*	*23.4%*	*16.3%*	*11.4%*
DPS	2,600	0	0	0	4,600	4,600	0	0	0	5,800	5,800	7,100	8,300	9,300
% payout ratio	*19.3%*	*0.0%*	*0.0%*	*0.0%*	*157.3%*	*30.0%*	*0.0%*	*0.0%*	*0.0%*	*127.8%*	*30.1%*	*29.8%*	*30.0%*	*30.2%*
Cash and cash equiv	**113**	**113**	**97**	**136**	**144**	**144**	**139**	**81**	**111**	**115**	**115**	**183**	**302**	**465**
Total assets	**1,600**	**1,695**	**1,654**	**1,664**	**1,699**	**1,699**	**1,726**	**1,738**	**1,800**	**1,857**	**1,857**	**2,058**	**2,286**	**2,536**
Total equity	1,177	1,200	1,195	1,236	1,268	1,268	1,315	1,321	1,377	1,425	1,425	1,618	1,839	2,081
Net debt	168	176	204	135	128	128	110	169	138	134	134	64	-58	-223
% ROE	*13.1%*	*13.9%*	*17.2%*	*13.6%*	*10.0%*	*13.5%*	*14.7%*	*16.7%*	*16.5%*	*13.9%*	*15.4%*	*16.8%*	*17.2%*	*16.9%*
D&A	**58**	**16**	**16**	**16**	**15**	**62**	**18**	**18**	**18**	**18**	**70**	**68**	**67**	**65**
CF from Operations	**200**	**39**	**63**	**110**	**49**	**262**	**69**	**43**	**82**	**55**	**248**	**318**	**363**	**399**
Capex	-84	-16	-26	-41	-41	-124	-51	-51	-51	-51	-205	-185	-165	-145
Free cash flow	**70**	**-12**	**57**	**69**	**8**	**122**	**17**	**-9**	**31**	**4**	**43**	**133**	**198**	**254**
CF from Investments	**-131**	**-50**	**-6**	**-41**	**-41**	**-139**	**-51**	**-51**	**-51**	**-51**	**-205**	**-185**	**-165**	**-145**
Dividend	-23	0	-29	0	0	-29	0	-49	0	0	-49	-62	-76	-89
CF from Finance	**-40**	**11**	**-73**	**-30**	**0**	**-92**	**-23**	**-49**	**0**	**0**	**-72**	**-65**	**-79**	**-92**
Net changes of cash	**29**	**1**	**-16**	**39**	**8**	**31**	**-5**	**-58**	**31**	**4**	**-29**	**68**	**119**	**163**

Source: Company data, UBS estimates

ISU Petasys (Jimmy Yoon) – Buy, PT Won190,000

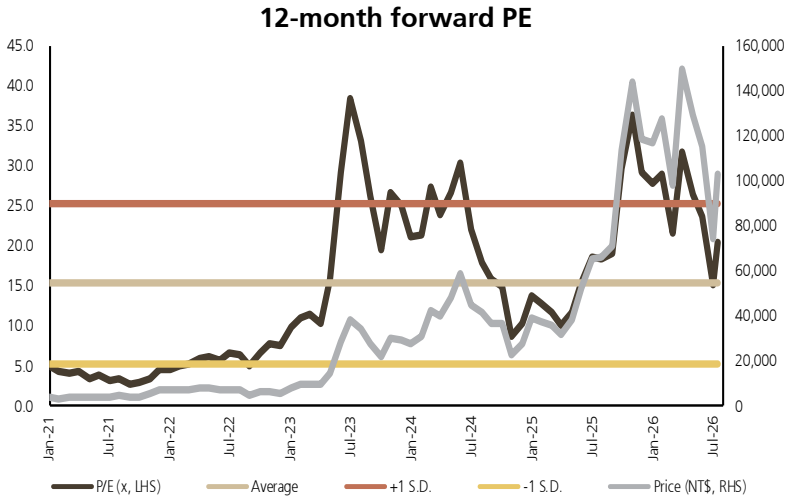
- **Our View**
  - At the current valuation, we believe the mid-LT growth profile of ISU's premium MLB demand is not yet fairly priced in. We expect ISU's blended ASP to record a robust 24% CAGR from Won3.8bn/km2 in 2024 to Won8.1bn/km2 by 2028E, led by an improved sales mix and AI accelerator TAM growth with more ASIC project wins.
- **Can ISU's ASP/OPM rise, driven by a richer sales mix from AI switch upgrades?**
  - Yes. On the back of robust capex growth for Ethernet AI switches (800G/1.6T) by major hyperscalers (ISU geared to Arista/Celestica/Google/NVDA) expected to grow from US$1bn in 2024 to US$9.2bn in 2028E, we expect ISU's revenue mix to improve with rising demand for high layer network switch MLBs. We expect the network switch MLB share revenue mix to grow to 60% in 2028E (50% in 2024), driving ASP/OPM.
- **Can ISU expand MLB addressable market in ASIC AI accelerators?**
  - Yes. While we believe ISU's slower pace of MLB capacity ramp vs more aggressive MLB competitors could raise concerns on relative wallet share loss, given 1) fast growing AI accelerator TAM (UBSe forecast for global GPU/ASIC unit growth of +32%/+68% YoY in 2026/27E) and associated MLB demand within which we see 2) ISU continuing to expand ASIC project wins (MSFT, Meta, AWS and also OAI into 2H26), we believe this should support meaningful YoY revenue growth and margin expansion.
- **Is there upside for ISU's capex/sales guidance?**
  - Yes. Even with ISU's revised-up / pulled-in MLB capacity ramp timing for ISU Korea, now expected to grow at 12% CAGR from 2024-27E, we see room for incremental upside capacity add / capex. In addition to the currently ramping Fab 5, ISU last year secured additional 10k pyeong of land in Daegu, Korea, with only 2k pyeong earmarked for capa add - implying 8k pyeong land available for incremental capacity add as needed with visibility on Fab 7, we believe potentially set to be decided by 2026YE.

- **PT Won190k –** We value ISU's shares based on 31.5x 2027E P/E, slightly above +1STD historical average of 28x given our LT EPS CAGR forecast of 24%.

ISU Petasys – Valuation

UBS price target	**190,000**
Current stock price (Won)	102,900
Number of shares (Mn)	73.4
Market cap. (Won bn)	7,554
Net debt (Won bn)	-66
Adjusted EV (Won bn)	7,488

	2025	2026E	2027E	2028E	2029E	2030E
Current Valuation						
EV/Revenues	6.85	4.48	3.17	2.41	1.83	1.58
P/Book	10.12	7.40	5.29	3.91	3.07	2.51
P/E	47.07	26.85	16.99	13.49	11.20	10.75
EV/EBITDA	31.79	19.01	11.58	9.04	7.25	6.67
UBS valuation (x)						
EV/Revenues	12.76	8.30	6.03	4.67	3.66	3.31
P/Book	18.68	13.67	9.76	7.22	5.66	4.64
P/E	86.91	49.58	31.36	24.90	20.67	19.85
EV/EBITDA	59.23	35.24	22.02	17.49	14.47	14.01
Other data						
EPS	2,186	4,117	6,058	7,629	9,191	9,573
% ROE	***21.2%***	***31.5%***	***36.1%***	***33.2%***	***30.6%***	***25.7%***
BVPS	10,170	13,902	19,465	26,327	33,560	40,938
Cash per share (Won)	2,299	1,892	5,547	5,552	8,369	9,930
FCF Per Share (Won)	208	-407	3,017	2,390	4,891	6,974
% FCF yield	*0.2%*	*-0.4%*	*2.9%*	*2.3%*	*4.8%*	*6.8%*

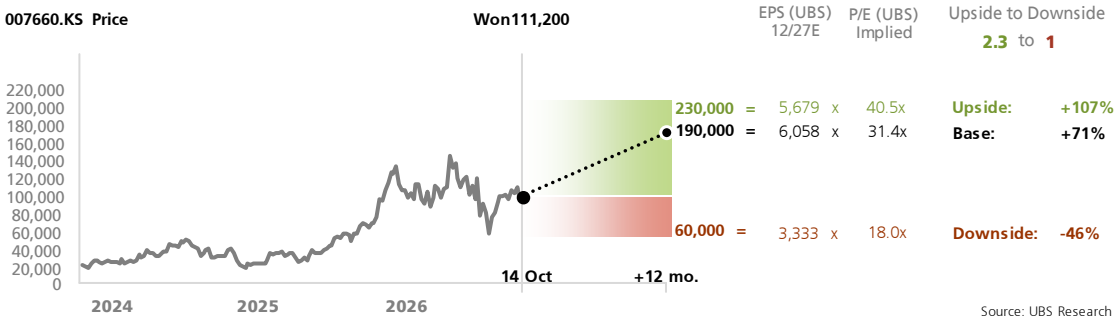


Source: Company data, UBS estimates

ISU Petasys – Key Charts

UBSe vs Consensus

	3Q26E			4Q26E			2026E			2027E			2028E		
(Won bn)	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**
Sales	436	419	4.0%	516	453	14.1%	1,672	1,577	6.1%	2,302	2,126	8.3%	2,976	2,609	14.1%
Op Profit	**96**	**92**	**3.8%**	**120**	**99**	**20.6%**	**359**	**335**	**7.3%**	**582**	**513**	**13.3%**	**726**	**639**	**13.6%**
% OP margin	21.9%	22.0%	-5bps	23.2%	21.9%	126bps	21.5%	21.2%	26bps	25.3%	24.1%	112bps	24.4%	24.5%	-10bps
EPS (Won)	1,034	927	11.6%	1,297	993	30.7%	4,117	3,657	12.6%	6,058	5,352	13.2%	7,629	6,759	12.9%



Value drivers	Sales growth (2025-30E CAGR)	OP growth (2025-30E CAGR)	OP margin (2026/27E)	ASP '000m²/Won bn (2026/27E)
Won230,000 upside	34%	39%	23%/29%	5.8/8.1
Won190,000 base	31%	35%	21%/25%	5.6/7.8
Won60,000 downside	22%	26%	18%/19%	5.2/6.8

Source: Company data, UBS estimates

ISU Petasys – Divisional Forecasts

(Won bn)	2025	1Q26	2Q26E	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues														
ISU Petasys Korea	930	276	308	356	434	1,374	461	472	486	505	1,923	2,539	3,300	3,630
ISU Petasys China	228	79	87	94	96	356	101	105	111	113	430	483	539	595
ISU Petasys US	22	7	8	8	9	31	9	10	10	10	39	44	48	53
Others (Adjustments)	-92	-21	-23	-23	-23	-89	-23	-23	-23	-23	-90	-90	-90	-90
Total	**1,088**	**340**	**380**	**436**	**516**	**1,672**	**549**	**564**	**584**	**606**	**2,302**	**2,976**	**3,797**	**4,189**
% YoY	*30%*	*35%*	*57%*	*47%*	*73%*	*54%*	*61%*	*49%*	*34%*	*17%*	*38%*	*29%*	*28%*	*10%*
% QoQ		*14%*	*12%*	*15%*	*19%*		*6%*	*3%*	*3%*	*4%*				
Operating profit														
ISU Petasys Korea	152	46	55	71	95	267	106	113	121	131	472	605	751	781
ISU Petasys China	40	17	17	19	18	71	19	19	22	24	84	93	92	92
ISU Petasys US	14	4	5	6	6	21	6	6	7	7	26	28	29	29
Others (Adjustments)	-1	0	0	0	0	0	0	0	0	0	0	0	0	0
Total	**205**	**67**	**77**	**96**	**120**	**359**	**131**	**139**	**150**	**162**	**582**	**726**	**871**	**902**
% OpM														
ISU Petasys Korea	*16.4%*	*16.6%*	*17.8%*	*20.0%*	*22.0%*	*19.5%*	*23.0%*	*24.0%*	*25.0%*	*26.0%*	*24.5%*	*23.8%*	*22.8%*	*21.5%*
ISU Petasys China	*17.5%*	*21.7%*	*19.5%*	*20.0%*	*19.0%*	*20.0%*	*19.0%*	*18.0%*	*20.0%*	*21.0%*	*19.5%*	*19.3%*	*17.0%*	*15.5%*
ISU Petasys US	*63.2%*	*64.6%*	*66.7%*	*67.0%*	*67.0%*	*66.4%*	*65.0%*	*65.0%*	*65.0%*	*65.0%*	*65.0%*	*65.0%*	*60.0%*	*55.0%*
Total	***18.8%***	***19.8%***	***20.3%***	***21.9%***	***23.2%***	***21.5%***	***23.9%***	***24.5%***	***25.7%***	***26.7%***	***25.3%***	***24.4%***	***23.0%***	***21.5%***

Source: Company data, UBS estimates

ISU Petasys – Forecasts

(Won bn)	2025	1Q26	2Q26E	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues	**1,088**	**340**	**380**	**436**	**516**	**1,672**	**549**	**564**	**584**	**606**	**2,302**	**2,976**	**3,797**	**4,189**
Gross profit	282	84	95	122	166	468	172	171	186	216	745	938	1,140	1,198
% GM	*25.9%*	*24.8%*	*25.1%*	*28.1%*	*32.1%*	*28.0%*	*31.3%*	*30.3%*	*31.9%*	*35.7%*	*32.4%*	*31.5%*	*30.0%*	*28.6%*
Operating profit	**205**	**67**	**77**	**96**	**120**	**359**	**131**	**139**	**150**	**162**	**582**	**726**	**871**	**902**
% OpM	*18.8%*	*19.8%*	*20.3%*	*21.9%*	*23.2%*	*21.5%*	*23.9%*	*24.5%*	*25.7%*	*26.7%*	*25.3%*	*24.4%*	*23.0%*	*21.5%*
Net profit post MI	160	50	60	76	95	281	100	105	114	125	445	560	675	703
EPS reported (Won)	**2,186**	**687**	**813**	**1,034**	**1,297**	**4,117**	**1,366**	**1,437**	**1,559**	**1,697**	**6,058**	**7,629**	**9,191**	**9,573**
Cash and cash equivalents	163	105	144	270	246	246	126	121	371	434	434	555	772	1,128
Total assets	**1,290**	**1,352**	**1,467**	**1,542**	**1,759**	**1,759**	**1,824**	**1,909**	**1,991**	**2,224**	**2,224**	**2,873**	**3,500**	**4,086**
Total liabilities	**533**	**555**	**607**	**606**	**728**	**728**	**693**	**711**	**679**	**786**	**786**	**934**	**1,034**	**1,075**
Total shareholders' equity	**756**	**797**	**860**	**936**	**1,031**	**1,031**	**1,131**	**1,198**	**1,313**	**1,437**	**1,437**	**1,939**	**2,467**	**3,008**
Depreciation & amortization	30	9	11	7	7	34	12	12	12	12	49	67	88	88
Cash flow from operations	**169**	**-76**	**27**	**170**	**19**	**139**	**-75**	**79**	**296**	**108**	**407**	**408**	**614**	**729**
Capex	-153	-33	-46	-45	-45	-169	-46	-46	-46	-46	-186	-232	-255	-217
Cash flow from investments	**-347**	**15**	**-6**	**137**	**137**	**284**	**136**	**136**	**136**	**136**	**542**	**496**	**472**	**511**
Dividends	-9	-0	-17	0	0	-17	-0	-38	0	0	-38	-59	-147	-162
Cash flow from financing	**262**	**-1**	**14**	**-181**	**-181**	**-349**	**-181**	**-219**	**-181**	**-181**	**-761**	**-782**	**-870**	**-884**
Net change in cash	**82**	**-58**	**39**	**126**	**-25**	**83**	**-120**	**-5**	**250**	**63**	**188**	**122**	**217**	**355**

Source: Company data, UBS estimates

LG Display (Jimmy Yoon) – Neutral, PT Won11,600

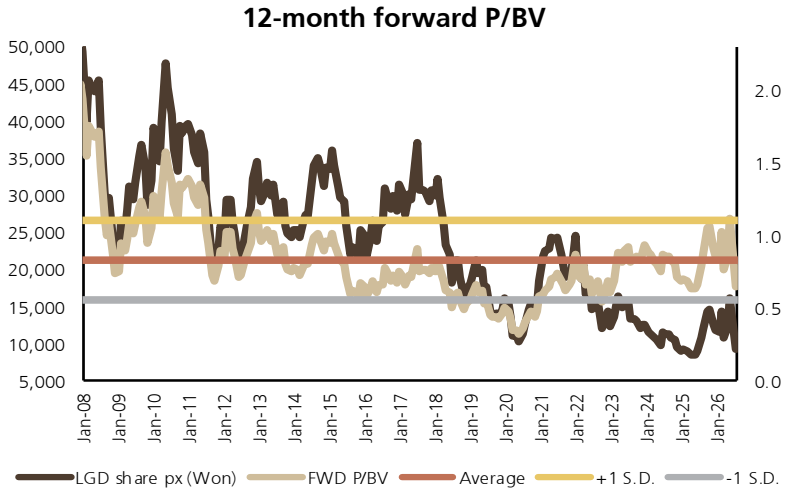
- **Our View**
  - While we expect LGD to further scale its RGB OLED business into 2026 on the back of more resilient build plans associated with the US smartphone OEM and foldable form factor launch adding to TAM, given elevated memory pricing and likely need for key smartphone OEM to control BOM costs, we see risk that LGD may face incremental ASP pressure into 2026. We also see YoY downside to PC unit shipments in 2026E potentially limiting IT LCD panel shipments. On the other hand, we believe WOLED panel shipments may remain relatively stable with ongoing cost reduction efforts (D&A roll over, fab UTR management) likely to support margins.
- **Can LGD further scale its RGB OLED business while securing profitability in 2026E?**
  - Yes, we believe LGD should be able to further scale its RGB OLED business in 2026 (UBSe mobile OLED units growing +12% YoY to c. 84m units) on the back of upside to build plans for the major US OEM as well as new foldable form factor expected to lift TAM. That said, the magnitude of OPM improvement remains less certain.
  - While end customers for key US smartphone OEM are typically less price sensitive vs Android, we still see potential risk of incremental pricing pressure into 2H26 in efforts to control BOM costs associated with rising memory prices.
- **Will LGD be able to expand OLED TV panel profits in 2026E?**
  - While global TV end-demand remains muted, we believe OLED penetration should continue to pick up modestly YoY in 2026E. Incremental penetration of OLED into gaming monitors (UBSe c. 20% of WOLED unit mix in CY26) should also support area shipments.
  - We forecast LGD's WOLED panel shipments to grow +15% YoY in 2026E to c. 7.5m units (was 6.5m units in '25). Margins should also be supported by cost reduction efforts (D&A roll over, fab capacity management etc.).
- **Will the IT LCD market supply/demand dynamics improve in 2026E?**
  - While noting some element of PC unit front-loading into 1Q26, with PC OEMs having started raising prices on the back of rising memory costs, we expect a more meaningful YoY slowdown into 2H26 (PC OEMs guiding down >10% YoY). We also expect demand for tablets to remain muted on the back of macro/inflationary pressure weighing in on consumer purchasing.

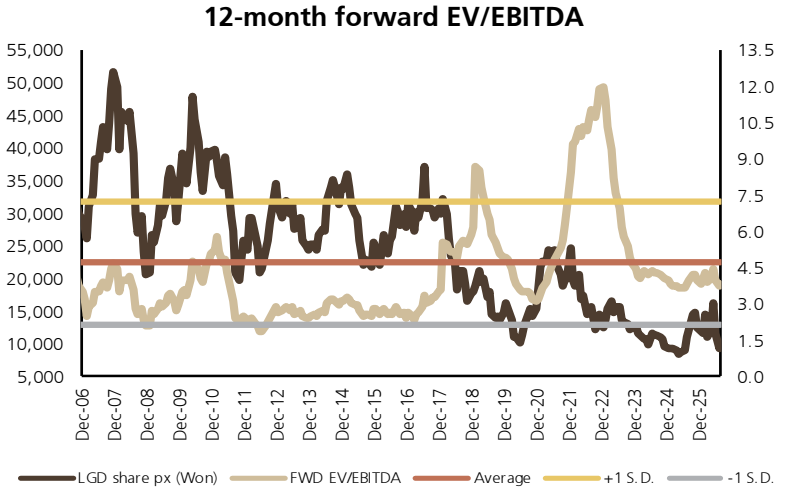
- **PT Won11,600 –** We value LGD shares at 0.81x NTM P/BV based on our LT ROE estimate of 7.6% and CoE of 9.5%.

LG Display (Jimmy Yoon) – Neutral, PT Won11,600

UBS price target	**11,600**
Current stock price (Won)	10,110
Number of shares (bn)	0.500
Market cap. (Won bn)	5,055
Debt (Won bn)	14,696
Cash & equivalents (Won bn)	1,451
Adjusted EV (Won bn)	18,300

	2025	2026E	2027E	2028E	2029E	2030E
Current Valuation						
EV/Revenues	0.71	0.72	0.72	0.71	0.71	0.71
P/Book	0.77	0.74	0.68	0.62	0.58	0.55
P/E	22.34	-10.53	8.69	6.22	6.50	6.85
EV/EBITDA	3.76	3.91	3.70	3.49	3.57	3.63
UBS valuation (x)						
EV/Revenues	0.74	0.75	0.75	0.74	0.74	0.74
P/Book	0.88	0.85	0.78	0.72	0.67	0.63
P/E	25.63	-12.08	9.98	7.14	7.46	7.86
EV/EBITDA	3.91	4.07	3.85	3.63	3.72	3.78
Other data						
% ROE	***2.8%***	***-6.0%***	***6.9%***	***9.0%***	***8.0%***	***7.2%***
Tangible book per share (Won)	12,722	13,012	14,655	16,380	17,855	19,250
Cash per share (Won)	3,144	3,659	6,352	5,656	5,684	5,681
FCF Per Share (Won)	5,494	919	9,386	3,209	2,856	2,794
% FCF yield	*54.3%*	*9.1%*	*92.8%*	*31.7%*	*28.2%*	*27.6%*



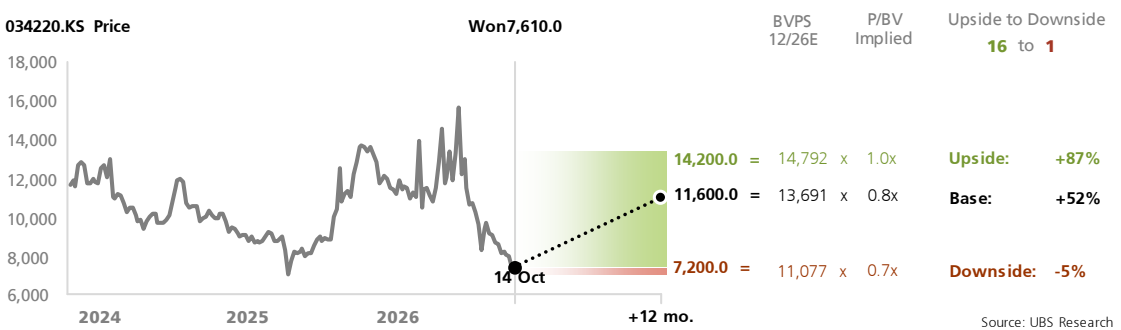


Source: Company data, UBS estimates

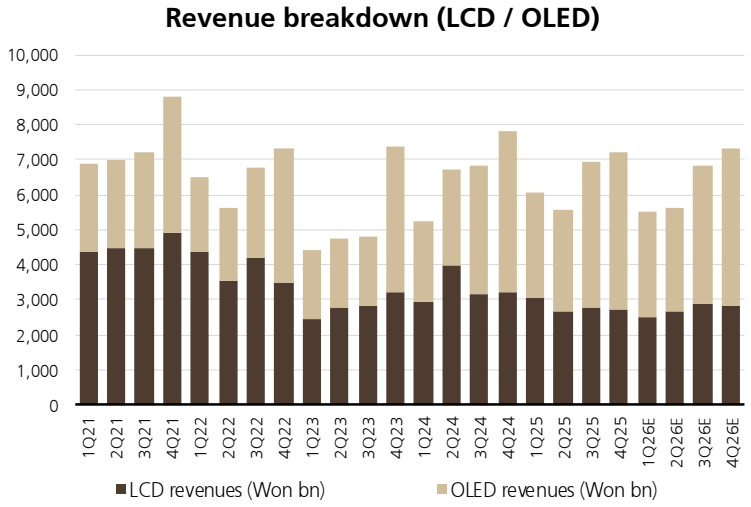
LG Display – UBSe vs. Consensus/ Key charts

UBS forecasts vs Consensus

	3Q26E			4Q26E			2026E			2027E			2028E		
(Won bn)	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**
Sales	6,836	6,820	*0.2%*	7,317	7,111	*2.9%*	25,299	25,123	*0.7%*	25,456	25,875	*-1.6%*	25,807	27,141	*-4.9%*
Op Profit	**386**	**493**	**-21.8%**	**432**	**569**	**-24.2%**	**857**	**1,132**	**-24.3%**	**1,191**	**1,498**	**-20.5%**	**1,342**	**1,571**	**-14.5%**
% OP margin	*5.6%*	*7.2%*	*-159bps*	*5.9%*	*8.0%*	*-211bps*	*3.4%*	*4.5%*	*-112bps*	*4.7%*	*5.8%*	*-111bps*	*5.2%*	*5.8%*	*-59bps*
EPS	453	467	-2.9%	537	672	-20.0%	-960	532	NM	1,163	1,637	-29.0%	1,625	2,129	-23.7%



Value drivers	OLED iPhone share 2026E	Large OLED panel unit shipments 2026E	Operating profit 2026E
Won14,200 upside	c. 36%	7.7mil	Won1.2trn
Won11,600 base	c. 34%	7.5mil	Won0.9trn
Won7,200 downside	c. 32%	7.0mil	Won0.6trn



Source: Bloomberg, Company data, UBS estimates

LG Display – Key Assumptions 1

	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E
Revenue breakdown (Won bn)													
Total Revenues	**25,810**	**5,534**	**5,612**	**6,836**	**7,317**	**25,299**	**5,551**	**5,632**	**6,812**	**7,462**	**25,456**	**25,807**	**25,871**
% YoY	*-3.0%*	*-8.8%*	*0.4%*	*-1.7%*	*1.6%*	*-2.0%*	*0.3%*	*0.3%*	*-0.3%*	*2.0%*	*0.6%*	*1.4%*	*0.2%*
% QoQ		*-23.1%*	*-78.3%*	*21.8%*	*7.0%*		*-24.1%*	*1.5%*	*21.0%*	*9.5%*			
LCD	**11,191**	**2,508**	**2,655**	**2,869**	**2,823**	**10,855**	**2,580**	**2,762**	**2,761**	**2,759**	**10,861**	**10,776**	**10,491**
% of total revenues	*43%*	*45%*	*47%*	*42%*	*39%*	*43%*	*46%*	*49%*	*41%*	*37%*	*43%*	*42%*	*41%*
OLED	**14,619**	**3,026**	**2,958**	**3,967**	**4,493**	**14,444**	**2,971**	**2,870**	**4,051**	**4,703**	**14,595**	**15,031**	**15,380**
% of total revenues	*57%*	*55%*	*53%*	*58%*	*61%*	*57%*	*54%*	*51%*	*59%*	*63%*	*57%*	*58%*	*59%*
Capacity and utilisation													
Total area capacity (sq.m.)	**25,892**	**5,813**	**5,813**	**5,813**	**5,813**	**23,252**	**6,259**	**6,259**	**6,259**	**6,259**	**25,036**	**26,427**	**27,318**
% YoY	*-21.1%*	*-31.2%*	*0.0%*	*0.0%*	*0.0%*	*-10.2%*	*7.7%*	*7.7%*	*7.7%*	*7.7%*	*7.7%*	*5.6%*	*3.4%*
% QoQ		*0.0%*	*0.0%*	*0.0%*	*0.0%*		*7.7%*	*0.0%*	*0.0%*	*0.0%*			
% Utilisation rate	*76.0%*	*63.3%*	*70.4%*	*72.6%*	*75.9%*	*70.6%*	*59.5%*	*64.6%*	*69.2%*	*74.2%*	*66.9%*	*66.0%*	*67.6%*
LCD capacity	**11,024**	**2,096**	**2,096**	**2,096**	**2,096**	**8,384**	**2,096**	**2,096**	**2,096**	**2,096**	**8,384**	**8,384**	**8,384**
OLED capacity	**14,868**	**3,717**	**3,717**	**3,717**	**3,717**	**14,868**	**4,163**	**4,163**	**4,163**	**4,163**	**16,652**	**18,043**	**18,935**
Net area shipment ('000 sq.m.)													
Total LCD	**9,435**	**1,503**	**1,538**	**1,424**	**1,372**	**5,837**	**1,334**	**1,407**	**1,381**	**1,343**	**5,465**	**5,339**	**5,281**
% YoY	*-44.5%*	*-60.3%*	*-24.9%*	*-26.3%*	*-17.6%*	*-38.1%*	*-11.3%*	*-8.5%*	*-3.0%*	*-2.1%*	*-6.4%*	*-2.3%*	*-1.1%*
% QoQ		*-9.8%*	*2.4%*	*-7.4%*	*-3.7%*		*-2.8%*	*5.5%*	*-1.8%*	*-2.8%*			
TV - LCD													
% YoY													
% QoQ		*0.0%*	*0.0%*	*0.0%*	*0.0%*		*0.0%*	*0.0%*	*0.0%*	*0.0%*			
IT - LCD	6,760	1,443	1,428	1,314	1,262	5,447	1,224	1,297	1,271	1,233	5,025	4,899	4,841
% YoY	*-9.6%*	*-11.1%*	*-21.4%*	*-24.2%*	*-20.4%*	*-19.4%*	*-15.2%*	*-9.2%*	*-3.3%*	*-2.3%*	*-7.7%*	*-2.5%*	*-1.2%*
% QoQ		*-9.0%*	*-1.0%*	*-8.0%*	*-4.0%*		*-3.0%*	*6.0%*	*-2.0%*	*-3.0%*			
Mobile and others - LCD	787	60	110	110	110	390	110	110	110	110	440	440	440
% YoY	*-25.6%*	*-78.3%*	*-52.2%*	*-45.0%*	*37.5%*	*-50.4%*	*83.3%*	*0.0%*	*0.0%*	*0.0%*	*12.8%*	*0.0%*	*0.0%*
% QoQ		*-25.0%*	*83.3%*	*0.0%*	*0.0%*		*0.0%*	*0.0%*	*0.0%*	*0.0%*			
Total OLED	**8,139**	**1,697**	**2,064**	**2,376**	**2,597**	**8,734**	**1,909**	**2,150**	**2,516**	**2,835**	**9,410**	**10,138**	**11,116**
% YoY	*5.5%*	*5.3%*	*4.9%*	*8.1%*	*10.0%*	*7.3%*	*12.5%*	*4.2%*	*5.9%*	*9.2%*	*7.7%*	*7.7%*	*9.6%*
% QoQ		*-28.2%*	*21.7%*	*15.1%*	*9.3%*		*-26.5%*	*12.7%*	*17.0%*	*12.7%*			
TV - OLED	6,650	1,384	1,781	1,977	2,069	7,210	1,580	1,853	2,080	2,264	7,777	8,378	9,305
% YoY	*2.9%*	*4.5%*	*5.1%*	*12.4%*	*10.4%*	*8.4%*	*14.2%*	*4.1%*	*5.2%*	*9.5%*	*7.9%*	*7.7%*	*11.1%*
% QoQ		*-26.1%*	*28.7%*	*11.0%*	*4.7%*		*-23.6%*	*17.3%*	*12.2%*	*8.9%*			
Implied OLED TV panel units ('000)	**6.5**	**1.4**	**1.8**	**2.1**	**2.2**	**7.5**	**1.6**	**1.9**	**2.2**	**2.4**	**8.1**	**8.6**	**9.4**
Mobile - OLED	1,489	313	283	399	528	1,524	329	297	436	570	1,633	1,760	1,811
% YoY	*19.1%*	*8.7%*	*3.3%*	*-9.0%*	*8.2%*	*2.4%*	*5.1%*	*4.8%*	*9.3%*	*8.0%*	*7.1%*	*7.8%*	*2.9%*
% QoQ		*-35.9%*	*-9.5%*	*41.0%*	*32.3%*		*-37.7%*	*-9.7%*	*47.0%*	*30.7%*			
Total area shipment	**17,372**	**3,199**	**3,603**	**3,800**	**3,968**	**14,571**	**3,242**	**3,557**	**3,897**	**4,178**	**14,875**	**15,477**	**16,397**
% YoY	*-29.7%*	*-40.8%*	*-9.9%*	*-3.7%*	*-1.5%*	*-16.1%*	*1.3%*	*-1.3%*	*2.6%*	*5.3%*	*2.1%*	*4.0%*	*5.9%*
% QoQ		*-20.5%*	*12.6%*	*5.5%*	*4.4%*		*-18.3%*	*9.7%*	*9.6%*	*7.2%*			

Source: Company data, UBS estimates

LG Display – Financial Statement Forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues	**25,810**	**5,534**	**5,612**	**6,836**	**7,317**	**25,299**	**5,551**	**5,632**	**6,812**	**7,462**	**25,456**	**25,807**	**25,871**	**25,724**
Gross profit	3,376	765	764	1,200	1,331	4,060	938	944	1,319	1,460	4,662	4,953	4,900	4,941
% GP margin	*13.1%*	*13.8%*	*13.6%*	*17.6%*	*18.2%*	*16.0%*	*16.9%*	*16.8%*	*19.4%*	*19.6%*	*18.3%*	*19.2%*	*18.9%*	*19.2%*
Operating profit	**517**	**147**	**-108**	**386**	**432**	**857**	**149**	**135**	**405**	**501**	**1,191**	**1,342**	**1,244**	**1,161**
% OP margin	*2.0%*	*2.7%*	*-1.9%*	*5.6%*	*5.9%*	*3.4%*	*2.7%*	*2.4%*	*6.0%*	*6.7%*	*4.7%*	*5.2%*	*4.8%*	*4.5%*
NI post MI	226	-571	-405	226	269	-480	9	-4	244	332	581	813	778	738
EPS	**453**	**-1,141**	**-809**	**453**	**537**	**-960**	**17**	**-8**	**489**	**665**	**1,163**	**1,625**	**1,555**	**1,475**
Cash and cash equiv	**1,572**	**1,524**	**1,451**	**1,735**	**1,829**	**1,829**	**1,796**	**2,375**	**2,579**	**3,176**	**3,176**	**2,828**	**2,842**	**2,841**
Total assets	**26,917**	**27,296**	**27,501**	**28,240**	**28,189**	**28,189**	**26,430**	**27,033**	**27,824**	**28,305**	**28,305**	**27,824**	**27,809**	**27,652**
Total equity	7,839	7,773	7,650	7,876	8,145	8,145	8,154	8,150	8,394	8,727	8,727	9,389	9,967	10,505
Net debt	11,161	12,288	11,945	11,661	11,567	11,567	10,600	10,021	9,817	9,220	9,220	8,568	8,054	7,555
Depreciation	**-3,603**	**-808**	**-807**	**-750**	**-760**	**-3,125**	**-750**	**-750**	**-800**	**-810**	**-3,110**	**-3,300**	**-3,320**	**-3,320**
Amortization	-751	-186	-173	-170	-165	-694	-160	-160	-160	-160	-640	-600	-560	-560
CF from Operations	**2,352**	**-122**	**1,157**	**985**	**844**	**2,863**	**1,817**	**1,529**	**1,154**	**1,497**	**5,996**	**4,502**	**4,514**	**4,499**
Capex	-2,107	-607	-650	-700	-750	-2,707	-850	-950	-950	-900	-3,650	-3,700	-3,800	-3,800
Free cash flow	**1,373**	**-664**	**515**	**285**	**94**	**230**	**967**	**579**	**204**	**597**	**2,346**	**802**	**714**	**699**
CF from Investments	**-979**	**-542**	**-642**	**-700**	**-750**	**-2,634**	**-850**	**-950**	**-950**	**-900**	**-3,650**	**-3,700**	**-3,800**	**-3,800**
CF from Finance	**-1,963**	**544**	**-588**	**0**	**0**	**-44**	**-1,000**	**0**	**0**	**0**	**-1,000**	**-1,150**	**-700**	**-700**

Source: Company data, UBS estimates

LG Electronics (Jimmy Yoon) – Neutral, PT Won168,000

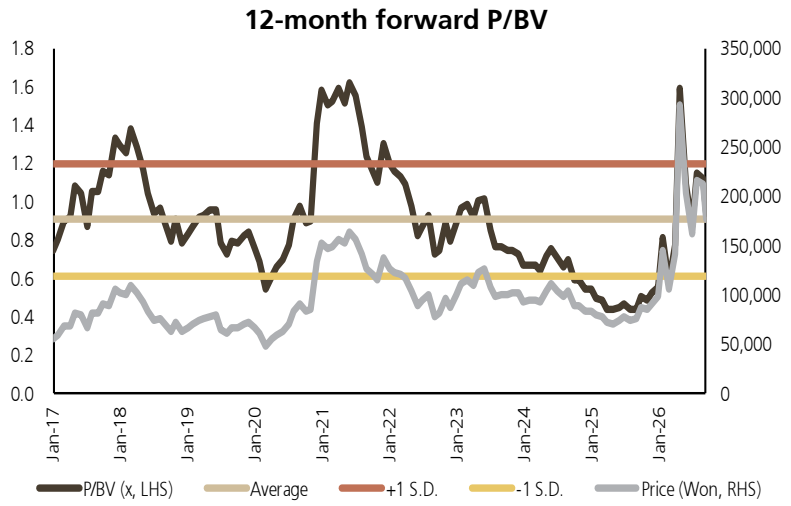
- **Our View**
  - We expect LGE may see incremental headwinds for its core HS and MS businesses into 2H26. This is in the context of inflationary pressure (semis cost pass through / macro impacting consumer purchasing) likely to dent demand for Appliances/TV. Off a 'leaner' cost base on the back of recent cost management efforts (both fixed and variable), the ongoing Middle East conflict should also continue to weigh on incremental logistics/freight costs for LGE. Mid-LT we see LGE's engagement in robotics/humanoid as well as AI DC cooling verticals as positive though earnings contribution and execution remain to be proven.
- **Can LGE's Home Appliance (HS) business record YoY revs/OP growth into 2026E?**
  - We believe LGE's HS business should be able see top-line grow +6% YoY in '26E on the back of its 'two-track' strategy (widened OLED line-up + premium LCD TV), incremental focus on growing B2B and online subscription-based HS revenues, as well as Global South strategy. That said, margin expansion remains uncertain in the context of incrementally higher logistics/maritime costs into 2H26. We hence forecast LGE's HS margins to come in at c. 5.4% (vs 4.9% in 2025).
- **Can LGE's Media Entertainment (MS) business turn around on a full-year basis in 2026E?**
  - We forecast LGE's MS business (TV+IT) to achieve c.2% OPM in 2026E (in line with mgmt guidance targeting profits) after having incurred Won(750)bn in losses in 2025 (reflective of one-offs). While both TV and IT demand remains uncertain in the context of inflationary pressure with semis cost pass through / macro impacting purchasing power, we believe recent cost reduction efforts (overhead) and moderated SG&A spend should help narrow full year losses.
- **Can Vehicle and Eco Solution businesses further scale up in OP contribution mid-LT?**
  - Yes, we forecast LGE's Vehicle and Eco Solution businesses to grow mid-LT to comprise c. 34%/29% of LGE (ex-LGI) OP vs only having comprised c. 4%/25% of OP in 2024. For VS, on the back of stable new order backlog / mix improvement, we forecast OP to grow at c. 13% CAGR from 2025-30E. For ES, though still relatively small in revenue contribution (UBSe c. 10%), strong revenue ramp from Ai data center chiller revenues should support both topline and OP expansion mid-LT.

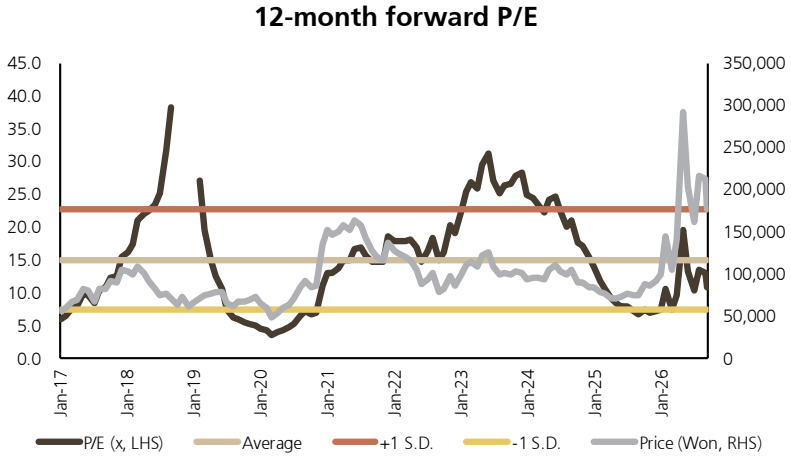
- **PT Won168,000 –** We value LGE shares at 1.03x NTM P/BV based on our LT ROE forecast of 9.8% and CoE of 9.5%.

LG Electronics (Jimmy Yoon) – Neutral, PT Won168,000

UBS price target	**168,000**
Current stock price (Won)	148,000
Number of shares (Mn)	180.1
Treasury shares	2.1
Market cap. (Won bn)	26,651
Net debt (Won bn)	2,642
Adjusted EV (Won bn)	29,293

	2025	2026E	2027E	2028E	2029E
Current Valuation					
EV/Revenues	0.33	0.30	0.28	0.27	0.26
P/Book	1.12	0.96	0.86	0.78	0.71
P/E	27.81	11.36	8.46	7.65	6.92
EV/EBITDA	4.82	3.66	3.57	3.42	3.29
UBS valuation (x)					
EV/Revenues	0.37	0.34	0.32	0.30	0.29
P/Book	1.27	1.09	0.98	0.89	0.80
P/E	31.57	12.89	9.61	8.68	7.85
EV/EBITDA	5.41	4.11	4.01	3.84	3.69
Other data					
EPS	5,321	12,993	17,484	19,347	21,390
% ROE	***4.0%***	***9.0%***	***10.7%***	***10.7%***	***10.7%***
Tangible book per share (Won)	132,331	154,731	171,212	189,546	209,914
Cash per share (Won)	46,834	11,155	20,881	32,438	46,429
FCF Per Share (Won)	9,225	44,169	21,421	23,261	25,704
% FCF yield	*6.2%*	*29.8%*	*14.5%*	*15.7%*	*17.4%*



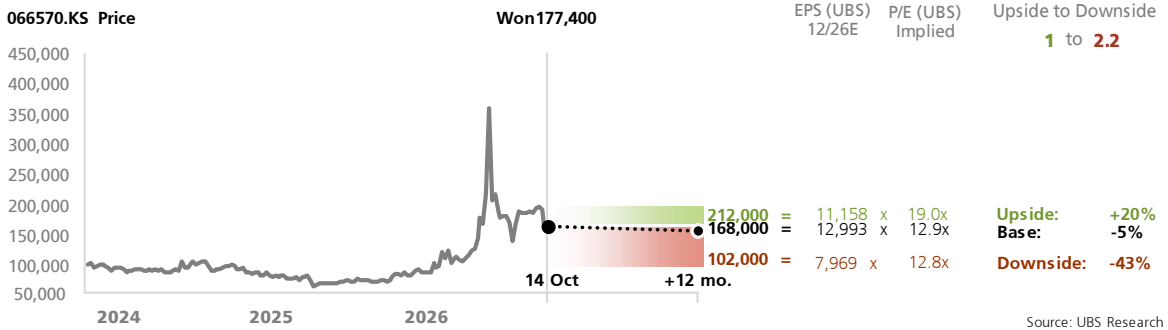


Source: Company data, UBS estimates

LG Electronics –Key charts

UBS forecasts vs Consensus

	4Q26E			1Q27E			2026E			2027E			2028E		
(Won bn)	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**
Sales	25,312	25,212	0.4%	25,404	25,497	-0.4%	97,557	96,388	1.2%	102,857	101,336	1.5%	108,195	103,124	4.9%
Op Profit	**349**	**406**	**-14.2%**	**1,789**	**1,782**	**0.4%**	**4,497**	**4,601**	**-2.3%**	**4,652**	**4,989**	**-6.8%**	**5,032**	**16,810**	**-70.1%**
% OP margin	1.4%	1.6%	-23bps	7.0%	7.0%	5bps	4.6%	4.8%	-16bps	4.5%	4.9%	-40bps	4.7%	16.3%	-1165bps
EPS (Won)	958	1,050	-8.8%	6,441	6,003	7.3%	12,993	13,640	-4.7%	17,484	16,822	3.9%	19,347	18,506	4.5%



Value drivers (2026/27E)	HS Division OPM (%)	MS Division OPM (%)	VS Rev Growth YoY (%)	ES Division OPM (%)	Operating Profit
Won212k upside	5.8%/5.1%	2.5%/2.2%	12%/11%	6.6%/7.6%	Won4.9tn/5.2tn
Won168k base	5.4%/3.8%	2.0%/1.8%	10%/7%	5.2%/6.9%	Won4.5tn/4.7tn
Won102k downside	4.5%/2.7%	BEP/-0.5%	7%/5%	4.0%/4.7%	Won4.1tn/4.0tn

56

Source: Bloomberg, Company data, UBS estimates

LG Electronics – Divisional Forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues														
Home Appliance Solution	26,126	6,943	7,076	7,173	6,523	27,715	7,360	7,288	7,460	6,784	28,891	30,199	31,417	32,838
Media Entertainment Solution	19,426	5,169	5,115	5,255	5,443	20,982	4,995	5,308	5,735	6,150	22,189	22,913	23,811	24,382
Vehicle Solution	11,136	3,064	3,026	3,189	2,964	12,244	3,279	3,207	3,444	3,142	13,073	13,921	14,579	15,088
Eco Solution	9,323	2,822	2,726	2,286	1,530	9,365	2,992	2,944	2,332	1,561	9,829	10,198	10,675	11,045
Others	1,840	304	465	527	360	1,656	253	473	475	301	1,501	1,369	1,257	1,163
LG Innotek	21,897	5,535	5,527	6,415	8,635	26,112	6,659	6,008	6,321	8,936	27,925	30,191	31,637	31,637
Internal transactions	547	111	109	153	144	516	133	118	151	149	550	595	623	623
Total	**89,201**	**23,727**	**23,827**	**24,692**	**25,312**	**97,557**	**25,404**	**25,111**	**25,617**	**26,725**	**102,857**	**108,195**	**112,752**	**115,529**
% YoY	*2%*	*4%*	*15%*	*13%*	*6%*	*9%*	*7%*	*5%*	*4%*	*6%*	*5%*	*5%*	*4%*	*2%*
% QoQ		*-1%*	*0%*	*4%*	*3%*		*0%*	*-1%*	*2%*	*4%*				
Operating profit														
Home Appliance Solution	1,279	570	686	316	-65	1,506	552	401	336	-204	1,085	1,094	1,133	1,155
Media Entertainment Solution	-751	372	219	-82	-92	417	329	141	-53	-11	406	310	233	12
Vehicle Solution	559	212	191	255	207	865	230	241	224	236	930	1,011	973	1,019
Eco Solution	647	249	236	114	-107	492	329	265	163	-78	679	752	838	857
Others	58	-26	-5	11	-15	-36	0	28	19	-12	36	36	37	37
LG Innotek	665	295	246	280	417	1,238	346	292	351	512	1,500	1,808	2,136	1,201
Internal transactions	-21	-3	-6	-1	-4	-14	-3	-7	-2	-5	-17	-21	-24	-11
Total	**2,478**	**1,674**	**1,579**	**895**	**349**	**4,497**	**1,789**	**1,375**	**1,041**	**447**	**4,652**	**5,032**	**5,374**	**4,292**
% OpM														
Home Appliance Solution	4.9%	8.2%	9.7%	4.4%	-1.0%	5.4%	7.5%	5.5%	4.5%	-3.0%	3.8%	3.6%	3.6%	3.5%
Media Entertainment Solution	-3.9%	7.2%	4.3%	-1.6%	-1.7%	2.0%	6.6%	2.7%	-0.9%	-0.2%	1.8%	1.4%	1.0%	0.1%
Vehicle Solution	5.0%	6.9%	6.3%	8.0%	7.0%	7.1%	7.0%	7.5%	6.5%	7.5%	7.1%	7.3%	6.7%	6.8%
Eco Solution	6.9%	8.8%	8.6%	5.0%	-7.0%	5.2%	11.0%	9.0%	7.0%	-5.0%	6.9%	7.4%	7.9%	7.8%
Others	3.2%	-8.5%	-1.1%	2.0%	-4.2%	-2.2%	0.1%	6.0%	4.0%	-4.0%	2.4%	2.6%	2.9%	3.2%
LG Innotek	3.0%	5.3%	4.4%	4.4%	4.8%	4.7%	5.2%	4.9%	5.6%	5.7%	5.4%	6.0%	6.8%	3.8%
Total	**2.8%**	**7.1%**	**6.6%**	**3.6%**	**1.4%**	**4.6%**	**7.0%**	**5.5%**	**4.1%**	**1.7%**	**4.5%**	**4.7%**	**4.8%**	**3.7%**

Source: Company data, UBS estimates

LG Electronics – Financial Statement Forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues	**89,201**	**23,727**	**23,827**	**24,692**	**25,312**	**97,557**	**25,404**	**25,111**	**25,617**	**26,725**	**102,857**	**108,195**	**112,752**	**115,529**
% YoY	*2%*	*4%*	*15%*	*13%*	*6%*	*9%*	*7%*	*5%*	*4%*	*6%*	*5%*	*5%*	*4%*	*2%*
% QoQ	*0%*	*-1%*	*0%*	*4%*	*3%*	*0%*	*0%*	*-1%*	*2%*	*4%*	*0%*	*0%*	*0%*	*0%*
Cost of goods sold	68,359	17,564	17,222	18,657	19,722	73,165	18,809	18,439	19,244	20,744	77,235	81,106	84,388	87,686
Gross profit	20,842	6,163	6,605	6,035	5,590	24,393	6,595	6,672	6,373	5,981	25,622	27,090	28,363	27,843
% GM	*23.4%*	*26.0%*	*27.7%*	*24.4%*	*22.1%*	*25.0%*	*26.0%*	*26.6%*	*24.9%*	*22.4%*	*24.9%*	*25.0%*	*25.2%*	*24.1%*
SG&A	18,363	4,489	5,026	5,139	5,241	19,896	4,807	5,297	5,332	5,534	20,969	22,058	22,990	23,552
Operating profit	**2,478**	**1,674**	**1,579**	**895**	**349**	**4,497**	**1,789**	**1,375**	**1,041**	**447**	**4,652**	**5,032**	**5,374**	**4,292**
% OpM	*2.8%*	*7.1%*	*6.6%*	*3.6%*	*1.4%*	*4.6%*	*7.0%*	*5.5%*	*4.1%*	*1.7%*	*4.5%*	*4.7%*	*4.8%*	*3.7%*
Non-operating profit	-642	-273	-340	79	94	-440	111	-30	100	146	327	408	570	739
Pretax profit	1,836	1,401	1,239	974	443	4,057	1,900	1,346	1,141	593	4,980	5,440	5,944	5,031
Tax expenses	-608	-396	-458	-268	-122	-1,244	-522	-370	-314	-163	-1,369	-1,496	-1,635	-1,383
Net profit	**961**	**816**	**668**	**630**	**219**	**2,333**	**1,188**	**862**	**751**	**328**	**3,129**	**3,463**	**3,828**	**3,166**
EPS reported (Won)	**5,321**	**4,541**	**3,708**	**3,521**	**1,223**	**12,993**	**6,637**	**4,818**	**4,198**	**1,831**	**17,484**	**19,347**	**21,390**	**17,690**
Depreciation & amortization	3,602	868	871	885	881	3,505	912	874	879	879	3,543	3,541	3,540	3,559
Cash flow from operations	**4,281**	**1,101**	**2,452**	**1,569**	**5,466**	**10,588**	**-2,079**	**1,261**	**1,903**	**5,566**	**6,650**	**6,980**	**7,417**	**6,893**
Capex	-2,615	-690	-553	-720	-720	-2,682	-704	-704	-704	-704	-2,816	-2,816	-2,816	-2,816
Cash flow from investments	**-3,014**	**-1,172**	**-761**	**-973**	**-973**	**-3,879**	**-957**	**-957**	**-957**	**-957**	**-3,830**	**-3,830**	**-3,830**	**-3,830**
Cash flow from financing activities	**-319**	**-326**	**-431**	**-225**	**-225**	**-1,207**	**-225**	**-405**	**-225**	**-225**	**-1,080**	**-1,081**	**-1,083**	**-1,085**
Total cash flow	**1,197**	**-138**	**1,438**	**371**	**4,268**	**5,939**	**-3,262**	**-101**	**720**	**4,383**	**1,741**	**2,069**	**2,504**	**1,979**
Cash and cash equivalents	8,770	8,632	10,070	10,441	14,708	14,708	11,447	11,346	12,066	16,449	16,449	18,518	21,022	23,001
Total assets	**68,620**	**71,235**	**73,696**	**75,618**	**79,025**	**79,025**	**77,100**	**77,512**	**79,413**	**83,333**	**83,333**	**87,976**	**92,857**	**96,866**
Total liabilities	**40,068**	**40,705**	**41,779**	**42,994**	**46,080**	**46,080**	**42,777**	**42,394**	**43,467**	**46,957**	**46,957**	**47,838**	**48,592**	**49,139**
Shareholders' equity	**28,552**	**30,529**	**31,918**	**32,624**	**32,945**	**32,945**	**34,322**	**35,118**	**35,946**	**36,376**	**36,376**	**40,138**	**44,265**	**47,727**

Source: Company data, UBS estimates

LG Innotek (Jimmy Yoon) – Buy, PT Won940k

- **Our View**
  - On the back of upside to near-term smartphone builds for LGI, as the major US OEM customer's build plan is expected to remain stable to up YoY into CY26, we see LGI as potentially better sheltered from the impact of rising memory costs impacting end demand (which we expect to be more centered on Android OEMs). ASP uplift from upgrades to camera module specs into 2H26 (variable aperture) should also be able to help offset potential incremental pricing pressure. We also expect LGI's substrate OP to grow 123%/105% YoY in CY26/27E on BT substrate recovery from better SP builds, but also ramping ABF revenues from new project/customer wins and scale up to >60% OP contribution by 2030E (from c.18% in 2025)

Will LGI be able to scale Optics revenue and improve profitability in 2026E?

  - Yes, we forecast LGI should be able to grow Optics revenue 12% YoY in 2026 on the back of 1/ smartphone builds for the major US OEM expected to remain stable to up YoY and 2/ ASP increases from camera spec upgrades for 2H26 refresh (UBSe featuring variable aperture) expected to help offset any potential incremental ASP pressure. We note that higher mix of Pro/ProMax models for which LGI has higher M/S, should potentially be positive for LGI's Optics margins as well.

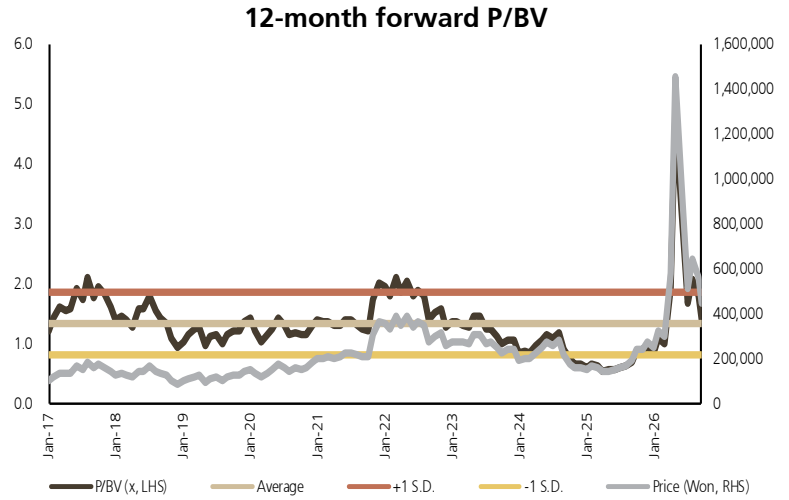
- **Can LGI scale up Substrate revenues and secure OPM above 20% mid- to long-term?**
  - Yes, we believe LGI's substrate margins should expand to c. 14%/20%/22% in 2026E/2027E/2028E (was 8% in 2025). This should be supported by continued top-line growth for BT substrates, driven by better-than-expected builds from the US smartphone OEM as well as strong YoY ramp in ABF substrate revenues, off a low base of Won50bn in 2025, to Won1.2tn in 2028E. This should be supported by volume ramp for PC CPU-related ABF applications before, over time, expanding into higher margin AI/server CPU related products.

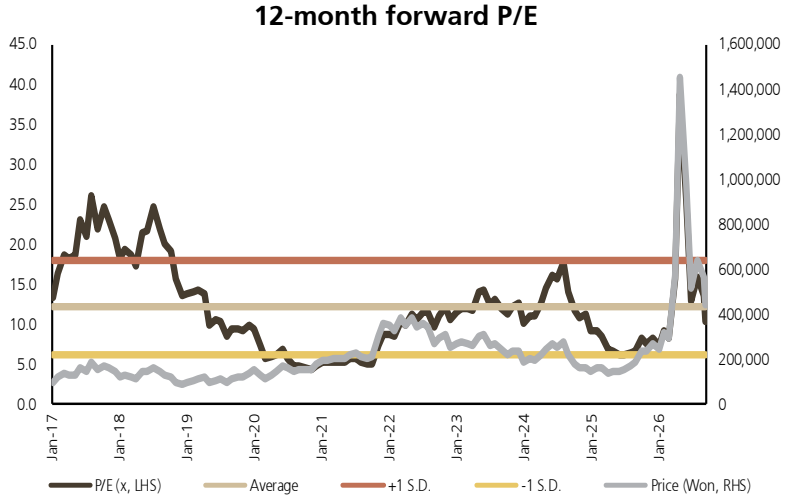
- **PT Won940k –** We value LGI shares at 2.9x 27E P/BV based on our LT ROE estimate of 16.8% and CoE of 8.2% and LTG of 4.0%

LG Innotek (Jimmy Yoon) – Buy, PT Won940k

UBS price target	**940,000**
Current stock price (Won)	519,000
Number of shares (Mn)	23.7
Market cap. (Won bn)	12,283
Net debt (Won bn)	831
Adjusted EV (Won bn)	13,114

	2025	2026E	2027E	2028E	2029E	2030E
Current Valuation						
EV/Revenues	0.60	0.52	0.44	0.37	0.31	0.25
P/Book	2.25	1.82	1.60	1.39	1.21	1.04
P/E	35.99	16.05	11.66	9.51	7.90	6.62
EV/EBITDA	6.88	5.38	4.31	3.39	2.50	1.85
UBS valuation (x)						
EV/Revenues	1.05	0.94	0.89	0.82	0.79	0.73
P/Book	4.07	3.30	2.90	2.52	2.18	1.89
P/E	65.19	29.07	21.12	17.23	14.32	11.99
EV/EBITDA	12.71	10.32	9.16	7.90	6.74	5.83
Other data						
EPS	14,419	32,334	44,506	54,558	65,665	78,367
% ROE	***5.9%***	***12.2%***	***14.6%***	***15.6%***	***16.3%***	***16.9%***
BVPS	230,982	284,659	324,664	373,223	430,388	497,755
Cash per share (Won)	56,255	66,214	90,140	95,512	117,136	131,710
FCF Per Share (Won)	30,439	1,674	-216	639	36,493	67,196
% FCF yield	*5.9%*	*0.3%*	*0.0%*	*0.1%*	*7.0%*	*12.9%*



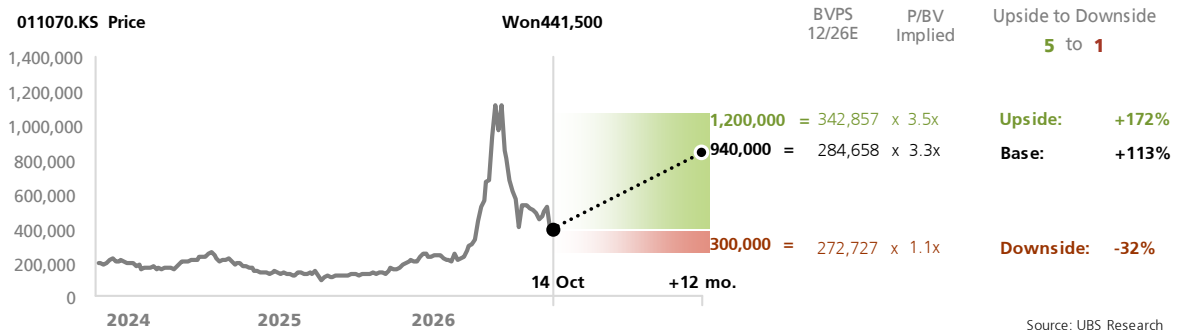


Source: Company data, UBS estimates

LG Innotek –Key charts

UBS forecasts vs Consensus

	3Q26E			4Q26E			2026E			2027E			2028E		
(Won bn)	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**
Sales	5,754	6,135	-6.2%	7,741	8,065	-4.0%	24,557	25,430	-3.4%	25,962	26,877	-3.4%	28,239	28,464	-0.8%
Op Profit	**170**	**235**	**-27.6%**	**355**	**403**	**-12.1%**	**1,066**	**1,255**	**-15.1%**	**1,365**	**1,456**	**-6.2%**	**1,683**	**1,726**	**-2.5%**
% OP margin	3.0%	3.8%	-87bps	4.6%	5.0%	-42bps	4.3%	4.9%	-60bps	5.3%	5.4%	-16bps	6.0%	6.1%	-11bps
EPS (Won)	4,627	7,586	-39.0%	10,857	12,195	-11.0%	32,334	37,987	-14.9%	44,506	45,409	-2.0%	54,558	53,562	1.9%



Value drivers (2026/27E)	iPhone Unit Builds	Optics Solution OPM	Package Solution Rev CAGR '25-30E	Package Solution OPM
Won1,200,000 upside	260m/267m	4.0%/4.4%	38%	14%/21%
Won940,000 base	254m/258m	3.7%/3.9%	32%	12%/18%
Won300,000 downside	245m/235m	3.2%/2.3%	23%	10%/14%

61

Source: Bloomberg, Company data, UBS estimates

LG Innotek – Divisional Forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues														
Optics Solution	18,319	4,611	4,518	4,722	6,652	20,503	5,182	4,513	4,790	6,686	21,171	22,172	21,554	22,068
Substrates & Materials	1,720	437	498	518	556	2,010	594	654	675	730	2,653	3,783	5,277	6,944
Automotive & Electronics	1,858	487	511	514	533	2,044	515	527	535	560	2,138	2,284	2,403	2,489
Total	**21,897**	**5,535**	**5,527**	**5,754**	**7,741**	**24,557**	**6,292**	**5,694**	**6,000**	**7,975**	**25,962**	**28,239**	**29,235**	**31,501**
% YoY	*3%*	*11%*	*40%*	*7%*	*2%*	*12%*	*14%*	*3%*	*4%*	*3%*	*6%*	*9%*	*4%*	*8%*
% QoQ		*-27%*	*0%*	*4%*	*35%*		*-19%*	*-10%*	*5%*	*33%*				
Mix %														
Optics Solution	*84%*	*83%*	*82%*	*82%*	*86%*	*83%*	*82%*	*79%*	*80%*	*84%*	*82%*	*79%*	*74%*	*70%*
Substrates & Materials	*8%*	*8%*	*9%*	*9%*	*7%*	*8%*	*9%*	*11%*	*11%*	*9%*	*10%*	*13%*	*18%*	*22%*
Automotive & Electronics	*8%*	*9%*	*9%*	*9%*	*7%*	*8%*	*8%*	*9%*	*9%*	*7%*	*8%*	*8%*	*8%*	*8%*
Operating profit														
Optics Solution	497	235	174	96	261	766	204	145	180	288	818	824	805	800
Substrates & Materials	121	40	53	64	83	239	97	112	132	136	478	781	1,132	1,514
Automotive & Electronics	47	21	19	10	11	60	15	15	21	18	70	77	77	79
Total	**665**	**295**	**246**	**170**	**355**	**1,066**	**316**	**273**	**333**	**442**	**1,365**	**1,683**	**2,014**	**2,393**
Mix %														
Optics Solution	*75%*	*79%*	*71%*	*56%*	*74%*	*72%*	*65%*	*53%*	*54%*	*65%*	*60%*	*49%*	*40%*	*33%*
Substrates & Materials	*18%*	*14%*	*21%*	*38%*	*23%*	*22%*	*31%*	*41%*	*40%*	*31%*	*35%*	*46%*	*56%*	*63%*
Automotive & Electronics	*7%*	*7%*	*8%*	*6%*	*3%*	*6%*	*5%*	*6%*	*6%*	*4%*	*5%*	*5%*	*4%*	*3%*
% OpM														
Optics Solution	*2.7%*	*5.1%*	*3.9%*	*2.0%*	*3.9%*	*3.7%*	*3.9%*	*3.2%*	*3.8%*	*4.3%*	*3.9%*	*3.7%*	*3.7%*	*3.6%*
Substrates & Materials	*7.0%*	*9.2%*	*10.5%*	*12.4%*	*14.8%*	*11.9%*	*16.3%*	*17.2%*	*19.6%*	*18.7%*	*18.0%*	*20.7%*	*21.4%*	*21.8%*
Automotive & Electronics	*2.5%*	*4.2%*	*3.7%*	*2.0%*	*2.0%*	*2.9%*	*3.0%*	*2.9%*	*3.9%*	*3.2%*	*3.3%*	*3.4%*	*3.2%*	*3.2%*
Total	***3.0%***	***5.3%***	***4.4%***	***3.0%***	***4.6%***	***4.3%***	***5.0%***	***4.8%***	***5.6%***	***5.5%***	***5.3%***	***6.0%***	***6.9%***	***7.6%***

Source: Company data, UBS estimates

LG Innotek – Financial Statement Forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues	**21,897**	**5,535**	**5,527**	**5,754**	**7,741**	**24,557**	**6,292**	**5,694**	**6,000**	**7,975**	**25,962**	**28,239**	**29,235**	**31,501**
% YoY	*3%*	*11%*	*40%*	*7%*	*2%*	*12%*	*14%*	*3%*	*4%*	*3%*	*6%*	*9%*	*4%*	*8%*
% QoQ		*-27%*	*0%*	*4%*	*35%*		*-19%*	*-10%*	*5%*	*33%*				
Gross profit	1,750	557	531	459	691	2,238	614	567	635	788	2,604	3,030	3,407	3,893
% GM	*8.0%*	*10.1%*	*9.6%*	*8.0%*	*8.9%*	*9.1%*	*9.8%*	*10.0%*	*10.6%*	*9.9%*	*10.0%*	*10.7%*	*11.7%*	*12.4%*
Operating profit	**665**	**295**	**246**	**170**	**355**	**1,066**	**316**	**273**	**333**	**442**	**1,365**	**1,683**	**2,014**	**2,393**
% OpM	*3.0%*	*5.3%*	*4.4%*	*3.0%*	*4.6%*	*4.3%*	*5.0%*	*4.8%*	*5.6%*	*5.5%*	*5.3%*	*6.0%*	*6.9%*	*7.6%*
Net profit post MI	341	229	170	109	257	765	254	209	252	338	1,053	1,291	1,554	1,855
EPS reported (Won)	**14,419**	**9,682**	**7,169**	**4,627**	**10,857**	**34,593**	**10,726**	**8,813**	**10,665**	**14,302**	**44,506**	**54,558**	**65,665**	**78,367**
Cash and cash equivalents	1,406	1,373	1,240	1,780	1,749	1,749	2,028	2,157	2,944	2,922	2,922	3,881	5,228	6,443
Total assets	**11,931**	**11,822**	**11,923**	**13,045**	**13,336**	**13,336**	**14,127**	**14,127**	**15,561**	**15,778**	**15,778**	**18,477**	**20,880**	**22,868**
Total liabilities	**6,168**	**5,741**	**5,552**	**6,564**	**6,599**	**6,599**	**7,136**	**7,034**	**8,216**	**8,094**	**8,094**	**9,644**	**10,694**	**11,087**
Shareholders' equity	**5,763**	**6,081**	**6,371**	**6,480**	**6,737**	**6,737**	**6,991**	**7,093**	**7,345**	**7,684**	**7,684**	**8,833**	**10,186**	**11,780**
Depreciation & amortization	1,150	240	248	342	341	1,170	288	287	290	289	1,153	1,238	1,410	1,565
Cash flow from operations	**1,331**	**300**	**98**	**970**	**199**	**1,567**	**443**	**398**	**1,050**	**242**	**2,133**	**2,260**	**2,772**	**3,117**
Capex	-611	-189	-135	-602	-602	-1,527	-535	-535	-535	-535	-2,138	-2,245	-1,909	-1,527
Cash flow from investments	**-796**	**-217**	**-167**	**-632**	**-632**	**-1,648**	**-565**	**-565**	**-565**	**-565**	**-2,260**	**-2,367**	**-2,030**	**-1,648**
Dividends paid	-49	0	-44	0	0	-44	0	-107	0	0	-107	-142	-201	-260
Cash flow from financing activities	**-453**	**-132**	**-72**	**202**	**402**	**400**	**402**	**295**	**302**	**302**	**1,300**	**1,065**	**605**	**-254**
Total cash flow	**77**	**-34**	**-132**	**540**	**-31**	**342**	**279**	**129**	**787**	**-21**	**1,174**	**958**	**1,348**	**1,215**

Source: Company data, UBS estimates

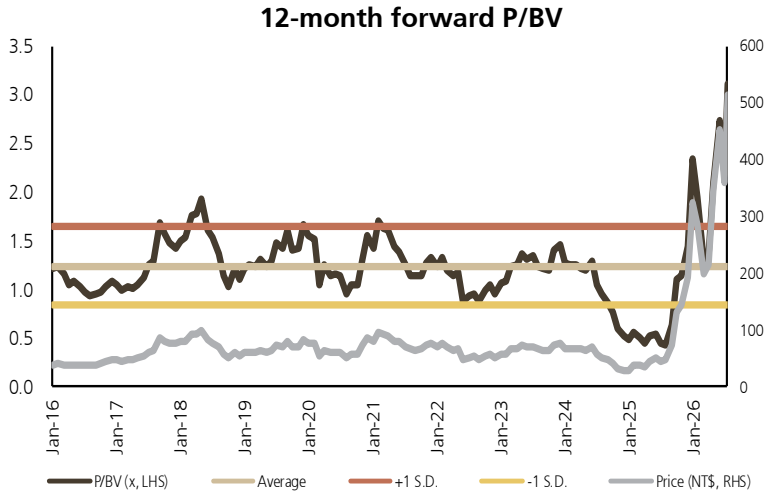
Nanya Tech (Nicolas Gaudois) – Buy, NT$640

- **UBS View**
  - We believe the strength of the unfolding DRAM upcycle as well as the tail of legacy products that Nanya can address as a niche player should provide support for the stock from here. In the near term, we note significant upside to consensus in 3Q26, as NTC benefits from pricing re-set post some of its 6m to 12m contracts expiring. Worth noting that its ASPs are moving up higher than global peers, albeit we believe temporarily so.
- **Will DRAM undersupply continue into 2028?**
  - Yes. Agentic AI is boosting further memory demand on multiple fronts beyond HBM, including DDR5/LPDDR5 for conventional servers and CPU head nodes in AI servers. We now forecast an acceleration of memory demand growth into 2027 with DRAM bit end-consumption growing 39% YoY vs 21% in 2026. As almost all of the incremental wafer capacity addition for DRAM is going to HBM, supply is not catching up with demand. The main risk remains affordability by customers as memory industry revenues are expected to reach US$1.71tn in 2027E.
- **Can NTC grow as a niche DRAM player?**
  - Yes. 2027 are likely to be years of more constrained growth. Subsequently though, we think its process tech migration to 1c nm and 1d nm will likely lead to a 30% increase in bits per wafer for each process tech migration (at comparable yields)
  - NTC is also pulling in its new fab ramp with equipment expected to move in 1Q27 and output mid-2027.
  - In the meantime, we also see a tail of DDR3/DDR4 products NTC can address for a while.
- **PT NT$640** – We value Nanya Tech shares at 2.63x NTM P/BV based on our LT ROE of 21.4% and CoE of 8.1%

Nanya Tech (Nicolas Gaudois) – Valuation

UBS price target (NT$)	**640.00**
Current stock price (NT$)	527.00
Number of shares (mil)	3,450
Market cap (NT$m)	1,818,254
Net debt (NT$m)	(284,510)
EV (NT$m)	1,533,745

	2025	2026E	2027E	2028E	2029E	2030E
Current valuation (x)						
EV/Revenues	23.03	3.80	2.34	2.09	3.69	3.66
P/Book	9.58	3.58	2.17	1.53	1.46	1.35
P/E	246.91	6.92	4.11	3.92	10.78	12.92
EV/EBITDA	78.56	4.74	2.75	2.58	6.31	6.80
Target price valuation (x)						
EV/Revenues	28.89	4.77	2.94	2.62	4.63	4.59
P/Book	11.63	4.35	2.63	1.85	1.77	1.64
P/E	299.85	8.41	4.99	4.76	13.09	4.76
EV/EBITDA	98.53	5.95	3.45	3.23	7.91	8.53
Other						
EPS	2.13	76.14	128.16	134.53	48.90	40.79
% ROE	*3.9%*	*77.2%*	*66.0%*	*45.7%*	*13.8%*	*10.9%*
Book value per share (NT$)	55.04	147.22	243.05	345.54	360.81	389.38
Cash per share (NT$)	18.74	100.82	144.23	216.04	235.05	238.63
Free cash flow per share (NT$)	1.44	61.79	75.15	103.86	52.64	15.80
% Free cash flow yield	*0.3%*	*11.7%*	*14.3%*	*19.7%*	*10.0%*	*3.0%*

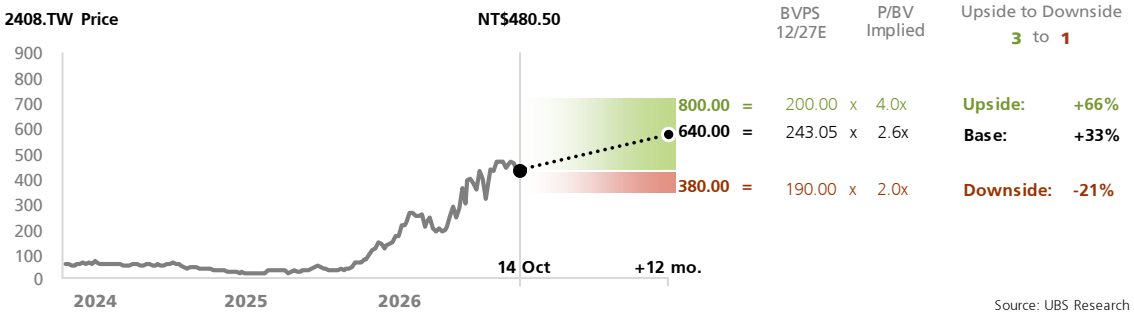


Source: Company data, UBS estimates

Nanya Tech – UBSe vs. Consensus/ Key charts

UBS forecasts vs Consensus

	4Q26E			1Q27E			2026E			2027E			2028E		
(NT$m)	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**
Sales	138,579	133,672	4%	139,875	142,479	-2%	402,817	391,819	3%	614,534	623,265	-1%	704,226	716,632	-2%
Operating Profit	112,308	107,365	5%	111,200	113,723	-2%	311,648	301,816	3%	472,847	491,832	-4%	503,881	520,481	-3%
% OP margin	*81.0%*	*80.3%*	*72bps*	*79.5%*	*79.8%*	*-32bps*	*77.4%*	*77.0%*	*34bps*	*76.9%*	*78.9%*	*-197bps*	*71.6%*	*72.6%*	*-108bps*
EPS (NT$)	**27.54**	**25.15**	**10%**	**27.49**	**26.59**	**3%**	**77.07**	**72.83**	**6%**	**116.96**	**116.75**	**0%**	**126.26**	**124.94**	**1%**



Source: UBS Research

Value drivers	DRAM ASP '26E/'27E	OPM % '26E/'27E
NT$800 upside	+390%/+65%	79%/80%
NT$640 base	+370%/+51%	77%/77%
NT$380 downside	+320%/+40%	72%/73%

Source: Company data, Visible Alpha, UBS estimates

Nanya Tech – DRAM forecasts

	2025	1Q26	2Q26	3Q26	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
DRAM revenues (NT$m)	**66,587**	**49,087**	**82,549**	**132,602**	**138,579**	**402,817**	**139,875**	**147,009**	**157,214**	**170,436**	**614,534**	**704,226**	**377,315**	**286,904**
% QoQ		*63%*	*68%*	*61%*	*5%*		*1%*	*5%*	*7%*	*8%*				
% YoY	*95%*	*583%*	*684%*	*606%*	*360%*	*505%*	*185%*	*78%*	*19%*	*23%*	*53%*	*15%*	*-46%*	*-24%*
DRAM revenues (US$m)	**2,158**	**1,579**	**2,647**	**4,144**	**4,352**	**12,722**	**4,385**	**4,609**	**4,929**	**5,343**	**19,266**	**22,078**	**11,829**	**8,995**
% QoQ		*62%*	*68%*	*57%*	*5%*		*1%*	*5%*	*7%*	*8%*				
% YoY	*97%*	*623%*	*677%*	*563%*	*347%*	*490%*	*178%*	*74%*	*19%*	*23%*	*51%*	*15%*	*-46%*	*-24%*
DRAM unit shipment (1Gb eq)	4,643	1,441	1,438	1,504	1,442	5,825	1,371	1,393	1,465	1,600	5,829	8,169	10,284	12,448
% QoQ		*-6%*	*0%*	*5%*	*-4%*		*-5%*	*2%*	*5%*	*9%*				
% YoY	*54%*	*123%*	*32%*	*10%*	*-6%*	*25%*	*-5%*	*-3%*	*-3%*	*11%*	*0%*	*40%*	*26%*	*21%*
DRAM ASP (US$ 1Gb eq)	0.46	1.10	1.84	2.75	3.02	2.18	3.20	3.31	3.36	3.34	3.31	2.70	1.15	0.72
% QoQ		*73%*	*68%*	*50%*	*10%*		*6%*	*3%*	*2%*	*-1%*				
% YoY	*28%*	*224%*	*487%*	*505%*	*376%*	*370%*	*192%*	*80%*	*22%*	*11%*	*51%*	*-18%*	*-57%*	*-37%*
Operating Profit (NT$m)	**5,242**	**30,102**	**60,828**	**108,411**	**112,308**	**311,648**	**111,200**	**115,654**	**120,453**	**125,541**	**472,847**	**503,881**	**137,565**	**14,503**
% OP margin	*7.9%*	*61.3%*	*73.7%*	*81.8%*	*81.0%*	*77.4%*	*79.5%*	*78.7%*	*76.6%*	*73.7%*	*76.9%*	*71.6%*	*36.5%*	*5.1%*

Source: Company data, UBS estimates

Nanya Tech – Financial Statement forecasts

(NT$m)	2025	1Q26	2Q26	3Q26	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues	**66,587**	**49,087**	**82,549**	**132,602**	**138,579**	**402,817**	**139,875**	**147,009**	**157,214**	**170,436**	**614,534**	**704,226**	**377,315**	**286,904**
% QoQ		*63%*	*68%*	*61%*	*5%*		*1%*	*5%*	*7%*	*8%*				
% YoY	*95%*	*583%*	*684%*	*606%*	*360%*	*505%*	*185%*	*78%*	*19%*	*23%*	*53%*	*15%*	*-46%*	*-24%*
COGS	51,600	15,761	16,932	18,431	20,062	71,186	21,818	23,540	27,990	35,166	108,514	157,603	189,353	214,348
Gross Profit	**14,986**	**33,325**	**65,617**	**114,170**	**118,517**	**331,631**	**118,057**	**123,469**	**129,225**	**135,269**	**506,020**	**546,623**	**187,962**	**72,556**
% GP margin	*22.5%*	*67.9%*	*79.5%*	*86.1%*	*85.5%*	*82.3%*	*84.4%*	*84.0%*	*82.2%*	*79.4%*	*82.3%*	*77.6%*	*49.8%*	*25.3%*
Operating Profit	**5,242**	**30,102**	**60,828**	**108,411**	**112,308**	**311,648**	**111,200**	**115,654**	**120,453**	**125,541**	**472,847**	**503,881**	**137,565**	**14,503**
% OP margin	*7.9%*	*61.3%*	*73.7%*	*81.8%*	*81.0%*	*77.4%*	*79.5%*	*78.7%*	*76.6%*	*73.7%*	*76.9%*	*71.6%*	*36.5%*	*5.1%*
Net Profit After MI	6,614	26,059	50,192	91,279	95,031	262,561	94,859	99,025	102,435	107,214	403,533	435,611	143,167	41,888
EPS	**2.13**	**8.41**	**14.66**	**26.46**	**27.54**	**77.07**	**27.49**	**28.70**	**29.69**	**31.07**	**116.96**	**126.26**	**41.50**	**12.14**
Cash and Cash Equivalents	58,074	86,281	217,603	271,869	343,750	343,750	393,892	342,596	401,854	466,381	466,381	685,957	752,234	675,265
Total Assets	**208,453**	**240,968**	**387,758**	**477,410**	**573,013**	**573,013**	**668,009**	**661,235**	**765,133**	**874,707**	**874,707**	**1,211,464**	**1,249,251**	**1,257,361**
Total Liabilities	**37,914**	**47,931**	**65,060**	**63,433**	**64,005**	**64,005**	**64,142**	**64,708**	**66,171**	**68,531**	**68,531**	**70,560**	**74,084**	**76,097**
Total Shareholders' Equity	**170,538**	**193,037**	**322,699**	**413,978**	**509,008**	**509,008**	**603,867**	**596,527**	**698,961**	**806,176**	**806,176**	**1,140,904**	**1,175,168**	**1,181,264**
Depreciation and Amortization	14,281	2,897	2,960	3,513	3,496	12,865	4,785	7,974	11,164	14,038	37,961	59,021	77,073	95,124
Cash Flow from Operations	**18,568**	**31,549**	**55,859**	**77,463**	**95,078**	**259,949**	**97,987**	**102,915**	**107,103**	**112,373**	**420,379**	**487,918**	**270,871**	**118,307**
Capital Expenditures	(13,245)	(2,851)	(2,587)	(23,196)	(23,196)	(51,831)	(47,846)	(47,846)	(47,846)	(47,846)	(191,382)	(167,459)	(95,691)	(159,485)
Cash Flow from Investments	**(14,110)**	**(2,814)**	**(4,242)**	**(23,196)**	**(23,196)**	**(53,450)**	**(47,846)**	**(47,846)**	**(47,846)**	**(47,846)**	**(191,382)**	**(167,459)**	**(95,691)**	**(159,485)**
Dividend	0	0	0	0	0	0	0	(106,366)	0	0	(106,366)	(100,883)	(108,903)	(35,792)
Cash Flow from Financing	**(6,317)**	**(1,547)**	**79,415**	**0**	**0**	**77,868**	**0**	**(106,366)**	**0**	**0**	**(106,366)**	**(100,883)**	**(108,903)**	**(35,792)**
Total Cash flow	**(3,829)**	**28,207**	**131,322**	**54,266**	**71,882**	**285,676**	**50,142**	**(51,296)**	**59,257**	**64,528**	**122,631**	**219,575**	**66,277**	**(76,969)**

Source: Company data, UBS estimates

Samsung Electro-Mechanics (Jimmy Yoon) – Buy, PT Won2.7m

- **Our View**
  - We expect SEMCO to record meaningful OP uplift into 2026E/2027E (+131%/+165% YoY) on AI-led demand pick-up for both MLCC and FC-BGA.
  - For MLCC, in addition to mix shift to non-IT (in particular AI server MLCCs) supporting blended ASP/margins, we see tight S/D likely prompting like-for-like ASP hikes into 2H26. We now forecast SEMCO's blended MLCC ASP to grow +20%/+47% YoY in 2026E/2027E.
  - For Substrates, with SEMCO expected to ramp ABF capacity (Vietnam/Korea) on new customer wins (ASIC AI accelerator) into 2027E/2028E along with potential ASP hike (material costs + S/D tightness), we forecast Substrate OP to inflect +292%/+143% YoY in 2026E/2027E.
- **Can SEMCO record strong YoY MLCC OP growth and expand margins in 2026E/2027E?**
  - Yes. We forecast SEMCO's component revenues to grow 35%/67% YoY in 2026/27E supported by demand from AI servers/xEVs. We forecast SEMCO's AI server MLCC revs to grow at c. 80% CAGR 2025-30E (>8x MLCC content for AI / >100x MLCC at rack level vs conventional) on scaling voltage/power envelope.
  - For supply, we expect relatively slower headline MLCC shipment growth given longer tact times / lower yields for high-end MLCCs. Net-net, this may prompt prolonged MLCC S/D tightness starting with high-end MLCCs beyond 2H26. In addition to mix, tight S/D, we believe, could result in like-for-like ASP hikes supporting SEMCO's blended MLCC ASP to grow +20%/+47% YoY in 2026/27E. We forecast SEMCO's component business earnings to grow +113%/+197% YoY in 2026/27E with OPM at c. 19%/34% (vs 12% OPM in 2025).
- **Can SEMCO scale Substrates revenues whilst securing margins in 2026E/2027E?**
  - Yes. We believe SEMCO should be able to expand its Substrates business OP margin to 18%/29% in 2026/27E (was 7% in 2025) on the back of new customer/project wins for FC-BGA (ABF) centered on AI ASICs / networking / LPU and server CPUs expected to ramp into 2H26, pivoting away from lower margin legacy PC CPU mix.
  - ABF price hikes (material cost pass through + tight S/D) should also bode well for margin upside. In particular for FC-BGA, we expect OP to grow at a 111% CAGR in 2025-30E, with earnings contribution growing from c. 17% in 2025 to c. 34% by 2030E.
- **PT Won2.7m –** We value SEMCO based on 48.9x 2027E EPS given our revised 2027-30E EPS CAGR of 37%

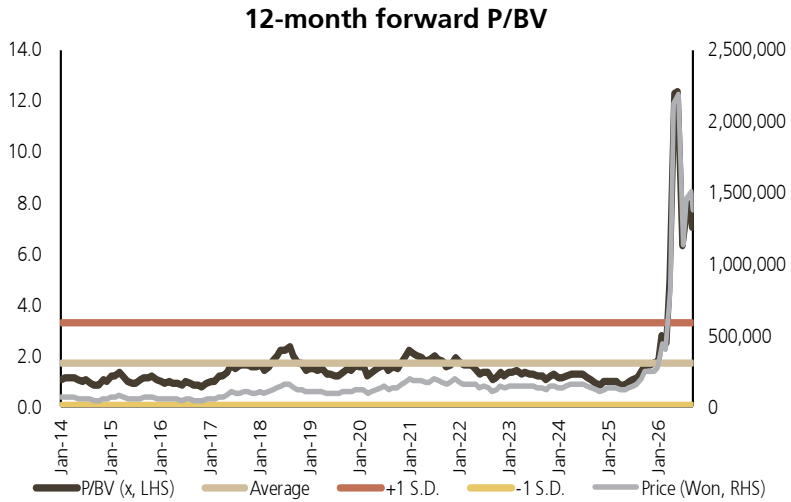
Source: Company data, UBS estimates

Samsung Electro-Mechanics (Jimmy Yoon) – Valuation

UBS price target	**2,700,000**
Current stock price (Won)	1,421,000
Number of shares (Mn)	77.6
Treasury shares	2.1
Market cap. (Won bn)	110,271
Net debt (Won bn)	-305
Adjusted EV (Won bn)	109,965

	2025	2026E	2027E	2028E	2029E	2030E
Current Valuation						
EV/Revenues	9.72	7.60	5.29	3.98	3.17	2.52
P/Book	11.56	9.26	6.93	4.94	3.57	2.66
P/E	156.17	67.08	25.73	16.19	12.22	9.94
EV/EBITDA	59.83	34.79	15.79	10.39	7.99	6.60
UBS valuation (x)						
EV/Revenues	18.49	14.46	10.06	7.57	6.03	4.80
P/Book	21.96	17.60	13.17	9.39	6.78	5.06
P/E	296.73	127.46	48.89	30.75	23.21	18.88
EV/EBITDA	113.83	66.20	30.05	19.77	15.20	12.56
Other data						
EPS	9,099	21,183	55,230	87,793	116,319	142,986
% ROE	***7.4%***	***15.4%***	***31.7%***	***36.4%***	***34.5%***	***31.1%***
Tangible book per share (Won)	122,960	153,411	204,957	287,646	398,199	534,098
Cash per share (Won)	26,131	26,131	26,131	26,131	26,131	26,131
FCF Per Share (Won)	3,840	-27,300	-20,740	12,027	60,572	95,977
% FCF yield	*0.3%*	*-1.9%*	*-1.5%*	*0.8%*	*4.3%*	*6.8%*

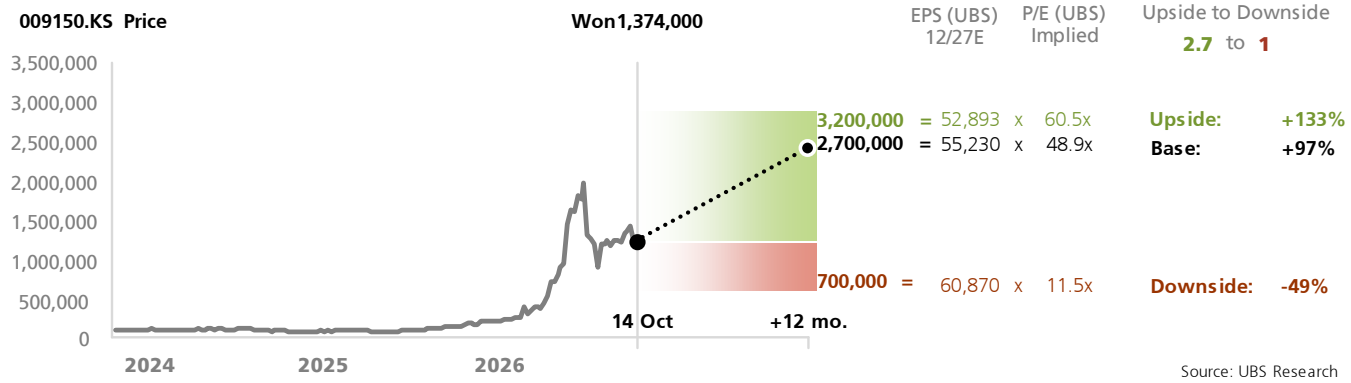
Source: Company data, UBS estimates



Samsung Electro-Mechanics – Key Charts

UBSe vs Consensus

	3Q26E			4Q26E			2026E			2027E			2028E		
(Won bn)	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**
Sales	3,816	3,751	1.7%	3,987	3,895	2.4%	14,469	14,312	1.1%	20,788	18,875	10.1%	27,622	24,431	13.1%
Op Profit	**646**	**584**	**10.6%**	**746**	**632**	**18.0%**	**2,113**	**1,905**	**10.9%**	**5,594**	**3,974**	**40.7%**	**8,793**	**6,116**	**43.8%**
% OP margin	16.9%	15.6%	136bps	18.7%	16.2%	248bps	14.6%	13.3%	129bps	26.9%	21.1%	585bps	31.8%	25.0%	680bps
EPS (Won)	6,337	6,035	5.0%	7,369	6,998	5.3%	21,180	20,168	5.0%	55,230	41,670	32.5%	87,793	64,089	37.0%



Source: UBS Research

Value drivers (2026/27E)	MLCC Blended ASP (% YoY)	Component (% OPM)	Substrate (% OPM)	Operating Profit
Won3,200,000 upside	+23%/+54%	21%/37%	20%/32%	Won2.2tn/5.9tn
Won2,700,000 base	+20%/+47%	19%/34%	18%/29%	Won2.1tn/5.6tn
Won700,000 downside	+16%/+28%	17%/25%	16%/20%	Won1.7tn/3.5tn

Source: Bloomberg, Company data, UBS estimates

Samsung Electro-Mechanics – Divisional Forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues														
Module Solution	3,814	1,076	1,036	931	968	4,011	1,077	918	1,056	933	3,984	4,143	4,141	4,211
Component Solution	5,198	1,408	1,649	1,956	2,023	7,037	2,349	2,840	3,310	3,240	11,739	16,004	19,912	25,128
Substrate Solution	2,302	725	772	929	996	3,422	1,078	1,187	1,337	1,463	5,065	7,474	10,664	14,221
Total	**11,314**	**3,209**	**3,457**	**3,816**	**3,987**	**14,469**	**4,504**	**4,946**	**5,702**	**5,636**	**20,788**	**27,622**	**34,717**	**43,561**
% YoY	*10%*	*17%*	*24%*	*32%*	*37%*	*28%*	*40%*	*43%*	*49%*	*41%*	*44%*	*33%*	*26%*	*25%*
% QoQ		*11%*	*8%*	*10%*	*4%*		*13%*	*10%*	*15%*	*-1%*				
Mix %														
Module Solution	*34%*	*34%*	*30%*	*24%*	*24%*	*28%*	*24%*	*19%*	*19%*	*17%*	*19%*	*15%*	*12%*	*10%*
Component Solution	*46%*	*44%*	*48%*	*51%*	*51%*	*49%*	*52%*	*57%*	*58%*	*57%*	*56%*	*58%*	*57%*	*58%*
Substrate Solution	*20%*	*23%*	*22%*	*24%*	*25%*	*24%*	*24%*	*24%*	*23%*	*26%*	*24%*	*27%*	*31%*	*33%*
Operating profit														
Module Solution	141	56	47	37	34	174	54	37	32	37	160	145	145	133
Component Solution	627	162	289	417	469	1,337	722	957	1,181	1,109	3,969	5,896	7,443	8,931
Substrate Solution	154	63	105	192	243	602	279	336	417	434	1,465	2,752	3,966	5,049
Total	**913**	**281**	**440**	**646**	**746**	**2,113**	**1,055**	**1,330**	**1,629**	**1,580**	**5,594**	**8,793**	**11,553**	**14,113**
% YoY	*24%*	*40%*	*107%*	*148%*	*211%*	*131%*	*276%*	*202%*	*152%*	*112%*	*165%*	*57%*	*31%*	*22%*
% QoQ		*17%*	*57%*	*47%*	*15%*		*41%*	*26%*	*22%*	*-3%*				
% OP margin														
Module Solution	*4%*	*5%*	*5%*	*4%*	*4%*	*4%*	*5%*	*4%*	*3%*	*4%*	*4%*	*4%*	*3%*	*3%*
Component Solution	*12%*	*11%*	*18%*	*21%*	*23%*	*19%*	*31%*	*34%*	*36%*	*34%*	*34%*	*37%*	*37%*	*36%*
Substrate Solution	*7%*	*9%*	*14%*	*21%*	*24%*	*18%*	*26%*	*28%*	*31%*	*30%*	*29%*	*37%*	*37%*	*36%*
Total	***8%***	***9%***	***13%***	***17%***	***19%***	***15%***	***23%***	***27%***	***29%***	***28%***	***27%***	***32%***	***33%***	***32%***

Source: Company data, UBS estimates

Samsung Electro-Mechanics – Financial Statement Forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues	**11,314**	**3,209**	**3,457**	**3,816**	**3,987**	**14,469**	**4,504**	**4,946**	**5,702**	**5,636**	**20,788**	**27,622**	**34,717**	**43,561**
Gross profit	2,277	660	845	1,096	1,244	3,845	1,588	1,908	2,300	2,286	8,081	12,098	15,706	19,322
% GM	*20.1%*	*20.6%*	*24.4%*	*28.7%*	*31.2%*	*26.6%*	*35.2%*	*38.6%*	*40.3%*	*40.5%*	*38.9%*	*43.8%*	*45.2%*	*44.4%*
Operating profit	**913**	**281**	**440**	**646**	**746**	**2,113**	**1,055**	**1,330**	**1,629**	**1,580**	**5,594**	**8,793**	**11,553**	**14,113**
% OpM	*8.1%*	*8.7%*	*12.7%*	*16.9%*	*18.7%*	*14.6%*	*23.4%*	*26.9%*	*28.6%*	*28.0%*	*26.9%*	*31.8%*	*33.3%*	*32.4%*
Net profit post MI	706	249	316	479	557	1,600	792	991	1,217	1,180	4,181	6,645	8,805	10,823
EPS reported (Won)	**9,099**	**3,211**	**4,068**	**6,337**	**7,369**	**21,183**	**10,465**	**13,088**	**16,083**	**15,594**	**55,230**	**87,793**	**116,319**	**142,986**
Cash and cash equivalents	2,701	3,243	3,314	2,955	1,796	1,796	1,498	1,382	1,380	2,030	2,030	4,114	6,922	12,311
Total assets	**14,596**	**15,660**	**16,566**	**17,423**	**18,697**	**18,697**	**20,021**	**21,468**	**23,461**	**25,355**	**25,355**	**33,889**	**41,637**	**51,396**
Total liabilities	**4,799**	**5,569**	**6,037**	**6,402**	**7,107**	**7,107**	**7,636**	**8,377**	**9,140**	**9,841**	**9,841**	**12,116**	**11,495**	**10,968**
Total shareholders' equity	**9,797**	**10,091**	**10,529**	**11,020**	**11,590**	**11,590**	**12,385**	**13,091**	**14,321**	**15,514**	**15,514**	**21,773**	**30,141**	**40,428**
Depreciation & amortization	925	241	255	270	270	1,048	339	340	345	345	1,369	1,791	2,215	2,544
Cash flow from operations	**1,490**	**479**	**581**	**369**	**562**	**1,991**	**544**	**924**	**740**	**1,492**	**3,699**	**6,443**	**9,011**	**11,027**
Capex	-1,192	-255	-588	-1,013	-2,197	-4,053	-1,317	-1,317	-1,317	-1,317	-5,269	-5,533	-4,426	-3,762
Free cash flow	**298**	**224**	**-7**	**-645**	**-1,635**	**-2,062**	**-774**	**-393**	**-578**	**174**	**-1,570**	**910**	**4,585**	**7,265**
Cash flow from investments	**-1,223**	**-263**	**-631**	**-1,016**	**-2,209**	**-4,120**	**-1,330**	**-1,330**	**-1,330**	**-1,330**	**-5,320**	**-5,583**	**-4,476**	**-3,813**
Dividends	-138	0	-138	0	0	-138	0	-298	0	0	-298	-428	-478	-578
Cash flow from financing	**418**	**310**	**116**	**288**	**488**	**1,201**	**488**	**290**	**588**	**488**	**1,854**	**1,224**	**-1,726**	**-1,826**
Net change in cash	**688**	**542**	**71**	**-359**	**-1,159**	**-905**	**-298**	**-115**	**-2**	**650**	**234**	**2,084**	**2,809**	**5,389**

Source: Company data, UBS estimates

SemiFive (Nicolas Gaudois) – Buy, PT Won38,000

- **Our View**
  - As one of the fastest-growing IC design service companies in the market, we believe SemiFive provides public investors an attractive opportunity to gain exposure to niche AI applications.
  - We forecast SemiFive's revenues to deliver a CAGR of 47% (2026E-30E) and OpM to improve materially over the next few years on the back of a strong competitive position within the Samsung Foundry ecosystem.
  - Recent disclosures, including a US$52m AI inference accelerator contract and a robotics AI chip development project, provide incremental evidence that SemiFive continues to win larger and more diversified AI-related projects despite investor concerns around order visibility earlier this year.
- **Could SemiFive sustain a 2026E-30E revenue CAGR of >40%?**
  - Yes. We believe SemiFive should be able to sustain a 2026E-30E CAGR of 47% as more projects in the design pipeline (recognized as Prototype revenues) enter mass production (recognized as Production revenues), which is a more earnings-accretive phase of a product's life cycle.
  - We forecast Production revenues to surpass Prototype in 2027E and account for 59% of total revenues by 2030E.
- **Could SemiFive's operating margin expand to mid-teens %?**
  - Yes. We expect SemiFive's operating margin to improve materially as the company gains leverage from: 1) growing scale, 2) more advanced node projects, 3) improving IP reusability and 4) efficiency gains in R&D.
  - We believe the operating margin trends of Alchip and Global Unichip, SemiFive's two largest regional peers, may be good indicators as to how SemiFive's operating margin may shape up over time.
  - All things considered, we model SemiFive's operating margin to break even in 2027E, reach 15.4% by 2030E and eventually stabilize around mid-teens % levels over the long term.
- **PT Won38k –** Our DCF-based price target for SemiFive uses a WACC of 9.9% and TGR of 3.0%. This implies 17x 2028E P/E, which we think is justified considering our 2028E-30E EPS CAGR estimate of 69%.

SemiFive (Nicolas Gaudois) – Valuation

UBS price target (Won)	**38,000**
Current stock price (Won)	**18,250**
Number of shares (mil, ex-treasury)	33.9
Treasury shares (mil)	0.0
Market cap (Won bn)	618
Net debt (Won bn)	(32)
Enterprise value (Won bn)	586

	2025	2026E	2027E	2028E	2029E	2030E
Current valuation						
EV/Revenues	4.8x	3.0x	1.6x	1.1x	0.8x	0.6x
P/BV	2.6x	2.8x	2.8x	2.4x	2.0x	1.5x
P/E	NM	NM	105.3x	16.8x	9.3x	5.9x
EV/EBITDA	NM	NM	29.2x	9.3x	5.6x	3.7x
Price target valuation						
EV/Revenues	10.4x	6.3x	3.3x	2.3x	1.7x	1.4x
P/BV	5.4x	5.9x	5.7x	5.0x	4.1x	3.2x
P/E	NM	NM	219.2x	34.9x	19.4x	12.2x
EV/EBITDA	NM	NM	62.6x	19.9x	12.0x	7.8x
Other						
EPS (Won)	-1,810	-438	173	1,089	1,960	3,107
% ROE	***-31.2%***	***-6.8%***	***2.6%***	***15.2%***	***23.1%***	***29.4%***
BVPS (Won)	7,099	6,462	6,636	7,675	9,305	11,822
CPS (Won)	2,345	570	800	1,810	3,179	5,135
FCFPS (Won)	-1,987	-1,487	230	1,059	1,699	2,546
% FCF yield	*-10.9%*	*-8.1%*	*1.3%*	*5.8%*	*9.3%*	*14.0%*

Source: Company data, UBS estimates

SemiFive – Revenue Forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Prototype	**70.9**	**28.0**	**26.1**	**28.5**	**26.9**	**109.4**	**30.3**	**33.2**	**39.0**	**41.8**	**144.3**	**184.1**	**226.4**	**261.0**
As % of total	*59%*	*59%*	*56%*	*57%*	*50%*	*55%*	*37%*	*37%*	*38%*	*40%*	*38%*	*33%*	*30%*	*28%*
% YoY	*7%*	*165%*	*65%*	*19%*	*30%*	*54%*	*8%*	*27%*	*37%*	*56%*	*32%*	*28%*	*23%*	*15%*
% QoQ		*36%*	*-7%*	*9%*	*-6%*		*13%*	*10%*	*17%*	*7%*				
Production	**19.6**	**3.5**	**4.5**	**8.7**	**12.2**	**28.9**	**34.8**	**39.8**	**46.4**	**44.7**	**165.7**	**277.8**	**428.3**	**546.9**
As % of total	*16%*	*7%*	*10%*	*17%*	*23%*	*15%*	*43%*	*44%*	*45%*	*43%*	*44%*	*50%*	*57%*	*59%*
% YoY	*105%*	*148%*	*-15%*	*3%*	*170%*	*47%*	*893%*	*784%*	*436%*	*266%*	*474%*	*68%*	*54%*	*28%*
% QoQ		*-22%*	*28%*	*93%*	*41%*		*185%*	*14%*	*17%*	*-4%*				
IP (Analog Bits)	**30.4**	**16.3**	**16.2**	**12.6**	**15.0**	**60.2**	**16.7**	**16.7**	**16.7**	**16.7**	**66.9**	**88.5**	**103.3**	**114.3**
As % of total	*25%*	*34%*	*35%*	*25%*	*28%*	*30%*	*20%*	*19%*	*16%*	*16%*	*18%*	*16%*	*14%*	*12%*
% YoY	*-15%*	*99%*	*141%*	*34%*	*148%*	*98%*	*2%*	*3%*	*32%*	*11%*	*11%*	*32%*	*17%*	*11%*
% QoQ		*170%*	*-1%*	*-22%*	*19%*		*11%*	*0%*	*0%*	*0%*				
Total	**121.0**	**47.9**	**46.8**	**49.8**	**54.1**	**198.5**	**81.8**	**89.7**	**102.1**	**103.3**	**376.9**	**550.4**	**758.0**	**922.2**
% YoY	*8%*	*137%*	*68%*	*19%*	*74%*	*64%*	*71%*	*92%*	*105%*	*91%*	*90%*	*46%*	*38%*	*22%*
% QoQ		*-573%*	*-2%*	*6%*	*9%*		*51%*	*10%*	*14%*	*1%*				

Source: Bloomberg, Company data, UBS estimates

SemiFive – Financial Statement Forecasts

(Won bn)	2025E	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues	**121.0**	**47.9**	**46.8**	**49.8**	**54.1**	**198.5**	**81.8**	**89.7**	**102.1**	**103.3**	**376.9**	**550.4**	**758.0**	**922.2**
Gross profit	19.2	20.3	9.6	15.9	17.8	63.7	25.4	29.6	33.7	33.1	121.7	184.6	261.5	336.8
% GM	*15.9%*	*42.3%*	*20.6%*	*32.0%*	*33.0%*	*32.1%*	*31.0%*	*33.0%*	*33.0%*	*32.0%*	*32.3%*	*33.5%*	*34.5%*	*36.5%*
Operating profit	**(53.4)**	**0.1**	**(11.0)**	**(2.7)**	**(0.8)**	**(14.5)**	**1.7**	**1.9**	**4.0**	**3.4**	**11.0**	**52.0**	**90.8**	**142.1**
% OpM	*-44.1%*	*0.2%*	*-23.6%*	*-5.5%*	*-1.5%*	*-7.3%*	*2.1%*	*2.2%*	*3.9%*	*3.3%*	*2.9%*	*9.4%*	*12.0%*	*15.4%*
Net profit before MI	**(56.2)**	**1.3**	**(12.2)**	**(2.7)**	**(1.2)**	**(14.8)**	**0.7**	**0.8**	**2.4**	**1.9**	**5.9**	**36.9**	**66.4**	**105.2**
EPS (Won)	**-1,810**	**40**	**-361**	**-80**	**-37**	**-438**	**20**	**25**	**72**	**58**	**173**	**1,089**	**1,960**	**3,107**
Cash and cash equivalents	79.4	47.4	31.7	19.4	19.3	19.3	13.2	17.3	24.7	27.1	27.1	61.3	107.6	173.8
Total assets	**309.3**	**319.7**	**348.7**	**326.1**	**329.2**	**329.2**	**348.2**	**355.9**	**369.2**	**372.3**	**372.3**	**425.3**	**492.4**	**579.2**
Total liabilities	**88.8**	**90.5**	**126.0**	**106.1**	**110.4**	**110.4**	**128.7**	**135.6**	**146.5**	**147.6**	**147.6**	**165.4**	**177.3**	**178.9**
Shareholders' equity	**220.5**	**229.2**	**222.8**	**220.1**	**218.8**	**218.8**	**219.5**	**220.3**	**222.7**	**224.7**	**224.7**	**259.9**	**315.1**	**400.3**
Depreciation & amortization	6.3	0.9	1.1	1.5	1.5	5.0	2.3	2.3	2.3	2.3	9.0	11.0	14.0	18.0
CF from operations	**(8.9)**	**(18.7)**	**(11.1)**	**(8.9)**	**3.4**	**(35.2)**	**(3.8)**	**6.3**	**9.7**	**4.6**	**16.8**	**47.9**	**72.5**	**106.2**
Capex	(2.2)	(2.5)	(5.5)	(3.5)	(3.5)	(15.0)	(2.3)	(2.3)	(2.3)	(2.3)	(9.0)	(12.0)	(15.0)	(20.0)
CF from investments	**(58.4)**	**(2.6)**	**(5.6)**	**(3.5)**	**(3.5)**	**(15.2)**	**(2.3)**	**(2.3)**	**(2.3)**	**(2.3)**	**(9.0)**	**(12.0)**	**(15.0)**	**(20.0)**
CF from financing	**129.5**	**(11.0)**	**0.9**	**0.0**	**0.0**	**(10.1)**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**(1.7)**	**(11.2)**	**(20.0)**
Total CF	**62.1**	**(32.0)**	**(15.7)**	**(12.4)**	**(0.1)**	**(60.1)**	**(6.1)**	**4.1**	**7.4**	**2.4**	**7.8**	**34.2**	**46.3**	**66.2**

Source: Bloomberg, Company data, UBS estimates

Simmtech (Jimmy Yoon) – Buy, PT Won195,000

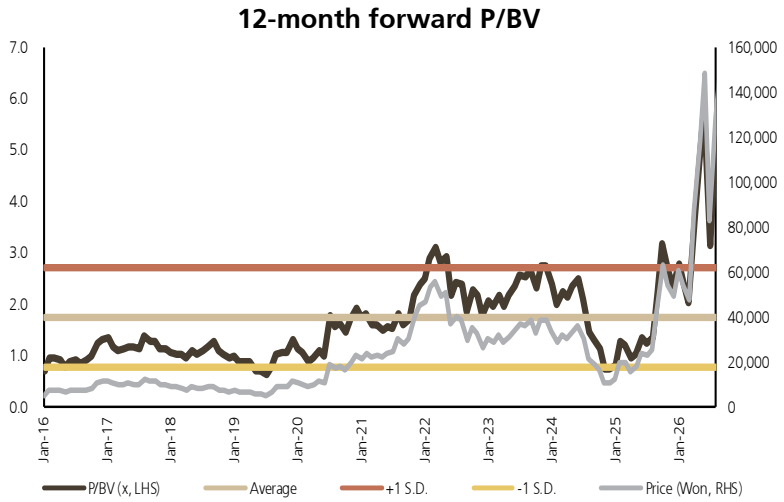
- **Our View**
  - We see Simmtech as a key beneficiary of memory-related PCB/Substrate demand pick up (UTR improvement) into 2H26/27 and in particular SOCAMM2/GDDR7 (higher margins), which should see UTR improve to c.87%/95% in 2026/27E (was 76% in 2025). Simmtech is also continuing to make progress in expanding out to higher ASP/margin non-memory verticals (SiP and FC-CSP modules). In addition to mix, Simmtech should also see potential ASP hikes on material cost pass through help lift margins to c. 12%/17% in 2026/27E (was 1% in 2025).
- **Can Simmtech benefit from growing SOCAMM2 adoption into 2H26/27E?**
  - Yes, we see Simmtech as a key beneficiary of SOCAMM2 ramp into 2H26/2027E on the back of Nvidia's Vera CPU expected to adopt 768GB of SOCAMM2. With Simmtech expected to secure c.55% of SOCAMM2 related PCB/MCP revenue TAM in 2026E, we forecast Simmtech to generate c.W224bn of related revenue in 2026E before seeing this grow to c.W453bn in 2027E. In the context of growing importance of server CPU within AI training/inference, we think wider SOCAMM or spec upgrades (density etc.) could offer incremental revenue/margin upside.
- **Can Simmtech expand the addressable market for non-memory semi packaging?**
  - Gradually yes. Simmtech currently generates c.20% of sales from non-memory semis packaging centered on FC-CSP (Solid state driver controllers and server buffer ICs) as well as SiP modules (wearables but also since 2Q25 SP RF-SiP). By leveraging cost efficiency and accumulated R&D efforts, Simmtech is aiming to further improve client/application mix - i.e. PMIC/MCU for auto as well as SoCs for other consumer products which should bode positively for margins given higher ASP.
- **Can Simmtech's OP margin recover above >10% with UTR improvement?**
  - Yes, having recorded Won12bn OP in CY25 (1% OPM), we forecast Simmtech's OP to see a sharp YoY inflection into 2026/27E to Won251bn/466bn as UTR rises from c. 76% in 2025 to c.87%/c.95% in 2026/27E led by memory-related PCB/Substrate demand pick-up. Increasing mix towards higher margin mSAP products (SiP, FC-CSP, GDDR modules etc.) as well as likely ASP increase on material cost pass-through should see margins recover to c. 12%/17% in 2026/27E.

- **PT Won195k –** We value Simmtech shares on 6.35x '27E P/BV, a c.10% discount vs global PCB/substrate peers on relatively smaller revenue scale and memory-centric exposure.

Simmtech (Jimmy Yoon) – Valuation

UBS price target	**195,000**
Current stock price (Won)	130,700
Number of shares (Mn)	37.3
Market cap. (Won bn)	4,881
Net debt (Won bn)	323
Adjusted EV (Won bn)	5,204

	2025	2026E	2027E	2028E	2029E	2030E
Current Valuation						
EV/Revenues	2.38	1.53	1.13	0.84	0.62	0.44
P/Book	8.94	6.20	4.25	3.03	2.26	1.78
P/E	-29.73	24.28	13.40	10.45	8.85	8.34
EV/EBITDA	55.03	15.11	9.02	6.80	5.32	4.54
UBS valuation (x)						
EV/Revenues	5.39	3.66	2.82	2.34	2.01	1.78
P/Book	13.33	9.24	6.35	4.52	3.38	2.66
P/E	-44.36	36.23	19.99	15.59	13.21	12.45
EV/EBITDA	77.91	22.02	13.51	10.83	9.27	8.80
Other data						
EPS	-4,396	5,382	9,753	12,505	14,767	15,665
% ROE	***-32.1%***	***29.5%***	***37.6%***	***33.9%***	***29.3%***	***23.9%***
BVPS	14,627	21,095	30,722	43,101	57,741	73,280
Cash per share (Won)	-5,319	4,066	9,179	12,244	14,938	15,563
FCF Per Share (Won)	-6,815	1,531	4,110	8,188	11,288	11,913
% FCF yield	*-5.2%*	*1.2%*	*3.1%*	*6.3%*	*8.6%*	*9.1%*

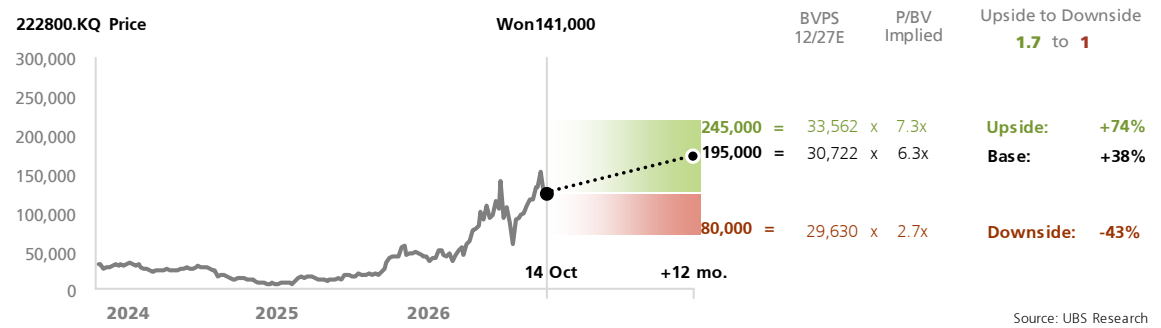


Source: Company data, UBS estimates

Simmtech – Key charts

UBS forecasts vs Consensus

	3Q26E			4Q26E			2026E			2027E			2028E		
(Won bn)	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**
Sales	555	554	0.1%	588	593	-0.9%	2,079	2,083	-0.2%	2,698	2,547	5.9%	3,257	2,951	10.4%
Op Profit	**82**	**79**	**3.2%**	**93**	**92**	**0.8%**	**251**	**250**	**0.4%**	**466**	**408**	**14.4%**	**600**	**566**	**5.9%**
% OP margin	14.7%	14.3%	44bps	15.8%	15.6%	25bps	12.1%	12.0%	7bps	17.3%	16.0%	128bps	18.4%	19.2%	-77bps
EPS (Won)	1,770	1,523	16.2%	2,028	1,862	8.9%	5,382	5,075	6.1%	9,753	8,820	10.6%	12,505	12,157	2.9%



Value drivers (2026E/27E)	Module PCB sales growth	Substrates sales growth	OP margin (%)
Won245,000 upside	62%/48%	46%/30%	14%/20%
Won195,000 base	59%/43%	43%/25%	12%/17%
Won80,000 downside	55%/35%	38%/19%	10%/12%

Source: Bloomberg, Company data, UBS estimates

Simmtech – SOCAMM revenue opportunity analysis (1)

	2026E	2027E	Notes
(1) SOCAMM Module PCB Revenue TAM			
VR200/300 + Standalone CPU Rack Assumptions			
RV200 NVL72 Server Racks	37,222	70,667	*Procurement Basis (+1.5Q Delay vs Rack Shipments)*
RV300 NVL144 Server Racks	0	22,111	*Procurement Basis (+1.5Q Delay vs Rack Shipments)*
Standalone CPU Racks	7,422	14,447	*Procurement Basis (+1.5Q Delay vs Rack Shipments)*
CPU Units			
VR200 NVL72 Rack Related	1,340,000	2,544,000	*Assuming 36x Vera CPUs Per NVL72 Rack*
VR300 NVL144 Rack Related	0	3,184,000	*Assuming 144x Vera CPUs Per NVL144 Rack*
Standalone CPU Rack Related	1,900,000	3,698,400	*Assuming 256x CPUs Per Vera CPU Rack*
Total CPU Units	**3,240,000**	**9,426,400**	
Implied SOCAMM Modules			
RV200 NVL72 Rack Related	10,720,000	20,352,000	*Assuming 8x SOCAMM2 Modules Per CPU*
RV300 NVL144 Rack Related	0	25,472,000	*Assuming 8x SOCAMM2 Modules Per CPU*
Standalone CPU Rack Related	15,200,000	29,587,200	*Assuming 8x SOCAMM2 Modules Per CPU*
Total SOCAMM Module Units (Use)	**25,920,000**	**75,411,200**	
Inventory Assumption (c. 2-3M)	6,480,000	18,852,800	
Total SOCAMM Module Units (Use + Inven	**32,400,000**	**94,264,000**	
Key Assumptions			
SOCAMM2 Per Module Area (mm2)	1,260	1,260	*Assuming 14 x 90mm*
Net Area + Yield Offset (%)	*40%*	*35%*	*Gross Area (c.15%) + HDI Production Yield Loss (c.20%)*
SOCAMM PCB Area Shipment (m2)	**57,154**	**160,343**	
ASP (US$/m2)	3,000	2,700	*Assuming c.3-4x ASP Premium vs RDIMM*
FX (USD/KRW)	1,460	1,450	
SOCAMM Module PCB TAM (US$m)	**171**	**433**	
SOCAMM Module PCB TAM (Won bn)	**250**	**628**	

Source: Company data, UBS estimates

Simmtech – SOCAMM revenue opportunity analysis (2)

(2) SOCAMM Substrate (MCP) Revenue TAM			
Implied SOCAMM Modules			
RV200 NVL72 Rack Related	10,720,000	20,352,000	*Assuming 8x SOCAMM2 Modules Per CPU*
RV300 NVL144 Rack Related	0	25,472,000	*Assuming 8x SOCAMM2 Modules Per CPU*
Standalone CPU Rack Related	15,200,000	29,587,200	*Assuming 8x SOCAMM2 Modules Per CPU*
Total SOCAMM Module Units	**25,920,000**	**75,411,200**	
Inventory Assumption (c. 2-3M)	6,480,000	15,082,240	
Total SOCAMM Module Units (Use + Inven	**32,400,000**	**90,493,440**	
Key Assumptions			
LPD5X Cubes / SOCAMM2 Module	4x	4x	*Assuming 4x LPD5X Cubes Per CPU*
LPD5X Substrate Area (mm2)	243	243	*Assuming 13.5 x 18mm*
Net Area + Yield Offset (%)	*40%*	*35%*	*Gross Area (c.15%) + mSAP Production Yield Loss (c.20%)*
LPD5X Substrate Area Shipment (m2)	**44,090**	**118,745**	
ASP (US$/m2)	2,400	2,200	*Assuming c.20% ASP Premium vs MCP Avg.*
FX (USD/KRW)	1,480	1,450	
LPDDR5X Substrate TAM (US$m)	**106**	**261**	
LPDDR5X Substrate TAM (Won bn)	**157**	**379**	

Summary (1) + (2)			
SOCAMM Module PCB Revenue TAM (Won	250	628	
LPD5X Substrate (MCP) Revenue TAM (Won	157	379	
SOCAMM Related TAM (Won bn)	**407**	**1,007**	
Simmtech Market Share (%)	*55%*	*45%*	*Guidance c. >50% M/S in 2026E*
Simmtech SOCAMM Related Revs (Won bn	**224**	**453**	

Source: Company data, UBS estimates

Simmtech – Revenue breakdown

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues														
Module PCB	347	97	128	155	172	552	179	192	207	214	792	1,044	1,261	1,421
Substrate	1,064	325	386	399	414	1,524	431	461	493	518	1,902	2,209	2,517	2,844
Others	-0	0	1	1	1	3	1	1	1	1	4	4	4	4
Total	**1,411**	**422**	**515**	**555**	**588**	**2,079**	**611**	**654**	**701**	**732**	**2,698**	**3,257**	**3,781**	**4,269**
% YoY	*15%*	*39%*	*51%*	*49%*	*49%*	*47%*	*45%*	*27%*	*26%*	*25%*	*30%*	*21%*	*16%*	*13%*
% QoQ		*7%*	*22%*	*8%*	*6%*		*4%*	*7%*	*7%*	*4%*				
Operating profit														
Module PCB	7	3	12	20	25	60	25	33	39	38	136	194	246	256
Substrate	5	11	51	61	68	191	73	81	89	88	331	406	472	506
Total	**12**	**14**	**63**	**82**	**93**	**251**	**98**	**113**	**128**	**126**	**466**	**600**	**718**	**762**
Operating margin %														
Module PCB	*2.1%*	*2.6%*	*9.5%*	*13.0%*	*14.5%*	*10.8%*	*14.0%*	*17.0%*	*19.0%*	*18.0%*	*17.1%*	*18.5%*	*19.5%*	*18.0%*
Substrate	*0.4%*	*3.4%*	*13.2%*	*15.4%*	*16.4%*	*12.6%*	*17.0%*	*17.5%*	*18.0%*	*17.0%*	*17.4%*	*18.4%*	*18.7%*	*17.8%*
Total	***0.8%***	***3.2%***	***12.2%***	***14.7%***	***15.8%***	***12.1%***	***16.1%***	***17.3%***	***18.3%***	***17.3%***	***17.3%***	***18.4%***	***19.0%***	***17.8%***

Source: Company data, UBS estimates

Simmtech – Financial Statement Forecasts

(Won bn)	2025	1Q26E	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2029E
Revenues	**1,411**	**422**	**515**	**555**	**588**	**2,079**	**611**	**654**	**701**	**732**	**2,698**	**3,257**	**3,781**	**4,269**
Gross profit	123	44	95	120	137	396	142	154	177	181	655	828	982	1,060
% GM	*8.7%*	*10.4%*	*18.5%*	*21.7%*	*23.3%*	*19.1%*	*23.2%*	*23.6%*	*25.3%*	*24.8%*	*24.3%*	*25.4%*	*26.0%*	*24.8%*
Operating profit	**12**	**14**	**63**	**82**	**93**	**251**	**98**	**113**	**128**	**126**	**466**	**600**	**718**	**762**
% OpM	*0.8%*	*3.2%*	*12.2%*	*14.7%*	*15.8%*	*12.1%*	*16.1%*	*17.3%*	*18.3%*	*17.3%*	*17.3%*	*18.4%*	*19.0%*	*17.8%*
Net profit post MI	-165	14	45	66	76	202	80	88	99	99	366	469	554	588
EPS reported (Won)	**-4,396**	**391**	**1,195**	**1,770**	**2,028**	**5,382**	**2,140**	**2,351**	**2,636**	**2,626**	**9,753**	**12,505**	**14,767**	**15,665**
Cash and cash equivalents	54	36	91	128	227	227	166	194	247	371	371	668	1,082	1,519
Total assets	**1,620**	**1,684**	**1,840**	**1,993**	**2,142**	**2,142**	**2,187**	**2,225**	**2,432**	**2,614**	**2,614**	**3,183**	**3,822**	**4,505**
Total liabilities	**1,044**	**1,086**	**1,190**	**1,277**	**1,351**	**1,351**	**1,315**	**1,268**	**1,377**	**1,461**	**1,461**	**1,566**	**1,656**	**1,755**
Total shareholders' equity	**576**	**597**	**650**	**716**	**792**	**792**	**872**	**956**	**1,055**	**1,153**	**1,153**	**1,617**	**2,167**	**2,749**
Depreciation & amortization	86	23	28	22	22	94	24	24	24	24	97	102	102	103
Cash flow from operations	**-199**	**-62**	**15**	**69**	**131**	**152**	**-12**	**80**	**102**	**173**	**344**	**459**	**560**	**583**
Capex	-56	-12	-22	-30	-30	-95	-47	-47	-47	-47	-190	-152	-137	-137
Cash flow from investments	**-19**	**-14**	**6**	**-2**	**-2**	**-12**	**-19**	**-19**	**-19**	**-19**	**-77**	**-39**	**-24**	**-24**
Dividends	-4	0	-4	-1	-1	-5	0	-4	-1	-1	-5	-5	-5	-5
Cash flow from financing	**144**	**58**	**34**	**-30**	**-30**	**32**	**-29**	**-33**	**-30**	**-30**	**-122**	**-122**	**-122**	**-122**
Net change in cash	**-74**	**-18**	**55**	**37**	**98**	**173**	**-60**	**28**	**53**	**124**	**145**	**297**	**413**	**437**

Source: Company data, UBS estimates

Soulbrain (Jimmy Yoon) – Buy, PT Won540,000

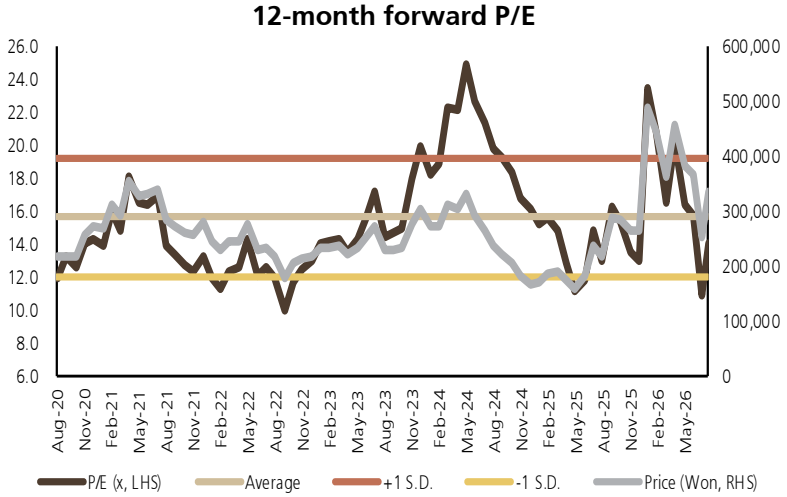
- **Our View**
  - Supported by Ai/HBM as well as conventional DDR demand, we see DRAM WFE spending at Samsung and SK Hynix trending on the upside into 2026. With Soulbrain's revenue growth tied closely with memory production as well as recovery in Samsung Foundry UTR, we expect semis material revenues to grow +31% YoY in 2026 supporting a sharp pickup in earnings of 53% YoY. While NAND spending remains limited, higher usage intensity from NAND tech migration (Samsung Xi'an) should offer some offset to wafer capacity loss.
- **Can Soulbrain grow semis material revenues >15% YoY in 2026/2027E?**
  - Yes. We believe Soulbrain will be able to record sequential revenue growth of +34%/+23% YoY in 2026/27E. This should be supported by semis WFE spending at Samsung/SK Hynix (+57%/+25% YoY in 2026/27E) leading to higher DRAM semis material (HF, Precursor, CMP) consumption as well as Samsung steadily increasing 3nm and below Foundry UTR throughout 2026E. Samsung Taylor ramp from 2H26 should further support Acetic etchant revenue growth. In the meantime, Soulbrain continues to make progress in gaining share for precursors in addition to development of High-K precursors for Samsung from 2027. Visibility on NAND flash WFE spend on the other hand remains limited which could for the time being 'cap' upside to semis material revenues.
- **Can Soulbrain maintain a >20% operating margin over the mid to longer term?**
  - Yes. While we expect Soulbrain to face ASP pressure for semis material shipments over time, we believe there could be ongoing etchant volume growth, particularly from: 1) HSN etchants with increased NAND flash double stack in the mix (vs. single) and, eventually, triple stack; and 2) GAA etchants, as Samsung Foundry gradually ramps up UTR and 3nm and below capacity, which could help offset ASP headwinds. The phase-out of margin-dilutive businesses such as thin glass over time and entry into higher margin products such as High-K precursors from 2027 should also help Soulbrain maintain a >20% operating margin in our view.

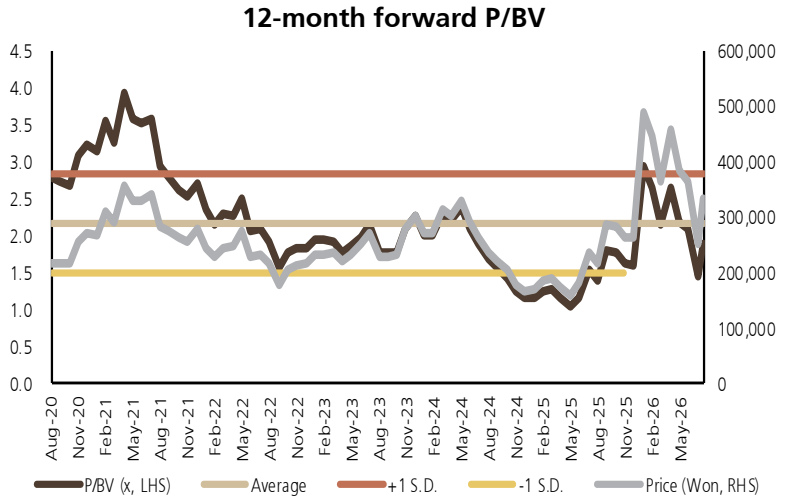
- **PT Won540k –** We value Soulbrain at 19.0x 2027E P/E based on our 14.9% 2027-31E EPS CAGR. This is at a c.15% discount to global etchant material peers.

Soulbrain (Jimmy Yoon) – Valuation

UBS price target	**540,000**
Current stock price (Won)	322,000
Number of shares (bn)	7.7
Treasury shares	0.1
Market cap. (Won bn)	2,466
Net debt (Won bn)	-237
Adjusted EV (Won bn)	2,229

	2025	2026E	2027E	2028E	2029E	2030E
Current Valuation						
EV/Revenues	2.41	1.78	1.44	1.20	1.09	0.99
P/Book	2.22	2.01	1.73	1.48	1.27	1.08
P/E	31.28	12.82	11.33	9.13	7.88	6.76
EV/EBITDA	11.23	8.11	6.23	5.18	4.57	3.97
UBS valuation (x)						
EV/Revenues	4.22	3.12	2.53	2.10	1.91	1.73
P/Book	3.73	3.37	2.91	2.48	2.12	1.82
P/E	52.46	21.50	18.99	15.31	13.22	11.34
EV/EBITDA	19.64	14.18	10.89	9.05	8.00	6.95
Other data						
EPS	10,294	25,122	28,431	35,280	40,841	47,607
% ROE	***7.4%***	***16.5%***	***16.4%***	***17.5%***	***17.3%***	***17.3%***
Tangible book per share (Won)	144,741	160,226	185,618	217,581	254,284	297,179
Cash per share (Won)	32,248	42,084	52,993	76,240	106,116	141,078
FCF Per Share (Won)	-13,530	18,783	16,526	27,547	35,176	41,062
% FCF yield	*-4.2%*	*5.8%*	*5.1%*	*8.6%*	*10.9%*	*12.8%*





Source: Company data, UBS estimates

Soulbrain – Key charts

UBS forecasts vs Consensus

	3Q26E			4Q26E			2026E			2027E			2028E		
(Won bn)	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**
Sales	331	326	*1.7%*	342	323	*6.0%*	1,249	1,235	*1.2%*	1,543	1,421	*8.6%*	1,855	1,714	*8.3%*
Op Profit	**59**	**54**	**9.9%**	**55**	**57**	**-3.6%**	**205**	**200**	**2.0%**	**287**	**266**	**8.0%**	**362**	**342**	**6.0%**
% OP margin	*17.9%*	*16.5%*	*134bps*	*16.0%*	*17.5%*	*-158bps*	*16.4%*	*16.2%*	*13bps*	*18.6%*	*18.7%*	*-11bps*	*19.5%*	*19.9%*	*-42bps*
EPS (Won)	4,539	4,128	9.9%	5,359	5,619	-4.6%	25,122	23,049	9.0%	28,431	26,790	6.1%	35,280	33,767	4.5%

Value drivers	Semis materials revs % YoY ('26E/'27E)	LiB and others revs % YoY ('26E/'27E)	Operating profit '26E/'27E
W640,000 upside	+44%/+32%	+26%/+12%	Won220bn/310bn
W540,000 base	+41%/+27%	+24%/+6%	Won205bn/287bn
W190,000 downside	+38%/+20%	+19%/-2%	Won174bn/24bn

87

Source: Bloomberg, Company data, UBS estimates

Soulbrain – Revenue breakdown

(Won bn)		2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Semis Material		**767**	**222**	**271**	**289**	**301**	**1,083**	**314**	**330**	**353**	**374**	**1,371**	**1,677**	**1,863**	**2,066**
	% YoY	*16.8%*	*32.9%*	*42.5%*	*42.7%*	*45.2%*	*41.2%*	*41.4%*	*21.9%*	*22.3%*	*24.0%*	*26.6%*	*22.3%*	*11.1%*	*10.9%*
	% QoQ		*6.9%*	*22.1%*	*6.6%*	*4.3%*		*4.1%*	*5.2%*	*7.0%*	*5.8%*				
Etching / Cleaning		**557**	**160**	**178**	**192**	**203**	**733**	**214**	**224**	**239**	**256**	**932**	**1,165**	**1,309**	**1,478**
	% YoY	*2.2%*	*15.3%*	*33.9%*	*37.2%*	*39.6%*	*31.6%*	*33.7%*	*25.8%*	*24.1%*	*26.3%*	*27.2%*	*25.0%*	*12.4%*	*12.9%*
	% QoQ		*10.0%*	*11.5%*	*8.1%*	*5.3%*		*5.4%*	*4.9%*	*6.6%*	*7.1%*				
Precursors		**42**	**12**	**14**	**15**	**16**	**56**	**17**	**18**	**19**	**20**	**73**	**93**	**104**	**115**
	% YoY	*2.9%*	*14.6%*	*38.5%*	*43.4%*	*40.0%*	*34.2%*	*38.5%*	*28.0%*	*26.0%*	*28.9%*	*29.9%*	*27.2%*	*11.8%*	*10.5%*
	% QoQ		*7.6%*	*16.0%*	*7.5%*	*4.3%*		*6.4%*	*7.2%*	*5.9%*	*6.7%*				
CMP Slurry		**83**	**21**	**25**	**27**	**28**	**101**	**29**	**31**	**34**	**35**	**128**	**155**	**176**	**195**
	% YoY	*16.3%*	*17.3%*	*31.7%*	*16.8%*	*23.1%*	*22.1%*	*36.0%*	*22.6%*	*24.9%*	*26.1%*	*27.0%*	*20.7%*	*13.7%*	*10.4%*
	% QoQ		*-5.8%*	*18.7%*	*8.0%*	*2.0%*		*4.0%*	*7.0%*	*10.0%*	*3.0%*				
Others (SFS+DNF)		**85**	**29**	**54**	**54**	**55**	**193**	**55**	**58**	**62**	**63**	**237**	**264**	**274**	**279**
	% YoY			*86.0%*	*90.8%*	*92.6%*	*126.0%*	*88.7%*	*7.1%*	*13.5%*	*13.5%*	*23.0%*	*11.1%*	*3.8%*	*1.9%*
	% QoQ		*1.0%*	*85.0%*	*1.0%*	*2.0%*		*-1.0%*	*5.0%*	*7.0%*	*2.0%*				
Display Materials		**65**	**14**	**13**	**13**	**13**	**52**	**12**	**13**	**13**	**13**	**51**	**51**	**49**	**47**
	% YoY	*-33.2%*	*-36.5%*	*-22.9%*	*-4.4%*	*-0.2%*	*-19.0%*	*-11.2%*	*-2.4%*	*0.6%*	*3.7%*	*-2.4%*	*-0.5%*	*-4.1%*	*-4.1%*
	% QoQ		*5.7%*	*-3.6%*	*0.0%*	*-2.0%*		*-6.0%*	*6.0%*	*3.0%*	*1.0%*				
LiB & Others		**92**	**28**	**28**	**29**	**28**	**114**	**29**	**31**	**31**	**31**	**121**	**127**	**134**	**140**
Total		**923**	**264**	**312**	**331**	**342**	**1,249**	**355**	**373**	**397**	**418**	**1,543**	**1,855**	**2,045**	**2,253**
	% YoY	*7.0%*	*25.9%*	*36.3%*	*37.4%*	*40.3%*	*35.3%*	*34.5%*	*19.7%*	*20.0%*	*21.9%*	*23.5%*	*20.2%*	*10.3%*	*10.1%*
	% QoQ		*8.1%*	*18.3%*	*6.2%*	*3.3%*		*3.7%*	*5.2%*	*6.4%*	*5.0%*				
% of Revenues															
Semis Material		*83.1%*	*84.1%*	*86.8%*	*87.2%*	*88.0%*	*86.7%*	*88.4%*	*88.4%*	*88.9%*	*89.5%*	*88.8%*	*90.4%*	*91.1%*	*91.7%*
Display Materials		*7.0%*	*5.1%*	*4.2%*	*3.9%*	*3.7%*	*4.2%*	*3.4%*	*3.4%*	*3.3%*	*3.2%*	*3.3%*	*2.7%*	*2.4%*	*2.1%*
Others (LiB etc.)		*9.9%*	*10.7%*	*9.0%*	*8.9%*	*8.3%*	*9.1%*	*8.2%*	*8.2%*	*7.8%*	*7.3%*	*7.9%*	*6.9%*	*6.5%*	*6.2%*

Source: Company data, UBS estimates

Soulbrain – Financial Statement Forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues	**923**	**264**	**312**	**331**	**342**	**1,249**	**355**	**373**	**397**	**418**	**1,543**	**1,855**	**2,045**	**2,253**
Gross profit	233	72	83	93	91	339	107	112	122	115	456	565	637	726
% GP margin	*25.2%*	*27.4%*	*26.8%*	*28.1%*	*26.5%*	*27.2%*	*30.1%*	*30.1%*	*30.6%*	*27.5%*	*29.5%*	*30.4%*	*31.1%*	*32.2%*
Operating profit	**134**	**45**	**46**	**59**	**55**	**205**	**69**	**68**	**80**	**70**	**287**	**362**	**419**	**493**
% OP margin	*14.5%*	*16.9%*	*14.8%*	*17.9%*	*16.0%*	*16.4%*	*19.4%*	*18.3%*	*20.2%*	*16.8%*	*18.6%*	*19.5%*	*20.5%*	*21.9%*
NI post MI	79	36	80	35	41	192	52	52	61	53	218	270	313	365
EPS	**10,294**	**4,715**	**10,509**	**4,539**	**5,359**	**25,122**	**6,820**	**6,758**	**7,922**	**6,931**	**28,431**	**35,280**	**40,841**	**47,607**
% EPS growth YoY	*-32.7%*	*17.7%*	*608.4%*	*33.7%*	*284.2%*	*144.0%*	*44.6%*	*-35.7%*	*74.5%*	*29.3%*	*13.2%*	*24.1%*	*15.8%*	*16.6%*
DPS	2,350	2,350	0	0	0	3,800	3,800	0	0	0	4,300	5,300	6,100	7,100
% payout ratio	*22.8%*	*49.8%*	*0.0%*	*0.0%*	*0.0%*	*15.1%*	*55.7%*	*0.0%*	*0.0%*	*0.0%*	*15.1%*	*15.0%*	*14.9%*	*14.9%*
Cash and cash equiv	**248**	**281**	**252**	**319**	**322**	**322**	**358**	**332**	**382**	**406**	**406**	**584**	**813**	**1,080**
Total assets	**1,447**	**1,492**	**1,600**	**1,638**	**1,679**	**1,679**	**1,717**	**1,745**	**1,811**	**1,872**	**1,872**	**2,128**	**2,415**	**2,752**
Total equity	1,112	1,152	1,149	1,185	1,227	1,227	1,281	1,305	1,367	1,422	1,422	1,666	1,947	2,276
Net debt	-146	-237	-236	-303	-307	-307	-356	-330	-380	-404	-404	-582	-811	-1,079
% ROE	*7.4%*	*12.8%*	*28.0%*	*11.9%*	*13.6%*	*16.5%*	*16.7%*	*16.0%*	*18.2%*	*15.2%*	*16.4%*	*17.5%*	*17.3%*	*17.3%*
D&A	**65**	**16**	**18**	**18**	**18**	**70**	**18**	**18**	**18**	**18**	**70**	**68**	**68**	**68**
CF from Operations	**147**	**34**	**59**	**99**	**46**	**238**	**81**	**35**	**82**	**56**	**255**	**311**	**369**	**414**
Capex	-120	-16	-22	-32	-42	-112	-32	-32	-32	-32	-128	-100	-100	-100
Free cash flow	**-104**	**60**	**13**	**67**	**3**	**144**	**49**	**3**	**50**	**24**	**127**	**211**	**269**	**314**
CF from Investments	**-251**	**26**	**-46**	**-32**	**-42**	**-94**	**-32**	**-32**	**-32**	**-32**	**-128**	**-100**	**-100**	**-100**
Dividend	-18	0	-18	0	0	-18	0	-29	0	0	-29	-33	-41	-47
CF from Finance	**109**	**-32**	**-46**	**0**	**0**	**-78**	**-14**	**-29**	**0**	**0**	**-43**	**-33**	**-41**	**-47**
Net changes of cash	**6**	**34**	**-30**	**67**	**3**	**75**	**35**	**-26**	**50**	**24**	**84**	**178**	**229**	**268**

Source: Company data, UBS estimates

Wonik IPS (Nicolas Gaudois) – Buy, PT Won155,000

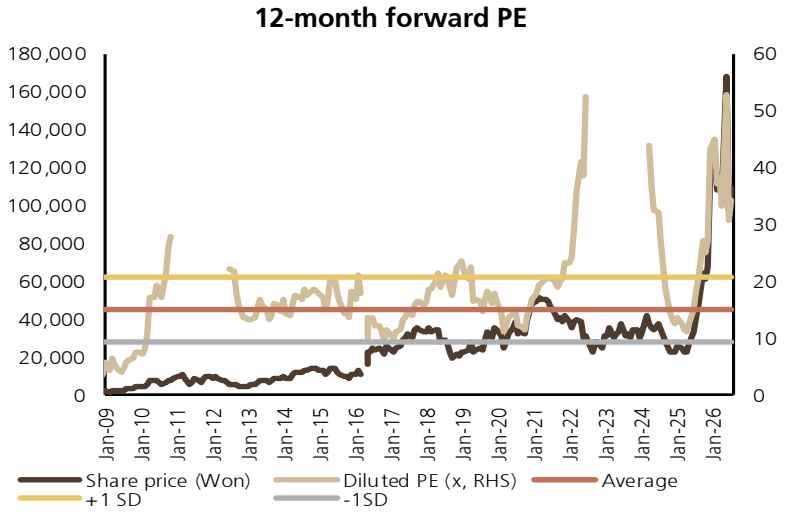
- **Our View**
  - Wonik IPS is highly geared towards semis WFE spending at Samsung and SK Hynix. HBM demand is likely to keep pressuring up DRAM capex over time, and require capacity addition every year. In NAND specifically, we believe tech migration will likely be a key driver. Given its natural gearing to NAND flash greenfield, any incremental NAND flash capacity addition should also present further upside.
- **Will Samsung and SK Hynix's semis WFE spending recover meaningfully in the next few years?**
  - Yes. We now forecast combined Samsung and SK Hynix memory WFE up +55%/+34%/+19% YoY in '26E/'27E/'28E. We believe further upside could come from High Bandwidth Memory (HBM) demand.
- **Will Wonik IPS see semis equipment revenues grow in 2026E/2027E?**
  - Yes. Over 80% of Wonik IPS revenues come from semis WFE and servicing. The largest portion of that is DRAM end-use, with Samsung the main customer (~65% of 2026E semis equipment revenues).
  - Going forward, we expect Wonik to continue to expand its TAM with: 1/ further process steps for Samsung's NAND V10 process (ramping slightly early); 2/ ALD shipping to SK Hynix starting 2026 for 1c nm DRAM; 3/incremental metal layers for Samsung Foundry 2nm.

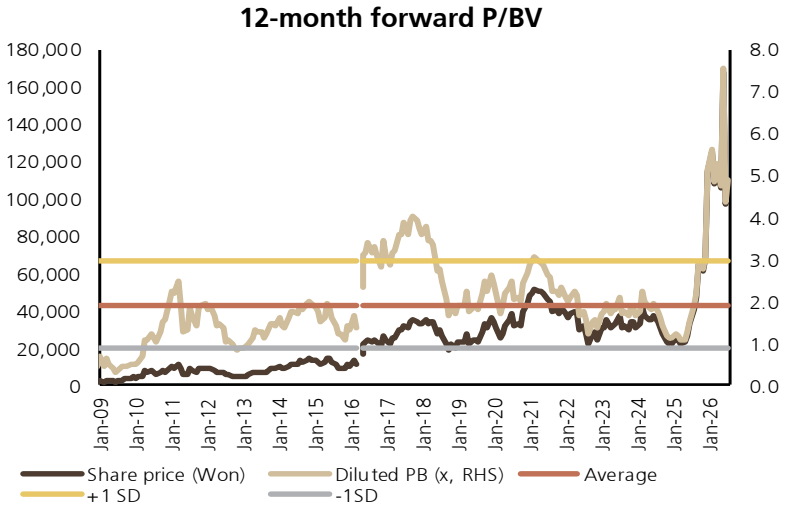
- **PT Won155k –** We value Wonik shares at 25x 2028E P/E based on our 2028E-30E EPS CAGR of 21%.

Wonik IPS (Nicolas Gaudois) – Valuation

UBS price target	**155,000**
Current stock price (Won)	129,000
Number of shares (mn, cum treasury shrs)	49.1
Treasury shares	0.4
Market cap. (Won bn)	6,332
Net debt (Won bn)	-136
EV (Won bn)	6,196

	2025	2026E	2027E	2028E	2029E	2030E
Current valuation (x)						
EV/Revenues	6.81	5.16	3.85	3.24	2.72	2.49
P/Book (diluted)	6.47	5.33	4.57	3.79	3.08	2.55
P/E (diluted)	75.35	29.29	28.59	20.81	15.60	14.19
EV/EBITDA	55.82	31.76	19.73	14.54	11.01	10.09
Target price valuation (x)						
EV/Revenues	8.29	6.28	4.69	3.94	3.31	3.03
P/Book (diluted)	8.00	6.58	5.61	4.65	3.76	3.11
P/E (diluted)	90.54	35.19	34.35	25.01	18.74	17.05
EV/EBITDA	67.95	38.66	24.02	17.70	13.41	12.28
Other						
EPS (diluted)	1,712	4,405	4,513	6,199	8,272	9,091
% ROE	***9.0%***	***20.1%***	***17.4%***	***20.1%***	***22.0%***	***19.9%***
Tangible book per share (Won, diluted)	19,367	23,561	27,612	33,364	41,208	49,878
Cash per share (Won, diluted)	2,205	4,309	6,440	13,150	18,785	26,307
Free cash flow per share (Won, diluted)	-450	2,299	2,631	7,210	6,135	8,022
% Free cash flow yield	*-0.3%*	*1.8%*	*2.0%*	*5.6%*	*4.8%*	*6.2%*



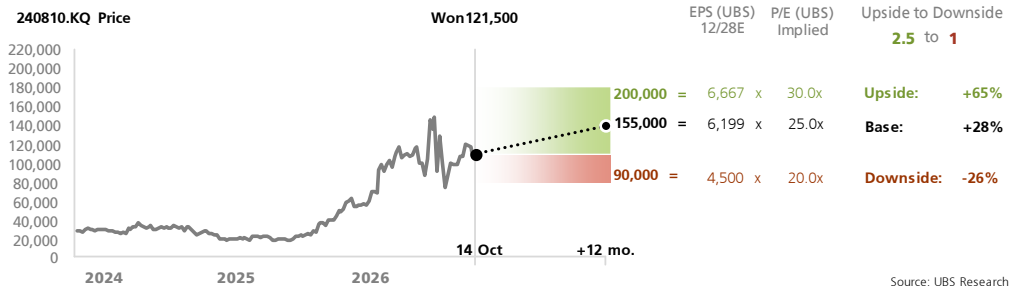


Source: Company data, UBS estimates

Wonik IPS – UBSe vs. Consensus

UBS forecasts vs Consensus

(Won bn)	3Q26E			4Q26E			2026E			2027E			2028E		
	UBSe	Cons	Delta	UBSe	Cons	Delta	UBSe	Cons	Delta	UBSe	Cons	Delta	UBSe	Cons	Delta
Sales	335	389	-14%	484	474	2%	1,200	1,250	-4%	1,609	1,623	-1%	1,913	1,887	1%
Operating Profit	**50**	**72**	**-30%**	**80**	**81**	**-2%**	**159**	**182**	**-13%**	**274**	**292**	**-6%**	**378**	**361**	**5%**
% OP margin	*14.9%*	*18.5%*	*-356bps*	*16.4%*	*17.1%*	*-67bps*	*13.2%*	*14.6%*	*-132bps*	*17.0%*	*18.0%*	*-96bps*	*19.8%*	*19.1%*	*62bps*
EPS (Won)	827	1,012	-18%	1,296	1,278	1%	4,405	4,381	1%	4,513	4,947	-9%	6,199	6,342	-2%



Value drivers	Semis revenues '27E/'28E (Won)	Display revenues '27E/'28E (Won)	Operating Profit '27E/'28E (Won)
Won200,000 upside	1.50tn/1.90tn	229bn/239bn	300bn/420bn
Won155,000 base	1.40tn/1.69tn	209bn/219bn	274bn/378bn
Won90,000 downside	1.20tn/1.40tn	189bn/199bn	240bn/330bn

Source: Bloomberg, Company data, UBS estimates

Wonik IPS – Revenue breakdown

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Total Sales	**909**	**165**	**216**	**335**	**484**	**1,200**	**344**	**356**	**443**	**465**	**1,609**	**1,913**	**2,278**	**2,490**
Semiconductors	**707**	**146**	**190**	**283**	**421**	**1,040**	**302**	**314**	**381**	**403**	**1,400**	**1,694**	**2,048**	**2,259**
% YoY	*23%*	*55%*	*-13%*	*44%*	*113%*	*47%*	*107%*	*65%*	*35%*	*-4%*	*35%*	*21%*	*21%*	*10%*
% QoQ		*-26%*	*30%*	*49%*	*49%*									
Memory	**563**	**109**	**157**	**208**	**263**	**737**	**221**	**226**	**286**	**291**	**1,024**	**1,252**	**1,546**	**1,697**
Samsung 3D NAND	175	7	33	68	82	189	56	56	56	56	224	240	404	404
Samsung DRAM	267	53	94	95	126	369	115	115	160	160	550	682	752	813
SK Hynix	75	25	20	30	35	110	30	35	40	45	150	200	240	280
Others	46	25	10	15	20	70	20	20	30	30	100	130	150	200
Non-Memory	**24**	**4**	**2**	**40**	**120**	**166**	**40**	**45**	**50**	**65**	**200**	**230**	**250**	**270**
Samsung Foundry	24	4	2	20	100	126	30	35	40	45	150	170	190	210
Parts and services	120	33	31	35	38	137	41	43	45	47	176	212	252	292
Displays	**203**	**19**	**27**	**52**	**63**	**160**	**42**	**42**	**62**	**62**	**209**	**219**	**230**	**231**
% YoY	*16%*	*-38%*	*8%*	*-27%*	*-19%*	*-21%*	*123%*	*59%*	*20%*	*-1%*	*30%*	*5%*	*5%*	*0%*
% QoQ		*-76%*	*41%*	*96%*	*21%*		*-33%*	*0%*	*48%*	*0%*				
Display Equipment	196	17	25	50	61	152	40	40	60	60	200	210	220	220
Parts and services	7	2	2	2	2	8	2	2	2	2	9	9	10	11

Source: Company data, UBS estimates

Wonik IPS – Financial Statement Forecasts

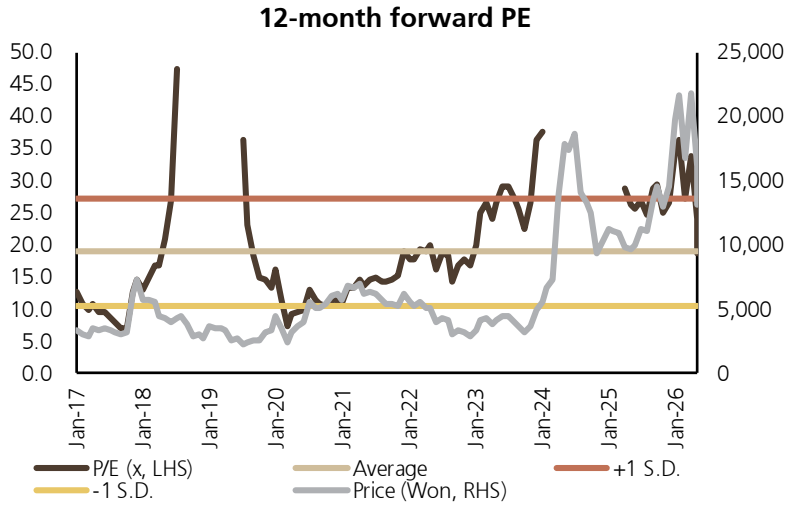
(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues	**910**	**165**	**216**	**335**	**484**	**1,200**	**344**	**356**	**443**	**465**	**1,609**	**1,913**	**2,278**	**2,490**
% QoQ		*-40%*	*31%*	*55%*	*44%*		*-29%*	*4%*	*24%*	*5%*				
% YoY	*22%*	*33%*	*-11%*	*25%*	*76%*	*32%*	*109%*	*65%*	*32%*	*-4%*	*34%*	*19%*	*19%*	*9%*
Gross profit	383	85	99	141	203	528	144	150	195	209	698	859	1,051	1,147
% GP margin	*42.1%*	*51.5%*	*45.7%*	*42.0%*	*42.0%*	*44.0%*	*42.0%*	*42.0%*	*44.0%*	*45.0%*	*43.4%*	*44.9%*	*46.1%*	*46.0%*
Operating profit	**74**	**11**	**18**	**50**	**80**	**159**	**51**	**50**	**91**	**82**	**274**	**378**	**507**	**558**
% OP margin	*8.1%*	*6.5%*	*8.5%*	*14.9%*	*16.4%*	*13.2%*	*14.8%*	*14.0%*	*20.6%*	*17.6%*	*17.0%*	*19.8%*	*22.2%*	*22.4%*
NI post MI	84	22	90	41	64	216	42	41	73	66	221	304	406	446
EPS	**1,712**	**449**	**1,832**	**827**	**1,296**	**4,405**	**848**	**833**	**1,492**	**1,339**	**4,513**	**6,199**	**8,272**	**9,091**
Cash and cash equiv	**107**	**153**	**136**	**331**	**210**	**210**	**372**	**334**	**276**	**313**	**313**	**640**	**914**	**1,280**
Total assets	**1,165**	**1,262**	**1,493**	**1,526**	**1,618**	**1,618**	**1,639**	**1,663**	**1,749**	**1,812**	**1,812**	**2,104**	**2,512**	**2,943**
Total equity	970	982	1,074	1,114	1,178	1,178	1,219	1,236	1,309	1,375	1,375	1,655	2,037	2,458
Net cash (debt)	107	153	136	331	210	210	372	334	276	313	313	640	914	1,280
% ROE	*9.0%*	*9.0%*	*35.0%*	*14.9%*	*22.2%*	*20.1%*	*13.9%*	*13.3%*	*23.0%*	*19.6%*	*17.4%*	*20.1%*	*22.0%*	*19.9%*
CF from Operations	**157**	**16**	**34**	**205**	**-112**	**143**	**174**	**-1**	**-46**	**49**	**176**	**403**	**358**	**450**
Capex	-26	-5	-4	-10	-10	-29	-12	-12	-12	-12	-48	-52	-60	-60
Free cash flow	**-22**	**45**	**-7**	**195**	**-122**	**112**	**162**	**-13**	**-58**	**37**	**128**	**351**	**298**	**390**
CF from Investments	**-179**	**30**	**-41**	**-10**	**-10**	**-31**	**-12**	**-12**	**-12**	**-12**	**-48**	**-52**	**-60**	**-60**
CF from Finance	**-6**	**-1**	**-10**	**0**	**0**	**-11**	**0**	**-24**	**0**	**0**	**-24**	**-24**	**-24**	**-24**
Net changes of cash	**-28**	**46**	**-17**	**195**	**-122**	**102**	**162**	**-38**	**-58**	**37**	**104**	**326**	**274**	**366**

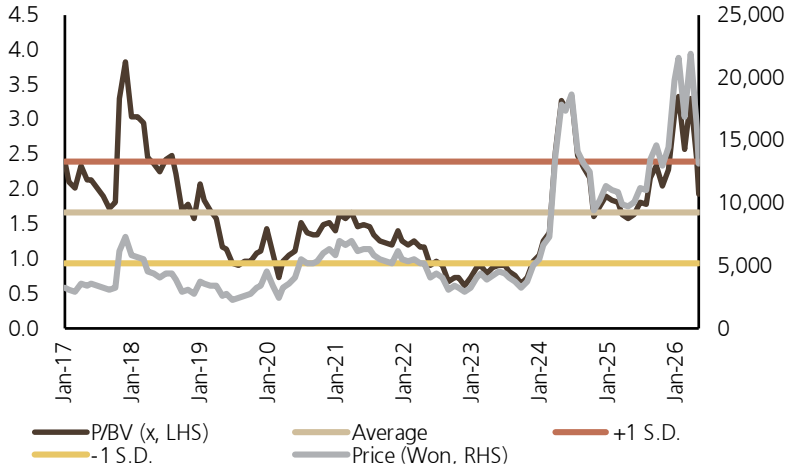
Source: Company data, UBS estimates

YC Corporation (Jimmy Yoon) – Buy, PT Won20,500

UBS price target	**20,500**
Current stock price (Won)	11,350
Number of shares (mil, ex-treasury shrs)	81.945
Treasury shares	2
Market cap. (Won bn)	930
Net debt (Won bn)	77
EV (Won bn)	1,007

	2025	2026E	2027E	2028E	2029E	2030E
Current valuation (x)						
EV/Revenues	3.69	2.51	1.92	1.59	1.26	1.12
P/Book	1.93	1.69	1.50	1.31	1.13	0.98
P/E	47.72	23.88	17.90	12.68	9.22	8.36
EV/EBITDA	36.73	14.64	10.44	8.12	6.32	5.82
Target price valuation (x)						
EV/Revenues	6.44	4.37	3.35	2.77	2.20	1.96
P/Book	3.49	3.05	2.71	2.37	2.03	1.77
P/E	86.20	43.12	32.32	22.91	16.66	15.10
EV/EBITDA	64.08	25.54	18.21	14.17	11.04	10.15
Other						
EPS	238	475	634	895	1,231	1,358
% ROE	*4.3%*	*7.6%*	*8.9%*	*11.0%*	*13.1%*	*12.5%*
Book per share (Won)	5,871	6,722	7,569	8,668	10,084	11,602
Cash per share (Won)	1,211	1,348	1,942	2,920	3,700	4,543
Free cash flow per share (Won)	632	200	1,111	1,072	900	996
% Free cash flow yield	*5.6%*	*1.8%*	*9.8%*	*9.4%*	*7.9%*	*8.8%*



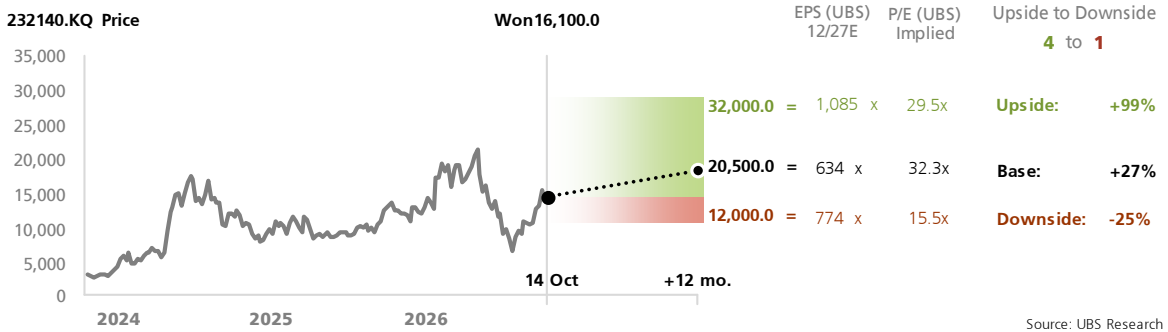


Source: Company data, UBS estimates

YC Corporation – UBSe vs. Consensus

UBS forecasts vs Consensus

	3Q26E			4Q26E			2026E			2027E			2028E		
(Won bn)	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**
Sales	119	118	1.3%	144	126	14.1%	401	382	5.0%	525	474	10.7%	633	589	7.5%
Operating profit	**18**	**18**	**4.0%**	**24**	**16**	**50.8%**	**56**	**55**	**3.2%**	**85**	**76**	**12.2%**	**112**	**106**	**5.9%**
% OP margin	15.5%	15.1%	40bps	17.0%	14.7%	227bps	14.0%	14.3%	-25bps	16.1%	15.9%	22bps	17.7%	18.0%	-26bps
EPS (Won)	136	136	0.0%	196	161	21.6%	475	429	10.9%	634	582	8.9%	895	767	16.7%



Value drivers (2026/27E)	KGSD Burn-in Qualification	HBM Testers	DRAM Testers	Operating profit
Won32,000 upside	Yes	82/108	23/37	Won76bn/115bn
Won20,500 base	No	65/77	18/27	Won57bn/85bn
Won12,000 downside	No	40/65	12/18	Won40bn/72bn

96

Source: Bloomberg, Company data, UBS estimates

YC Corporation – Revenue Forecasts

(Won bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	1Q28E	2Q28E	3Q28E	4Q28E	2028E	2029E	2030E
Total Revenue (Won bn)	**273**	**33**	**105**	**119**	**144**	**401**	**127**	**142**	**126**	**129**	**525**	**150**	**176**	**165**	**143**	**633**	**799**	**898**
% YoY	*29%*	*-14%*	*31%*	*79%*	*66%*	*47%*	*283%*	*35%*	*6%*	*-10%*	*31%*	*17%*	*24%*	*30%*	*11%*	*21%*	*26%*	*12%*
% QoQ		*-62%*	*215%*	*14%*	*20%*		*-11%*	*11%*	*-11%*	*2%*		*16%*	*18%*	*-6%*	*-13%*			
Wafer Tester Equipment	178	4	71	82	107	263	86	97	78	76	337	93	116	102	82	393	509	546
Equipment Parts (incl. SEMCNS)	78	27	31	33	32	124	36	39	42	46	164	49	52	53	52	207	243	289
Servicing	17	3	3	4	5	14	5	6	6	7	24	8	8	9	9	33	47	63
Operating Profit (Won bn)	**17**	**-2**	**15**	**18**	**24**	**56**	**21**	**26**	**19**	**20**	**85**	**24**	**34**	**29**	**26**	**112**	**147**	**161**
OP Margin (%)	*6%*	*-5%*	*14%*	*15%*	*17%*	*14%*	*16%*	*18%*	*15%*	*15%*	*16%*	*16%*	*19%*	*17%*	*18%*	*18%*	*18%*	*18%*
Wafer Tester Revenue																		
DRAM (MT6133)	56	2	7	14	18	41	16	18	14	14	62	21	18	18	14	71	113	131
NAND (MT6122)	10	1	1	2	2	7	4	6	5	6	20	6	5	5	2	18	30	25
HBM (MT8311)	113	0	63	66	86	215	66	73	59	56	254	66	92	79	66	304	366	389
Total Tester Revenue (Won bn)	**179**	**4**	**71**	**82**	**107**	**263**	**86**	**97**	**78**	**76**	**337**	**93**	**116**	**102**	**82**	**393**	**509**	**546**
Tester Shipments																		
DRAM (MT6133)	25	1	3	6	8	18	7	8	6	6	27	9	8	8	6	31	49	57
NAND (MT6122)	8	1	1	2	2	6	3	5	4	5	17	5	4	4	2	15	25	21
HBM (MT8311)	39	0	19	20	26	65	20	22	18	17	77	20	28	24	20	92	111	118
Total Shipments (Units)	**72**	**2**	**23**	**28**	**36**	**89**	**30**	**35**	**28**	**28**	**121**	**34**	**40**	**36**	**28**	**138**	**185**	**196**
Tool ASP																		
DRAM (MT6133)	2.2	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3
NAND (MT6122)	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
HBM (MT8311)	2.9	3.1	3.3	3.3	3.3	3.3	3.3	3.3	3.3	3.3	3.3	3.3	3.3	3.3	3.3	3.3	3.3	3.3
Blended ASP (Won bn)	**2.5**	**1.8**	**3.1**	**2.9**	**3.0**	**3.0**	**2.9**	**2.8**	**2.8**	**2.7**	**2.8**	**2.7**	**2.9**	**2.8**	**2.9**	**2.8**	**2.8**	**2.8**

Source: Bloomberg, Company data, UBS estimates

YC Corporation – Financial Statement Forecasts

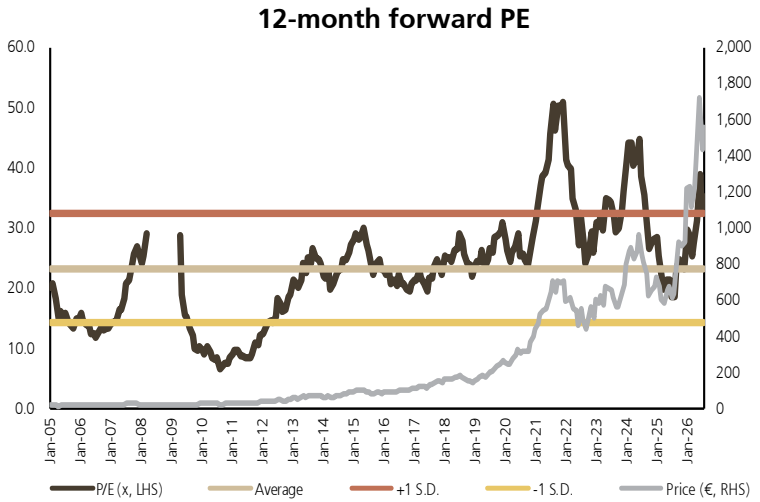
(Wonbn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues	**273**	**33**	**105**	**119**	**144**	**401**	**127**	**142**	**126**	**129**	**525**	**633**	**799**	**898**
Gross profit	72	19	30	36	43	128	41	48	43	43	175	209	259	282
% GM	*26.3%*	*56.2%*	*28.5%*	*30.0%*	*30.0%*	*31.8%*	*32.0%*	*34.0%*	*34.0%*	*33.0%*	*33.3%*	*33.0%*	*32.4%*	*31.5%*
Operating profit	**17**	**(2)**	**15**	**18**	**24**	**56**	**21**	**26**	**19**	**20**	**85**	**112**	**147**	**161**
% OpM	*6.2%*	*-4.5%*	*14.3%*	*15.5%*	*17.0%*	*14.0%*	*16.1%*	*18.0%*	*14.9%*	*15.3%*	*16.1%*	*17.7%*	*18.4%*	*18.0%*
Net profit post MI	**19**	**1**	**11**	**11**	**16**	**39**	**13**	**16**	**11**	**12**	**52**	**73**	**101**	**111**
EPS reported (Won)	**238**	**15**	**128**	**136**	**196**	**475**	**153**	**193**	**138**	**150**	**634**	**895**	**1,231**	**1,358**
Cash and cash equivalents	98	71	70	96	110	110	93	95	130	159	159	239	303	372
Total assets	**607**	**656**	**652**	**682**	**711**	**711**	**685**	**708**	**719**	**738**	**738**	**826**	**948**	**1,075**
Total liabilities	**131**	**154**	**138**	**152**	**160**	**160**	**116**	**121**	**116**	**118**	**118**	**116**	**122**	**125**
Shareholders' equity	**476**	**502**	**(0)**	**530**	**551**	**551**	**569**	**587**	**603**	**620**	**620**	**710**	**826**	**951**
Depreciation & amortization	10	3	3	3	3	12	3	3	3	3	12	12	12	12
Cash flow from operations	**41**	**(28)**	**18**	**31**	**19**	**40**	**25**	**11**	**39**	**33**	**109**	**106**	**92**	**99**
Capex	(16)	(5)	(4)	(4)	(4)	(18)	(4)	(4)	(4)	(4)	(18)	(18)	(18)	(18)
Cash flow from investments	**10**	**(8)**	**(6)**	**(4)**	**(4)**	**(24)**	**(4)**	**(4)**	**(4)**	**(4)**	**(18)**	**(18)**	**(18)**	**(18)**
Dividends paid	0	0	0	0	0	0	0	(4)	0	0	(4)	(5)	(7)	(10)
Cash flow from financing	**28**	**9**	**(14)**	**0**	**0**	**(5)**	**(39)**	**(4)**	**0**	**0**	**(42)**	**(8)**	**(10)**	**(13)**
Total cash flow	**79**	**(27)**	**(1)**	**26**	**14**	**12**	**(17)**	**2**	**35**	**29**	**49**	**80**	**64**	**69**

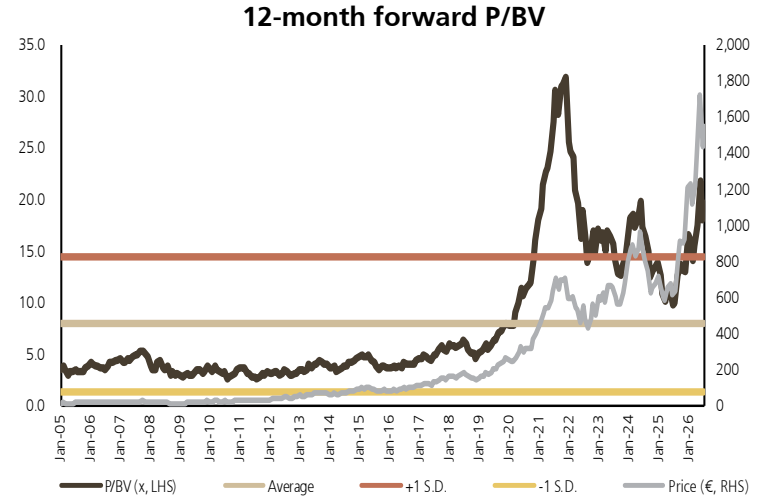
Source: Bloomberg, Company data, UBS estimates

ASML (Francois Bouvignies, Nicolas Gaudois) – Buy, PT €2,400

Price target (€)	**2,400.0**
Current stock price (€)	1,701.7
Number of shares (m)	393.5
Market cap. (€m)	669,611
EV (€m)	669,751

	2025	2026E	2027E	2028E	2029E	2030E
Current valuation						
EV/Revenues	20.50	15.07	11.00	8.19	8.12	7.64
P/Book	46.23	32.26	18.26	10.67	7.71	5.93
P/E	70.19	40.31	26.83	18.21	17.85	16.42
EV/EBITDA	54.51	34.13	23.05	15.72	15.16	14.11
UBS valuation						
EV/Revenues	28.91	21.25	15.52	11.56	11.45	10.77
P/Book	65.21	45.50	25.75	15.05	10.87	8.36
P/E	99.00	56.85	37.84	25.68	25.17	23.15
EV/EBITDA	76.87	48.14	32.50	22.17	21.39	19.90
Other						
% ROE	***49.5%***	***71.2%***	***71.2%***	***64.3%***	***44.6%***	***36.5%***
Tangible book per share (€)	36.8	52.8	93.2	159.5	220.8	287.1
Cash per share (€)	33.9	55.7	109.5	196.4	274.5	358.2
Free cash flow per share (€)	28.17	54.20	74.48	110.07	107.24	114.63
% Free cash flow yield	*1.7%*	*3.2%*	*4.4%*	*6.5%*	*6.3%*	*6.7%*





Source: Company data, UBS estimates

ASML – EUV Shipment by Customer

0.33 NA EUV shipments

	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Logic													
TSMC	10	15	15	20	22	19	13	20	24	34	46	47	47
Samsung Foundry	2	4	9	11	14	4	2	2	2	4	6	6	6
Intel	3	3	2	2	9	7	8	5	5	8	13	15	15
GF													
Others	1	2		1			1	2	2	4	3		
DRAM													
Samsung	1	1	4	7	5	6	6	5	13	16	19	19	19
SK Hynix	1	1	1	1	3	4	10	8	13	16	20	20	20
Micron	1			1	1	2	2	2	6	8	10	10	10
Total	**19**	**26**	**31**	**43**	**54**	**42**	**42**	**44**	**65**	**90**	**117**	**117**	**117**

0.55 NA EUV shipments

	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Customer								
TSMC		1	2	1	2	2	5	8
Samsung			1	1	2	5	7	7
Intel	1	2	2	2	1	2	2	2
SK Hynix			1	1	1	3	5	5
Micron								
Other		1			1			
Total	**1**	**4**	**6**	**5**	**7**	**7**	**13**	**18**

Source: Company data, UBS estimates

ASML – EUV Orders by Customer

0.33 NA EUV orders

	1Q22	2Q22	3Q22	4Q22	2022	1Q23	2Q23	3Q23	4Q23	2023	1Q24	2Q24	3Q24	4Q24	2024	1Q25	2Q25	3Q25	4Q25	2025
TSMC	2	5	11	8	26	4	3	1	2	10	1	10	5	7	23		7	8	9	24
Samsung Foundry		15	2	3	20	2	1		3	6					0			2	1	3
Samsung DRAM	1	1		4	6	1	1		1	3	1	1	2	2	6	2	1	3	9	15
Intel	1	5	6	2	14	1	1	1	7	10	1				1					
SK Hynix	1	1			2				14	14				5	5	3		2	12	17
Micron														1	1				1	1
Other																	1			
Total	**5**	**27**	**19**	**17**	**68**	**8**	**6**	**2**	**27**	**43**	**3**	**11**	**7**	**15**	**36**	**5**	**9**	**15**	**32**	**61**

0.55 NA EUV orders

	1Q22	2Q22	3Q22	4Q22	2022	1Q23	2Q23	3Q23	4Q23	2023	1Q24	2Q24	3Q24	4Q24	2024	1Q25	2Q25	3Q25	4Q25	2025
TSMC	1		1	1	3	1	3			4					0					0
Samsung Foundry	1		1	1	3	1	1			2					0					0
Samsung DRAM	1				1				2	2					0					0
Intel	1		1		2					0					0					0
SK Hynix	1				1					0					0					0
Micron					0					0					0					0
Other					0					0					0		1			1
Total	**5**	**0**	**3**	**2**	**10**	**2**	**4**	**0**	**2**	**8**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**1**

Source: Company data, UBS estimates

ASML – Unit / ASP Forecasts

	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q25E	2Q25E	3Q25E	4Q25E	2027E	2028E	2029E	2030E
Unit sales (units)														
i-line	48	9	9	17	14	49	15	15	15	15	60	72	75	79
KrF Dry	72	24	30	32	35	121	30	30	35	35	130	136	143	135
ArF Dry	14	4	8	9	15	36	4	8	10	10	33	41	43	45
ArF Immersion	119	14	22	37	50	123	35	40	40	45	160	200	150	160
EUV - 0.33 NA	44	14	15	16	20	65	20	22	24	24	90	117	117	117
EUV - 0.55 NA (High-NA)	4	2	1	1	1	5	1	2	2	1	6	12	19	22
New tools	**301**	**67**	**85**	**112**	**135**	**399**	**105**	**117**	**126**	**130**	**479**	**578**	**547**	**559**
Used tools	27	12	5	8	7	32	7	7	7	7	28	28	28	28
Product sales	**328**	**79**	**90**	**120**	**142**	**431**	**112**	**124**	**133**	**137**	**507**	**606**	**575**	**587**
ASP (EUR mn)														
Total ASP	**74.6**	**79.5**	**72.9**	**69.8**	**77.4**	**74.7**	**90.2**	**92.3**	**91.4**	**89.4**	**90.8**	**109.3**	**113.6**	**115.6**
% YoY Change	*45%*	*7%*	*-1%*	*-10%*	*5%*	*0%*	*13%*	*27%*	*31%*	*16%*	*22%*	*20%*	*4%*	*2%*
New tool ASP	**78.6**	**91.8**	**74.9**	**72.5**	**79.2**	**78.5**	**92.9**	**94.4**	**92.9**	**90.8**	**92.7**	**111.1**	**115.5**	**117.5**
% YoY Change	*43%*	*20%*	*-8%*	*-10%*	*3%*	*0%*	*1%*	*26%*	*28%*	*15%*	*18%*	*20%*	*4%*	*2%*

Source: Company data, UBS estimates

ASML – Financial Statement Forecasts

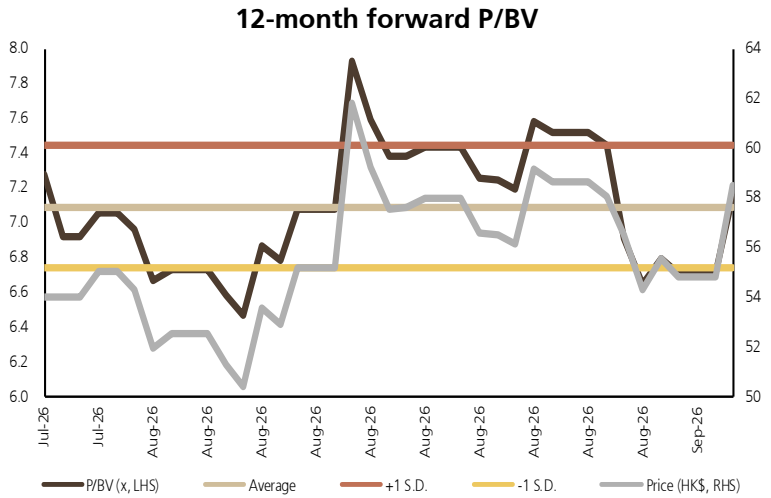
(EUR mn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues	**32,667**	**8,767**	**9,327**	**11,662**	**14,691**	**44,447**	**13,538**	**15,049**	**16,016**	**16,264**	**60,867**	**81,742**	**82,507**	**87,716**
% YoY	*15.6%*	*13.2%*	*21.3%*	*55.2%*	*51.2%*	*36.1%*	*54.4%*	*61.4%*	*37.3%*	*10.7%*	*36.9%*	*34.3%*	*0.9%*	*6.3%*
% QoQ		*-9.8%*	*6.4%*	*25.0%*	*26.0%*		*-7.9%*	*11.2%*	*6.4%*	*1.6%*				
Gross profit	17,298	4,655	5,045	6,710	8,586	24,997	7,644	8,584	9,362	9,622	35,213	48,808	49,678	53,226
% GP margin	*53.0%*	*53.1%*	*54.1%*	*57.5%*	*58.4%*	*56.2%*	*56.5%*	*57.0%*	*58.5%*	*59.2%*	*57.9%*	*59.7%*	*60.2%*	*60.7%*
Operating profit	**11,342**	**3,168**	**3,466**	**5,183**	**6,986**	**18,803**	**5,994**	**6,924**	**7,759**	**7,942**	**28,620**	**41,885**	**42,324**	**45,536**
% OP margin	*34.7%*	*36.1%*	*37.2%*	*44.4%*	*47.6%*	*42.3%*	*44.3%*	*46.0%*	*48.4%*	*48.8%*	*47.0%*	*51.2%*	*51.3%*	*51.9%*
Net incom adj.	9,426	2,660	2,885	4,834	5,824	16,203	5,060	5,842	6,543	6,697	24,141	35,284	35,653	38,351
EPS	**24.24**	**6.90**	**7.49**	**12.61**	**15.22**	**42.22**	**13.25**	**15.33**	**17.20**	**17.64**	**63.42**	**93.45**	**95.34**	**103.66**
% EPS growth YoY	*29.0%*	*16.2%*	*31.7%*	*137.9%*	*108.0%*	*74.1%*	*92.1%*	*104.5%*	*36.5%*	*15.9%*	*50.2%*	*47.3%*	*2.0%*	*8.7%*
DPS	7.50	2.70	1.88	1.88	3.51	9.97	2.97	1.69	1.69	3.86	10.22	14.43	16.47	18.92
% payout ratio	*30.9%*	*39.1%*	*25.1%*	*14.9%*	*23.1%*	*23.6%*	*22.4%*	*11.0%*	*9.8%*	*21.9%*	*16.1%*	*15.4%*	*17.3%*	*18.3%*
Cash and cash equiv	**12,916**	**7,970**	**6,672**	**8,031**	**17,221**	**17,221**	**21,398**	**25,700**	**28,587**	**38,375**	**38,375**	**72,567**	**103,323**	**136,240**
Total assets	**50,567**	**20,477**	**20,666**	**21,955**	**22,075**	**67,517**	**22,195**	**22,315**	**22,435**	**22,555**	**86,916**	**131,131**	**169,918**	**211,824**
Total equity	6,691	6,943	6,235	12,013	12,013	12,013	12,013	12,013	12,013	12,013	12,013	12,013	12,013	12,013
Net cash (debt)	10,613	5,671	5,597	10,772	19,962	19,962	24,139	28,441	31,328	41,116	41,116	75,308	106,064	138,981
CF from Operations	**12,659**	**-2,186**	**1,703**	**11,798**	**11,380**	**22,695**	**6,866**	**6,783**	**4,880**	**11,780**	**30,309**	**44,311**	**44,199**	**47,508**
Capex	-1,574	-402	-299	-413	-250	-1,365	-250	-250	-250	-250	-1,000	-1,000	-2,000	-2,400
Free cash flow	**11,085**	**-2,588**	**1,404**	**11,384**	**11,130**	**21,330**	**6,616**	**6,533**	**4,630**	**11,530**	**29,309**	**43,311**	**42,199**	**45,108**
CF from Investments	**-3,778**	**-488**	**-926**	**-4,687**	**-250**	**-6,350**	**-250**	**-250**	**-250**	**-250**	**-1,000**	**-1,000**	**-2,000**	**-2,400**
Purchase of shares	-5,950	-1,000	-1,078	-4,071	-1,221	-7,370	-1,100	-1,100	-1,100	-1,100	-4,400	-4,400	-5,280	-5,610
Dividend	-2,550	-617	-1,039	-721	-719	-3,095	-1,339	-1,131	-643	-642	-3,755	-4,719	-6,163	-6,580
CF from Finance	**-8,671**	**-2,275**	**-2,080**	**-5,756**	**-1,940**	**-12,050**	**-2,439**	**-2,231**	**-1,743**	**-1,742**	**-8,155**	**-9,119**	**-11,443**	**-12,190**

Source: Company data, UBS estimates

CXMT (Jimmy Yu, Nicolas Gaudois) – Buy, PT Rmb80

UBS price target (Rmb)	**80.0**
Current stock price (Rmb)	57.9
Current shares outstanding (bn)	67.9
Treasury / non-common shares (bn)	0.0
Market cap. (Rmb bn)	3,927.8
Debt (Rmb bn)	146.5
Cash & equivalents (Rmb bn)	72.6
Adjusted EV (Rmb bn)	4,001.6

	2026E	2027E	2028E	2029E	2030E
Current valuation					
EV/Revenues	9.76x	5.59x	4.27x	6.22x	6.19x
P/Book	12.11x	6.22x	3.94x	3.49x	3.05x
P/E (ex-treasury shares)	19.55x	10.67x	8.24x	13.34x	14.14x
P/E (including treasury shares)	18.44x	10.67x	8.24x	13.34x	14.14x
EV/EBITDA	10.83x	6.01x	4.57x	7.00x	7.21x
UBS valuation					
EV/Revenues	13.49x	7.72x	5.91x	8.61x	8.56x
P/Book	16.75x	8.59x	5.44x	4.82x	4.22x
P/E (ex-treasury shares)	27.03x	14.76x	11.40x	18.45x	19.55x
P/E (including treasury shares)	25.50x	14.76x	11.40x	18.45x	19.55x
EV/EBITDA	14.97x	8.31x	6.32x	9.68x	9.97x
Other					
% ROE	***105.5%***	***77.0%***	***58.5%***	***27.7%***	***23.0%***
Book value per share (Rmb)	4.78	9.31	14.70	16.58	18.94
Cash per share (Rmb)	5.13	10.08	17.16	19.50	22.51
Free cash flow per share (Rmb)	2.05	3.99	5.93	3.59	3.36
% Free cash flow yield	*3.5%*	*6.9%*	*10.3%*	*6.2%*	*5.8%*



Source: Company data, UBS estimates

CXMT – DRAM Forecast

	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
DRAM revenues (Rmbm)	**7,891**	**8,583**	**22,972**	**60,235**	**403,563**	**709,348**	**929,305**	**635,729**	**638,898**
% YoY		*9.7%*	*166.1%*	*155.6%*	*563.6%*	*74.7%*	*30.7%*	*-31.4%*	*0.5%*
DRAM unit shipment (1Gb eq m)	4,939	7,384	12,470	20,834	34,038	48,207	61,366	71,200	80,340
% YoY		*49.5%*	*68.9%*	*67.1%*	*63.4%*	*41.6%*	*27.3%*	*16.0%*	*12.8%*
12-inch wafers out (k wafers)	105	144	186	233	278	351	431	523	601
% YoY		*36.9%*	*29.6%*	*24.8%*	*19.6%*	*26.3%*	*22.8%*	*21.2%*	*14.9%*
DRAM ASP (US$ 1Gb eq m)	0.24	0.16	0.26	0.40	1.74	2.15	2.18	1.25	1.08
% YoY		*-30.6%*	*56.8%*	*56.1%*	*330.3%*	*23.5%*	*1.5%*	*-42.5%*	*-13.4%*
Gross profit (Rmbm)				**22,124**	**352,491**	**639,221**	**837,991**	**519,411**	**491,798**
% GP margin				***36.7%***	***87.3%***	***90.1%***	***90.2%***	***81.7%***	***77.0%***
Manufacturing Cost/Unit (1Gb US$)	0.31	0.35	0.26	0.24	0.21	0.20	0.20	0.22	0.24
% YoY		*12.4%*	*-24.2%*	*-8.2%*	*-13.4%*	*-3.2%*	*0.9%*	*7.0%*	*8.9%*
DDR revenues (Rmbm)	**7,891**	**8,583**	**22,972**	**59,744**	**402,381**	**703,016**	**905,200**	**569,002**	**505,275**
% YoY		*8.8%*	*167.6%*	*160.1%*	*573.5%*	*74.7%*	*28.8%*	*-37.1%*	*-11.2%*
DDR unit shipment (1Gb eq m)	4,939	7,384	12,470	20,834	34,038	48,207	61,366	71,200	80,340
% YoY		*49.5%*	*68.9%*	*67.1%*	*63.4%*	*41.6%*	*27.3%*	*16.0%*	*12.8%*
DDR ASP (US$ 1Gb eq m)	0.24	0.16	0.26	0.40	1.74	2.15	2.18	1.25	1.08
% YoY		*-30.6%*	*56.8%*	*56.1%*	*330.3%*	*23.5%*	*1.5%*	*-42.5%*	*-13.4%*
HBM revenues (Rmbm)				**491**	**1,182**	**6,332**	**24,105**	**66,727**	**133,623**
% YoY					*140.6%*	*435.9%*	*280.7%*	*176.8%*	*100.3%*
HBM unit shipment (1Gb eq m)				53	122	403	1,364	3,473	6,343
% YoY					*130.1%*	*231.5%*	*238.0%*	*154.7%*	*82.6%*
HBM ASP (US$ 1Gb eq m)				1.30	1.43	2.31	2.60	2.83	3.10
% YoY					*9.8%*	*61.7%*	*12.6%*	*8.7%*	*9.7%*

Source: Company data, UBS estimates

CXMT – Capex/Financial Statement Forecast

Capex Forecast

	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Semis Capex (US$m)	4,313	7,350	10,340	4,604	12,523	17,859	16,199	16,551	16,862
% YoY		*70.4%*	*40.7%*	*-55.5%*	*172.0%*	*42.6%*	*-9.3%*	*2.2%*	*1.9%*
New capacity added (k wpm)	**45**	**45**	**35**	**62**	**90**	**80**	**80**	**80**	
% YoY		*0.0%*	*-22.2%*	*77.1%*	*44.4%*	*-11.2%*	*0.6%*	*0.0%*	
Implied capex per wafer capacity added (US$ m/k wafer)	163	230	132	202	200	204	207	211	
% YoY			41%	-43%	54%	-1%	2%	2%	2%

Financial Statement Forecast

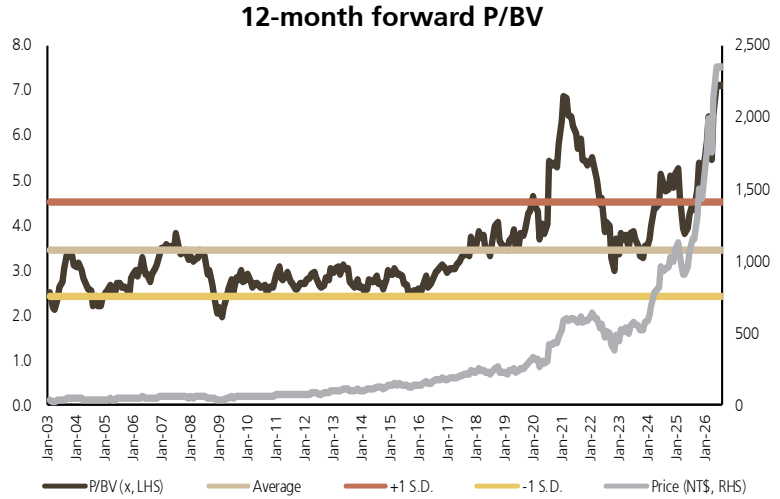
(Won bn)	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Revenues	**8,287**	**9,087**	**24,178**	**61,799**	**410,094**	**716,451**	**937,081**	**642,832**	**646,000**
Gross profit	-259	-175	1,349	25,334	358,560	645,861	845,345	526,091	498,478
% GP margin	*-3.1%*	*-1.9%*	*5.6%*	*41.0%*	*87.4%*	*90.1%*	*90.2%*	*81.8%*	*77.2%*
Operating profit	**-8,191**	**-19,028**	**-8,488**	**7,667**	**334,434**	**616,483**	**812,448**	**493,108**	**464,886**
% OP margin	*-98.8%*	*-209.4%*	*-35.1%*	*12.4%*	*81.6%*	*86.0%*	*86.7%*	*76.7%*	*72.0%*
NI post MI	-8,626	-16,752	-7,870	5,316	206,307	367,581	476,562	294,017	277,425
EPS	**-0.30**	**-0.13**	**0.03**	**3.14**	**5.42**	**7.03**	**4.34**	**4.09**	
% EPS growth YoY				*9553.4%*	*72.8%*	*29.6%*	*-38.3%*	*-24.5%*	
DPS				0.94	1.63	2.46	1.73	1.64	
% payout ratio				*30.0%*	*30.0%*	*35.0%*	*40.0%*	*40.0%*	
Cash and cash equiv	**32,072**	**31,374**	**42,699**	**51,990**	**327,400**	**663,394**	**1,144,510**	**1,303,539**	**1,507,873**
Total assets	**147,694**	**192,756**	**271,599**	**336,785**	**689,428**	**1,130,739**	**1,694,131**	**1,909,819**	**2,173,427**
Total equity	64,216	65,113	104,257	154,100	524,076	981,127	1,570,380	1,814,727	2,083,466
Net cash (debt)	-36,951	-69,766	-86,390	-94,479	203,400	561,894	1,075,510	1,267,039	1,483,873
% ROE	*-16.8%*	*-38.3%*	*-18.6%*	*3.8%*	*105.5%*	*77.0%*	*58.5%*	*27.7%*	*23.0%*
Depreciation	**5,243**	**10,555**	**14,875**	**24,680**	**33,700**	**47,004**	**58,899**	**70,717**	**81,423**
CF from Operations	**-1,957**	**-7,272**	**6,897**	**36,520**	**323,202**	**601,288**	**798,309**	**531,407**	**509,726**
Capex	35,431	43,658	71,230	49,739	85,158	121,443	110,159	112,553	114,663
CF from Investments	**-26,949**	**-44,002**	**-72,415**	**-86,823**	**-91,930**	**-132,520**	**-124,307**	**-122,950**	**-125,135**
Dividend						-60,274	-110,386	-166,928	-117,757
CF from Finance	**38,971**	**50,807**	**77,214**	**46,121**	**44,138**	**-132,774**	**-192,886**	**-249,428**	**-180,257**
Net changes of cash	**10,176**	**-669**	**11,396**	**-4,710**	**275,409**	**335,994**	**481,116**	**159,029**	**204,334**

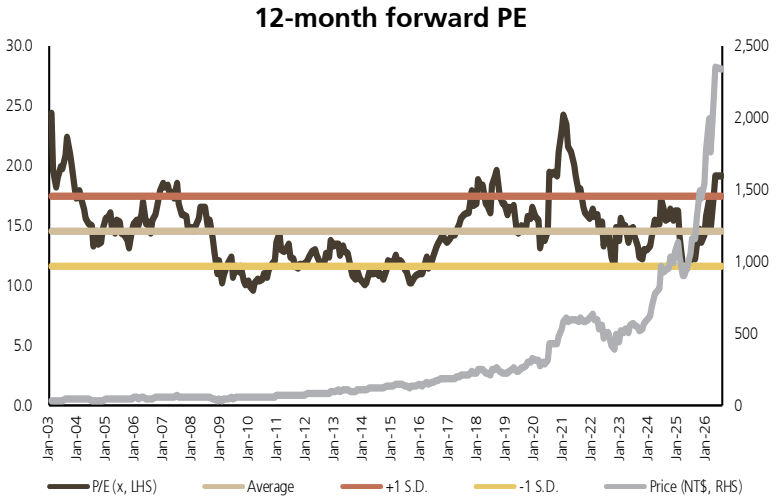
Source: Company data, UBS estimates

TSMC (Sunny Lin, Randy Abrams, Nicolas Gaudois) – Buy PT NT$3,650

Price target (TWD)	**3,650**
Current stock price (TWD)	2,440
Number of shares (bn)	25.9
Market cap. (TWDbn)	63,265
Net debt (TWDbn)	-1,514
EV (TWDbn)	61,751

	2025	2026E	2027E	2028E	2029E	2030E
Current valuation						
EV/Revenues	16.21	11.00	7.80	5.84	4.77	4.02
P/Book	11.59	8.20	5.92	4.29	3.21	2.46
P/E	36.83	21.78	16.03	11.86	9.59	8.02
EV/EBITDA	23.94	15.18	10.92	8.09	6.54	5.49
UBS valuation						
EV/Revenues	24.45	16.59	11.76	8.80	7.19	6.06
P/Book	17.33	12.27	8.86	6.41	4.81	3.69
P/E	55.09	32.58	23.98	17.73	14.35	12.00
EV/EBITDA	36.10	22.89	16.47	12.21	9.86	8.28
Other						
% ROE	***35.1%***	***44.1%***	***42.9%***	***42.0%***	***38.3%***	***34.8%***
Tangible book per share (TWD)	209.6	296.5	411.2	568.1	758.4	989.5
Cash per share (TWD)	106.8	133.2	172.6	240.4	360.6	529.6
Free cash flow per share (TWD)	38.70	41.90	67.49	106.75	173.90	231.94
% Free cash flow yield	*1.6%*	*1.7%*	*2.8%*	*4.4%*	*7.1%*	*9.5%*





Source: Company data, UBS estimates

TSMC – UBSe vs Consensus / Key Charts

UBSe vs Consensus

	UBSe			Consensus			Difference		
(NT$m)	**2026E**	**2027E**	**2028E**	**2026F**	**2027F**	**2028F**	**2026E**	**2027E**	**2028E**
Revenue	**5,614,326**	**7,921,336**	**10,578,173**	**5,473,474**	**7,565,728**	**9,527,866**	3%	5%	11%
- YoY chg	47%	41%	34%	44%	38%	26%			
Gross profit	**3,736,635**	**5,279,963**	**7,098,535**	**3,648,290**	**5,075,511**	**6,368,013**	2%	4%	11%
- Gross margin	66.6%	66.7%	67.1%	66.7%	67.1%	66.8%			
Operating profit	**3,308,636**	**4,649,402**	**6,275,612**	**3,210,186**	**4,484,151**	**5,639,038**	3%	4%	11%
- Operating margin	58.9%	58.7%	59.3%	58.6%	59.3%	59.2%			
Net profit	**2,905,391**	**3,947,441**	**5,336,884**	**2,803,055**	**3,838,392**	**4,855,556**	4%	3%	10%
- Net margin	51.7%	49.8%	50.5%	51.2%	50.7%	51.0%			
EPS (NT$)	**112.05**	**152.23**	**205.81**	**108.02**	**147.67**	**186.54**	4%	3%	10%
- YoY chg	69%	36%	35%	63%	37%	26%			

Key operating metrics

	Q126	Q226	Q326E	Q426E	Q127E	Q227E	Q327E	Q427E	2023	2024	2025	2026E	2027E	2028E
Capacity (WPQ/Y, kpcs 8" eq.)	**10,495**	**10,663**	**10,663**	**10,798**	**10,859**	**11,095**	**11,196**	**11,331**	**36,849**	**38,233**	**40,528**	**42,620**	**44,480**	**47,349**
Sequential change (%)	*0%*	*2%*	*0%*	*1%*	*1%*	*2%*	*1%*	*1%*	*6%*	*4%*	*6%*	*5%*	*4%*	*6%*
Shipment (WPQ/Y, kpcs 8" eq.)	**9,392**	**9,756**	**10,071**	**11,400**	**11,438**	**11,753**	**12,211**	**12,284**	**27,005**	**29,050**	**33,802**	**40,618**	**47,685**	**54,226**
Sequential change (%)	*5%*	*4%*	*3%*	*13%*	*0%*	*3%*	*4%*	*1%*	*-21%*	*8%*	*16%*	*20%*	*17%*	*14%*
Utilization rate (%)	**89%**	**91%**	**94%**	**106%**	**105%**	**106%**	**109%**	**108%**	**73%**	**76%**	**83%**	**95%**	**107%**	**115%**
ASP (US$)	**3,263**	**3,483**	**3,931**	**4,011**	**4,117**	**4,287**	**4,498**	**4,739**	**2,235**	**2,694**	**3,112**	**3,691**	**4,417**	**5,177**
Sequential change (%)	*0%*	*7%*	*13%*	*2%*	*3%*	*4%*	*5%*	*5%*	*14.8%*	*20.5%*	*15.5%*	*18.6%*	*19.7%*	*17.2%*
FX rate (NT$)	31.6	31.6	32.0	31.8	31.8	31.8	31.8	31.8	31.2	32.1	31.1	31.8	31.8	31.8
Other Revenue (NT$m)	165,991	196,673	228,332	261,702	269,553	288,421	323,112	342,499	279,218	379,846	536,501	852,698	1,223,585	1,651,445
Total Revenue (NT$m)	**1,134,103**	**1,270,380**	**1,494,243**	**1,715,599**	**1,767,067**	**1,890,762**	**2,069,663**	**2,193,843**	**2,161,736**	**2,894,308**	**3,809,054**	**5,614,326**	**7,921,336**	**10,578,173**
Sequential change (%)	*8%*	*12%*	*18%*	*15%*	*3%*	*7%*	*9%*	*6%*	*-5%*	*34%*	*32%*	*47%*	*41%*	*34%*
Capex (US$m)	**11,046**	**15,691**	**13,847**	**22,397**	**22,500**	**22,500**	**22,500**	**22,500**	**30,425**	**29,760**	**40,739**	**63,000**	**90,000**	**105,000**
Capex/revenue (%)	*31%*	*39%*	*30%*	*42%*	*40%*	*38%*	*35%*	*33%*	*44%*	*33%*	*33%*	*36%*	*36%*	*32%*
Depreciation (NT$mn)	**5,237**	**6,283**	**6,886**	**7,162**	**7,520**	**8,225**	**8,861**	**9,397**	**16,736**	**20,843**	**21,638**	**25,533**	**34,004**	**45,740**
Gross margin (%)	**66.2%**	**67.7%**	**66.4%**	**66.0%**	**66.4%**	**67.0%**	**66.5%**	**66.6%**	**54.4%**	**56.1%**	**59.9%**	**66.6%**	**66.7%**	**67.1%**
Operating margin (%)	58.1%	60.3%	58.7%	58.7%	58.8%	59.2%	58.4%	58.5%	42.6%	45.7%	50.8%	58.9%	58.7%	59%

Source: Company data, Visible Alpha, UBS estimates

TSMC – Revenue from Key Customers

TSMC Revenue	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2023	2024	2025	2026E	2027E
Apple	$6,906	$8,158	$11,928	$15,156	$17,753	$17,521	$19,393	$20,926	$28,712	$35,077	25.3%	21.5%	17.1%	16.7%	14.6%
AMD	$1,302	$1,232	$3,148	$5,562	$7,572	$7,750	$8,336	$11,523	$16,796	$31,700	11.2%	9.2%	9.4%	9.8%	13.2%
Qualcomm	$1,915	$1,994	$3,187	$4,280	$6,574	$6,314	$7,910	$9,448	$8,880	$9,506	9.1%	8.8%	7.7%	5.2%	3.9%
NVIDIA	$2,302	$2,024	$2,802	$2,987	$3,144	$6,097	$10,865	$22,973	$35,008	$56,010	8.8%	12.0%	18.8%	20.3%	23.3%
Mediatek	$1,423	$1,424	$2,683	$5,114	$3,962	$3,379	$4,412	$5,166	$6,156	$12,996	4.9%	4.9%	4.2%	3.6%	5.4%
Broadcom	$3,314	$2,517	$2,686	$3,770	$4,919	$3,719	$4,024	$5,281	$13,762	$24,472	5.4%	4.5%	4.3%	8.0%	10.2%
Intel	$1,092	$1,120	$1,792	$2,453	$3,110	$3,274	$7,615	$9,489	$9,964	$7,640	4.7%	8.4%	7.8%	5.8%	3.2%
Marvell	$544	$514	$568	$970	$1,293	$1,362	$1,505	$2,333	$3,536	$5,640	2.0%	1.7%	1.9%	2.1%	2.3%
Renesas / Intersil / Dialog	$267	$229	$267	$1,087	$1,292	$1,090	$1,021	$1,069	$1,445	$1,601	1.6%	1.1%	0.9%	0.8%	0.7%
NXP	$700	$526	$569	$911	$1,167	$1,158	$1,149	$1,273	$1,435	$1,555	1.7%	1.3%	1.0%	0.8%	0.6%
TSMC sales from its top 10	**$20,600**	**$19,836**	**$29,013**	**$41,361**	**$52,780**	**$51,663**	**$66,229**	**$89,481**	**$125,694**	**$186,196**	**75%**	**73%**	**73%**	**73%**	**77%**
TSMC YoY at its top 10	**0%**	**-4%**	**46%**	**43%**	**28%**	**-2%**	**28%**	**35%**	**40%**	**48%**					
TSMC sales from its top 30	**$31,048**	**$31,220**	**$41,181**	**$50,572**	**$61,790**	**$60,194**	**$74,555**	**$99,095**	**$136,485**	**$198,516**	**87%**	**83%**	**81%**	**79%**	**82%**
TSMC YoY at its top 30	**7%**	**1%**	**32%**	**23%**	**22%**	**-3%**	**24%**	**33%**	**38%**	**45%**					
Rest of TSMC/Inventory	$2,404	$3,838	$4,240	$6,235	$14,261	$9,104	$15,629	$22,936	$35,727	$42,179	**13%**	**17%**	**19%**	**21%**	**18%**
Rest of TSMC YoY	**-31%**	**60%**	**10%**	**47%**	**129%**	**-36%**	**72%**	**47%**	**56%**	**18%**					
TSMC Sales (UBS Estimate)	**$33,452**	**$35,059**	**$45,422**	**$56,808**	**$76,051**	**$69,298**	**$90,184**	**$122,031**	**$172,213**	**$240,695**	**100%**	**100%**	**100%**	**100%**	**100%**
TSMC Overall YoY	**3%**	**5%**	**30%**	**25%**	**34%**	**-9%**	**30%**	**35%**	**41%**	**40%**					

Source: Company data, Visible Alpha, UBS estimates

TSMC – Financial Statement Forecasts

(TWD bn)	2025	1Q26	2Q26	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Revenues	**3,809**	**1,134**	**1,270**	**1,494**	**1,716**	**5,614**	**1,767**	**1,891**	**2,070**	**2,194**	**7,921**	**10,578**	**12,948**	**15,358**
% YoY	*31.6%*	*35.1%*	*36.0%*	*50.9%*	*64.0%*	*47.4%*	*48.8%*	*38.5%*	*27.9%*	*27.9%*	*41.1%*	*33.5%*	*22.4%*	*18.6%*
% QoQ		*8.4%*	*12.0%*	*17.6%*	*14.8%*		*7.0%*	*9.5%*	*6.0%*	*3.0%*				
Gross profit	2,281	751	860	992	1,133	3,737	1,174	1,267	1,377	1,462	5,280	7,099	8,678	10,285
% GP margin	*59.9%*	*66.2%*	*67.7%*	*66.4%*	*66.0%*	*66.6%*	*66.4%*	*67.0%*	*66.5%*	*66.6%*	*66.7%*	*67.1%*	*67.0%*	*67.0%*
Operating profit	**1,936**	**659**	**767**	**876**	**1,007**	**3,309**	**1,039**	**1,119**	**1,208**	**1,284**	**4,649**	**6,276**	**7,699**	**9,110**
% OP margin	*50.8%*	*58.1%*	*60.3%*	*58.7%*	*58.7%*	*58.9%*	*58.8%*	*59.2%*	*58.4%*	*58.5%*	*58.7%*	*59.3%*	*59.5%*	*59.3%*
Net income Adj.	1,715	573	707	763	864	2,906	889	928	1,033	1,098	3,949	5,338	6,597	7,886
EPS	**66.26**	**22.08**	**27.25**	**29.43**	**33.29**	**112.05**	**34.28**	**35.79**	**39.82**	**42.35**	**152.23**	**205.81**	**254.36**	**304.08**
% EPS growth YoY	*46.4%*	*58.3%*	*77.4%*	*68.7%*	*70.7%*	*69.1%*	*55.2%*	*31.3%*	*35.3%*	*27.2%*	*35.9%*	*35.2%*	*23.6%*	*19.5%*
DPS	22.00	7.00	7.00	7.50	8.00	29.50	8.50	10.00	11.00	11.00	40.50	53.00	66.00	76.00
% payout ratio	*33.2%*	*31.7%*	*25.7%*	*25.5%*	*24.0%*	*26.3%*	*24.8%*	*27.9%*	*27.6%*	*26.0%*	*26.6%*	*25.8%*	*25.9%*	*25.0%*
Cash and cash equiv	**2,768**	**3,036**	**3,134**	**3,622**	**3,452**	**3,452**	**3,669**	**3,870**	**4,133**	**4,475**	**4,475**	**6,234**	**9,349**	**13,731**
Total assets	**7,933**	**8,661**	**9,376**	**10,053**	**10,808**	**10,808**	**11,517**	**12,281**	**13,157**	**14,068**	**14,068**	**18,460**	**23,711**	**30,024**
Total equity	5,461	5,932	6,474	7,043	7,712	7,712	8,393	9,101	9,874	10,687	10,687	14,753	19,689	25,681
Net cash (debt)	-1,775	-2,019	-2,152	-2,638	-2,469	-2,469	-2,696	-2,897	-3,159	-3,502	-3,502	-5,271	-8,396	-12,787
% ROCE	***28.0%***	***27.3%***	***23.6%***	***23.0%***	***19.1%***	***19.1%***	***17.3%***	***15.8%***	***14.5%***	***13.5%***	***13.5%***	***10.0%***	***8.3%***	***7.1%***
CF from Operations	**2,275**	**699**	**783**	**920**	**684**	**3,086**	**1,114**	**1,089**	**1,153**	**1,256**	**4,612**	**6,107**	**8,007**	**9,512**
Capex	-1,272	-349	-496	-443	-712	-2,000	-716	-716	-716	-716	-2,862	-3,339	-3,498	-3,498
Free cash flow	**1,003**	**350**	**288**	**477**	**-28**	**1,086**	**399**	**373**	**437**	**541**	**1,750**	**2,768**	**4,509**	**6,014**
CF from Investments	**-1,144**	**-357**	**-493**	**-267**	**-712**	**-1,829**	**-716**	**-716**	**-716**	**-716**	**-2,862**	**-3,339**	**-3,498**	**-3,498**
Dividend	-467	-130	-156	-156	-182	-622	-182	-194	-207	-220	-804	-1,141	-1,478	-1,763
CF from Finance	**-440**	**-120**	**-185**	**-160**	**-141**	**-606**	**-182**	**-172**	**-175**	**-198**	**-727**	**-1,009**	**-1,394**	**-1,632**
Net changes of cash	**690**	**222**	**106**	**493**	**-169**	**651**	**217**	**201**	**262**	**343**	**1,023**	**1,759**	**3,115**	**4,382**

Source: Company data, UBS estimates

Accton (Jerry Su) – Buy, PT NT$3,688

- **Leading beneficiary of AI networking upgrades:** Accton is a key supplier of cloud switches and AI networking systems to major hyperscalers and social-media customers. The ongoing ramp of Trainium 3 and 800G cloud switching should support continued growth, while the transition toward 1.6T networking, switch trays and optical circuit switches expands its addressable content. Component constraints continue to limit near-term fulfilment, but underlying customer demand remains robust.
- **Content growth supports longer-term earnings expansion:** We expect Trainium 3 ASPs to increase further from Q426 into 2027E as selected designs adopt liquid cooling. Accton's content opportunity should rise again with Trainium 4 from its more complex board design, further liquid-cooling adoption and potential CPO/NPO integration. These technology transitions increase both system complexity and Accton's value-add beyond traditional switch assembly.
- **Well positioned for AI cluster scaling:** Accton is working with a major social-media customer on Nvidia Spectrum-X networking platforms, with potential revenue contribution from 2027E. Continued migration to 800G/1.6T, broader switch-tray supply and the introduction of OCS should reinforce Accton's position as a key beneficiary of rising networking intensity within AI clusters.

Accton – Key Financials

(NT$M)		2025	2026E	2027E	2028E
Revenue		**248,320**	**414,336**	**570,275**	**774,038**
	%YoY	125%	67%	38%	36%
Gross Profit		**44,886**	**78,630**	**107,657**	**147,312**
	GP margin %	18%	19%	19%	19%
Operating Profit		**32,135**	**60,477**	**84,373**	**117,624**
	OP margin %	13%	15%	15%	15%
Net Profit		**26,304**	**48,670**	**67,404**	**94,759**
	NP margin %	11%	12%	12%	12%
EPS (NT$)		47.1	87.1	120.6	169.5

Accton – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	**17.9**	**22.6**	**16.3**	**11.6**
P/B (x)	**8.2**	**12.2**	**8.6**	**6.1**
EV/EBITDA (x)	12.9	16.5	11.6	8.1
DPS (NT$)	11.0	15.0	52.2	78.4
Dividend Yield (Net) (%)	1%	1%	3%	4%
Dividend Payout Ratio (%)	23%	17%	43%	46%
Equity FCF Yield (%)	6%	2%	5%	6%

Source: Company data, UBS estimates

Advantest (Atsuhiro Kinoshita) – Buy, PT ¥43,600

- **Highly competitive in logic testers:** Advantest is set to strongly benefit from upside in server CPU and AI ASIC volumes, combined with lengthening testing cycle times.
- **High logic exposure provides leverage to packaging complexity:** Advantest has very high exposure to logic semiconductors, at more than 70%. In addition to Known Good Die screening at the wafer stage, integrated verification and system-level testing after packaging are becoming increasingly complex. Testing cycle times for advanced packaging sever CPU like AMD Venice could converge with AI GPUs. In addition, AI ASICs testing cycle times are also getting closer to those of AI GPUs.
- **Market share concerns overdone:** Market repeatedly focuses on concerns about share erosion by Teradyne, but at present there is still no clear evidence of significant market share loss. Advantest may experience some share loss for HBM and TPU, gain for Trainium, stead for Nvidia. Should Intel move more widely to external testers, this could provide further upside.

Advantest – Key Financials

(¥ bn)		2025	2026E	2027E	2028E
Revenue		**780**	**955**	**1,714**	**2,162**
	%YoY	60%	22%	79%	26%
Gross Profit		**445**	**606**	**1,151**	**1,446**
	GP margin %	57%	63%	67%	67%
Operating Profit		**228**	**390**	**756**	**919**
	OP margin %	29%	41%	44%	42%
Net Profit		**161**	**290**	**570**	**690**
	NP margin %	21%	30%	33%	32%
EPS (¥)		219.7	399.0	787.9	955.8

Advantest – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	**33.1**	**38.8**	**49.0**	**40.4**
P/B (x)	**10.5**	**16.8**	**29.3**	**19.5**
EV/EBITDA (x)	20.8	26.7	35.1	29.2
DPS (¥)	39.0	79.8	116.9	141.3
Dividend Yield (Net) (%)	1%	1%	0%	0%
Dividend Payout Ratio (%)	18%	20%	15%	15%
Equity FCF Yield (%)	5%	2%	1%	2%

112

Source: Company data, UBS estimates

AEM Holdings (Joseph Koh) – Buy, PT S$16

- **Expected to continue to gain share:** Key is the potential adoption of AEM's test platforms for more test insertions by its key HPC customers. It retains 100% share at Intel for customized test handlers for its high density modular test platform (system level test, burn-in). AEM is supporting IFS. Main risk would be Intel externalizing and hence, Advantest. In addition, AEM has secured 66% share at AMD for configurables, consumables, as well as handlers (both server CPU and AI GPUs).
- **Stand to benefit from increased test complexity:** Decreased test coverage from final test is driving portion of test content into alternative insertions. AEM offers solutions for those insertions through its high parallel test solution.
- **Strong near-term outlook:** AEM raised its 2026E full-year revenue guidance range by ~15% to S$630-680m, implying a 55-75% HoH revenue ramp in 2H26 and a 2026 full-year revenue growth of 58-70% YoY. The company also introduced fresh 2026 EPS guidance of S$24.5-27.5cents, implying a 50-81% HoH EPS growth. AEM flagged that its guidance upgrade is backed by solid order backlog which has now exceeded S$400m, giving it good topline visibility at least until end-2026 in our view, and the company disclosed that it is seeing incremental demand from both of its key HPC customers.

AEM Holdings – Key Financials

(S$m)		2025	2026E	2027E	2028E
Revenue		**399**	**641**	**950**	**1,193**
	%YoY	5%	60%	48%	26%
Gross Profit		**103**	**217**	**342**	**438**
	GP margin %	26%	34%	36%	37%
Operating Profit		**22**	**103**	**176**	**232**
	OP margin %	6%	16%	19%	19%
Net Profit		**17**	**83**	**142**	**187**
	NP margin %	4%	13%	15%	16%
EPS (S$)		0.1	0.3	0.4	0.6

AEM Holdings – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	**28.3**	**39.3**	**22.9**	**17.4**
P/B (x)	**1.0**	**5.6**	**4.6**	**3.8**
EV/EBITDA (x)	9.7	24.2	15.4	11.8
DPS (S$)	0.0	0.1	0.1	0.1
Dividend Yield (Net) (%)	1%	1%	1%	1%
Dividend Payout Ratio (%)	24%	22%	20%	20%
Equity FCF Yield (%)	23%	4%	1%	3%

113

Source: Company data, UBS estimates

All Ring (Sunny Lin) – Buy, PT NT$1,700

- **Dominant positioning in critical CoWoS process steps:** All Ring remains a key enabler of advanced packaging through its leadership in underfill, heat sink attachment and inspection solutions. The company maintains 100% market share in substrate underfill and SOIC-related underfill processes, while continuing to gain share from incumbents. We believe its proprietary dispensing know-how and self-developed pump technology create meaningful barriers to entry.
- **Content gains from larger AI die and packaging complexity:** The shift toward larger AI accelerators is increasing equipment intensity in advanced packaging. Increasing die size on Blackwell, Rubin and ASIC platforms is reducing throughput and driving incremental equipment demand. All Ring is particularly well positioned in graphite-based TIM and heat sink attachment solutions, where Rubin and Blackwell adoption is creating new content opportunities.
- **CPO remains a meaningful long-term growth driver:** All Ring continues to expand its footprint in CPO, with products already shipping for coupling, protection, cleaning and inspection processes. Engagement with multiple customers includes a major US advanced packaging customer. While industry adoption has been pushed out slightly due to throughput constraints, we believe improving production readiness and rising optical packaging complexity should support meaningful revenue contribution beginning in 2027-28E.

All Ring – Key Financials

(NT$M)		2025	2026E	2027E	2028E
Revenue		**5,366**	**9,385**	**14,954**	**23,493**
	%YoY	-3%	75%	59%	57%
Gross Profit		**2,913**	**5,053**	**8,050**	**12,804**
	GP margin %	54%	54%	54%	55%
Operating Profit		**1,611**	**3,331**	**5,732**	**9,444**
	OP margin %	30%	35%	38%	40%
Net Profit		**1,485**	**2,950**	**4,643**	**7,615**
	NP margin %	28%	31%	31%	32%
EPS (NT$)		15.3	29.6	46.5	76.3

All Ring – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	**23.0**	**43.1**	**27.4**	**16.7**
P/B (x)	**4.7**	**13.3**	**10.4**	**7.6**
EV/EBITDA (x)	20.2	36.5	21.3	12.8
DPS (NT$)	10.7	21.2	33.4	54.8
Dividend Yield (Net) (%)	3%	2%	3%	4%
Dividend Payout Ratio (%)	70%	70%	70%	70%
Equity FCF Yield (%)	4%	2%	2%	4%

114

Source: Company data, UBS estimates

ASE (Sunny Lin) – Buy, PT NT$835

- **Stronger IC ATM outlook across LEAP and mainstream:** ASE guided IC ATM sales to grow 35% in 2026, above our pre-earnings estimate of 29%. Packaging is showing robust upside, which we believe is lifted by both LEAP and mainstream demand, and management is now guiding mainstream to grow in line with testing in 2026.
- **1) LEAP sales guidance is raised to US$3.7bn in 2026, and to more than double in 2027**. We forecast ASE's LEAP sales to grow to US$7.5bn in 2027, of which US$2.5bn is from full process CoWoS. We forecast ASE CoWoS capacity to reach 70k wpm end of 2027, from 20k end of 2026.
- **2) Mainstream packaging is also recovering** toward tight utilisation in Q2-Q3. We believe ASE may raise blended ASP of wirebond and flip chip by at least 10% in H226.
- **Faster margin expansion in 2027-28:** ASE raised guidance for IC ATM GM to exceed 30% in Q426 vs prior guidance of 29-30%. The stronger ramp of LEAP especially the higher margin full process CoWoS service remains a key driver. We estimate LEAP as a whole GM may reach close to 40% in 2027. Meanwhile, improving mainstream business with favorable pricing should further fuel the margin improvement. We forecast IC ATM gross margin to reach 33.8% in 2027E and 34.6% in 2028E from 28.4% in 2026E.

ASE – Key Financials

(NT$M)		2025	2026E	2027E	2028E
Revenue		**645,388**	**836,080**	**1,072,054**	**1,286,842**
	%YoY	8%	30%	28%	20%
Gross Profit		**114,193**	**178,550**	**270,247**	**335,761**
	GP margin %	18%	21%	25%	26%
Operating Profit		**50,756**	**99,036**	**176,530**	**225,665**
	OP margin %	8%	12%	16%	18%
Net Profit		**40,658**	**83,081**	**136,200**	**170,487**
	NP margin %	6%	10%	13%	13%
EPS (NT$)		9.2	18.5	30.2	37.8

ASE – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	**18.5**	**38.6**	**23.6**	**18.8**
P/B (x)	**2.2**	**7.2**	**6.0**	**5.2**
EV/EBITDA (x)	7.2	17.7	10.6	8.3
DPS (NT$)	6.1	12.3	20.2	25.3
Dividend Yield (Net) (%)	4%	2%	3%	4%
Dividend Payout Ratio (%)	65%	65%	65%	65%
Equity FCF Yield (%)	-3%	-7%	-4%	-2%

115

Source: Company data, UBS estimates

Delta Electronics (Randy Abrams) – Key Call Buy, PT NT$2,750

- **Leading supplier in AI power infrastructure:** Delta is a key beneficiary of AI infrastructure demand through its grid-to-chip portfolio across server power, distribution and liquid cooling. Our bottom-up model estimates a 69% CAGR in AI server power TAM to US$91.1bn in 2030E, with Delta positioned to sustain approximately 40% share.
- **Power architecture transition drives content expansion:** Rack power rises from approximately 100kW for Blackwell to around 250kW for Vera Rubin, making traditional low-voltage distribution increasingly inefficient. The transition toward 400V and eventually 800V HVDC should materially increase Delta's content opportunity, particularly for Rubin Ultra and subsequent platforms.
- **Demand remains ahead of supply:** Power-semiconductor constraints continue to limit shipments, but underlying demand remains robust and capacity investment is accelerating. Delta raised capex by 40% to NT$70bn, supporting continued growth as accelerator volumes and rack-level power density rise.

Delta Electronics – Key Financials

(NT$m)		2025	2026E	2027E	2028E
Revenue		**554,885**	**782,854**	**1,098,155**	**1,419,750**
	%YoY	32%	41%	40%	29%
Gross Profit		**190,157**	**284,489**	**408,437**	**529,624**
	GP margin %	34%	36%	37%	37%
Operating Profit		**83,932**	**144,948**	**231,901**	**306,335**
	OP margin %	15%	19%	21%	22%
Net Profit		**59,806**	**107,420**	**175,341**	**233,771**
	NP margin %	11%	14%	16%	16%
EPS (Rmb)		23.0	41.4	67.5	90.0

Delta Electronics – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	26.3	45.6	27.9	20.9
P/B (x)	5.9	13.5	9.8	7.4
EV/EBITDA (x)	14.1	27.7	17.9	13.4
DPS (Rmb)	7.0	11.6	21.2	33.8
Dividend Yield (Net) (%)	1%	1%	1%	2%
Dividend Payout Ratio (%)	30%	28%	31%	38%
Equity FCF Yield (%)	3%	1%	2%	4%

116

Source: Company data, UBS estimates

Hon Hai Precision (Randy Abrams) – Buy, PT NT$340

- **Expanding role in AI infrastructure:** Hon Hai remains one of the leading AI rack suppliers and continues to deepen its engagement with Nvidia, hyperscalers and neocloud customers. Its role as a reference-design partner should support early participation in Vera Rubin and improve its ability to win new accounts. The company remains confident in approaching 50% share of the Rubin rack opportunity. We believe its early NPI involvement and manufacturing scale provide a sustainable advantage as AI systems move toward rack-scale architectures.
- **Automation and vertical integration driving value-add:** Higher automation and increasing vertical integration across power, cooling, sockets, midplanes and board assembly should expand Hon Hai's content per rack. The company has achieved full automation at the board level and internally supplies more than 70% of midplane and compute-board content. It is also targeting approximately 50% internal supply of non-chip rack content, with selected components generating margins above 20%. This should support operating-margin improvement despite the high component pass-through associated with AI racks.
- **Multiple growth vectors into 2027E:** Beyond GPU racks, Hon Hai is expanding into ASIC systems, 800G+/CPO top-of-rack switches and LPU racks. It targets 40% ASIC rack share while expecting another year of healthy double-digit general-server growth. Higher rack ASPs and increasing internal content should support earnings growth as Vera Rubin begins to scale. Broader exposure across compute, networking and rack components reduces dependence on any single AI platform.

Hon Hai Precision – Key Financials

(NT$M)		2025	2026E	2027E	2028E
Revenue		**8,103,105**	**11,073,700**	**14,962,477**	**17,412,642**
	%YoY	18%	37%	35%	16%
Gross Profit		**498,161**	**667,762**	**895,126**	**979,622**
	GP margin %	6%	6%	6%	6%
Operating Profit		**259,223**	**390,989**	**493,224**	**583,813**
	OP margin %	3%	4%	3%	3%
Net Profit		**189,354**	**259,472**	**328,304**	**385,885**
	NP margin %	2%	2%	2%	2%
EPS (NT$)		13.6	18.5	23.4	27.5

Hon Hai Precision – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	**13.8**	**13.6**	**10.7**	**9.1**
P/B (x)	**1.5**	**1.9**	**1.7**	**1.5**
EV/EBITDA (x)	7.8	7.2	5.7	4.6
DPS (NT$)	5.8	7.2	9.3	11.7
Dividend Yield (Net) (%)	3%	3%	4%	5%
Dividend Payout Ratio (%)	43%	39%	40%	43%
Equity FCF Yield (%)	2%	3%	4%	19%

117

Source: Company data, UBS estimates

JCET Group (Jimmy Yu) – Buy, PT Rmb135

- **Beneficiary of domestic and overseas AI demand:** JCET is a leading Chinese OSAT with exposure to AI accelerators, SoCs, DSPs, server PMICs and other infrastructure-related semiconductors. Its combination of domestic and overseas customers provides diversified exposure to the global semiconductor recovery. Higher AI content increases demand for more complex packaging and testing, improving the value of JCET's service offering. We continue to view the company as a major beneficiary of both global AI demand and China semiconductor localization.
- **Improving product mix supports profitability:** Advanced packaging and higher-value AI applications should become a greater portion of revenue, supporting utilization and operating leverage. Continued product-mix improvement, cost control and operational-efficiency gains provide additional margin support. Strong demand across domestic and overseas operations should help absorb capacity as the semiconductor cycle strengthens. We raised our 2026-28E EPS forecasts by 16-18% to reflect these structural earnings drivers.
- **Localization provides a multiyear growth runway:** China's continued push toward semiconductor self-sufficiency should drive demand for domestic assembly and testing services. JCET is well placed to participate as local AI, power and compute chips move into higher-volume production.

JCET Group – Key Financials

(Rmbm)		2025	2026E	2027E	2028E
Revenue		**38,871**	**46,748**	**55,998**	**64,903**
	%YoY	8%	20%	20%	16%
Gross Profit		**5,395**	**7,036**	**9,203**	**11,426**
	GP margin %	14%	15%	16%	18%
Operating Profit		**1,802**	**2,879**	**4,495**	**6,218**
	OP margin %	5%	6%	8%	10%
Net Profit		**1,565**	**2,691**	**4,107**	**5,635**
	NP margin %	4%	6%	7%	9%
EPS (Rmb)		0.9	1.5	2.3	3.1

JCET Group – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	**41.8**	**42.7**	**28.0**	**20.4**
P/B (x)	**2.3**	**3.7**	**3.3**	**2.9**
EV/EBITDA (x)	12.5	17.8	14.1	11.5
DPS (Rmb)	0.1	0.3	0.5	0.6
Dividend Yield (Net) (%)	0%	0%	1%	1%
Dividend Payout Ratio (%)	11%	20%	20%	20%
Equity FCF Yield (%)	-3%	-3%	-2%	2%

118

Source: Company data, UBS estimates

KYEC (Sunny Lin) – Buy, PT NT$325

- **Sales growth to accelerate in Q4 after softer Q3:** We forecast KYEC's Q3 sales to grow high single digits QoQ, below consensus estimate of 16-17% QoQ. Q3 gross margin may also decline to 38-39% from 39.5% in Q2, due to higher electricity and labor cost, and 1% dilution from the new Singapore fab. The major Cloud AI accelerators are ramping up in Q3, albeit still in early stage. We expect multiple Cloud AI accelerator programs to ramp meaningfully in Q4, to lift sales growth to 15-20% QoQ. We estimate Q4 GM to also rise to 42-43% with higher utilization and lower utility cost.
- **Larger Cloud AI TAM to support 2027E sales growth of 45%:** Looking ahead, KYEC has planned clean room construction through end of 2027, reflecting the rising testing demand from probing and final test from growing units of server CPU, accelerators, networking etc. Final test times are increasing for Rubin. HBM de-spec may not be negative as allows more SiP volumes. On top of Nvidia, KYEC is exposed to TPUs (both Broadcom and MTK). For CPUs, it has exposure to Grace, Vera and Axion (for which it is adding capacity).
- **Diversifying footprint in Singapore and the US:** Given the geopolitical dynamics, KYEC is also enhancing its supply resilience by diversifying its production base. The new Singapore fab just started mass production in Q226. KYEC also recently announced its plan for the US expansion. In terms of implication to profitability, our view is the dilution should be manageable if KYEC could negotiate with key clients for tool consignment.

KYEC – Key Financials

(NT$m)		2025	2026E	2027E	2028E
Revenue		**34,934**	**47,663**	**69,243**	**84,810**
	%YoY	30%	36%	45%	22%
Gross Profit		**12,522**	**19,160**	**28,622**	**34,999**
	GP margin %	36%	40%	41%	41%
Operating Profit		**9,003**	**13,174**	**21,360**	**27,154**
	OP margin %	26%	28%	31%	32%
Net Profit		**7,963**	**11,184**	**18,107**	**22,448**
	NP margin %	23%	23%	26%	26%
EPS (Rmb)		6.5	9.1	14.8	18.3

KYEC – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	21.1	32.4	20.0	16.1
P/B (x)	3.3	5.8	5.0	4.3
EV/EBITDA (x)	10.8	15.5	10.7	9.0
DPS (Rmb)	5.9	5.9	5.9	11.9
Dividend Yield (Net) (%)	4%	2%	2%	4%
Dividend Payout Ratio (%)	90%	65%	40%	65%
Equity FCF Yield (%)	-11%	-8%	-8%	-3%

119

Source: Company data, UBS estimates

MediaTek (Sunny Lin) – Buy, PT NT$7,300

- **Google TPU roadmap remains a major growth driver:** We forecast TPU revenue of US$18.0bn/US$43.5bn/US$52.5bn in 2027/28/29E, supported by V8 and a materially larger V9 platform from 2028E. MediaTek targets 15-20% share in 2027 and remains on track for US$12-16bn of revenue, with potential upside if additional customer demand can be fulfilled. V9 is nearing design finalization and is expected to generate its largest revenue contribution in 2028. Its position as the sole supplier for the customer-defined V9 design strengthens visibility beyond the current V8 cycle.
- **EMIB-T progress reduces V9 execution risk:** Several rounds of EMIB-T validation have been completed, with one remaining ahead of tape-out. The architecture supports the larger reticle size required for V9, while additional substrate partners have been qualified to improve supply resilience. A potential CoWoS solution could provide further flexibility if V9 volume exceeds initial EMIB-T capacity. MediaTek's experience across both EMIB-T and CoWoS should improve its ability to support increasingly complex AI accelerators.
- **Custom AI silicon creates upside beyond TPU:** MediaTek is actively engaged in Nvidia's NVLink Fusion ecosystem and other custom AI infrastructure programs. Its existing Nvidia relationships across PCs and automotive, along with pre-qualification through NVLink Fusion, provide an advantage in winning customized silicon projects. Smaller programs could begin contributing from 2027, broadening revenue beyond the Google TPU roadmap. SpaceX Dojo 3 and Tesla AI6 offer further long-term upside, although both remain less mature than the TPU opportunity.

MediaTek – Key Financials

(NT$M)		2025	2026E	2027E	2028E
Revenue		**595,966**	**673,973**	**1,248,996**	**2,139,567**
	%YoY	12%	13%	85%	71%
Gross Profit		**283,080**	**310,833**	**560,341**	**938,344**
	GP margin %	47%	46%	45%	44%
Operating Profit		**103,470**	**107,089**	**316,482**	**661,915**
	OP margin %	17%	16%	25%	31%
Net Profit		**105,319**	**107,707**	**282,719**	**578,726**
	NP margin %	18%	16%	23%	27%
EPS (NT$)		66.1	67.5	177.2	362.6

MediaTek – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	**20.8**	**73.3**	**27.9**	**13.7**
P/B (x)	**5.5**	**17.7**	**13.7**	**9.5**
EV/EBITDA (x)	16.0	58.1	22.6	11.2
DPS (NT$)	53.5	54.1	142.0	290.7
Dividend Yield (Net) (%)	4%	1%	3%	6%
Dividend Payout Ratio (%)	81%	80%	80%	80%
Equity FCF Yield (%)	6%	1%	3%	5%

120

Source: Company data, UBS estimates

Murata Mfg. (Shingo Hirata) – Buy, PT ¥13,200

- **Strong growth in AI server MLCCs:** x2 in current FY, x2 again next FY. In line with SEMCO. SEMCO and Murata control 80% of the server MLCC market.
- **Price increases likely to come in earnest:** Murata acknowledges that if competition raises pricing, they will follow suite. They will balance the outcome vs market share. For FY, blended pricing to increase 10-15%.
- **LTAs coming through, pulling in capacity increases:** LTAs cover 1-2 years. Murata is pulling in capacity increase to 17% this FY and 17% again next FY vs 10% normal cadence.
- **Latest outlook/guidance:**1) The FY3/27 sales guidance for data centres was raised 14.0% from the previous plan to ¥370.6bn, up 110.0% Y/Y. The largest upward revision was for MLCCs, followed by power supplies and inductors/EMI.

Murata Mfg. – Key Financials

(¥ bn)		2025	2026E	2027E	2028E
Revenue		**1,743**	**1,831**	**2,028**	**2,399**
	%YoY	6%	5%	11%	18%
Gross Profit		**718**	**775**	**870**	**1,205**
	GP margin %	41%	42%	43%	50%
Operating Profit		**280**	**282**	**459**	**750**
	OP margin %	16%	15%	23%	31%
Net Profit		**232**	**234**	**360**	**581**
	NP margin %	13%	13%	18%	24%
EPS (¥)		124.6	128.4	199.5	325.1

Murata Mfg. – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	**22.6**	**21.5**	**42.4**	**26.0**
P/B (x)	**2.0**	**1.8**	**5.4**	**4.9**
EV/EBITDA (x)	10.4	9.7	23.2	15.6
DPS (¥)	57.0	65.0	70.0	90.0
Dividend Yield (Net) (%)	2%	2%	1%	1%
Dividend Payout Ratio (%)	46%	51%	35%	28%
Equity FCF Yield (%)	5%	4%	2%	3%

121

Source: Company data, UBS estimates

NAURA Technology (Jimmy Yu) – Buy, PT Rmb905

- **Leading beneficiary of China equipment localization:** NAURA is China's largest domestic WFE supplier, with broad exposure across etch, deposition and epitaxy. Memory-equipment localization already exceeds 50%, or approximately 60% excluding lithography, while longer overseas lead times and NAURA's cost advantage support further share gains.
- **Memory expansion remains stronger for longer:** China's memory capacity addition is estimated at approximately 140-160k wpm in 2026, with further growth possible in 2027 as new clean rooms become available. New memory orders for NAURA have roughly doubled, providing strong shipment visibility as component constraints ease.
- **Technology leadership supports sustained growth:** NAURA leads domestically in metal ALD and holds more than 70% local share in epitaxy, while its latest high-aspect-ratio etch tool reached production installation ahead of a major competitor. New opportunities in advanced packaging, hybrid bonding and future memory architectures further expand its addressable market.

NAURA Technology – Key Financials

(Rmbm)		2025	2026E	2027E	2028E
Revenue		**39,353**	**50,611**	**71,881**	**98,107**
	%YoY	31%	29%	42%	36%
Gross Profit		**15,782**	**20,166**	**28,644**	**38,926**
	GP margin %	40%	40%	40%	40%
Operating Profit		**4,887**	**6,527**	**11,726**	**17,990**
	OP margin %	12%	13%	16%	18%
Net Profit		**5,522**	**6,804**	**11,588**	**17,283**
	NP margin %	14%	13%	16%	18%
EPS (Rmb)		7.6	9.4	16.0	23.8

NAURA Technology – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	**46.8**	**65.7**	**38.6**	**25.9**
P/B (x)	**6.8**	**10.2**	**8.1**	**6.3**
EV/EBITDA (x)	41.1	50.0	30.3	20.4
DPS (Rmb)	0.8	0.9	1.9	3.1
Dividend Yield (Net) (%)	0%	0%	0%	1%
Dividend Payout Ratio (%)	10%	10%	12%	13%
Equity FCF Yield (%)	0%	0%	0%	2%

122

Source: Company data, UBS estimates

Tokyo Electron (Atsuhiro Kinoshita) – Buy, PT ¥18,720

- **Order outlook strengthening across the board:** 1/ Leading edge logic orders strengthened in June /July (TSMC, Intel) and have remained strong since. 2/ DRAM investment is also seeing pull-ins, and accelerated cadence of capacity deployment. 3/ TEL retains meaningful exposure to China, including Memory, which is also seeing strength.
- **Upside to WFE outlook:** We expect upside to current outlook of US$190bn+ for CY27 (after US$150bn for CY26). TEL does point out to the strain its supply chain could be placed under should the WFE market upsizes further.
- **Well engaged re price increases:** TEL has now engaged with all major customers in re-price increases for tools (like for like), going beyond covering cost increases / FX. This implies double digit price increases in order to reach 50% gross profit margin target.
- **Making progress for cryogenic etch:** One NAND customer will adopt for MP in CY27 for >400L NAND (likely Samsung V10 – which is 430L).

Tokyo Electron – Key Financials

(¥ bn)		2025	2026E	2027E	2028E
Revenue		**2,432**	**2,444**	**3,387**	**4,446**
	%YoY	33%	0%	39%	31%
Gross Profit		**1,146**	**1,108**	**1,596**	**2,127**
	GP margin %	47%	45%	47%	48%
Operating Profit		**697**	**625**	**1,012**	**1,378**
	OP margin %	29%	26%	30%	31%
Net Profit		**545**	**457**	**780**	**1,064**
	NP margin %	22%	19%	23%	24%
EPS (¥)		238.0	199.3	344.1	471.5

Tokyo Electron – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	**23.6**	**29.7**	**35.2**	**25.7**
P/B (x)	**6.9**	**6.5**	**12.1**	**10.5**
EV/EBITDA (x)	16.5	18.6	24.3	18.4
DPS (¥)	118.4	57.5	165.4	225.6
Dividend Yield (Net) (%)	2%	1%	1%	2%
Dividend Payout Ratio (%)	50%	29%	48%	48%
Equity FCF Yield (%)	3%	2%	1%	2%

123

Source: Company data, UBS estimates

Asustek (Randy Abrams) – Neutral, PT NT$910

- **Leveraging its Nvidia ecosystem into AI servers:** Asustek is using its strong gaming and graphics-card franchise and close Nvidia relationship to expand into AI servers. AI servers could reach 30-35% of sales in 2026E and 40-45% in 2027E, providing a meaningful topline driver. Its access to Nvidia accelerators and established enterprise channels should support continued customer additions. The company targets another doubling of server revenue in 2027, albeit from a smaller scale than the leading server ODMs.
- **Server growth comes with margin trade-offs:** Asustek lacks the manufacturing scale and integrated rack capabilities of larger ODM peers, limiting its ability to capture value beyond accelerator and system pass-through. AI servers carry higher working-capital requirements and lower margins than its traditional gaming and commercial products. Strong revenue growth therefore does not translate proportionately into gross-profit growth. We expect the server business to remain dilutive to corporate-average margins as it scales.
- **Margin normalization likely:** The recent gross- and operating-margin strength benefited from passing higher component costs to less price-sensitive gaming and commercial customers. This pricing benefit should fade once memory prices stabilize, while a larger server mix creates an additional margin headwind.

Asustek – Key Financials

(NT$M)		2025	2026E	2027E	2028E
Revenue		**688,936**	**978,392**	**1,214,538**	**1,323,425**
	%YoY	26%	42%	24%	9%
Gross Profit		**95,090**	**139,300**	**151,883**	**159,014**
	GP margin %	14%	14%	13%	12%
Operating Profit		**33,907**	**57,581**	**52,297**	**53,756**
	OP margin %	5%	6%	4%	4%
Net Profit		**39,263**	**55,936**	**51,996**	**53,206**
	NP margin %	6%	6%	4%	4%
EPS (NT$)		52.9	75.3	70.0	71.6

Asustek – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	**12.0**	**12.8**	**13.7**	**13.4**
P/B (x)	**1.8**	**2.4**	**2.3**	**2.2**
EV/EBITDA (x)	11.5	10.6	11.5	11.2
DPS (NT$)	34.0	42.0	53.7	49.0
Dividend Yield (Net) (%)	5%	4%	6%	5%
Dividend Payout Ratio (%)	64%	56%	77%	68%
Equity FCF Yield (%)	3%	4%	4%	4%

124

Source: Company data, UBS estimates

Compal (Randy Abrams) – Neutral, PT NT$40

- **Rapid transition toward AI servers:** Compal is using its manufacturing scale, tier-one customer relationships and new Taiwan and US capacity to diversify from notebooks into AI servers. Servers represented approximately 7-8% of sales in Q226 and are targeted to reach 10% in 2026 and 30-40% in 2027. Nvidia systems should begin ramping from Q426, alongside smaller AMD AI volumes.
- **Broader customer engagement is emerging:** Initial server growth is led by neocloud customers, while Compal is engaging US hyperscalers through motherboard programs that could progress toward full systems from 2027. The company is also conducting Vera rack proof-of-concept work and preparing for Dell AI server opportunities. New facilities in Texas and Taiwan provide capacity for board assembly through full-rack integration.
- **Execution and margins remain the key risks:** Compal remains a late entrant and has lower R&D scale and rack-integration experience than the leading server ODMs. Server margins are currently above the corporate average, but larger tier-one and hyperscale programs could introduce more component pass-through and pricing pressure. Ramp-up costs, substantial capex and weak notebook volumes create offsetting earnings headwinds. Our Neutral rating is unchanged until stronger server scale translates into clearer and more sustainable margin improvement.

Compal – Key Financials

(NT$M)		2025	2026E	2027E	2028E
Revenue		**757,513**	**970,173**	**1,277,284**	**1,490,681**
	%YoY	-17%	28%	32%	17%
Gross Profit		**42,747**	**46,694**	**61,118**	**72,719**
	GP margin %	6%	5%	5%	5%
Operating Profit		**10,636**	**12,457**	**19,419**	**24,053**
	OP margin %	1%	1%	2%	2%
Net Profit		**5,967**	**9,177**	**12,292**	**15,354**
	NP margin %	1%	1%	1%	1%
EPS (NT$)		1.4	2.1	2.8	3.5

Compal – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	**22.9**	**17.3**	**12.9**	**10.3**
P/B (x)	**1.0**	**1.2**	**1.1**	**1.1**
EV/EBITDA (x)	8.3	9.3	8.4	7.9
DPS (NT$)	1.4	1.1	0.0	1.7
Dividend Yield (Net) (%)	4%	3%	0%	5%
Dividend Payout Ratio (%)	102%	52% -		48%
Equity FCF Yield (%)	16%	-15%	-34%	17%

125

Source: Company data, UBS estimates

Desay Battery (Jimmy Yu) – Sell, PT Rmb21

- **Premium smartphone exposure provides some resilience:** Desay should regain allocation for two Pro models in the upcoming product cycle and has exposure to additional premium-device programs. UBS forecasts iPhone builds of 260-270mn units and Apple sell-in growth of 9% in 2026, providing relative support versus the broader smartphone market. Android demand remains weaker as memory inflation drives customer volume reductions. The product-allocation recovery therefore provides support but does not resolve Desay's longer-term competitive issues.
- **Core battery positioning remains challenged:** Competitors already offer integrated cell-and-pack solutions, while Desay remains behind in internal cell capability. The company plans to invest up to Rmb2.5bn in consumer soft-pack battery-cell capacity, with Phase 1 targeting 60mn units by the end of 2027. This could strengthen vertical integration over time, but current customer-order visibility remains limited.
- **ESS improvement insufficient to offset smartphone pressure:** ESS profitability has improved on high utilization, but H226E revenue is expected to remain broadly flat HoH. The business provides earnings diversification but remains too small to offset structural pressure in smartphone batteries. Re-rating requires clearer evidence of customer wins for the combined cell-and-pack offering. Until then, we expect the core-business downturn to outweigh the ESS turnaround.

Desay Battery – Key Financials

(Rmbm)		2025	2026E	2027E	2028E
Revenue		**22,399**	**24,869**	**26,883**	**28,014**
	%YoY	7%	11%	8%	4%
Gross Profit		**2,020**	**2,440**	**2,631**	**2,853**
	GP margin %	9%	10%	10%	10%
Operating Profit		**364**	**728**	**780**	**1,009**
	OP margin %	2%	3%	3%	4%
Net Profit		**292**	**531**	**527**	**689**
	NP margin %	1%	2%	2%	2%
EPS (Rmb)		0.8	1.4	1.4	1.8

Desay Battery – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	**31.9**	**16.2**	**16.4**	**12.5**
P/B (x)	**1.4**	**1.2**	**1.2**	**1.1**
EV/EBITDA (x)	10.7	8.0	7.4	6.0
DPS (Rmb)	0.2	0.4	0.4	0.5
Dividend Yield (Net) (%)	1%	2%	2%	2%
Dividend Payout Ratio (%)	30%	30%	30%	30%
Equity FCF Yield (%)	2%	12%	1%	5%

126

Source: Company data, UBS estimates

INNOLUX Corporation (Jerry Su) – Sell, PT NT$26

- **Legacy displays remain structurally challenged:** Innolux retains meaningful exposure to notebooks, monitors and TVs, where end demand and industry economics remain weak. Notebook demand is expected to fall following H126 pull-ins, while monitor softness is prompting a shift toward higher-end products. Its focus on sub-50-inch TV panels provides some resilience because competition is less intense in this category. Nevertheless, legacy display weakness should remain the main constraint on operating performance.
- **Asset-light strategy improves business quality:** Following the sale of three display fabs, Innolux is redirecting capacity from lower-margin commodity panels toward semiconductor, automotive and specialized non-display applications. The company intends to monetize and repurpose existing assets rather than construct additional display fabs. This should improve capital discipline and reduce exposure to the most commoditized panel segments.
- **The transformation will take time:** Higher-value non-display businesses are not yet large enough to offset pressure in legacy displays. Operating performance is likely to remain weaker than H126, although asset-disposal gains could support a positive full-year bottom line. The asset-light strategy improves the longer-term direction but does not create an immediate earnings inflection.

INNOLUX Corporation – Key Financials

(NT$M)		2025	2026E	2027E	2028E
Revenue		**226,724**	**255,598**	**264,326**	**272,184**
	%YoY	5%	13%	3%	3%
Gross Profit		**18,657**	**34,803**	**33,940**	**35,287**
	GP margin %	8%	14%	13%	13%
Operating Profit		**-4,160**	**3,584**	**3,658**	**4,835**
	OP margin %	-2%	1%	1%	2%
Net Profit		**250**	**23,760**	**4,819**	**5,608**
	NP margin %	0%	9%	2%	2%
EPS (NT$)		0.0	3.0	0.6	0.7

INNOLUX Corporation – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	**431.5**	**17.7**	**87.5**	**75.2**
P/B (x)	**0.5**	**1.8**	**1.8**	**1.8**
EV/EBITDA (x)	3.3	11.7	11.9	11.6
DPS (NT$)	1.0	1.0	1.0	1.0
Dividend Yield (Net) (%)	7%	2%	2%	2%
Dividend Payout Ratio (%)	3197%	34%	166%	142%
Equity FCF Yield (%)	12%	8%	4%	4%

127

Source: Company data, UBS estimates

Largan (Randy Abrams) – Sell, PT NT$5,000

- **CPO opportunity remains attractive but early:** Largan is applying its precision-optics and volume-manufacturing expertise to fiber-array-unit solutions for optical interconnects. Its approach combines a prism micro-lens array with V-groove and fiber components to couple light from the PIC into the fiber. Early feedback is positive on precision and Largan's record of improving manufacturing yields. The opportunity could become meaningful if the company converts its production-oriented FA program into volume orders.
- **Qualification and yield improvement will take time:** The sample line begins in late Q3, followed by several quarters of yield improvement, integrator acceptance and customer qualification. Largan remains behind incumbent solutions already designed into 2026-27 platforms, while its targeted Gen 3 optics opportunity may not scale until 2028. Incumbents such as TFC and vertically integrated competitors including Coherent retain meaningful advantages.
  Valuation appears ahead of fundamentals: Current valuation appears to price in a high probability of commercialization despite a broad range of potential qualification and market-share outcomes. Alternative optical approaches, including vertically integrated and meta-lens solutions, add further uncertainty. The re-rating has materially outpaced evidence of volume adoption.

Largan Corporation – Key Financials

(NT$M)		2025	2026E	2027E	2028E
Revenue		**61,148**	**67,869**	**77,189**	**83,044**
	%YoY	3%	11%	14%	8%
Gross Profit		**30,837**	**33,718**	**38,405**	**42,004**
	GP margin %	50%	50%	50%	51%
Operating Profit		**23,558**	**25,937**	**29,796**	**34,945**
	OP margin %	39%	38%	39%	42%
Net Profit		**23,613**	**24,642**	**28,019**	**32,818**
	NP margin %	39%	36%	36%	40%
EPS (NT$)		176.9	188.2	214.0	250.7

Largan Corporation – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	**13.3**	**33.4**	**29.4**	**25.1**
P/B (x)	**1.6**	**4.2**	**3.9**	**3.6**
EV/EBITDA (x)	6.3	19.4	16.6	14.8
DPS (NT$)	97.5	80.0	95.4	107.4
Dividend Yield (Net) (%)	4%	1%	2%	2%
Dividend Payout Ratio (%)	55%	43%	45%	43%
Equity FCF Yield (%)	3%	4%	3%	4%

128

Source: Company data, UBS estimates

Maxscend (Jimmy Yu) – Neutral, PT Rmb80.2

- **Building a broader specialty-semiconductor platform:** Maxscend operates a 12-inch specialty-process platform covering SOI, SiGe, BCD and IPD, with capacity reaching approximately 9k wafers per month. The company is extending beyond its core RF business into optical communications, where its existing materials and process capabilities could provide synergies. Optical communications offers an attractive structural TAM as AI networking demand expands. However, product details, customer engagements and commercialization timing remain undisclosed.
- **RF progress is offset by industry pressure:** Maxscend's L-PAMiD has secured design wins with multiple leading handset brands and entered large-scale shipment. These wins demonstrate improving product reliability and delivery capability in higher-value RF modules. Nevertheless, weak smartphone demand and intensifying RF front-end competition continue to pressure utilization and pricing.
- **Optical communications remain too early to underwrite:** The strategic move could provide a second growth pillar, but current investment is running ahead of visible revenue. There is not yet enough detail on products, customers or qualification milestones to estimate the earnings opportunity reliably. Our Neutral rating is unchanged pending clearer evidence of design wins and a defined commercialization schedule.

Maxscend – Key Financials

(Rmbm)		2025	2026E	2027E	2028E
Revenue		**3,726**	**3,827**	**4,625**	**5,836**
	%YoY	-17%	3%	21%	26%
Gross Profit		**942**	**597**	**882**	**1,467**
	GP margin %	25%	16%	19%	25%
Operating Profit		**-307**	**-673**	**-333**	**267**
	OP margin %	-8%	-18%	-7%	5%
Net Profit		**-293**	**-629**	**-254**	**289**
	NP margin %	-8%	-16%	-5%	5%
EPS (Rmb)		-0.5	-1.1	-0.4	0.5

Maxscend – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	**-141.2**	**-61.9**	**-159.0**	**139.8**
P/B (x)	**4.2**	**3.2**	**3.2**	**3.2**
EV/EBITDA (x)	98.6	217.1	61.1	28.5
DPS (Rmb)	0.0	0.0	0.0	0.0
Dividend Yield (Net) (%)	0%	0%	0%	0%
Dividend Payout Ratio (%)	-	-	-	-
Equity FCF Yield (%)	-2%	0%	-1%	1%

129

Source: Company data, UBS estimates

Pegatron (Randy Abrams) – Neutral, PT NT$95

- **Server growth is diversifying the business:** Pegatron's server contribution increased from 4% of sales in Q1 to 13% in Q2, supported by HGX systems and an extended GB300 lifecycle. Additional customers in Southeast Asia should help 2026E server revenue approach NT$100bn, close to the scale of notebooks. The server business could surpass notebooks in 2027 as customer and platform coverage expands. This provides meaningful diversification from slower-growth smartphones and consumer electronics.
- **Subsidiaries provide an earnings buffer:** Subsidiaries should account for approximately 60% of group gross profit in 2026-27E, led by Kinsus and an improving substrate environment. Their greater exposure to semiconductor components reduces dependence on Pegatron's lower-margin assembly operations. Rising substrate utilization could provide additional support to group earnings. This mix benefit partially offsets the margin dilution from faster server growth.
- **Margin expansion remains modest:** Server gross margin is likely between Pegatron's corporate average and ODM peers at approximately 4-5%. Higher server mix increases revenue but also introduces substantial component pass-through, limiting the benefit to operating profit. Component inflation and only mild growth across smartphones, tablets and consoles create further pressure.

Pegatron – Key Financials

(NT$M)		2025	2026E	2027E	2028E
Revenue		**1,117,160**	**1,284,542**	**1,538,317**	**1,618,138**
	%YoY	-1%	15%	20%	5%
Gross Profit		**42,501**	**55,150**	**67,380**	**72,613**
	GP margin %	4%	4%	4%	4%
Operating Profit		**11,609**	**18,529**	**24,150**	**27,838**
	OP margin %	1%	1%	2%	2%
Net Profit		**14,403**	**15,678**	**19,091**	**22,917**
	NP margin %	1%	1%	1%	1%
EPS (NT$)		5.4	5.8	7.1	8.5

Pegatron – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	**14.9**	**15.4**	**12.6**	**10.5**
P/B (x)	**1.0**	**1.1**	**1.1**	**1.0**
EV/EBITDA (x)	6.9	5.7	4.4	3.7
DPS (NT$)	4.5	4.0	4.0	4.3
Dividend Yield (Net) (%)	6%	4%	4%	5%
Dividend Payout Ratio (%)	83%	68%	56%	50%
Equity FCF Yield (%)	1%	11%	11%	6%

130

Source: Company data, UBS estimates

Realtek (Sunny Lin) – Neutral, PT NT$690

- **Broad exposure across connectivity markets supports resilient demand:** Realtek's portfolio spans networking, PC, automotive, IoT and consumer applications, providing diversification despite weaker consumer electronics demand. While restocking in PCs and consumer products is slowing, demand from networking, automotive and home-appliance categories remains relatively stable.
- **Pricing power and mix upgrade help offset cost inflation:** We expect H226 sales and profitability to benefit from June price increases and a more favorable product mix. Ongoing pricing negotiations should allow Realtek to pass through rising foundry, packaging and memory-related costs, supporting a sequential improvement in gross margin through H226.
- **Supply tightness should support H2 earnings resilience:** Despite softer end demand in selected consumer categories, customers continue to build inventory amid tightening backend and memory supply. We believe stable demand from non-consumer segments and continued supply-chain tightness should help sustain earnings momentum into H226, although weaker demand could eventually limit cost pass-through capability.

Realtek – Key Financials

(NT$m)		2025	2026E	2027E	2028E
Revenue		**122,706**	**152,854**	**180,042**	**198,772**
	%YoY	8%	25%	18%	10%
Gross Profit		**61,374**	**73,821**	**86,836**	**96,629**
	GP margin %	50%	48%	48%	49%
Operating Profit		**14,360**	**17,165**	**20,792**	**22,367**
	OP margin %	12%	11%	12%	11%
Net Profit		**14,753**	**17,311**	**20,712**	**22,194**
	NP margin %	12%	11%	12%	11%
EPS (Rmb)		28.6	33.7	40.3	43.2

Realtek – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	18.7	22.4	18.7	17.5
P/B (x)	5.3	8.4	7.4	6.7
EV/EBITDA (x)	12.3	15.5	13.2	12.4
DPS (Rmb)	23.6	27.6	33.1	35.5
Dividend Yield (Net) (%)	4%	4%	4%	5%
Dividend Payout Ratio (%)	82%	82%	82%	82%
Equity FCF Yield (%)	8%	5%	4%	5%

131

Source: Company data, UBS estimates

Xiaomi (Jimmy Yu) – Neutral, PT HK$30.4

- **Integrated ecosystem expanding into premium EVs:** Xiaomi is extending its smartphone and IoT ecosystem into the more profitable premium EV segment, with auto gross margin targeted at approximately 20%. Its installed user base, retail reach and device ecosystem provide differentiation versus pure-play EV entrants.
- **AI and proprietary silicon deepen differentiation:** Xiaomi plans to invest approximately Rmb60bn in AI during 2026-28 while expanding its in-house silicon capabilities across smartphones, ADAS and smart cockpits. The objective is tighter hardware-software integration rather than near-term component-cost savings.
- **International expansion provides longer-term upside:** Germany is the first target for overseas vehicle expansion from 2H27, followed by the Middle East, Africa and Southeast Asia. We remain constructive on the Human × Car × Home ecosystem, but smartphone weakness, memory inflation and elevated investment requirements support our Neutral rating.

Xiaomi – Key Financials

(Rmbm)		2025	2026E	2027E	2028E
Revenue		**457,287**	**435,009**	**509,916**	**603,423**
	%YoY	25%	-5%	17%	18%
Gross Profit		**101,806**	**89,942**	**106,627**	**128,542**
	GP margin %	22%	21%	21%	21%
Operating Profit		**39,628**	**21,522**	**32,515**	**48,269**
	OP margin %	9%	5%	6%	8%
Net Profit		**39,167**	**23,797**	**29,637**	**43,112**
	NP margin %	9%	5%	6%	7%
EPS (Rmb)		1.5	0.9	1.1	1.6

Xiaomi – Valuation Metrics

Metric	2025	2026E	2027E	2028E
P/E (x)	**30.5**	**24.4**	**19.3**	**13.3**
P/B (x)	**4.2**	**2.0**	**1.9**	**1.7**
EV/EBITDA (x)	23.6	17.2	12.6	8.8
DPS (Rmb)	0.0	0.0	0.0	0.0
Dividend Yield (Net) (%)	0%	0%	0%	0%
Dividend Payout Ratio (%)	-	-	-	-
Equity FCF Yield (%)	1%	1%	2%	4%

132

Source: Company data, UBS estimates

Valuation Method and Risk Statement

Investing in the technology sector involves a high degree of risk. Rapid technological changes, increasing competition and exposure to macroeconomic cycles are among the many risks faced by investors in technology stocks. Moreover, it is extremely difficult to project the financial results of tech companies since their operating models are highly volatile and unpredictable. Finally, valuing technology stocks can prove challenging, as neither traditional nor non-traditional valuation measures have provided much insight into how these stocks trade. Valuation methodologies in the sector include target price-book-multiples based on the long term average ROE and the cost of equity.

Required Disclosures

This document has been prepared by UBS Securities Pte. Ltd., an affiliate of UBS AG. UBS AG, its subsidiaries, branches and affiliates, including former Credit Suisse AG and its subsidiaries, branches and affiliates are referred to herein as "UBS".

For information on the ways in which UBS manages conflicts and maintains independence of its UBS Global Research product; historical performance information; certain additional disclosures concerning UBS Global Research recommendations; and terms and conditions for certain third party data used in research report, please visit https://www.ubs.com/disclosures. Unless otherwise indicated, information and data in this report are based on company disclosures including but not limited to annual, interim, quarterly reports and other company announcements. The figures contained in performance charts refer to the past; past performance is not a reliable indicator of future results. Additional information will be made available upon request. UBS Securities Co. Limited is licensed to conduct securities investment consultancy businesses by the China Securities Regulatory Commission. UBS acts or may act as principal in the debt securities (or in related derivatives) that may be the subject of this report. This recommendation was finalized on: 16 October 2026 07:42 AM GMT. UBS has designated certain UBS Global Research department members as Derivatives Research Analysts where those department members publish research principally on the analysis of the price or market for a derivative, and provide information reasonably sufficient upon which to base a decision to enter into a derivatives transaction. Where Derivatives Research Analysts co-author research reports with Equity Research Analysts or Economists, the Derivatives Research Analyst is responsible for the derivatives investment views, forecasts, and/or recommendations. **Quantitative Research Review:** UBS Global Research publishes a quantitative assessment of its analysts' responses to certain questions about the likelihood of an occurrence of a number of short term factors in a product known as the 'Quantitative Research Review'. Views contained in this assessment on a particular stock reflect only the views on those short term factors which are a different timeframe to the 12-month timeframe reflected in any equity rating set out in this note. For the latest responses, please see the Quantitative Research Review Addendum at the back of this report, where applicable. For previous responses please make reference to (i) previous UBS Global Research reports; and (ii) where no applicable research report was published that month, the Quantitative Research Review which can be found at https://neo.ubs.com/quantitative, or contact your UBS sales representative for access to the report or the Quantitative Research Team on ubs-quant-answers@ubs.com. A consolidated report which contains all responses is also available and again you should contact your UBS sales representative for details and pricing or the Quantitative Research team on the email above.

Analyst Certification:

Each research analyst primarily responsible for the content of this research report, in whole or in part, certifies that with respect to each security or issuer that the analyst covered in this report: (1) all of the views expressed accurately reflect his or her personal views about those securities or issuers and were prepared in an independent manner, including with respect to UBS, and (2) no part of his or her compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed by that research analyst in the research report.

UBS Global Research: Global Equity Rating Definitions

12-Month Rating	Definition	Coverage[1]	IB Services[2]
Buy	FSR is > 6% above the MRA.	55%	24%
Neutral	FSR is between -6% and 6% of the MRA.	39%	20%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 30 September 2026.
1:Percentage of companies under coverage globally within the 12-month rating category.
2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

KEY DEFINITIONS:Forecast Stock Return (FSR) is defined as expected percentage price appreciation plus gross dividend yield over the next 12 months. In some cases, this yield may be based on accrued dividends. **Market Return Assumption (MRA)** is defined as the one-year local market interest rate plus 5% (a proxy for, and not a forecast of, the equity risk premium). **Under Review (UR)** Stocks may be flagged as UR by the analyst, indicating that the stock's price target and/or rating are subject to possible change in the near term, usually in response to an event that may affect the investment case or valuation. **Equity Price Targets** have an investment horizon of 12 months.

EXCEPTIONS AND SPECIAL CASES:UK and European Investment Fund ratings and definitions are: Buy: Positive on factors such as structure, management, performance record, discount; **Neutral:** Neutral on factors such as structure, management, performance record, discount; **Sell:** Negative on factors such as structure, management, performance record, discount. **Core Banding Exceptions (CBE):** Exceptions to the standard +/-6% bands may be granted by the Investment Review Consultation (IRC). Factors considered by the IRC include the stock's volatility and the credit spread of the respective company's debt. As a result, stocks deemed to be very high or low risk may be subject to higher or lower bands as they relate to the rating. When such exceptions apply, they will be identified in the Company Disclosures table in the relevant research piece.

Research analysts contributing to this report who are employed by any non-US affiliate of UBS Securities LLC are not registered/qualified as research analysts with FINRA. Such analysts may not be associated persons of UBS Securities LLC and therefore are not subject to the FINRA restrictions on communications with a subject company, public appearances, and trading securities held by a research analyst account. The name of each affiliate and analyst employed by that affiliate contributing to this report, if any, follows.

UBS AG, Singapore Branch: Jimmy Yoon, Nicolas Gaudois.**UBS Securities Pte. Ltd., Seoul Branch:** Luke Yoo.

Company Disclosures

Company Name	Reuters	12-month rating	Price	Price date
ACM Research (Shanghai)	688082.SS	Buy	Rmb265.00	15 Oct 2026
AEM Holdings[13,28a]	AEM.SI	Buy	S$8.80	15 Oct 2026
ASE Industrial[28a,16b]	3711.TW	Buy	NT$671.00	16 Oct 2026
ASML[28a,16b]	ASML.AS	Buy	€1,464.2	15 Oct 2026
ASPEED Technology Inc[28a]	5274.TWO	Buy	NT$15,945	15 Oct 2026
Accton Technology Corporation[28a]	2345.TW	Buy	NT$1,980	16 Oct 2026
Advanced Micro-Fabrication Equipment	688012.SS	Buy	Rmb297.67	15 Oct 2026
Advantest[28b]	6857.T	Buy	¥34,530	16 Oct 2026
Alchip Technologies[13,28a]	3661.TW	Buy	NT$3,685	16 Oct 2026
All Ring Tech	6187.TWO	Buy	NT$1,390	15 Oct 2026
Alps Alpine	6770.T	Neutral	¥2,141	16 Oct 2026
Asustek Computer Inc.[28a]	2357.TW	Neutral	NT$994.00	16 Oct 2026
Avary Holding Shenzhen	002938.SZ	Buy	Rmb75.26	15 Oct 2026
CXMT Corporation	688825.SS	Buy	Rmb43.59	15 Oct 2026
China Resources Microelectronics	688396.SS	Buy	Rmb48.62	15 Oct 2026
Chroma ATE[28a]	2360.TW	Buy	NT$2,000	16 Oct 2026
Compal Electronics[1,5,6]	2324.TW	Neutral	NT$36.70	16 Oct 2026

Company Name	Reuters	12-month rating	Price	Price date
DI Corporation	003160.KS	Buy	Won40,500	16 Oct 2026
Delta Electronics[4,6,28a]	2308.TW	Buy	NT$1,635	16 Oct 2026
Disco[28a]	6146.T	Neutral	¥53,850	16 Oct 2026
Eugene Technology	084370.KQ	Buy	Won129,300	16 Oct 2026
Giga-Byte Technology[1,2,4,5,6,28a]	2376.TW	Neutral	NT$357.50	16 Oct 2026
Global Unichip[28a]	3443.TW	Buy	NT$7,480	16 Oct 2026
GlobalWafers	6488.TWO	Buy	NT$923.00	15 Oct 2026
Gudeng Precision Industrial[13]	3680.TWO	Buy	NT$513.00	15 Oct 2026
Hanmi Semiconductor	042700.KS	Sell	Won210,000	16 Oct 2026
Hansol Chemical	014680.KS	Buy	Won232,500	16 Oct 2026
Hon Hai Precision[4,5,6,28a]	2317.TW	Buy	NT$244.00	16 Oct 2026
Hua Hong Grace Semiconductor[16a]	1347.HK	Neutral	HK$76.55	15 Oct 2026
INNOLUX Corporation[1,5,6,28a]	3481.TW	Sell	NT$48.25	16 Oct 2026
ISU Petasys	007660.KS	Buy	Won99,400	16 Oct 2026
Inventec[28a]	2356.TW	Neutral	NT$59.00	16 Oct 2026
JCET Group	600584.SS	Buy	Rmb57.50	15 Oct 2026
Kelington Group Berhad	KELG.KL	Buy	RM8.50	15 Oct 2026
King Yuan Electronics[28a]	2449.TW	Buy	NT$275.00	16 Oct 2026
Kioxia Holdings Corp[28a]	285A.T	Buy	¥13,375	16 Oct 2026
Kyocera[28a]	6971.T	Buy	¥3,629	16 Oct 2026
LG Display[16b]	034220.KS	Neutral	Won7,490	16 Oct 2026
LG Electronics[8,28a]	066570.KS	Neutral	Won169,700	16 Oct 2026
LG Innotek	011070.KS	Buy	Won406,500	16 Oct 2026
Largan Precision[28a]	3008.TW	Sell	NT$6,400	16 Oct 2026
Lenovo Group[2,5,6,28a,16a]	0992.HK	Buy	HK$25.98	15 Oct 2026
Maxscend Microelectronics	300782.SZ	Neutral	Rmb55.23	15 Oct 2026
MediaTek Inc.[28a]	2454.TW	Buy	NT$4,250	16 Oct 2026
Murata Manufacturing[28b]	6981.T	Buy	¥6,900	16 Oct 2026
NAURA Technology Group	002371.SZ	Buy	Rmb610.00	15 Oct 2026

Company Name	Reuters	12-month rating	Price	Price date
Nan Ya Printed Circuit Board	8046.TW	Neutral	NT$1,355	16 Oct 2026
Nanya Technology	2408.TW	Buy	NT$447.50	16 Oct 2026
Nidec[28a]	6594.T	Neutral	¥1,800	16 Oct 2026
Partron Co Ltd	091700.KQ	Neutral	Won4,795	16 Oct 2026
Pegatron[28a]	4938.TW	Neutral	NT$102.00	16 Oct 2026
Quanta[2,4,5,6,28a]	2382.TW	Buy	NT$322.00	16 Oct 2026
Realtek Semiconductor Corp[28a]	2379.TW	Neutral	NT$778.00	16 Oct 2026
SCREEN Holdings Co., Ltd.[5,6,28a]	7735.T	Neutral	¥13,815	16 Oct 2026
SFA Engineering	056190.KQ	Buy	Won27,700	16 Oct 2026
SK Hynix[8,16b]	000660.KS	Buy	Won1,272,000	16 Oct 2026
SMIC[16a]	0981.HK	Buy	HK$52.15	15 Oct 2026
SUMCO Corporation[28a]	3436.T	Buy	¥2,703	16 Oct 2026
Samsung Electro-Mechanics[28a]	009150.KS	Buy	Won1,188,000	16 Oct 2026
Samsung Electronics[2,6,8,28a]	005930.KS	Buy	Won215,500	16 Oct 2026
SemiFive[2,4,6,18]	490470.KQ	Buy	Won16,350	16 Oct 2026
Shenzhen Desay Battery Technology	000049.SZ	Sell	Rmb20.65	15 Oct 2026
Simmtech	222800.KQ	Buy	Won126,600	16 Oct 2026
Soulbrain	357780.KQ	Buy	Won336,000	16 Oct 2026
Taiwan Semiconductor Manufacturing[28b,16b]	2330.TW	Buy	NT$2,420	16 Oct 2026
Tokyo Electron[28a]	8035.T	Buy	¥10,630	16 Oct 2026
UMC[28a,16b]	2303.TW	Buy	NT$133.00	16 Oct 2026
Wiwynn[28a]	6669.TW	Buy	NT$1,845	16 Oct 2026
Wonik IPS	240810.KQ	Buy	Won109,000	16 Oct 2026
Wonik Materials	104830.KQ	Buy	Won37,200	16 Oct 2026
Xiaomi[28a,16a]	1810.HK	Neutral	HK$27.08	15 Oct 2026
YC Corporation	232140.KQ	Buy	Won14,820	16 Oct 2026

Source: UBS Global Research; LSEG Eikon. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.

1. UBS is acting as manager/co-manager, underwriter, placement or sales agent in regard to an offering of securities of this company/entity or one of its affiliates.
2. UBS has acted as manager/co-manager in the underwriting or placement of securities of this company/entity or one of its affiliates within the past 12 months.
4. Within the past 12 months, UBS has received compensation for investment banking services from this company/entity or one of its affiliates.
5. UBS expects to receive or intend to seek compensation for investment banking services from this company/entity within the next three months.

6. This company/entity is, or within the past 12 months has been, a client of UBS Securities LLC and/or its affiliates, and investment banking services are being, or have been, provided.
8. The equity analyst covering this company, a member of his or her team, or one of their household members has a long common stock position in this company.
13. UBS beneficially owned 1% or more of a class of this company's common equity securities as of last month's end (or the prior month's end if this report is dated less than 10 days after the most recent month's end).
16a. UBS Securities Hong Kong Limited is a market maker in the Hong Kong-listed securities of this company.
16b. UBS Securities LLC makes a market in the securities and/or ADRs of this company.
18. UBS Securities Pte. Ltd. - Seoul Branch has acted as a Joint Lead Manager for initial listing of this company/entity within the past 12 months.
28a. UBS holds a long position of 0.5% or more of the listed shares of this company.
28b. UBS holds a short position of 0.5% or more of the listed shares of this company.

Unless otherwise indicated, please refer to the Valuation and Risk sections within the body of this report. For a complete set of disclosure statements associated with the companies discussed in this report, including information on valuation and risk, please contact UBS Securities LLC, 11 Madison Avenue, New York, NY 10010, USA, Attention: Investment Research.

Company profile and fee and risk statement under the Japanese Financial Instruments & Exchange Law

Company Name etc: UBS Securities Japan Co., Ltd., Financial Instruments & Exchange Firm, Kanto Local Financial Bureau (Kinsho) No.2633

Associated Memberships: Japan Securities Dealers' Association, the Financial Futures Association of Japan, and Type II Financial Instruments Firms Association

UBS Securities Japan Co., Ltd. will receive a brokerage fee based on an individual contract and no standard upper limit or calculating method. For the trading of domestic stocks, consumption tax is added to the fee. For the trading of foreign stock, fee on the foreign stock exchange or foreign tax may be charged in addition to the domestic fee and tax. Those amounts may vary depending on the jurisdiction. There is a risk that a loss may occur due to a change in the price of the stock in the case of trading stocks, and that a loss may occur due to the exchange rate in the case of trading foreign stocks. There is a risk that a loss may occur due to a change in the price or performance of the properties in the portfolio in the case of trading REITs.

UBS Securities Japan Co., Ltd. will only receive the purchasing amounts for trading unlisted bonds (JGBs, municipals, government guaranteed bonds, corporate bonds) when UBS Securities Japan Co., Ltd. is the counterparty. There is a risk that a loss may occur due to a change in the price of the bond caused by the fluctuations in the interest rates, and that a loss may occur due to the exchange rate in the case of trading foreign bonds.

For the detailed information of commissions and risks associated with individual financial instrument transactions, please carefully read the documents provided prior to the execution of the contracts or other materials for customers, etc. before executing any transaction. If you need any clarification, please consult with our sales person in charge.

The Disclaimer relevant to Global Wealth Management clients follows the Global Research Disclaimer.

UBS Global Research Disclaimer

This document has been prepared by UBS Securities Pte. Ltd., an affiliate of UBS AG. UBS AG, its subsidiaries, branches and affiliates, including former Credit Suisse AG and its subsidiaries, branches and affiliates are referred to herein as "UBS".

Any opinions expressed in this document may change without notice and are only current as of the date of publication. Different areas, groups, and personnel within UBS may produce and distribute separate research products independently of each other. For example, research publications from **UBS CIO** are produced by UBS Global Wealth Management. **UBS Global Research** is produced by UBS Investment Bank. Research methodologies and rating systems of each separate research organization may differ, for example, in terms of investment recommendations, investment horizon, model assumptions, and valuation methods. As a consequence, except for certain economic forecasts (for which UBS CIO and UBS Global Research may collaborate), investment recommendations, ratings, price targets, and valuations provided by each of the separate research organizations may be different, or inconsistent. You should refer to each relevant research product for the details as to their methodologies and rating system. Not all clients may have access to all products from every organization. Each research product is subject to the policies and procedures of the organization that produces it.

This document is provided solely to recipients who are expressly authorized by UBS to receive it. If you are not so authorized you must immediately destroy the document.

UBS Global Research is provided to our clients through UBS Neo, and in certain instances, UBS.com and any other system or distribution method specifically identified in one or more communications distributed through UBS Neo or UBS.com (each a system) as an approved means for distributing UBS Global Research. It may also be made available through third party vendors and distributed by UBS and/or third parties via e-mail or alternative electronic means.

All UBS Global Research is available on UBS Neo. Please contact your UBS sales representative if you wish to discuss your access to UBS Neo. Where UBS Global Research refers to "UBS Evidence Lab Inside" or has made use of data provided by UBS Evidence Lab and you would like to access that data please contact your UBS sales representative. UBS Evidence Lab data is available on UBS Neo. The level and types of services provided by UBS Global Research and UBS Evidence Lab to a client may vary depending upon various factors such as a client's individual preferences as to the frequency and manner of receiving communications, a client's risk profile and investment focus and perspective (e.g., market wide, sector specific, long-term, short-term, etc.), the size and scope of the overall client relationship with UBS Global Research and UBS Evidence Lab and legal and regulatory constraints. UBS HOLT and UBS Pharma Values are offerings of UBS Global Research. HOLT Lens is a corporate performance platform offering that provides an objective accounting-led framework for comparing and valuing companies and is available to clients of UBS Global Research; for further details and pricing please contact your UBS Sales representative. In particular, HOLT has a variety of warranted prices based on the scenario chosen; please mail UBS Securities LLC, 11 Madison Avenue, New York, NY 10010, USA, Attention: Investment Research, if you are interested in the warranted price on a particular company, again subject to commercial considerations. UBS Pharma Values is an analytical tool that involves the creation of a number of individual product net present value calculations, based on published forecasts of sales for pharmaceuticals, and is available to clients of UBS Global Research; for further details and pricing please contact your UBS Sales representative. For all other specific disclaimers, please see https://www.ubs.com/disclosures.

When you receive UBS Global Research through a system, your access and/or use of such UBS Global Research is subject to this UBS Global Research Disclaimer and to the UBS Neo Platform Use Agreement (the "Neo Terms") together with any other relevant terms of use governing the applicable System.

When you receive UBS Global Research via a third party vendor, e-mail or other electronic means, you agree that use shall be subject to this UBS Global Research Disclaimer, the Neo Terms and where applicable the UBS Investment Bank terms of business (https://www.ubs.com/global/en/investment-bank/regulatory.html) and to UBS's Terms of Use/Disclaimer (https://www.ubs.com/global/en/legalinfo2/disclaimer.html). In addition, you consent to UBS processing your personal data and using cookies in accordance with our Privacy Statement (https://www.ubs.com/global/en/legalinfo2/privacy.html) and cookie notice (https://www.ubs.com/global/en/legal/privacy/users.html).

If you receive UBS Global Research, whether through a System or by any other means, you agree that you shall not copy, revise, amend, create a derivative work, provide to any third party, or in any way commercially exploit any UBS research provided via UBS Global Research or otherwise, and that you shall not extract data from any research or estimates provided to you via UBS Global Research or otherwise, without the prior written consent of UBS. You agree not to use UBS Global Research in any artificial intelligence system, without the prior written consent of UBS.

In certain circumstances you may receive UBS Global Research otherwise than in the capacity of a client of UBS and you understand and agree that under these circumstances (i) the UBS Global Research is provided to you for information purposes only; (ii) for the purposes of receiving it you are not intended to be and will not be treated as a "client" of UBS for any legal or regulatory purpose; (iii) the UBS Global Research must not be relied on or acted upon for any purpose; and (iv) such content is subject to the relevant disclaimers that follow.

This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject UBS to any registration or licensing requirement within such jurisdiction.

This document is a general communication and is educational in nature; it is not an advertisement nor is it a solicitation or an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Nothing in this document constitutes a representation that any investment strategy or recommendation is suitable or appropriate to an investor's individual circumstances or otherwise constitutes a personal recommendation. By providing this document, none of UBS or its representatives has any responsibility or authority to provide or have provided investment advice in a fiduciary capacity or otherwise. Investments involve risks, and investors should exercise prudence and their own judgment in making their investment decisions. None of UBS or its representatives is suggesting that the recipient or any other person take a specific course of action or any action at all. The recipient should carefully read this document in its entirety and not draw inferences or conclusions from the rating alone. By receiving this document, the recipient acknowledges and agrees with the intended purpose described above and further disclaims any expectation or belief that the information constitutes investment advice to the recipient or otherwise purports to meet the investment objectives of the recipient. The financial instruments described in the document may not be eligible for sale in all jurisdictions or to certain categories of investors.

Options, structured derivative products and futures (including OTC derivatives) are not suitable for all investors. Trading in these instruments is considered risky and may be appropriate only for sophisticated investors. Prior to buying or selling an option, and for the complete risks relating to options, you must receive a copy of "The Characteristics and Risks of Standardized Options." You may read the document at https://www.theocc.com/publications/risks/riskchap1.jsp or ask your salesperson for a copy. Various theoretical explanations of the risks associated with these instruments have been published. Supporting documentation for any claims, comparisons, recommendations, statistics or other technical data will be supplied upon request. Past performance is not necessarily indicative of future results. Transaction costs may be significant in option strategies calling for multiple purchases and sales of options, such as spreads and straddles. Because of the importance of tax considerations to many options transactions, the investor considering options should consult with his/her tax advisor as to how taxes affect the outcome of contemplated options transactions.

Mortgage and asset-backed securities may involve a high degree of risk and may be highly volatile in response to fluctuations in interest rates or other market conditions. Foreign currency rates of exchange may adversely affect the value, price or income of any security or related instrument referred to in the document. For investment advice, trade execution or other enquiries, clients should contact their local sales representative.

UBS notes that no globally accepted framework or definition (legal, regulatory or otherwise) currently exists, nor is there a market consensus as to what constitutes an "ESG" (Environmental, Social or Governance) or an equivalent-label, or as to what precise attributes are required for the Information (as defined below) to be defined as ESG or equivalently-labelled. Any information, data or other content including from a third party source contained, referred to herein or used for whatsoever purpose by UBS or a third party ("Information"), in relation to any actual or potential ESG objective, issue or consideration is not intended to be relied upon for ESG classification, regulatory regime or industry initiative purposes ("ESG Regimes"). Nothing in these materials is intended to convey, suggest or indicate that UBS considers or represents any product, service, person or body mentioned in these materials as meeting or qualifying for any ESG classification, labelling or similar standards that may exist under the ESG Regimes. UBS has not conducted any assessment of compliance with ESG Regimes. Parties are reminded to make their own assessments for these purposes.

The value of any investment or income may go down as well as up, and investors may not get back the full (or any) amount invested. Past performance is not necessarily a guide to future performance. Neither UBS nor any of its directors, employees or agents accepts any liability for any loss (including investment loss) or damage arising out of the use of all or any of the Information.

Prior to making any investment or financial decisions, any recipient of this document or the information should take steps to understand the risk and return of the investment and seek individualized advice from his or her personal financial, legal, tax and other professional advisors that takes into account all the particular facts and circumstances of his or her investment objectives.

Any prices stated in this document are for information purposes only and do not represent valuations for individual securities or other financial instruments. There is no representation that any transaction can or could have been effected at those prices, and any prices do not necessarily reflect UBS's internal books and records or theoretical model-based valuations and may be based on certain assumptions. Different assumptions by UBS or any other source may yield substantially different results.

No representation or warranty, either expressed or implied, is provided in relation to the accuracy, completeness or reliability of the information contained in any materials to which this document relates (the "Information"), except with respect to Information concerning UBS. The Information is not intended to be a complete statement or summary of the securities, markets or developments referred to in the document. UBS does not undertake to update or keep current the Information. Any statements contained in this report attributed to a third party represent UBS's interpretation of the data, information and/or opinions provided by that third party either publicly or through a subscription service, and such use and interpretation have not been reviewed by the third party. In no circumstances may this document or any of the Information (including any forecast, value, index or other calculated amount ("Values")) be used for any of the following purposes:

(i) valuation or accounting purposes;

(ii) to determine the amounts due or payable, the price or the value of any financial instrument or financial contract; or

(iii) to measure the performance of any financial instrument including, without limitation, for the purpose of tracking the return or performance of any Value or of defining the asset allocation of portfolio or of computing performance fees.

By receiving this document and the Information you will be deemed to represent and warrant to UBS that you will not use this document or any of the Information for any of the above purposes or otherwise rely upon this document or any of the Information.

UBS has policies and procedures, which include, without limitation, independence policies and permanent information barriers, that are intended, and upon which UBS relies, to manage potential conflicts of interest and control the flow of information within divisions of UBS and among its subsidiaries, branches and affiliates. UBS does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For further information on the ways in which UBS Global Research manages conflicts and maintains independence of its research products, historical performance information and certain additional disclosures concerning UBS Global Research recommendations, please visit https://www.ubs.com/disclosures.

UBS Global Research will initiate, update and cease coverage solely at the discretion of UBS Global Research Management, which will also have sole discretion on the timing and frequency of any published research product. The analysis contained in this document is based on numerous assumptions. All material information in relation to published research reports, such as valuation methodology, risk statements, underlying assumptions (including sensitivity analysis of those assumptions), ratings history etc. as required by the Market Abuse Regulation, can be found on UBS Neo. Different assumptions could result in materially different results.

UBS Global Research may utilise artificial intelligence tools ("AI Tools") in the preparation of this document. Notwithstanding any such use of AI Tools, this document has undergone human review.

The analyst(s) responsible for the preparation of this document may interact with trading desk personnel, sales personnel and other parties for the purpose of gathering, applying and interpreting market information. UBS relies on information barriers to control the flow of information contained in one or more areas within UBS into other areas, units, groups or affiliates of UBS. The compensation of the analyst who prepared this document is determined exclusively by UBS Global Research management and senior management (not including investment banking). Analyst compensation is not based on investment banking revenues; however, compensation may relate to the revenues of UBS and/or its divisions as a whole, of which investment banking, sales and trading are a part, and UBS as a whole.

For financial instruments admitted to trading on an EU regulated market: UBS (excluding UBS Securities LLC) acts as a market maker or liquidity provider (in accordance with the interpretation of these terms under English law or, if not carried out by UBS in the UK the law of the relevant jurisdiction in which UBS determines it carries out the activity) in the financial instruments of the issuer save that where the activity of liquidity provider is carried out in accordance with the definition given to it by the laws and regulations of any other EU jurisdictions, such information is separately disclosed in this document. For financial instruments admitted to trading on a non-EU regulated market: UBS may act as a market maker save that where this activity is carried out in the US in accordance with the definition given to it by the relevant laws and regulations, such activity will be specifically disclosed in this document. UBS may have issued a warrant the value of which is based on one or more of the financial instruments referred to in the document. UBS and its affiliates and employees may have long or short positions, trade as principal and buy and sell in instruments or derivatives identified herein; such transactions or positions may be inconsistent with the opinions expressed in this document.

Within the past 12 months UBS may have received or provided investment services and activities or ancillary services as per MiFID II which may have given rise to a payment or promise of a payment in relation to these services from or to this company.

Please note that all transactions conducted by UBS are consistent with sanctions regulations imposed by Switzerland, the European Union, the United Nations, the United Kingdom and the United States, per UBS' global sanctions policy. UBS opinion as to future investment worthiness assumes no new sanctions are imposed.

US persons are prohibited from purchasing or selling securities of certain companies designated as being associated with the Chinese Military in accordance with the amended US Presidential Executive Order 13959.

United Kingdom: This material is distributed by UBS AG, London Branch to persons who are eligible counterparties or professional clients. UBS AG, London Branch is authorised by the Prudential Regulation Authority and subject to regulation by the Financial Conduct Authority and limited regulation by the Prudential Regulation Authority. **Europe:** Except as otherwise specified herein, these materials are distributed by UBS Europe SE, a subsidiary of UBS AG, to persons who are eligible counterparties or professional clients (as detailed in the Bundesanstalt fur Finanzdienstleistungsaufsicht (BaFin) Rules and according to MIFID) and are only available to such persons. The information does not apply to, and should not be relied upon by, retail clients. UBS Europe SE is authorised by the European Central Bank (ECB) and regulated by the BaFin and the ECB. **Germany, Luxembourg, the Netherlands, Belgium and Ireland:** Where an analyst of UBS Europe SE has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE. In all cases it is distributed by UBS Europe SE and UBS AG, London Branch. **Turkey:** Distributed by UBS AG, London Branch. No information in this document is provided for the purpose of offering, marketing and sale by any means of any capital market instruments and services in the Republic of Turkey. Therefore, this document may not be considered as an offer made or to be made to residents of the Republic of Turkey. UBS AG, London Branch is not licensed by the Turkish Capital Market Board under the provisions of the Capital Market Law (Law No. 6362). Accordingly, neither this document nor any other offering material related to the instruments/services may be utilized in connection with providing any capital market services to persons within the Republic of Turkey without the prior approval of the Capital Market Board. However, according to article 15 (d) (ii) of the Decree No. 32, there is no restriction on the purchase or sale of the securities abroad by residents of the Republic of Turkey. **Poland:** Distributed by UBS Europe SE (spolka z ograniczona odpowiedzialnoscia) Oddzial w Polsce. Where an analyst of UBS Europe SE (spolka z ograniczona odpowiedzialnoscia) Oddzial w Polsce has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE (spolka z ograniczona odpowiedzialnoscia) Oddzial w Polsce. **Russia:** Prepared and distributed by UBS Bank (OOO). Should not be construed as an individual Investment Recommendation for the purpose of the Russian Law - Federal Law #39-FZ ON THE SECURITIES MARKET Articles 6.1-6.2.**Switzerland:** Distributed by UBS AG to persons who are institutional investors only. UBS AG is regulated by the Swiss Financial Market Supervisory Authority (FINMA). **Italy:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, Italy Branch. Where an analyst of UBS Europe SE, Italy Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, Italy Branch. **France:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, France Branch. Where an analyst of UBS Europe SE, France Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, France Branch. **Spain:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, Spain Branch. Where an analyst of UBS Europe SE, Spain Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, Spain Branch. **Sweden:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, Sweden Branch. Where an analyst of UBS Europe SE, Sweden Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, Sweden Branch. **South Africa:** Distributed by UBS South Africa (Pty) Limited (Registration No. 1995/011140/07), an authorised user of the JSE and an authorised Financial Services Provider (FSP 7328). **Saudi Arabia:** This document has been issued by UBS AG (and/or any of its subsidiaries, branches or affiliates), a public company limited by shares, incorporated in Switzerland with its registered offices at Aeschenvorstadt 1, CH-4051 Basel and Bahnhofstrasse 45, CH-8001 Zurich. This publication has been approved by UBS Saudi Arabia (a subsidiary of UBS AG), a Saudi closed joint stock company incorporated in the Kingdom of Saudi Arabia under commercial register number 1010257812 having its registered office at Tatweer Towers, P.O. Box 75724, Riyadh 11588, Kingdom of Saudi Arabia. UBS Saudi Arabia is authorized and regulated by the Capital Market Authority to

conduct securities business under license number 08113-37. **UAE / Dubai:** The information distributed by UBS AG Dubai Branch is only intended for Professional Clients and/or Market Counterparties, as classified under the DFSA rulebook. No other person should act upon this material/communication. The information is not for further distribution within the United Arab Emirates. UBS AG Dubai Branch is regulated by the DFSA in the DIFC. UBS is not licensed to provide banking services in the UAE by the Central Bank of the UAE, nor is it licensed by the UAE Securities and Commodities Authority. **Israel:** This Material is distributed by UBS AG, London Branch. UBS Securities Israel Ltd is a licensed Investment Marketer that is supervised by the Israel Securities Authority (ISA). UBS AG, London Branch and its affiliates incorporated outside Israel are not licensed under the Israeli Advisory Law. UBS may engage among others in issuance of Financial Assets or in distribution of Financial Assets of other issuers for fees or other benefits. UBS AG, London Branch and its affiliates may prefer various Financial Assets to which they have or may have an Affiliation (as such term is defined under the Israeli Advisory Law). Nothing in this Material should be considered as investment advice under the Israeli Advisory Law. This Material is being issued only to and/or is directed only at persons who are Eligible Clients within the meaning of the Israeli Advisory Law, and this Material must not be furnished to, relied on or acted upon by any other persons. **United States:** Distributed to US persons by either UBS Securities LLC or by UBS Financial Services Inc., subsidiaries of UBS AG; or by a group, subsidiary or affiliate of UBS AG that is not registered as a US broker-dealer (a **'non-US affiliate'**) to major US institutional investors only. UBS Securities LLC or UBS Financial Services Inc. accepts responsibility for the content of a report prepared by another non-US affiliate when distributed to US persons by UBS Securities LLC or UBS Financial Services Inc. All transactions by a US person in the securities mentioned in this report must be effected through UBS Securities LLC or UBS Financial Services Inc., and not through a non-US affiliate. UBS Securities LLC is not acting as a municipal advisor to any municipal entity or obligated person within the meaning of Section 15B of the Securities Exchange Act (the "Municipal Advisor Rule"), and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of the Municipal Advisor Rule. **Canada:** Distributed by UBS Securities Canada Inc., a registered investment dealer in Canada and a Member-Canadian Investor Protection Fund, or by another affiliate of UBS AG that is registered to conduct business in Canada or is otherwise exempt from registration. **Brazil:** Except as otherwise specified herein, this Material is prepared by UBS BB Corretora de Câmbio, Títulos e Valores Mobiliários S.A. (UBS BB CCTVM) to persons who are eligible investors residing in Brazil, which are considered to be Professional Investors (Investidores Profissionais), as designated by the applicable regulation, mainly the CVM Resolution No. 30 from the 11th of May 2021 (determines the duty to verify the suitability of products, services and transactions with regards to the client´s profile). UBS BB CCTVM is a subsidiary of UBS BB Servicos de Assessoria Financeira e Participacoes S.A. ("UBS BB"). UBS BB is an association between UBS AG and Banco do Brasil (through its subsidiary BB – Banco de Investimentos S.A.), of which UBS AG is the majority owner and which provides investment banking services and coverage in Brazil, Argentina, Chile, Paraguay, Peru and Uruguay. UBS BB CCTVM is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman: 0800-940-0266/ https://www.ubs.com/br/pt/ubsbb-investment-bank/ombudsman.html. UBS may hold relevant financial and commercial interest in relation to the company subject to this Research report. **Hong Kong:** Distributed by UBS Securities Asia Limited. Please contact local licensed persons of UBS Securities Asia Limited in respect of any matters arising from, or in connection with, the analysis or document **Singapore:** Distributed by UBS Securities Pte. Ltd. [Co. Reg. No.: 198500648C] or UBS AG, Singapore Branch. Please contact UBS Securities Pte. Ltd., an exempt financial adviser under the Singapore Financial Advisers Act (Cap. 110); or UBS AG, Singapore Branch, an exempt financial adviser under the Singapore Financial Advisers Act (Cap. 110) and a wholesale bank licensed under the Singapore Banking Act (Cap. 19) regulated by the Monetary Authority of Singapore, in respect of any matters arising from, or in connection with, the analysis or document. The recipients of this document represent and warrant that they are accredited and institutional investors as defined in the Securities and Futures Act (Cap. 289). **Japan:** Distributed by UBS Securities Japan Co., Ltd. to professional investors (except as otherwise permitted). Where this report has been prepared by UBS Securities Japan Co., Ltd., UBS Securities Japan Co., Ltd. is the author, publisher and distributor of the report. Distributed by UBS AG, Tokyo Branch to Professional Investors (except as otherwise permitted) in relation to foreign exchange and other banking businesses when relevant. **Australia:** Recipients through a superannuation fund: Distributed by UBS AG, Australia Branch (ABN 47 088 129 613 and holder of Australian Financial Service Licence No. 231087). For all other recipients: Distributed by UBS Securities Australia Limited (ABN 62 008 586 481 and holder of Australian Financial Services Licence No. 231098). This document contains general information and/or general advice only and does not constitute personal financial product advice. As such, the Information in this document has been prepared without taking into account any investor's objectives, financial situation or needs, and investors should, before acting on the Information, consider the appropriateness of the Information, having regard to their objectives, financial situation and needs. If the Information contained in this document relates to the acquisition, or potential acquisition of a particular financial product by a 'Retail' client as defined by section 761G of the Corporations Act 2001 where a Product Disclosure Statement would be required, the retail client should obtain and consider the Product Disclosure Statement relating to the product before making any decision about whether to acquire the product. **New Zealand:** Distributed by UBS New Zealand Ltd. UBS New Zealand Ltd is not a registered bank in New Zealand. You are being provided with this publication or material because you have indicated to UBS that you are a "wholesale client" within the meaning of clause 4 of schedule 5 of the Financial Markets Conduct Act 2013 of New Zealand (Permitted Client). This publication or material is not intended for clients who are not Permitted Clients (non-permitted Clients). If you are a non-permitted Client you must not rely on this publication or material. If despite this warning you nevertheless rely on this publication or material, you hereby (i) acknowledge that you may not rely on the content of this publication or material and that any recommendations or opinions in such this publication or material are not made or provided to you, and (ii) to the maximum extent permitted by law (a) indemnify UBS and its associates or related entities (and their respective Directors, officers, agents and Advisors) (each a 'Relevant Person') for any loss, damage, liability or claim any of them may incur or suffer as a result of, or in connection with, your unauthorised reliance on this publication or material and (b) waive any rights or remedies you may have against any Relevant Person for (or in respect of) any loss, damage, liability or claim you may incur or suffer as a result of, or in connection with, your unauthorised reliance on this publication or material. **Korea:** Distributed in Korea by UBS Securities Pte. Ltd., Seoul Branch. This report may have been edited or contributed to from time to time by affiliates of UBS Securities Pte. Ltd., Seoul Branch. This material is intended for professional/institutional clients only and not for distribution to any retail clients. **Malaysia:** This material is authorized to be distributed in Malaysia by UBS Securities Malaysia Sdn. Bhd (Capital Markets Services License No.: CMSL/A0063/2007). This material is intended for professional/institutional clients only and not for distribution to any retail clients. **India:** Distributed by UBS Securities India Private Ltd. (Corporate Identity Number U67120MH1996PTC097299) Unit 2901, Level 29 Altimus, Pandurang Budhkar Road, Worli, Mumbai – 400 018. It provides brokerage services bearing SEBI Registration Number: INZ000259830; Merchant Banking services bearing SEBI Registration Number: INM000013101; and Research Analyst services bearing SEBI Registration Number: INH000001204. Name of Compliance Officer Mr. Parameshwaran Shivaramakrishnan, Phone : +912261556151, Email : parameshwaran.s@ubs.com, Name of Grievance Officer Parameshwaran Shivaramakrishnan, Phone : +912261556151, Email: ol-ubs-sec-compliance@ubs.com Registration granted by SEBI, and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. UBS may have debt holdings or positions in the subject Indian company/companies. UBS may have financial interests (e.g. loan/derivative products, rights to or interests in investments, etc.) in the subject Indian company / companies from time to time. Within the past 12 months, UBS may have received compensation for non-investment banking securities-related services and/or non-securities services from the subject Indian company/companies. The subject company/companies may have been a client/clients of UBS during the 12 months preceding the date of distribution of the research report with respect to investment banking and/or non-investment banking securities-related services and/or non-securities services. With regard to information on associates, please refer to the Annual Report at: https://www.ubs.com/global/en/about_ubs/investor_relations/annualreporting.html The Research Annual Compliance Report for UBS Securities India Private Limited is available on www.ubs.com/ubssi under Research tab. **Taiwan:** Except as otherwise specified herein, this material may not be distributed in Taiwan. Information and material on securities/instruments that are traded in a Taiwan organized exchange is deemed to be issued and distributed by UBS Securities Pte. LTD., Taipei Branch to professional institutional investors and/or persons permitted under applicable regulation. Unless permitted by applicable Taiwan laws and regulations, this material is for reference and information only and should not constitute "recommendation" to clients or recipients in Taiwan for the covered companies or any companies mentioned in this document. No portion of the document may be reprinted, reproduced or quoted by the press or any other person without authorisation from UBS. **Indonesia:** This report is being distributed by PT UBS Sekuritas Indonesia and is delivered by its licensed employee(s), including marketing/sales person, to its client. PT UBS Sekuritas Indonesia, having its registered office at Sequis Tower Level 22 unit 22-1,Jl.Jend. Sudirman, kav.71, SCBD lot 11B, Jakarta 12190. Indonesia, is a subsidiary company of UBS AG and licensed under Capital Market Law no. 8 year 1995, a holder of broker-dealer and underwriter licenses issued by the Capital Market and Financial Institution Supervisory Agency (now Otoritas Jasa Keuangan/OJK). PT UBS Sekuritas Indonesia is also a member of Indonesia Stock Exchange and supervised by Otoritas Jasa Keuangan (OJK). Neither this report nor any copy hereof may be distributed in Indonesia or to any Indonesian citizens except in compliance with applicable Indonesian capital market laws and regulations. This report is not an offer of securities in Indonesia and may not be distributed within the territory of the Republic of Indonesia or to Indonesian citizens in circumstance which constitutes an offering within the meaning of Indonesian capital market laws and regulations.

The disclosures contained in research documents produced by UBS AG, London Branch or UBS Europe SE shall be governed by and construed in accordance with English law.

UBS specifically prohibits the redistribution of this document in whole or in part without the written permission of UBS and in any event UBS accepts no liability whatsoever for any redistribution of this document or its contents or the actions of third parties in this respect. Images may depict objects or elements that are protected by third party copyright, trademarks and other intellectual property rights. © UBS 2026. The key symbol and UBS are among the registered and unregistered trademarks of UBS. All rights reserved.

Global Wealth Management Disclaimer

You receive this document in your capacity as a client of UBS Global Wealth Management. This publication has been distributed to you by UBS Switzerland AG (regulated by FINMA in Switzerland) or its affiliates (**"UBS"**) with whom you have a banking relationship with. The full name of the distributing affiliate and its competent authority can be found in the country-specific disclaimer at the end of this document. UBS may utilise artificial intelligence tools ("AI Tools") in the preparation of this document. Notwithstanding any such use of AI Tools, this document has undergone human review.

The date and time of the first dissemination of this publication is the same as the date and time of its publication.

Risk information:

You agree that you shall not copy, revise, amend, create a derivative work, provide to any third party, or in any way commercially exploit any UBS research, and that you shall not extract data from any research or estimates, without the prior written consent of UBS.

This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject UBS to any registration or licensing requirement within such jurisdiction.

This document is **for your information only**; it is not an advertisement nor is it a solicitation or an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Nothing in this document constitutes a representation that any investment strategy or recommendation is suitable or appropriate to an investor's individual circumstances or otherwise constitutes a personal recommendation. By providing this document, none of UBS or its representatives has any responsibility or authority to provide or have provided investment advice in a fiduciary capacity or otherwise. Investments involve risks, and investors should exercise prudence and their own judgment in making their investment decisions. None of UBS or its representatives is suggesting that the recipient or any other person take a specific course of action or any action at all. By receiving this document, the recipient acknowledges and agrees with the intended purpose described above and further disclaims any expectation or belief that the information constitutes investment advice to the recipient or otherwise purports to meet the investment objectives of the recipient. The financial instruments described in the document may not be eligible for sale in all jurisdictions or to certain categories of investors.

Options, derivative products and futures are not suitable for all investors, and trading in these instruments is considered risky. Mortgage and asset-backed securities may involve a high degree of risk and may be highly volatile in response to fluctuations in interest rates or other market conditions. Foreign currency rates of exchange may adversely affect the value, price or income of any security or related instrument referred to in the document. For investment advice, trade execution or other enquiries, clients should contact their local sales representative.

The value of any investment or income may go down as well as up, and investors may not get back the full (or any) amount invested. Past performance is not necessarily a guide to future performance. Neither UBS nor any of its directors, employees or agents accepts any liability for any loss (including investment loss) or damage arising out of the use of all or any of the information (as defined below).

Prior to making any investment or financial decisions, any recipient of this document or the information should take steps to understand the risk and return of the investment and seek individualized advice from his or her personal financial, legal, tax and other professional advisors that takes into account all the particular facts and circumstances of his or her investment objectives.

Any prices stated in this document are for information purposes only and do not represent valuations for individual securities or other financial instruments. There is no representation that any transaction can or could have been effected at those prices, and any prices do not necessarily reflect UBS's internal books and records or theoretical model-based valuations and may be based on certain assumptions. Different assumptions by UBS or any other source may yield substantially different results.

No representation or warranty, either expressed or implied, is provided in relation to the accuracy, completeness or reliability of the information contained in any materials to which this document relates (the "Information"), except with respect to Information concerning UBS. The Information is not intended to be a complete statement or summary of the securities, markets or developments referred to in the document. UBS does not undertake to update or keep current the Information. Any opinions expressed in this document may change without notice and may differ or be contrary to opinions expressed by other business areas or groups, personnel or other representative of UBS. Any statements contained in this report attributed to a third party represent UBS's interpretation of the data, information and/or opinions provided by that third party either publicly or through a subscription service, and such use and interpretation have not been reviewed by the third party. In no circumstances may this document or any of the Information (including any forecast, value, index or other calculated amount ("Values")) be used for any of the following purposes: (i) valuation or accounting purposes; (ii) to determine the amounts due or payable, the price or the value of any financial instrument or financial contract; or (iii) to measure the performance of any financial instrument including, without limitation, for the purpose of tracking the return or performance of any Value or of defining the asset allocation of portfolio or of computing performance fees.

By receiving this document and the Information you will be deemed to represent and warrant to UBS that you will not use this document or any of the Information for any of the above purposes or otherwise rely upon this document or any of the Information.

UBS has policies and procedures, which include, without limitation, independence policies and permanent information barriers, that are intended, and upon which UBS relies, to manage potential conflicts of interest and control the flow of information within divisions of UBS (including between Global Wealth Management and UBS Global Research) and among its subsidiaries, branches and affiliates. For further information on the ways in which UBS manages conflicts and maintains independence of its research products, historical performance information and certain additional disclosures concerning UBS research recommendations, please visit https://www.ubs.com/research-methodology.

Research will initiate, update and cease coverage solely at the discretion of research management, which will also have sole discretion on the timing and frequency of any published research product. The analysis contained in this document is based on numerous assumptions. Different assumptions could result in materially different results.

The analyst(s) responsible for the preparation of this document may interact with trading desk personnel, sales personnel and other parties for the purpose of gathering, applying and interpreting market information. UBS relies on information barriers to control the flow of information contained in one or more areas within UBS into other areas, units, groups or affiliates of UBS. The compensation of the analyst who prepared this document is determined exclusively by research management and senior management (not including investment banking). Analyst compensation is not based on investment banking revenues; however, compensation may relate to the revenues of UBS and/or its divisions as a whole, of which investment banking, sales and trading are a part, and UBS's subsidiaries, branches and affiliates as a whole.

For financial instruments admitted to trading on an EU regulated market: UBS AG, its affiliates or subsidiaries (excluding UBS Securities LLC) acts as a market maker or liquidity provider (in accordance with the interpretation of these terms in the UK) in the financial instruments of the issuer save that where the activity of liquidity provider is carried out in accordance with the definition given to it by the laws and regulations of any other EU jurisdictions, such information is separately disclosed in this document. For financial instruments admitted to trading on a non-EU regulated market: UBS may act as a market maker save that where this activity is carried out in the US in accordance with the definition given to it by the relevant laws and regulations, such activity will be specifically disclosed in this document. UBS may have issued a warrant the value of which is based on one or more of the financial instruments referred to in the document. UBS and its affiliates and employees may have long or short positions, trade as principal and buy and sell in instruments or derivatives identified herein; such transactions or positions may be inconsistent with the opinions expressed in this document.

Options and futures are not suitable for all investors, and trading in these instruments is considered risky and may be appropriate only for sophisticated investors. Prior to buying or selling an option, and for the complete risks relating to options, you must receive a copy of "Characteristics and Risks of Standardized Options". You may read the document at https://www.theocc.com/about/publications/character-risks.jsp or ask your financial advisor for a copy.

Investing in structured investments involves significant risks. For a detailed discussion of the risks involved in investing in any particular structured investment, you must read the relevant offering materials for that investment. Structured investments are unsecured obligations of a particular issuer with returns linked to the performance of an underlying asset. Depending on the terms of the investment, investors could lose all or a substantial portion of their investment based on the performance of the underlying asset. Investors could also lose their entire investment if the issuer becomes insolvent. UBS does not guarantee in any way the obligations or the financial condition of any issuer or the accuracy of any financial information provided by any issuer. Structured investments are not traditional investments and investing in a structured investment is not equivalent to investing directly in the underlying asset. Structured investments may have limited or no liquidity, and investors should be prepared to hold their investment to maturity. The return of structured investments may be limited by a maximum gain, participation rate or other feature. Structured investments may include call features and, if a structured investment is called early, investors would not earn any further return and may not be able to reinvest in similar investments with similar terms. Structured investments include costs and fees which are generally embedded in the price of the investment. The tax treatment of a structured investment may be complex and may differ from a direct investment in the underlying asset. UBS and its employees do not provide tax advice. Investors should consult their own tax advisor about their own tax situation before investing in any securities.

Important Information About Sustainable Investing Strategies: Sustainable investing strategies aim to consider and incorporate environmental, social and governance (ESG) factors into investment process and portfolio construction. Strategies across geographies approach ESG analysis and incorporate the findings in a variety of ways. Incorporating ESG factors or sustainable investing considerations may inhibit the portfolio manager's ability to participate in certain investment

opportunities that otherwise would be consistent with its investment objective and other principal investment strategies. The returns on a portfolio incorporating ESG factors or sustainable investing considerations may be lower or higher than portfolios where ESG factors, exclusions, or other sustainability issues are not considered by the portfolio manager, and the investment opportunities available to such portfolios may differ.

Within the past 12 months UBS Switzerland AG, its affiliates or subsidiaries may have received or provided investment services and activities or ancillary services as per MiFID II which may have given rise to a payment or promise of a payment in relation to these services from or to this company.

Disclosures: If you require detailed information on disclosures of interest or conflict of interest as required by Market Abuse Regulation please contact the mailbox MAR_disclosure_twopager@ubs.com. Please note that e-mail communication is unsecured.

External Asset Managers / External Financial Consultants: In case this research or publication is provided to an External Asset Manager or an External Financial Consultant, UBS expressly prohibits that it is redistributed by the External Asset Manager or the External Financial Consultant and is made available to their clients and/or third parties.

Argentina: All securities that will be offered to you by UBS are not authorized by the Argentine Securities and Exchange Commission (CNV) and are not subject to its reporting, periodic information requirements, or oversight. Furthermore, the CNV has not reviewed or endorsed the information provided in any securities offering document, nor the accuracy of any accounting, financial, economic data, or any other information disclosed therein, which remains the sole responsibility of the respective issuer and the other parties involved. **Australia:** This document is provided by UBS Switzerland AG. UBS Switzerland AG does not hold an Australian Financial Services Licence (AFSL) and relies on an exemption to provide financial services to persons in Australia. This document is intended only for distribution to wholesale clients under the Corporations Act 2001 (Cth). UBS Switzerland AG is a related body corporate of UBS AG, Australia Branch and UBS Securities Australia Limited. This document may be distributed to clients by those entities, but it is provided by UBS Switzerland AG and is not provided under any of the other entities' AFSL. The information in this document is general in nature and is not intended to address the objectives, financial situation or needs of any particular individual or entity. Each recipient should consider their own objectives, financial situation or needs before acting on the advice and obtain the relevant Product Disclosure Statement (if required) before making any decision whether to acquire any product. In Australia, UBS entities, other than UBS AG, Australia Branch, are not authorized deposit-taking institutions for the purposes of the Banking Act 1959 (Cth.) and their obligations do not represent deposits or other liabilities of UBS AG, Australia Branch. UBS AG, Australia Branch does not guarantee or otherwise provide assurance in respect of the obligations of such UBS entities or the funds. An investor is exposed to investment risk including possible delays in repayment and loss of income and principal invested, as relevant. If you do not wish to receive marketing materials from UBS, please contact your UBS representative or the contact details listed in the Australia Privacy Notice: https://www.ubs.com/global/en/legal/privacy.html. Your personal data will be processed in accordance with this notice. **Bahrain**: This report is being distributed by UBS AG, Bahrain Branch, duly licensed and regulated by the Central Bank of Bahrain (CBB) as an Investment Business Firm - Category 2 (Branch). Related financial services or products are only made available to Accredited Investors, as defined by the CBB, and are not intended for any other persons. UBS AG, Bahrain Branch is a Foreign Branch of UBS AG, Zurich/Switzerland and is located on Level 21, East Tower, Bahrain World Trade Centre, Manama, Kingdom of Bahrain. **Brazil**: This report is distributed in Brazil by UBS BB Corretora de Câmbio, Títulos e Valores Mobiliários S.A. or its affiliates ("UBS"). Pursuant to CVM Resolution No. 20/2021, of February 25, 2021, the author(s) of the report hereby certify(ies) that the views expressed in this report solely and exclusively reflect their personal opinions and have been prepared independently, including with respect to UBS. Part of the author(s)'s compensation is based on various factors, including the total revenues of the relevant UBS Group entity of which they are in employment of, but no part of the compensation has been, is, or will be related to the specific recommendations or views expressed in this report. In addition, UBS declares that: UBS has provided, and/or may in the future provide investment banking, brokerage, asset management, commercial banking and other financial services to the subject company/companies or its affiliates, for which they have received or may receive customary fees and commissions, and which constituted or may constitute relevant financial or commercial interests in relation to the subject company/companies or the subject securities. **Canada**: The information contained herein is not, and under no circumstances is to be construed as, a prospectus, an advertisement, a public offering, an offer to sell securities described herein, solicitation of an offer to buy securities described herein, in Canada or any province or territory thereof. Any offer or sale of the securities described herein in Canada will be made only under an exemption from the requirements to file a prospectus with the relevant Canadian securities regulators and only by a dealer properly registered under applicable securities laws or, alternatively, pursuant to an exemption from the dealer registration requirement in the relevant province or territory of Canada in which such offer or sale is made. Under no circumstances is the information contained herein to be construed as investment advice in any province or territory of Canada and is not tailored to the needs of the recipient. To the extent that the information contained herein references securities of an issuer incorporated, formed or created under the laws of Canada or a province or territory of Canada, any trades in such securities must be conducted through a dealer registered in Canada or, alternatively, pursuant to a dealer registration exemption. No securities commission or similar regulatory authority in Canada has reviewed or in any way passed upon these materials, the information contained herein or the merits of the securities described herein and any representation to the contrary is an offence. In Canada, this publication is distributed by UBS Investment Management Canada Inc. **China**: This report and any offering material such as term sheet, research report, other product or service documentation or any other information (the "Material") sent with this report was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the material erroneously, UBS asks that you kindly delete it and inform UBS immediately. This report is prepared by UBS Switzerland AG or its offshore subsidiary or affiliate (collectively as "UBS Offshore"). UBS Offshore is an entity incorporated out of China and is not licensed, supervised or regulated in China to carry out banking or securities business. The recipient should not contact the analysts or UBS Offshore which produced this report for advice as they are not licensed to provide securities investment advice in China. UBS Investment Bank (including Research) has its own wholly independent research and views which at times may vary from the views of UBS Global Wealth Management. The recipient should not use this document or otherwise rely on any of the information contained in this report in making investment decisions and UBS takes no responsibility in this regard. **Denmark**: This publication is not intended to constitute a public offer under Danish law. It is distributed only for information purposes to clients of UBS Europe SE, Denmark Branch, filial af UBS Europe SE, with place of business at Sankt Annae Plads 13, 1250 Copenhagen, Denmark, registered with the Danish Commerce and Companies Agency, under No. 38 17 24 33. UBS Europe SE, Denmark Branch, filial af UBS Europe SE is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the Danish Financial Supervisory Authority (Finanstilsynet), to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Egypt**: Securities or other investment products are not being offered or sold by UBS to the public in Egypt and they have not been and will not be registered with the Egyptian Financial Regulatory Authority (FRA). **France**: This publication is not intended to constitute a public offer under French law, it does not constitute a personal recommendation as it is distributed only for information purposes to clients of UBS Europe SE Succursale de France (a branch of UBS Europe SE), having its registered office at 39, rue du Colisée, 75008 Paris, France, registered with the "Registre du Commerce et des Sociétés" of Paris under N°844 425 629. UBS Europe SE Succursale de France is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the French "Autorité de contrôle prudentiel et de résolution" and "Autorité des marchés financiers", to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Germany**: This publication is not intended to constitute a public offer under German law. It is distributed only for information purposes to clients of UBS Europe SE, Germany, with place of business at Bockenheimer Landstrasse 2-4, 60306 Frankfurt am Main. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin") and supervised jointly by the European Central Bank, the German Central Bank (Deutsche Bundesbank) and BaFin, to which this publication has not been submitted for approval. **Hong Kong SAR**: This publication is distributed to clients of UBS AG Hong Kong Branch by UBS AG Hong Kong Branch, a licensed bank under the Hong Kong Banking Ordinance and a registered institution under the Securities and Futures Ordinance. UBS AG Hong Kong Branch is incorporated in Switzerland with limited liability. The contents of this material have not been reviewed by any regulatory authority in Hong Kong. **India**: UBS Securities India Private Ltd. (Corporate Identity Number U67120MH1996PTC097299) 2/F, 3 North Avenue, Maker Maxity, Bandra Kurla Complex, Bandra (East), Mumbai (India) 400051. Phone: +912261556000. It provides brokerage services bearing SEBI Registration Number: INZ000259830; Merchant Banking services bearing SEBI Registration Number: INM000013101 and Research Analyst services bearing SEBI Registration Number: INH000001204. Name of Compliance Officer Mr. Parameshwaran Shivaramakrishnan, Phone: +912261556151, Email: parameshwaran.s@ubs.com, Name of Grievance Officer Parameshwaran Shivaramakrishnan, Phone: +912261556151, Email: ol-ubs-sec-compliance@ubs.com. Registration granted by SEBI, and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. UBS AG, its affiliates or subsidiaries may have debt holdings or positions in the subject Indian company/companies. UBS AG, its affiliates or subsidiaries may have financial interests (e.g. loan/derivative products, rights to or interests in investments, etc.) in the subject Indian company/companies from time to time. Within the past 12 months, UBS AG, its affiliates or subsidiaries may have received compensation for non-investment banking securities-related services and/or non-securities services from the subject Indian company/companies. The subject company/companies may have been a client/clients of UBS AG, its affiliates or subsidiaries during the 12 months preceding the date of distribution of the research report with respect to investment banking and/or non-investment banking securities-related services and/or non-securities services. With regard to information on associates, please refer to the Annual Report at: https://www.ubs.com/global/en/about_ubs/investor_relations/annualreporting.html. The Research Annual Compliance report for UBS Securities India Private Limited is available on www.ubs.com/ubssi under Research tab. **Indonesia**: This communication and any offering material term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed without the consent of UBS. None of the Material has been registered or filed under the prevailing laws and with any financial or regulatory authority in your jurisdiction. The Material may not have been approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your

jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or products or futures contracts). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including futures contracts) nor is it an offer to buy or to sell any securities or products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Israel**: UBS is a premier global financial firm offering wealth management, asset management and investment banking services from its headquarters in Switzerland and its operations in over 50 countries worldwide to individual, corporate and institutional investors. This publication is intended for information only and is not intended as an offer to buy or solicitation of an offer. Furthermore, this publication is not intended as an investment advice. Nothing contrary to the above, no action has been, or will be, taken in Israel that would permit an offering of the product(s) mentioned in this document or a distribution of this document to the public in Israel. In particular, this document has not been reviewed or approved by the Israeli Securities Authority. The product(s) mentioned in this document is/are being offered to a limited number of sophisticated investors who qualify as one of the investors listed in the first supplement to the Israeli Securities Law, 5728-1968. This document may not be reproduced or used for any other purpose, nor be furnished to any other person other than those to whom copies have been sent. Anyone who purchases the product(s) mentioned herein shall do so for its own benefit and for its own account and not with the aim or intention of distributing or offering the product(s) to other parties. Anyone who purchases the product(s) shall do so in accordance with its own understanding and discretion and after it has received any relevant financial, legal, business, tax or other advice or opinion required by it in connection with such purchase(s). The word "advice" and/or any of its equivalent terms shall be read and construed in conjunction with the definition of the term "investment marketing" as defined under the Israeli Regulation of Investment Advice, Investment Marketing and Portfolio Management Law. The Swiss laws and regulations require a number of mandatory disclosures to be made in independent financial research reports or recommendations. Pursuant to the Swiss Financial Market Infrastructure Act and the Financial Market Infrastructure Ordinance-FINMA, banks must disclose the percentage of voting rights they hold in companies being researched, if these holdings are equal to or exceed the statutory thresholds. In addition, the Directives on the Independence of Financial Research, issued by the Swiss Bankers Association, mandate a number of disclosures, including the disclosure of potential conflicts of interest, the participation within previous 12 months in any securities issues on behalf of the company being researched, as well as the fact that remuneration paid to the financial analysts is based generally upon the performance of (i) the new issues department or investment banking; or (ii) securities trading performance (including proprietary trading) or sales. **Italy**: This publication is not intended to constitute a public offer under Italian law. It is distributed only for information purposes to clients of UBS Europe SE, Succursale Italia, with place of business at Via del Vecchio Politecnico, 3-20121 Milano. UBS Europe SE, Succursale Italia is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the Bank of Italy (Banca d'Italia) and the Italian Financial Markets Supervisory Authority (CONSOB - Commissione Nazionale per le Società e la Borsa), to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Japan**: This report is solely distributed in Japan by UBS SuMi TRUST Wealth Management Co., Ltd, Financial Instruments Dealer, Director-General of Kanto Local Finance Bureau (Kinsho) No. 3233, a member of the Japan Securities Dealers Association, Financial Futures Association of Japan, Investment Management Association of Japan. UBS SuMi TRUST Wealth Management Co., Ltd will not distribute or forward this report outside Japan. **Jersey**: UBS AG, Jersey Branch, is regulated and authorized by the Jersey Financial Services Commission for the conduct of banking, funds and investment business. Where services are provided from outside Jersey, they will not be covered by the Jersey regulatory regime. UBS AG, Jersey Branch is a branch of UBS AG a public company limited by shares, incorporated in Switzerland whose registered offices are at Aeschenvorstadt 1, CH-4051 Basel and Bahnhofstrasse 45, CH 8001 Zurich. UBS AG, Jersey Branch's principal place of business is 1, IFC Jersey, St Helier, Jersey, JE2 3BX. **Luxembourg**: This publication is not intended to constitute a public offer under Luxembourg law. It is distributed only for information purposes to clients of UBS Europe SE, Luxembourg Branch ("UBS Luxembourg"), R.C.S. Luxembourg n° B209123, with registered office at 33A, Avenue J. F. Kennedy, L-1855 Luxembourg. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea (HRB n° 107046), with registered office at Bockenheimer Landstrasse 2-4, D-60306 Frankfurt am Main, Germany, duly authorized by the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin") and subject to the joint prudential supervision of BaFin, the European Central Bank and the central bank of Germany (Deutsche Bundesbank). UBS Luxembourg is furthermore supervised by the Luxembourg prudential supervisory authority (Commission de Surveillance du Secteur Financier), in its role as host member state authority. This publication has not been submitted for approval to any public supervisory authority. **Malaysia**: This communication and any offering material term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or derivatives products). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or derivatives products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Mexico**: This information is distributed by UBS Asesores México, S.A. de C.V. ("UBS Asesores"), an affiliate of UBS Switzerland AG, incorporated as a non-independent investment advisor under the Mexican regulation due to the relation with a Foreign Bank. UBS Asesores is registered under number 30060-001-(14115)-21/06/2016 and subject to the supervision of the Mexican Banking and Securities Commission ("CNBV") exclusively regarding the rendering of (i) portfolio management services, (ii) securities investment advisory services, analysis and issuance of individual investment recommendations, and (iii) anti-money laundering and terrorism financing matters. This UBS publication or any material related thereto is addressed only to Sophisticated or Institutional Investors located in Mexico. Research reports only reflect the views of the analysts responsible for the report. The compensation of the analyst(s) who prepared this report is determined exclusively by research management and senior management of any entity of UBS Group to which such analyst(s) render(s) services. **Monaco**: This document is not intended to constitute a public offering or a comparable solicitation under the Principality of Monaco laws, but might be made available for information purposes to clients of UBS (Monaco) S.A., a regulated bank having its registered office at 2 avenue de Grande Bretagne 98000 Monaco operating under a banking license granted by the "Autorité de Contrôle Prudentiel et de Résolution" (ACPR) and the Monegasque government which authorizes the provision of banking services in Monaco. UBS (Monaco) S.A. is also licensed by the "Commission de Contrôle des Activités Financières" (CCAF) to provide investment services in Monaco. The latter has not approved this publication. **Philippines:** This communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or derivatives products). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or derivatives products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Portugal**: UBS Switzerland AG is not licensed to conduct banking and financial activities in Portugal nor is UBS Switzerland AG supervised by the portuguese regulators (Bank of Portugal "Banco de Portugal" and Portuguese Securities Exchange Commission "Comissão do Mercado de Valores Mobiliários"). **Qatar**: UBS Qatar LLC is licensed by the Qatar Financial Centre Authority and authorized by the QFC Regulatory Authority, with QFC no. 01169, and has its registered office at Building No. 36, 4th Floor, Abdulla Bin Thani Street, Doha Design District 1, Msheireb Downtown Doha, State of Qatar. This material is strictly intended for Eligible Counterparties and/or Business Customers only as classified under the QFCRA's Customer and Investor Protection Rules 2019. No other person should act upon this material. **Russia**: This document or information contained therein is for information purposes only and constitutes neither a public nor a private offering, is not an invitation to make offers, to sell, exchange or otherwise transfer any financial instruments in the Russian Federation to or for the benefit of any Russian person or entity and does not constitute an advertisement or offering of securities in the Russian Federation within the meaning of Russian securities laws. The information contained herein is not an "individual investment recommendation" as defined in Federal Law of 22 April 1996 No 39-FZ "On Securities Market" (as amended) and the financial instruments and operations specified herein may not be suitable for your investment profile or your investment goals or expectations. The determination of whether or not such financial instruments and operations are in your interests or are suitable for your investment goals, investment horizon or the acceptable risk level is your responsibility. We assume no liability for any losses connected with making any such operations or investing into any such financial instruments and we do not recommend to use such information as the only source of information for making an investment decision. **Saudi Arabia**: This material is for marketing and information purposes by UBS Saudi Arabia only. It has been distributed by UBS Saudi Arabia or reviewed and approved by UBS Saudi Arabia if it was issued by UBS Switzerland AG, by UBS AG, or any of its subsidiaries or branches. UBS Switzerland AG is a public company limited by shares, incorporated in Switzerland with its registered offices at Bahnhofstrasse 45, CH-8001 Zurich. UBS Saudi Arabia (a subsidiary of UBS AG) is a Saudi closed joint stock company incorporated in the Kingdom of Saudi Arabia with a paid capital of 110,000,000 Saudi Riyals under commercial register number 1010257812 having its registered office at Laysen Valley, P.O. Box 75724, Riyadh 11588, Kingdom of Saudi Arabia. UBS Saudi Arabia is authorized and regulated by the Capital Market Authority to conduct securities business under license number 08113-37. This document is distributed only under such circumstances as may be permitted by

applicable laws and regulations. The information and – if any - opinions contained in this document have been compiled or arrived at based upon information obtained from sources believed to be reliable and in good faith, but is not guaranteed as being accurate, nor is it a complete statement or summary of the securities, markets or developments referred to in the document. All such information and opinions are subject to changes without notice. Past performance of investments (whether simulated or actual) is not necessarily an indicator of future results. Forward looking data is based on estimates that are subject to changes. Should the currency of a financial product or service not match your reference currency, performance may rise or fall due to currency fluctuations. UBS Switzerland AG and / or other members of the UBS Group may have a position in and may make a purchase and/or sale of any of the securities or other financial instruments mentioned in this document While statements that constitute "forward-looking statements", including, but not limited to, statements relating to our future business development represent UBS judgments and future expectations concerning the development of our business, a number of risks, uncertainties and other important factors could cause actual developments and results to differ materially from our expectations. This document may not be forwarded, reproduced, redistributed or republished for any purpose without the written permission of UBS Saudi Arabia. Source for all data and charts (if not indicated otherwise) is UBS Switzerland AG. **Singapore**: Where applicable, this material is distributed in Singapore by UBS AG, Singapore Branch, which is licensed by the Monetary Authority of Singapore under the Banking Act 1970 to carry on banking business. UBS AG is incorporated in Switzerland with limited liability. UBS AG has a branch registered in Singapore (UEN S98FC5560C). This material has been prepared and issued for distribution in Singapore to institutional investors, accredited investors and expert investors (each as defined under the Financial Advisers Regulations (the "FAR")) only. By virtue of your status as an institutional investor, accredited investor, or expert investor, UBS AG is exempted from complying with certain requirements under the Financial Advisers Act 2001 (the "FAA"), the FAR and the relevant Notices and Guidelines issued thereunder, in respect of any financial advisory service which UBS AG may provide to you. These include exemptions from complying with: Section 34 of the FAA (pursuant to Regulation 33(1) of the FAR); Section 36 of the FAA (pursuant to Regulation 34(1) of the FAR); and Section 45 of the FAA (pursuant to Regulation 35(1) of the FAR). Singapore recipients and clients of UBS AG, Singapore Branch should contact UBS AG, Singapore Branch for any matters arising from, or in connection with, this material. Where applicable, this communication and any offering material term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved or endorsed by any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or products or futures contracts). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Spain:** This report is distributed in Spain by UBS AG, Sucursal en España, authorized under number 1460 in the Register by the Banco de España. **Sweden**: This publication is not intended to constitute a public offer under Swedish law. It is distributed only for information purposes to clients of UBS Europe SE, Sweden Bankfilial, with place of business at Regeringsgatan 38, 11153 Stockholm, Sweden, registered with the Swedish Companies Registration Office under Reg. No 516406-1011. UBS Europe SE, Sweden Bankfilial is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the Swedish supervisory authority (Finansinspektionen), to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Taiwan**: This material is provided by UBS AG, Taipei Branch in accordance with laws of Taiwan, in agreement with or at the request of clients/prospects. **Thailand**: This communication and any offering material, term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication were done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or derivatives products). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or derivatives products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Türkiye**: The information in this document is not provided for the purpose of offering, marketing or sale of any capital market instrument or service in the Republic of Türkiye. Therefore, this document may not be considered as an offer made, or to be made, to residents of the Republic of Türkiye in the Republic of Türkiye. UBS Switzerland AG is not licensed by the Capital Markets Board of Türkiye (the CMB) under the provisions of the Capital Market Law (Law No. 6362). Accordingly, neither this document nor any other offering material related to the instrument/service may be utilized in connection with providing any capital market services to persons within the Republic of Türkiye without the prior approval of the CMB. However, according to article 15 (d) (ii) of the Decree No. 32 residents of the Republic of Türkiye are allowed to purchase or sell the financial instruments traded in financial markets outside of the Republic of Türkiye. Further to this, pursuant to article 9 of the Communiqué on Principles Regarding Investment Services, Activities and Ancillary Services No. III-37.1, investment services provided abroad to residents of the Republic of Türkiye based on their own initiative are not restricted. **United Arab Emirates (UAE) / DIFC / Abu Dhabi**: UBS is not licensed in the UAE by the Central Bank of the UAE nor by the Emirates' Securities and Commodities Authority and does not undertake banking activities in the UAE. This document is provided for your information only and does not constitute financial advice. **DIFC**: UBS AG Dubai Branch is regulated by the DFSA in the DIFC. This material is strictly intended for Professional Clients and/or Market Counterparties only as classified under the DFSA rulebook. It should not be distributed to Retail Clients. The Investment Research is provided for information purposes only and is not a recommendation or offer to buy/sell/hold a particular investment. The investment research may be out of date. You should seek investment advice before acting on the basis of the Investment Research. **Abu Dhabi**: UBS AG Abu Dhabi Branch is licensed and regulated by the Financial Services Regulatory Authority ("FSRA") of the Abu Dhabi Global Market. This material is intended solely for professional clients or market counterparties, as defined in the rules of the FSRA. It is not directed at, nor intended for, retail clients or any person who does not meet the criteria of a professional client or market counterparty. **United Kingdom**: This document is issued by UBS Wealth Management, a division of UBS AG which is authorised and regulated by the Financial Market Supervisory Authority in Switzerland. In the United Kingdom, UBS AG is authorised by the Prudential Regulation Authority and is subject to regulation by the Financial Conduct Authority and limited regulation by the Prudential Regulation Authority. Details about the extent of regulation by the Prudential Regulation Authority are available from us on request. A member of the London Stock Exchange. This publication is distributed to retail clients of UBS Wealth Management. **Ukraine**: UBS is a premier global financial services firm offering wealth management services to individual, corporate and institutional investors. UBS is established in Switzerland and operates under Swiss law and in over 50 countries and from all major financial centers. UBS is not registered and licensed as a bank/financial institution under Ukrainian legislation and does not provide banking and other financial services in Ukraine. UBS has not made, and will not make, any offer of the mentioned products to the public in Ukraine. No action has been taken to authorize an offer of the mentioned products to the public in Ukraine and the distribution of this document shall not constitute financial services for the purposes of the Law of Ukraine "On Financial Services and Financial Companies" dated 14 December 2021. Any offer of the mentioned products shall not constitute an investment advice, public offer, circulation, transfer, safekeeping, holding or custody of securities in the territory of Ukraine. Accordingly, nothing in this document or any other document, information or communication related to the mentioned products shall be interpreted as containing an offer, a public offer or invitation to offer or to a public offer, or solicitation of securities in the territory of Ukraine or investment advice under Ukrainian law. Electronic communication must not be considered as an offer to enter into an electronic agreement or other electronic instrument within the meaning of the Law of Ukraine "On Electronic Commerce" dated 3 September 2015. This document is strictly for private use by its holder and may not be passed on to third parties or otherwise publicly distributed. **USA**: Distributed to US persons only by UBS Financial Services Inc. or UBS Securities LLC, subsidiaries of UBS AG. UBS Switzerland AG, UBS Europe SE, UBS Bank, S.A., UBS BB Corretora de Câmbio, Títulos e Valores Mobiliários S.A., UBS Asesores México, S.A. de C.V., UBS SuMi TRUST Wealth Management Co., Ltd., UBS Wealth Management Israel Ltd. and UBS Menkul Degerler AS are affiliates of UBS AG. **UBS Financial Services Inc. accepts responsibility for the content of a report prepared by a non-US affiliate when it distributes reports to US persons. All transactions by a US person in the securities mentioned in this report should be effected through a US-registered broker dealer affiliated with UBS, and not through a non-US affiliate. The contents of this report have not been and will not be approved by any securities or investment authority in the United States or elsewhere. UBS Financial Services Inc. is not acting as a municipal advisor to any municipal entity or obligated person within the meaning of Section 15B of the Securities Exchange Act (the "Municipal Advisor Rule") and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of the Municipal Advisor Rule.** For information on the ways in which UBS Securities LLC manages conflicts and maintains independence of its UBS Global Research product; historical performance information; certain additional disclosures concerning UBS Global Research recommendations; and terms and conditions for certain third party data used in research report, please visit https://www.ubs.com/disclosures.

© UBS 2026. The key symbol and UBS are among the registered and unregistered trademarks of UBS. All rights reserved.